كلية علوم الحاسب والمعلومات
قسم تقنية المعلومات

King Saud University College of Computer and Information Sciences

Department of Information Technology

IT497: Graduation Project -2- Report

2nd Semester 1447 H

Munir

Product Release - 2

Prepared by

NAME	ID
Fajer Alamro	444200800
Aljawharah Alsubaie	444201140
Rama Alomair	444200662
Mashael Alammar	444200786

Supervised by: Dr. Nora Alhammad

Second Semester 1447 H Spring 2026

Table of Contents

List of Tables

List of Figures

Munir

Fajer Alamro [1], Aljawharah Alsubaie [2], Rama Alomair [3] and Mashael Alammar [4]

[1] Information Technology Department, College of Computer and Information Sciences, King Saud University, Riyadh, Saudi Arabia; 444200800@student.ksu.ed.sa

[2] Information Technology Department, College of Computer and Information Sciences, King Saud University, Riyadh, Saudi Arabia; 444201140@student.ksu.ed.sa

[3] Information Technology Department, College of Computer and Information Sciences, King Saud University, Riyadh, Saudi Arabia; 444200662@student.ksu.ed.sa

[4] Information Technology Department, College of Computer and Information Sciences, King Saud University, Riyadh, Saudi Arabia; 444200786@student.ksu.ed.sa

Abstract (English):

Munir is an AI-powered assistive system developed to support blind and visually impaired individuals in safely navigating their environment and accessing visual information independently. With the increasing dependence on visual cues in daily life, there is a strong need for intelligent assistive technologies that enhance autonomy and personal safety. Munir consists of smart glasses integrated with a mobile application developed using Flutter. The system employs multiple specialized technologies to perform its functions: machine learning models are used for face recognition and currency identification, an external API is integrated for object detection, scene description, and Optical Character Recognition for reading printed text, and GPS-based services are implemented to enable an emergency SOS feature for real-time location sharing. Audio feedback is provided to communicate results to users in an accessible manner. The mobile application acts as the main interface for managing system interactions and user preferences. The system supports both Arabic and English languages to ensure accessibility for a wider range of users. The system was evaluated based on identification accuracy, responsiveness, usability, and user feedback. Evaluation results demonstrate high identification accuracy, fast responsiveness, and strong user satisfaction. Findings indicate that Munir significantly improves users' independence, situational awareness, and sense of security, demonstrating that integrated wearable and AI technologies can effectively address real-world accessibility challenges for visually impaired individuals.

Abstract (Arabic):

منير هو نظام مساعد ذكي قائم على تقنيات الذكاء الاصطناعي، طُوِّر لدعم الأشخاص المكفوفين وضعاف البصر في التنقل الآمن والوصول إلى المعلومات المرئية بصورة مستقلة. ويتكوّن النظام من نظارة ذكية متصلة بتطبيق يعمل على هاتف محمول، وقد طُوِّر هذا التطبيق باستخدام إطار العمل فلتر، ويُعد الواجهة الرئيسة للتحكم في النظام وإدارة إعداداته، مع دعم اللغتين العربية والإنجليزية. ويعتمد النظام على مجموعة من التقنيات المتقدمة، تشمل نماذج تعلم الآلة للتعرّف على الوجوه وتمييز العملات، وواجهة برمجة تطبيقات خارجية للكشف عن العناصر وتحليل المشاهد، وتقنية التعرّف الضوئي على الحروف لقراءة النصوص المطبوعة، بالإضافة إلى خدمات تحديد المواقع العالمية لتفعيل ميزة الاستغاثة ومشاركة الموقع الجغرافي في الوقت الفعلي. كما يقدّم النظام المعلومات للمستخدم من خلال إرشادات صوتية واضحة وسهلة الوصول. وقد أظهرت نتائج التقييم ارتفاعًا في دقة الاكتشاف وسرعة الاستجابة وسهولة الاستخدام، إلى جانب رضا المستخدمين عن أداء النظام، مما يؤكد فاعلية منير في تعزيز استقلالية الأشخاص ذوي الإعاقة البصرية، وزيادة وعيهم بما يحيط بهم، وتعزيز شعورهم بالأمان أثناء التنقل.

Keywords: Assistive Technology, Visual Impairment, Smart Glasses, Machine Learning, Accessibility, Independent Living, IOT, Wearable Technology.

CHAPTER 1: INTRODUCTION

Chapter 1: Introduction

In the age of digital transformation, assistive technologies play a critical role in enhancing independence, safety, and social inclusion for the blind and visually impaired individuals [1]. They represent a significant and vital group within society.

Despite this community's importance, many daily tasks remain challenging and often require constant assistance. Most current solutions are fragmented and limited in scope [2], which leaves major accessibility gaps and fails to address the broader needs of the blind and visually impaired users. These limitations highlight the urgent need for an integrated, comprehensive approach.

To bridge this gap, we present **Munir**, a next-generation assistive solution in the form of smart glasses paired with a mobile application. Powered by artificial intelligence, **Munir** enables intuitive audio-based interaction with the surrounding environment by combining text reading, face recognition, color detection, currency identification, and emergency support, offering a reliable, inclusive, and user-centered experience that prioritizes autonomy and security.

This document provides a comprehensive overview of the project by outlining the core problem and the challenges faced by the blind and visually impaired individuals, along with the proposed solution [Chapter 1]. It then presents the background [Chapter 2] and the technical foundation established through a literature review [Chapter 3]. This is followed by a detailed system design and development [Chapter 4], covering the methodology, system requirements, system design, data design, interface design, and implementation. The document continues with system evaluation [Chapter 5], including experimental results, user acceptance testing, quality attributes (NFR testing), and discussion. It concludes with conclusions and future work [Chapter 6], acknowledgements [Chapter 7], references [Chapter 8], and supporting appendices [Chapter 9].

1.1 Problem

In this section, we present the problem by outlining the key obstacles faced by blind and visually impaired individuals that our project seeks to address.

Individuals with visual impairments continue to face critical challenges in daily life, including reading text, recognizing faces, identifying currency and detecting colors. According to the 2022 Population and Housing Census, the number of individuals with visual impairments in Saudi Arabia reached 181,728, representing 0.56% of the population [3]. Globally, the World Health Organization, at least 2.2 billion people worldwide have a vision impairment [4]. These figures emphasize the magnitude of the issue and the urgent need for more inclusive technological interventions.

Despite the availability of assistive technologies, many remain fragmented and functionally limited. Devices often address only a single capability without offering comprehensive, integrated experience. This lack of integration not only reduces usability but also increases costs, as users must rely on multiple devices or subscriptions.

An examination of current assistive technologies shows that important needs remain unmet. For instance, OrCam MyEye [5] offers text reading and product identification but remains prohibitively expensive for many users and lacks integrated features such as comprehensive currency identification, emergency support systems, and customizable reminders within a single affordable platform. Similarly, Envision Glasses [6], while tailored for accessibility, still do not offer critical features such as an integrated emergency alert system capable of sharing location data with trusted contacts, or a built-in reminders and notification system to assist with daily task management. These gaps demonstrate how existing solutions fail to fully meet the complex and diverse needs of blind and visually impaired individuals, particularly in areas of safety, independence, and daily life organization.

1.2 Solution

In this section, we present our proposed solution, an integrated assistive system named **Munir**, designed to address the previously discussed challenges faced by blind and visually impaired individuals.

Munir is an assistive solution consisting of smart glasses integrated with a mobile application, designed specifically to empower individuals who are blind and visually impaired. The solution enables non-visual interaction with the surrounding environment through AI-powered, audio-based feedback, offering practical support in everyday life.

The system integrates multiple artificial intelligence techniques such as Optical Character Recognition (OCR) for reading text, Face Recognition for identifying familiar individuals, and Text-to-Speech (TTS) for delivering audio feedback. These capabilities are accessible through smart glasses integration, allowing users to interact with the system hands-free via voice commands. The glasses' built-in camera and microphone enable real-time capture and voice control, while the mobile application processes the AI functions and delivers audio feedback, supporting features like object detection, scene description and currency identification. Complementing this hands-free interaction, the mobile application serves as a control hub, enabling users or caregivers to manage face recognition profiles, configure reminders, and set emergency contacts. In critical situations, the system can also send GPS-based alerts to pre-saved contacts.

To address the challenges associated with OrCam MyEye [5], **Munir** focuses on accessibility and independence. Likewise, unlike Envision Glasses [6], which offer limited capabilities, **Munir** extends support through its emergency alerts, and caregiver integration via the mobile app.

The impact of **Munir** extends to both local and global communities. Locally, the system addresses the needs of visually impaired individuals in Saudi Arabia by supporting Saudi currency and aligning with accessibility goals outlined in Vision 2030, contributing to greater inclusion and independence. Globally, **Munir** addresses a universal accessibility challenge by providing an English-based, scalable assistive solution that promotes accessibility-first design principles and enhances quality of life for visually impaired users worldwide.

By consolidating multiple assistive technologies into one integrated platform, **Munir** delivers a reliable, inclusive, and forward-looking solution that enhances safety, independence, and quality of life in ways current market solutions have yet to achieve.

1.3 Objectives

In this section outlines the product, project, and learning objectives associated with the development of the **Munir**. These objectives define the goals and intended outcomes to be achieved upon the project's completion.

- **Product (customer focus-value):**

  Recognizing the challenges blind and visually impaired users face in daily tasks, **Munir** is designed to provide the following key features:

  - log in, log out, and register as a new user.
  - Text reading.
  - Face recognition for social interaction.
  - Object detection.
  - Scene description
  - Currency identification for Saudi Riyal banknote denominations.
  - Customizable reminders & notifications
  - Emergency alerts with GPS location sharing.

- **Project (solution focus-plan):**

  To develop **Munir**, we will utilize specific tools and follow a structured process:

  Steps:

  - Domain analysis to understand challenges faced by blind and visually impaired individuals.
  - Requirements engineering to specify system features (OCR, face recognition, reminders, etc.).
  - Research state-of-the-art methods for text recognition, face recognition, Currency identification and scene description.
  - Design the user interfaces of the mobile application.
  - Select the appropriate IDEs and programming languages.
  - Train AI models for face recognition and currency identification.
  - Test and validate model accuracy and performance.
  - Develop the Android mobile application and configure Firebase services.

- Integrate the AI models and smart glasses with the application.
- Conduct system testing, usability evaluation, and refinement.

Tools & Programming Languages:

- Python.
- Flutter.
- Visual Studio Code.
- Android Studio.
- Firebase.
- Git & GitHub.
- Jira (project management).
- Figma & Canva (UI/UX design).
- Google Colab.

- **Learning (student focus):**

By completing the **Munir** project, we will gain new skills and knowledge, including:

- Developing Android mobile apps with Firebase integration.
- Integrating hardware and software for wearable devices.
- Performing data preprocessing, feature extraction, model training, and testing for AI-powered assistive systems.
- Applying machine learning and computer vision techniques for tasks like face recognition, currency identification and scene description.
- Designing and developing mobile applications using Flutter, with a focus on creating accessible and user-friendly interfaces for blind and visually impaired users.
- Understanding the complete software development lifecycle (requirements engineering, designing, developing, testing, deploying).
- Enhancing our skills in project management and problem-solving within real-world technical projects.

These objectives guide the project's direction, ensuring both functionality and team growth. In the next section, we will define the scope of the **Munir**, outlining its boundaries.

1.4 Scope

In this section outlines the scope of the **Munir** application, The scope defines the boundaries of the project and highlights the intended functionality to be delivered upon completion.

Munir will be developed as an Android mobile application integrated with smart glasses to support blind and visually impaired users. The application will include features such as text reading, face recognition, object detection, scene description, currency identification, reminders and notifications and emergency alerts with GPS location sharing.

AI models for face recognition will be trained using publicly available datasets [7], enabling the system to accurately detect and recognize individuals in real time. Similarly, AI models for currency identification will be developed using publicly available datasets [8], focusing exclusively on Saudi Riyal paper banknotes (5, 10, 50, 100, 200, and 500 SR), excluding metallic coins from the currency identification scope. the system to distinguish between different Saudi banknote denominations with high accuracy and assist users in handling financial tasks with confidence. Text-to-Speech (TTS) will be developed through an API that provides a complete pipeline for speech synthesis, managing authentication, processing structured requests, retrieving audio securely, and ensuring synchronized playback for natural and reliable feedback. Object detection and scene description will be implemented using a cloud-based vision API, enabling the system to identify specific objects in front of the user and provide detailed spatial descriptions of surrounding environments. Object detection delivers concise, actionable identification of individual items, while scene description narrates the full environment including the positions of elements relative to the user, supporting independent navigation and spatial awareness. To ensure privacy, captured images are securely stored in the cloud with encryption, giving users protected and controlled access to their personal data. The mobile app will act as a hub for managing user profiles and settings, while the smart glasses capture visual and audio input through the integrated camera and microphone. The project will be limited to English and Arabic language support in its initial phase.

1.5 Vision Statement

Munir is designed **for** blind and visually impaired individuals **who** face challenges with reading, recognizing people, detecting object, describing scenes and currency, managing reminders, and ensuring personal safety. **The** AI-powered assistive smart glasses application provides independence through features such as text reading, face recognition, object detection, scene description, and currency identification, as well as built-in reminders and emergency alerts. **Unlike** existing alternatives such as OrCam, MyEye and Envision Glasses, **Munir** offers GPS-based emergency alerts, caregiver integration, and comprehensive support, delivering a solution that enhances autonomy, safety, and overall quality of life.

1.6 Development Approach

The development of **Munir** followed a systematic methodology to ensure a user-centered, reliable, and effective assistive system. The process began with understanding user needs through interviews and surveys with visually impaired individuals, identifying their main challenges and requirements. Based on these insights, system requirements were collected and analyzed, forming a clear foundation for the solution.

Next, the design phase involved creating system architecture, user interface prototypes, and defining the interactions between smart glasses and mobile applications. After the design was established, the focus shifted to AI model development. Relevant datasets were collected for face recognition and Currency identification. These datasets underwent preprocessing to ensure all images are clean, standardized, and properly formatted. Then, feature extraction was performed to prepare the data for model training, followed by training the AI models using three different algorithms. The models were subsequently evaluated to ensure accuracy, robustness, and reliability.

Once the AI models were trained, they were integrated into the mobile application, along with additional features such as reminders, emergency alerts, and API Integrations for OCR, object detection, and scene description. After completing the application features, the smart glasses were connected to the system and tested to ensure proper real-time interaction. Finally, the complete system was evaluated with real users in scenario-based experiments, ensuring accuracy, usability, and real-time performance, to verify that **Munir** met user needs and provided a seamless, hands-free assistive experience.

CHAPTER 2:
BACKGROUND

Chapter 2: Background

To fully understand the **Munir** project, it is essential to first explore the broader context of visual impairment challenges and assistive technologies. This chapter provides the necessary domain knowledge and theoretical background required for understanding the project, including key concepts in assistive technology, artificial intelligence techniques used in the solution, and the hardware components that enable the system.

2.1 Visual Impairment and Assistive Technologies

In this section, we examine the global and regional landscape of visual impairment and introduce the role of assistive technologies in addressing accessibility challenges.

The World Health Organization defines visual impairment as an eye condition that affects the visual system and its functions [9]. Visual impairment manifests in various forms, including complete blindness (total vision loss), low vision (significantly reduced acuity), peripheral vision loss (tunnel vision), central vision loss (affecting reading and facial recognition), night blindness (difficulty in low-light conditions), and blurred vision. Each type presents unique challenges requiring tailored assistive solutions.

Globally, over 2.2 billion people are affected, including 1 billion cases that are preventable or unaddressed [9]. In Saudi Arabia, recent studies (2024 –2025) report prevalence rates of up to 9.3%, with rural areas showing higher levels ranging from 13.9% to 32.1% [10]. These challenges highlight the growing importance of assistive technologies aimed at improving independence and quality of life for individuals with visual impairments.

Assistive technology refers to tools, devices, or software designed to improve the functional abilities of individuals with disabilities [11]. For blind and visually impaired individuals, such technologies support:

- **Navigation and mobility:** safe movement in different environments
- **Information access:** converting visual content into accessible formats
- **Communication:** enabling interaction with people and digital systems
- **Independence:** reducing reliance on others for daily tasks

Assistive solutions for blind and visually impaired have evolved from simple aids, such as white canes, to advanced devices powered by artificial intelligence and machine learning, reflecting their growing role in enhancing accessibility and independence [12].

2.2 Smart Glasses Technology for Assistive Applications

In this section, we introduce the concept of smart glasses and their relevance to assistive technology applications, followed by a detailed overview of the Omi Glass Dev Kit, which serves as the hardware foundation for the **Munir** project.

2.2.1 Overview of Smart Glasses

Smart glasses are wearable computing devices that combine traditional eyewear with digital technology. They typically incorporate components such as cameras, microphones, speakers, processors, and wireless connectivity. For assistive applications, smart glasses serve as an ideal platform because they provide hands-free operation while maintaining the user's natural head orientation for environmental scanning [13].

2.2.2 The Omi Glass Dev Kit

The **Munir** project leverages the Omi Glass Dev Kit as its hardware foundation [14]. This developer-friendly and open-source smart glasses platform is designed for lifelogging, AI-powered recall, and accessibility functions.

Key Features:

- Built on the XIAO ESP32S3 microcontroller, offering dual-core processing, Wi-Fi, Bluetooth, 8 MB PSRAM, and SD card support.
- Powered by six 150 mAh batteries, ensuring extended usage time.
- Equipped with an integrated camera and microphone for real-time environmental sensing.
- Fully open-source design, enabling customization at the hardware, firmware, and software levels.

- Developer support through Arduino IDE compatibility and 3D-printable hardware mounts.

The Omi Glass Dev Kit comes with pre-built, open-source firmware provided by the Omi open-source project, which handles low-level hardware operations such as camera capture and BLE communication. The open-source nature of the firmware allowed the **Munir** team to adapt and integrate it with the **Munir** system, enabling the device to send captured images and voice commands to the paired mobile application for processing.

2.3 Artificial Intelligence for Assistive Functions

In this section, we explore the key artificial intelligence technologies that power **Munir's** assistive features, including OCR, face recognition, text-to-speech, and computer vision.

2.3.1 Optical Character Recognition (OCR)

Optical Character Recognition (OCR) refers to the process of converting printed text into machine-readable digital data [15]. This technology plays a pivotal role in assistive solutions for blind and visually impaired individuals, as it enables access to materials such as books, menus, and signage by converting text into formats that can be rendered audibly through Text-to-Speech (TTS) systems. In this way, OCR directly supports greater independence in daily activities that rely on reading.

Contemporary OCR technologies leverage deep learning architectures, particularly convolutional neural networks (CNNs) and recurrent neural networks (RNNs), to achieve higher accuracy and robustness under diverse real-world conditions. Widely adopted implementations include **Tesseract OCR**, an open-source solution for text extraction, and commercial cloud-based services such as **Google Vision API** and **Microsoft Azure OCR**, both of which provide multilingual recognition, layout analysis, and scalability for large-scale applications [16]. [17] [18].

For **Munir**, OCR serves as a foundational component that bridges visual text recognition with auditory feedback. Its integration ensures that users can seamlessly interpret textual information in their environment, thereby enhancing autonomy and reducing reliance on external assistance.

2.3.2 Face Recognition Technology

Face recognition enables identification of individuals by analyzing facial features extracted from digital images [19]. The process typically involves three steps:

1. **Face detection:** locating the face within an image.
2. **Feature extraction:** generates unique numerical representations (embeddings) of the facial characteristics
3. **Matching/verification:** comparing the embedding with a stored database [20].

Common frameworks include OpenCV, Mediapipe, and FaceNet, which are widely used for embedding generation and recognition. For accessibility, face recognition provides crucial social support by helping users identify familiar individuals. However, challenges such as variable lighting, occlusion, and dataset bias must be addressed to ensure fairness and reliability [21].

2.3.3 Text-to-Speech (TTS) Technology

Text-to-Speech (TTS) technology transforms written text into spoken words, allowing blind and visually impaired users to access information that would otherwise only be available in print or on screens. This makes it possible to listen to messages, or even labels scanned by a camera, rather than having to rely on someone else for assistance.

A TTS system generally works in two steps: **text analysis**, where the computer breaks down sentences and understands how they should be spoken, and **speech synthesis**, where the system generates the actual voice output.

Recent advances in neural TTS have made these systems sound much more natural and human-like. Modern models such as “Tacotron” and “WaveNet” can produce speech with intonation, rhythm, and even emotional tone [22]. This is important because it makes listening less tiring and more engaging for the user. Many systems also support multiple languages, which are essential for accessibility in diverse communities.

In the context of **Munir**, TTS enables the system to deliver real-time audio feedback to the user, conveying information about recognized text, recognized individuals, identified currency, or detected colors.

2.3.4 Computer Vision for Scene Description, Object Detection & Currency Identification

Computer vision is a core component of Munir, enabling the system to interpret visual information and support users in navigating daily life. One key capability is scene description, where the camera captures the surrounding environment and generates a natural language description of what is present in detail, such as identifying people, furniture, or obstacles with their specific positions. This helps blind and visually impaired users build awareness of their surroundings without relying on sighted assistance [23].

Another important feature is object detection, which enables the system to locate and identify specific items within the camera's field of view. Whether recognizing a cup on a table, a door handle, or a piece of clothing, object detection provides real-time feedback that supports independence in everyday tasks [24].

Handling money can also present challenges when banknotes feel like the touch. Through currency identification, the camera scans the banknote and identifies its value, delivering the result to the user through audio feedback. This reduces uncertainty and promotes financial independence in everyday situations [25].

2.4 Android Mobile Application Development for Accessibility

In this section, we outline the key design principles and standards that guide the development of accessible mobile applications for blind and visually impaired users.

Key considerations include:

- **Audio-first** interface design to ensure all functionality is accessible through audio feedback delivered via the mobile device's speaker or Bluetooth-connected audio devices, enabling navigation without visual input in compliance with Android TalkBack requirements.
- **Voice command** integration to enable hands-free operation and reduce reliance on touch-based navigation.
- **Android TalkBack** compatibility with properly labelled UI elements and content descriptions, to enable seamless navigation through Android's native screen reader.

- **Haptic feedback** to provide tactile confirmation of button presses, successful actions, and critical alerts. [26].
- **Scalable text and high contrast color** schemes meeting WCAG 2.1 AA contrast ratios to improve readability for users with partial vision in different lighting conditions.

These principles ensure that the application is not only functional but also usable for a wide spectrum of blind and visually impaired individuals.

2.5 GPS and Emergency Alert Systems

In this section, we discuss how GPS technology and emergency alert systems contribute to safety and independence for blind and visually impaired users.

Global Positioning System (GPS) technology enhances safety and independence for blind and visually impaired users through navigation support and real-time location awareness provided by Android's Location Services API [27]. Emergency alert systems integrate GPS location tracking with messaging technologies such as SMS or Firebase Cloud Messaging, converting coordinates into navigable links via mapping services like Google Maps to enable rapid communication with caregivers during critical situations

2.6 Privacy and Data Security in AI-Powered Assistive Technology

In this section, we address the critical privacy and security considerations associated with AI-powered assistive devices that process sensitive personal information.

AI-powered assistive devices that process sensitive personal data such as facial images, and location information raise significant privacy concerns regarding unauthorized access, surveillance, and data misuse [28]. To address these risks, industry best practices emphasize multi-layered security measures including data encryption during storage and transmission following National Institute of Standards and Technology (NIST) standards [29], secure authentication protocols, and protected cloud infrastructure to maintain system integrity [30].

Regulatory frameworks such as the General Data Protection Regulation (GDPR) establish legal requirements for lawful, fair, and transparent data processing, mandating explicit user consent and purpose limitation when handling biometric and location data [31]. For blind and visually impaired users who depend on assistive technology for daily independence, these security and privacy measures are essential for building trust and ensuring ethical, responsible AI implementation.

CHAPTER 3:
LITERATURE REVIEW

Chapter 3: Literature Review

In this chapter, we examine existing assistive technologies for blind and visually impaired individuals, analyzing their strengths and limitations to establish the context for **Munir's** development and highlight the gaps it addresses.

In recent years, assistive technologies for blind and visually impaired individuals have gained increasing attention due to their potential to enhance independence, safety, and social participation. Several solutions have been developed, including wearable devices and mobile applications. Current offerings vary significantly in their assistance approach from audio-based AI feedback that announces information through voice output to visual enhancement systems that magnify images for users with remaining sight. However, many remain fragmented, focusing on a single capability such as text reading or object recognition, while others create accessibility barriers through high costs or complex dependencies. This section provides an overview of existing products and highlights how **Munir** addresses gaps in current solutions.

3.1 Competitive Product Analysis

Munir competes with existing assistive technologies designed to improve daily functioning for blind and visually impaired users. While these solutions provide valuable support, most lack integration of multiple functionalities, autonomy, or regional optimization. [**Table 1**] provides a detailed comparison of these solutions with **Munir** across key features, processing architecture, cost, and functionality.

- **Envision Glasses**

Figure 1: Envision Glasses Logo

Envision Glasses Home Edition offers AI-powered capabilities including text reading (OCR), face recognition, scene description, and object detection [6]. The device operates using a hybrid processing model with audio feedback for all features through built-in speakers. At a price of approximately 2,499 USD (≈9,400 SAR), Envision Glasses represents a significant investment for users seeking advanced capabilities.

However, Envision Glasses has significant limitations compared to comprehensive assistive solutions. First, the device lacks integrated emergency alert functionality with GPS location sharing to notify family members during critical situations. Second, it provides no automated reminder systems for medication, appointments, or daily tasks with audio notifications. Most critically, the high purchase price of approximately 9,400 SAR places the device beyond the reach of many blind and visually impaired individuals who could benefit from assistive technology, creating a significant accessibility barrier for users seeking affordable solutions.

- **OrCam MyEye**

Figure 2: OrCam MyEye logo

OrCam MyEye 3 Pro offers text reading, face recognition, product and currency identification, and object detection with complete audio feedback [5]. This device operates entirely offline using onboard local AI processing with no internet connection or subscriptions required. The device magnetically attaches to any user's existing glasses frame. At a price of approximately $4,250 USD (≈16,000 SAR).

However, OrCam MyEye has significant limitations compared to comprehensive assistive solutions. First, the device is not integrated into dedicated smart glasses, requiring users to attach an external device to their existing eyewear rather than having a unified wearable solution. Second, OrCam lacks integrated emergency alert functionality with GPS location sharing to notify family members during critical situations. Third, it provides no automated reminder systems for medication, appointments, or daily tasks with audio notifications. Fourth, while the device supports international currencies, it does not specifically support Saudi Riyal (SAR) identification, limiting its utility for users in Saudi Arabia and the Gulf region. Most critically, the high purchase price of approximately 16,000 SAR places the device beyond the reach of many blinds and visually impaired individuals who could benefit from assistive technology, creating the most significant accessibility barrier among all competing solutions.

- **Microsoft Seeing AI**

Figure 3: Microsoft Seeing AI logo

Seeing AI is a free mobile application offering text reading (OCR), face recognition, object detection, scene description, and barcode scanning with complete audio feedback [32]. Available on both iOS and Android platforms with full mobile app integration, it removes financial barriers without any purchase cost. The application narrates the environment and recognizes objects through voice announcements. Currently, Seeing AI supports currency identification for US Dollars, Canadian Dollars, British Pounds, and Euros only, and does not support Saudi Riyal (SAR) recognition.

However, Seeing AI has significant limitations compared to comprehensive assistive solutions. First, the application is not integrated into dedicated smart glasses, preventing hands-free operation and requiring users to continuously hold their smartphone to access visual information. Second, the application lacks integrated emergency alert functionality with GPS location sharing to notify family members during critical situations. Third, it provides no automated reminder systems for medication, appointments, or daily tasks with audio notifications. Most critically, while Seeing AI excels in affordability, the lack of Saudi Riyal (SAR) support significantly limits its practical utility for users in Saudi Arabia who need to identify local currency for independent financial transactions.

- **Smart Vision Glasses (SHG)**

Figure 4: Smart Vision Glasses (SHG) logo

Smart Vision Glass (SVG), developed by SHG Technologies in India, is a dedicated smart glasses device with mobile app integration designed to provide assistive technology for individuals with visual impairments in the Indian market [33]. The lightweight tool mounted on a spectacle frame features Braille-coded keys and connects to an Android smartphone app to deliver four primary functions: Things Around You for object recognition, and reading for OCR text assistance, walking assistance for navigation support, and face recognition for familiar individuals through real-time voice interface. At approximately ₹50,400 INR (≈2,520 SAR), SVG positions itself as a cost-effective alternative in the assistive technology market targeting developing economies.

However, SVG has significant limitations compared to comprehensive assistive solutions. First, while the device offers basic tap-to-call emergency functionality, it lacks integrated GPS location sharing to automatically notify family members or emergency contacts with the user's real-time coordinates during critical situations, removing a vital safety layer particularly important for elderly users or those navigating unfamiliar environments. Second, SVG lacks automated reminder systems for medication schedules, appointments, or daily tasks with audio notifications, eliminating structured time-management support crucial for users managing chronic health conditions or complex daily routines. Third, the device lacks scene description capability, preventing users from receiving spatial narrations of their full surrounding environment through spoken feedback, a feature essential for independent navigation and orientation in unfamiliar spaces. Fourth, SVG provides no currency identification for Saudi Riyal (SAR), supporting only Indian Rupee and Kuwaiti Dinar denominations and thereby eliminating the ability for Saudi users to independently handle financial transactions in their local currency. Most critically, at approximately 2,520 SAR, SVG exceeds **Munir's** affordability threshold of less than 1,400 SAR by 80% while delivering 30% fewer features (7 functions versus **Munir's** 11 capabilities), creating a significant cost barrier for many visually impaired individuals who require maximum functionality within limited budgets.

- **NuEyes Smart Glasses**

Figure 5: NUEYES logo

NuEyes Pro 4 are dedicated smart glasses with mobile app integration designed to provide vision enhancement for individuals with low vision [34]. The lightweight wireless headset offers up to 12x magnification, multiple color and contrast modes, voice-controlled zoom, and OCR text-to-speech functionality with audio feedback. At approximately $5,995 USD (≈22,500 SAR), NuEyes Pro 4 represents one of the most expensive options among assistive smart glasses.

However, NuEyes has significant limitations compared to comprehensive assistive solutions. First, the device lacks face recognition systems for identifying familiar individuals through audio announcements, limiting social independence for blind and visually impaired users. Second, NuEyes provides no AI-powered object detection or scene description with audio feedback, preventing users from independently identifying specific objects or receiving spatial narrations of their surrounding environment. Third the device offers no currency identification capability, particularly for Saudi Riyal (SAR) or other regional currencies essential for independent financial transactions. Fourth, NuEyes lacks integrated emergency alert functionality with GPS location sharing to automatically notify family members during critical situations. Fifth, it provides no automated reminder systems for medication, appointments, or daily tasks with audio notifications. Most critically, the high purchase price of approximately 22,500 SAR places the device far beyond the affordability threshold of 1,400 SAR, creating the highest cost barrier among all competing assistive technologies and limiting accessibility for many blinds and visually impaired individuals who could benefit from such solutions.

As summarized in [Table 1], **Munir** addresses critical gaps in existing assistive technology solutions through its integrated approach, comprehensive feature set, and affordability.

Munir combines dedicated smart glasses with complete audio feedback for all AI features, delivering text reading, face recognition, object detection, scene description, Saudi Riyal currency identification, GPS-based emergency alerts, and automated reminders within a single hands-free system.

Among competing solutions, **SVG** provides object detection, text reading, face recognition, and basic tap-to-call emergency dialing, but lacks scene description, Saudi Riyal support, GPS-based location sharing, and automated reminders. **NuEyes** focuses exclusively on vision magnification and OCR, offering no face recognition, object detection, scene description, currency identification, or safety features despite its premium pricing. **OrCam MyEye** supports text reading, face recognition, and object detection but lacks scene description, emergency alerts, reminders, and dedicated SAR currency support. **Envision Glasses** offers the broadest feature set among competitors including scene description and object detection, yet still lacks GPS-based emergency alerts, automated reminders, and SAR currency identification. **Seeing AI** provides scene description, object detection, and face recognition at no cost, but requires continuous smartphone handling and offers no dedicated smart glasses, emergency alerts, reminders, or regional currency support.

With a projected cost of less than 1,400 SAR (≈$373 USD), **Munir** delivers these capabilities at dramatically lower costs: 16× lower than NuEyes (22,500 SAR), 11× lower than OrCam MyEye (16,000 SAR), 7× lower than Envision Glasses (9,400 SAR), and 1.8× lower than SVG (2,520 SAR). While Seeing AI is free, its lack of dedicated hardware and safety features limits its practical utility for users who require comprehensive, hands-free assistance.

These features position **Munir** as an affordable, comprehensive assistive technology platform combining dedicated smart glasses hardware, AI-powered audio feedback, regional currency support, and integrated safety features specifically designed for blind and visually impaired users.

Table 1: Competitor's Summary

Application / Feature	Envision Glasses	OrCam MyEye	Seeing AI	Smart Vision Glasses (SHG)	NuEyes	Munir (Proposed)
Dedicated Smart Glasses	✓			✓	✓	✓
Mobile Application	✓	✓	✓	✓	✓	✓
Audio Feedback	✓	✓	✓	✓	✓	✓
OCR (Text Reading)	✓	✓	✓	✓	✓	✓
Face Recognition	✓	✓	✓	✓		✓
Object Detection	✓	✓	✓			✓
Scene Description	✓		✓			✓
Saudi Riyal currency identification	✓					✓
Emergency Alerts (GPS)						✓
Reminders & Notifications						✓
Purchase Price (<1,400 SAR)			✓			✓

CHAPTER 4:
SYSTEM DESIGN AND DEVELOPMENT

Chapter 4: System Design and Development

This chapter presents the design and development of the **Munir** system, an intelligent assistive application designed to help blind and visually impaired individuals perform daily tasks more independently and effectively. It describes the methodology followed throughout the project, including requirements analysis, system design, data design, interface design, and implementation.

The project began by identifying the target users and studying their needs to ensure that the system would be accessible and user-friendly. To elicit requirements, interviews (see Appendix A) and questionnaires (see Appendix B) were conducted to understand users' expectations, daily challenges, and experiences with existing assistive technologies. The collected data provided valuable insights that guided the definition of both functional and non-functional requirements.

User interactions were modelled using a Use Case Diagram to illustrate the main system features and accessibility workflows. A product backlog was also developed to organize and prioritize the identified requirements and support iterative development.

Based on these requirements, the system architecture and component-level design were created to define the structure of the application and the relationships among its modules. The chapter further presents the data design, interface design, and implementation details, including the technologies and tools used to build the complete **Munir** system.

4.1 Methodology

In developing the **Munir** software, an Agile software development approach was adopted to ensure flexibility, continuous improvement, and effective collaboration throughout the development process. Agile is based on iterative development and continuous feedback, which enables adaptation to changing requirements. The project was divided into short development cycles known as sprints, which enabled incremental delivery of features and continuous refinement of the system. Regular reassessment of plans helped maintain the project timeline while accommodating evolving requirements, ultimately leading to the successful delivery of functional software releases.

To organize and manage the development process, we followed the Scrum framework, which provided a clear structure for roles, events, and artifacts. Within the team, the product owner was responsible for defining the product vision and managing the product backlog, ensuring that the most valuable features were prioritized. The Scrum master facilitated Scrum activities, supported the team in following Agile principles, and removed obstacles that could hinder progress. The development team worked collaboratively to design, implement, and test features, delivering a working product increment at the end of each sprint.

Throughout the project, we conducted Scrum events to guide our workflow and maintain alignment. At the beginning of each sprint, sprint planning sessions were held to select user stories from the product backlog and define sprint objectives. Daily stand-up meetings were conducted to synchronize team efforts, discuss progress, and identify challenges. At the end of each sprint, sprint review meetings were held to demonstrate completed features, gather feedback from the product owner, and discuss potential improvements. Additionally, sprint retrospectives allowed the team to reflect on their performance, identify areas for improvement, and implement adjustments to enhance future sprints.

The Scrum framework also introduced key artifacts that supported transparency and effective project tracking. The product backlog contained a prioritized list of user stories that guided the overall development process. The sprint backlog included the selected tasks for each sprint, while the increment represented the working version of the **Munir** system delivered at the end of every sprint. These artifacts helped ensure clarity, accountability, and continuous progress throughout the project lifecycle.

To support our Agile and Scrum practices, we utilized Jira [1] and GitHub [2] as essential development tools. Jira was used to manage the product backlog, plan sprints, track tasks, and visualize progress, helping the team stay organized and meet deadlines. GitHub served as the version control system, enabling collaborative code development, maintaining version history, and facilitating code reviews through pull requests. Automated testing was implemented using GitHub Actions to ensure code quality and reliability. Together, these tools enhanced communication, coordination, and efficiency, allowing the team to remain agile and focused on delivering a high-quality software solution.

4.2 System Requirements

4.2.2 System Users

In this section, we define the primary and secondary user groups for **Munir**, outlining their characteristics, technical expertise levels, and specific needs to ensure the system is accessible and usable for all stakeholders.

The primary users of **Munir** are blind and visually impaired individuals. They come from different educational backgrounds and age groups, and the product does not assume any specific academic level. Their technical expertise also varies; some users may have only beginner-level experience with smartphones, mainly limited to basic calling or texting, while others might be more familiar with accessibility features such as screen readers.

[1] https://student-team-sxuclzqg.atlassian.net/jira/software/projects/GP/summary

[2] https://github.com/aljawharah-alsubaie/2025_GP_1

The primary users of **Munir** are blind and visually impaired individuals. They come from different educational backgrounds and age groups, and the product does not assume any specific academic level. Their technical expertise also varies; some users may have only beginner-level experience with smartphones, mainly limited to basic calling or texting, while others might be more familiar with accessibility features such as screen readers. For this reason, **Munir** is designed to be simple, accessible, and intuitive, ensuring that users with limited technical knowledge can easily perform essential tasks such as reading text, recognizing faces, identifying colors and currencies, setting reminders, and sending emergency notifications.

The secondary users include caregivers, family members, and trusted contacts. They generally possess at least a basic educational level and have enough technical comfort to operate mobile applications. Their main role is to support the primary users when needed, for example, by managing reminders and to receive emergency notifications with the user's GPS location. While their technical expertise may vary, the system is designed so that no specialized technical skills are required, only general familiarity with smartphone functions and messaging applications.

Munir is designed to be user-friendly and fully accessible not only to blind and visually impaired individuals but also to their caregivers. This ensures inclusiveness for both groups, regardless of educational background or technical expertise. Users only need a basic understanding of smartphone operations, with no advanced technical skills required.

4.2.2 Requirements Elicitation and Analysis

In this section, we present the findings from our requirements elicitation process, which involved interviews and questionnaires with blind and visually impaired individuals to gather insights into their daily challenges, needs, and preferences for assistive technology.

During the requirements elicitation for the **Munir** project, blind and visually impaired individuals were the primary stakeholders, as they provided valuable insights based on their daily challenges and experiences with assistive technologies.

Information was mainly collected through interviews and questionnaires [Table 2]. By combining these two approaches, we ensured that **Munir's** design was guided by real-world user feedback and aligned with the actual needs of its target users.

Table 2: Requirements Elicitation Methods

Method	No. Of Question	No. Of Responses
Interview	10	5
Questionnaire	14	25

A) Interviews findings

After interviewing five participants, three of them were completely blind and two had extremely limited vision, with three females and two males, we found that all faced daily challenges that could be supported by assistive technology. Despite differences in vision and gender, participants shared similar experiences regarding difficulties with reading text, recognizing people, detecting colors and identifying currencies, navigating safely, and managing daily tasks independently.

Participants described several common difficulties, including reading printed documents or bills, distinguishing colors, performing daily tasks without assistance, and managing social interactions. While some had experience with screen readers, voice-based applications, or AI tools, these solutions were often insufficient, cumbersome, or limited in functionality, particularly for outdoor mobility or multi-step tasks. participants with extremely limited vision occasionally relied on zooming in on text or using digital magnifiers, but this method was slow, inconvenient, and at times inadequate, especially in low-light conditions. Color detection also posed challenges, affecting daily routines like selecting clothing or matching outfits.

The interviews highlighted user preferences for interaction with assistive technology. Participants valued clear voice feedback or vibration alerts, with some preferring voice, others vibration, and some a combination depending on the situation. This flexibility allows users to choose the method that best suits their environment and privacy needs.

One participant emphasized the usefulness of reminders, noting that spoken alerts for tasks, medications, or appointments help them stay organized and avoid forgetting important activities. Participants also stressed the importance of emergency support; in situations where they might get lost or have difficulty describing their location, being able to share their GPS position with trusted contacts would make it much easier to receive timely assistance.

Overall, the interviews showed that participants highly value independence, safety, and confidence in daily life. Their challenges and preferences align directly with assistive features such as text reading for documents and product labels, face recognition to recognize people, color detection to determine the color of clothing or objects, Currency identification to handle cash transactions independently, reminders and notifications for tasks, and emergency alerts with GPS. These features address everyday needs and provide a strong foundation for designing effective assistive solutions that support both blind and visually impaired users.

Participants also proposed additional capabilities aimed at improving safety and mobility. These included smart-glasses functions that detect obstacles in real time such as notifying users of a closed door or other barriers and step-by-step movement guidance to support independent navigation. These insights will inform future development of **Munir**, reinforcing its goal of enhancing both safety and independence for users. The full interview questions and answers can be found in **(Appendix A).**

B) Questionnaire findings

The questionnaire was distributed to 25 participants in collaboration with organizations supporting blind and visually impaired individuals, such as “Nuqtat Tahawol” [35] and “Ru’ya” [36], to gain insights into their demographics, daily challenges, use of assistive technologies, and openness to adopting smart glasses. The majority of respondents were female (96%) and most fell within the age group of 18 to 30 years (68%), followed by 16% aged 31 to 60, while smaller proportions were under 18 (16%). Regarding the type of impairment, low vision was the most common condition, reported by 72% of participants, followed by total blindness (24%), and Stargardt disease (4%).

When asked about their experience with assistive technologies, a significant majority (68%) indicated that they had never used such tools. This low adoption rate can be attributed to several factors: the high cost of existing devices making them financially inaccessible, limited awareness and availability of assistive technologies in the region, complexity of use requiring technical expertise. The remainder had limited exposure through screen readers, magnifiers, or similar solutions. This reflects a clear gap in the adoption of existing technologies and highlights the potential for more accessible and user-friendly alternatives. Moreover, participants identified several daily activities that were most affected by their condition, including reading printed text (80%), recognizing faces (76%), distinguishing currency (52%), and managing reminders or tasks (28%). In addition, both requesting assistance during emergencies and distinguishing colors were reported by (24%) of the participants as particularly challenging activities. A few respondents also mentioned that they encounter significant difficulties in daily practices such as cooking and even in leisure activities like playing football, which further demonstrates the wide impact of visual impairment on multiple aspects of life.

The frequency of these difficulties varied among participants: 36% reported facing challenges on a moderate basis, 24% occasionally, and 20% frequently. Furthermore, the perceived impact of visual impairment on independence and quality of life was rated as medium by 44% of participants, while others reported either mild or severe effects.

To cope with these challenges, 56% relied on assistive technology and another 40% sought help from others, whereas smaller groups reported avoiding tasks (8%). These findings indicate that while technology plays an important role, social support remains a critical factor in managing daily life.

An overwhelming majority expressed strong openness to adopting smart glasses integrated with a mobile application and headphones. Specifically, (76%) stated they would be likely or very likely to use such a device if it included the required features. Additionally, (52%) strongly believed that connecting the glasses to a mobile application would significantly improve their ability to manage daily tasks. When rating potential features, reading printed texts, currency identification, and emergency alerts were consistently identified as the most valuable.

Participants also provided valuable suggestions for additional functionalities. These included GPS navigation, health sensors, a "Find my device" option, instant translation, and enhanced text recognition. Many emphasized the importance of simplicity, affordability, and customization to accommodate different levels of digital literacy. Educational applications were highlighted, particularly for students who require accessible tools to participate effectively in learning environments.

These insights provide valuable guidance for the future development of the **Munir**, highlighting the critical role of both functionality and privacy in designing the smart glasses and mobile application. The full questionnaire can be found in **(Appendix B).**

4.2.3 User Interaction

In this section, the use case diagram for the **Munir** System is presented in [**Figure 6**], illustrating user interactions with core features. Several important points to consider are outlined below:

- **Authentication-Dependent Password Management*:***

The **Change Password** and **Reset Password** use cases are exclusively accessible to users who registered via email-based authentication. Users authenticated through Google Auth Service cannot access this functionality, as their credentials are managed entirely within Google's ecosystem.

- **Interpretation of Management Use Cases:**

In this diagram, use cases prefixed with **Manage** denote comprehensive data management functionality. Specifically, they encompass operations for **adding new entries, updating existing records**, and **deleting existing entries**. Accordingly, use cases such as Manage Registered Faces**,** Manage Reminders**,** and Manage Emergency Contacts provide users with full control over their respective datasets.

- **Device-Specific Use Case Accessibility:**

While the **Munir** System operates across both the mobile application and the smart glasses, not all use cases are accessible from both devices. Core AI-powered features — including **Read Text**, **Identify Currency**, **Recognize Person**, **Detect Object**, and **Describe Scene** — as well as Send Emergency Alerts, are accessible through both the smart glasses and the mobile application, allowing users to interact with their environment hands-free via the glasses or directly through the app.

Figure 6: User Interactions with Munir System

4.2.4 Roadmap and Product Backlog

A. Product Roadmap

A project roadmap presents a high-level outline of the product's vision, goals, and planned progress over time. It serves as a strategic guide that illustrates when major features or milestones are expected to be delivered. For our software product, the roadmap is divided into two main releases and six sprints, offering a clear timeline for development, as shown in [Figure 7].

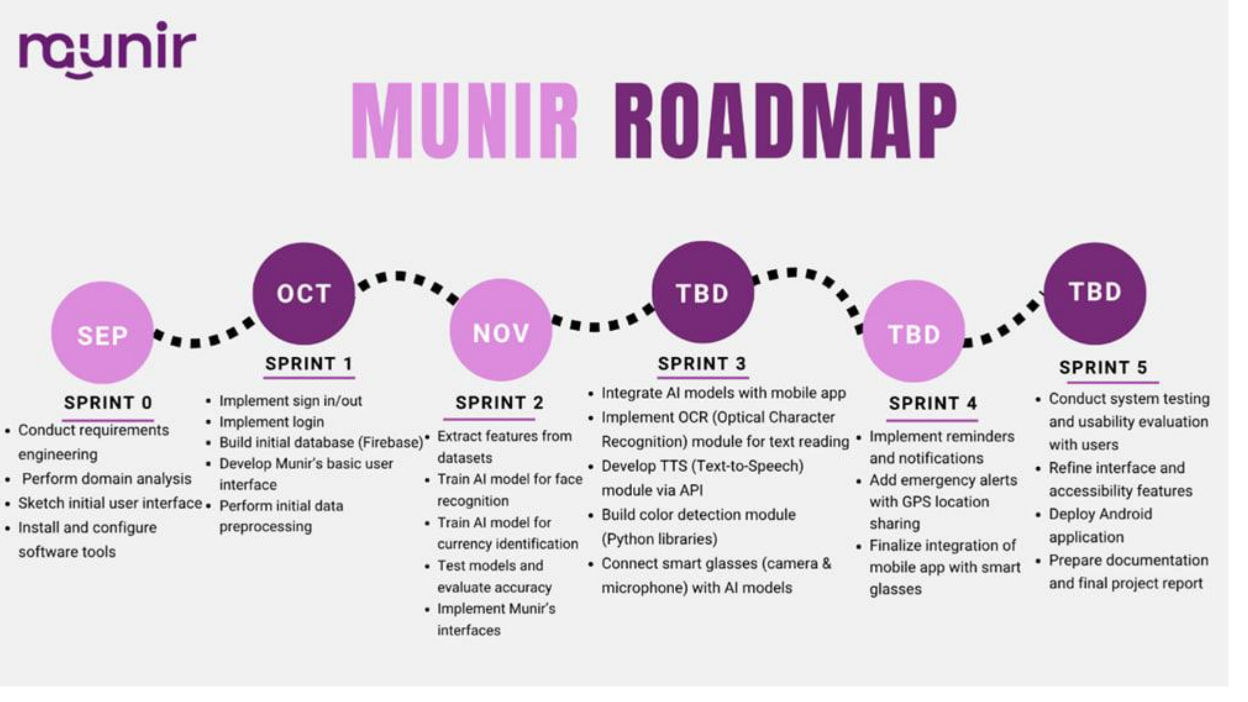


Figure 7: Munir Roadmap

B) Product Backlog

All user stories and acceptance criteria in the product backlog were written following the principles outlined in *Essential Scrum* by Kenneth S. Rubin, ensuring they are user-centered, testable, and aligned with agile values [37].

- Functional Requirements

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
GP-1	As a “Munir” user, I want to create an account using my name, email, and password, so that I can sign-up into “Munir” app	3	Feature	Done	1. As a “Munir” user, if I click the “Create Account” button, then there are two options: fill out the sign-up form, or “Sign up with Google” 2. As a “Munir” user, if I choose to fill out the sign-up form, then I should be able to enter my username, email, and password in the required fields 3. As a “Munir” user, if I fill out sign-up form with all required valid information, then my information should be stored in the database, and I should receive a confirmation message that my account has been created successfully 4. As a “Munir” user, if I submit the sign-up form without filling in all required fields, then I should receive an error indicating the missing fields 5. As a “Munir” user, if I try to sign up with an email that is already associated with an existing account, then I should receive an error message indicating that the account already exists

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
					6. As a "Munir" user, if I submit the sign-up form with an invalid format (e.g. incorrect email format), then I should receive an error message explaining the issue 7. As a "Munir" user, if I sign up using my email, then I should receive a verification email containing a verification link. My account should remain inactive until I verify my email address
GP-2	As a "Munir" user, I want to sign-up using my Google account, so that I can access the application without creating a new password	3	Feature	Done	1. As a "Munir" user, if I click the "Create Account" button, then there are two options: fill out the sign-up form, or "Sign up with Google" 2. As a "Munir" user, if I click the "Sign up with Google" option, then I should be redirected to the Google account selection screen 3. As a "Munir" user, if I select a Google account that is not already registered in Munir, then a new account should be created using my Google email, and I should be signed in successfully 4. As a "Munir" user, if I cancel the Google sign-up process or deny permissions, then I should be redirected back to the Create Account screen without creating a new account
GP-3	As a "Munir" user, I want to login to my account, so that I can use the application	3	Feature	Done	1. As a "Munir" user, if I click the Login button, then there are two options: fill out the login form, or "Login with Google" 2. As a "Munir" user, if I choose to fill out the login form, then I should be able to enter my email and password in the required fields 3. As a "Munir" user, if I submit the login form, then my credentials should be validated against the database

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
					4. As a "Munir" user, if my credentials are correct, then I should be granted access to the homepage with confirmation message 5. As a "Munir" user, if my credentials are correct but my email address is not verified, then I should receive a message prompting me to verify my email before logging in 6. As a "Munir" user, if my credentials are incorrect, then I should receive an error message
GP-4	As a "Munir" user, I want to login using my Google account, so that I can access the application without creating a new password	3	Feature	Done	1. As a "Munir" user, if I click the login button, then there are two options: fill out the login form, or "Login with Google" 2. As a "Munir" user, if I choose "Login with Google," then Google authentication dialog should appear to select or add an account. 3. As a "Munir" user, if I successfully authenticate with my Google account, then I should be granted access to the homepage with confirmation message. 4. As a "Munir" user, if I cancel the Google login process or deny permissions, then the authentication dialog should close.
GP-5	As a "Munir" user, I want to reset my password if I forget it, so that I can regain access to my account	3	Feature	Done	1. As a "Munir" user, if I click "Forgot Password" on the login page, then I should be prompted to enter my registered email address 2. As a "Munir" user, if I submit a registered email, then I should receive a confirmation message and receive an email containing a secure reset link 3. As a "Munir" user, if I submit an unregistered email, then I should receive an error message indicating that no account is associated with that email

ID	**PBI** (user story)	**Size** (Story points)	**Type** (Feature, defect, technical work, knowledge acquisition)	**Status** (to do, in progress, or done)	**Acceptance Criteria** (The conditions of satisfaction that must be met for that item to be accepted)
					4. As a "Munir" user, if I open the reset link from the email, then I should be taken to a "Reset your Password" screen so that I can set my new password 5. As a "Munir" user, if I click "Save" with a valid new password, then my password should be updated, and I should receive a success message and be updated in database
GP-6	As a "Munir" user, I want to edit my account information, so that I have my account with updated information	3	Feature	Done	1. As a "Munir" user, if I click on my profile page, then my information fields (name and email) should be pre-filled if applicable 2. As a "Munir" user, if I update the "Name" field and click the "Save" button, then the updated information should be stored in the database, and a confirmation message should appear indicating that the changes have been successfully saved 3. As a "Munir" user, if I enter invalid data in the "Name" field (e.g., including numbers), then I should receive an inline validation message explaining the issue 4. As a "Munir" user, if I attempt to interact with the "Email" field, then I should receive a message indicating that the email address is read-only and cannot be changed
GP-7	As a "Munir" user, I want to logout from my account, so that I can keep my information secure	1	Feature	Done	1. As a "Munir" user, if I navigate to the Settings page, then the "Log Out" button should be a clearly accessible option 2. As a "Munir" user, if I click the "Log Out" button, then I should be prompted with a confirmation message 3. As a "Munir" user, if I confirm the log out action, then I should be logout of my account and redirected to the login page

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
					4. As a "Munir" user, if I cancel the logout action, then I should remain signed in and stay on the Settings page
GP-8	As a "Munir" user, I want to be able to delete my account, so that I can remove all my personal data from the app and discontinue using the service	2	Feature	Done	1. As a "Munir" user, if I navigate to the Settings page and open the "Account Info" section, then the "Delete My Account" button should be a clearly accessible option 2. As a "Munir" user, if I click on the "Delete My Account" button, then I should be prompted with a confirmation dialog asking me to confirm the deletion of my account 3. As a "Munir" user who signed up with email/password, if I confirm the account deletion, then I should be required to enter my current password before the system proceeds with deleting my account 4. As a "Munir" user who logged in with Google, if I confirm the account deletion, then a Google authentication dialog should appear to verify my identity before the system proceeds with deleting my account 5. As a Munir user, if I confirm the account deletion, then I should be required to enter my current password before the system proceeds with deleting my account 6. As a "Munir" user, if I enter my password correctly, then my account and all associated personal data should be permanently removed from the app, and I should receive a confirmation message

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
					7. As a "Munir" user, if I enter my password incorrectly during the deletion process, then my account and all associated personal data should remain unchanged, and I should receive an error message 8. As a "Munir" user, if I cancel the deletion, then my account and all associated personal data should remain unchanged, and I should be redirected back to the Account Info page
GP-9	As a "Munir" user, I want to add a person's face and name, so that the system can later recognize and identify them	3	Feature	Done	1. As a "Munir" user, if I navigate to "Face Recognition" from the Homepage, then I should have access to two options: "Face Management" and "Identify Person" 2. As a "Munir" user, if I select "Face Management," then a list of all previously added persons should be available 3. As a "Munir" user, if I click on the Add button, I should be redirected to the "Add Person" page where I can enter the person's name and either upload a face photo or take a new photo 4. As a "Munir" user, if I choose to upload a photo, I should be able to upload images for all required face orientations (front, left, right) so that the system can recognize the person correctly 5. As a "Munir" user, if a required face image is uploaded with the wrong direction (e.g., a left-facing image is provided instead of the required right-facing image), I should receive an error message

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
					6. As a "Munir" user, if the three uploaded orientation photos (front, left, right) contain different people (e.g., the front photo is a different person than the left and right photos), I should receive an error message indicating that the faces do not belong to the same person and must be retaken 7. As a "Munir" user, if any of the uploaded orientation photos contains more than one face, I should receive an error message indicating that multiple faces were detected and that each photo must contain only one person 8. As a “Munir” user, if I choose to take a new photo, I should be able to generate and preview the face in front, right, and left orientations before saving 9. As a “Munir” user, if my face is outside the camera frame, I should receive an error message to reposition my face within the camera frame 10. As a “Munir” user, if my face is facing a direction other than the required orientation, I should receive an error message 11. As a “Munir” user, if I try to add a person without entering a name or uploading a photo, then the “Add Person” button should remain disabled until both fields are completed 12. As a “Munir” user, if the person is added successfully, then their name and photo should be saved in the database and immediately appear in the Face Management list

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
GP-10	As a “Munir” user, I want to edit the details of stored faces (e.g., name or photo), so that I can keep the information updated	3	Feature	Done	1. As a “Munir” user, if I navigate to “Face Recognition” from the Homepage, then I should have access to two options: “Face Management” and “Identify Person” 2. As a “Munir” user, if I select “Face Management,” then a list of all previously added persons should be available 3. As a "Munir" user, if I click the edit button, I should be redirected to the "Edit Person" page where I can update the name and/or modify the face images by uploading a new photo or take a new one 4. As a "Munir" user, if I replace any face image during editing, the same validation rules should apply, incorrect face orientation, mismatched identities across the three photos, or multiple faces in a single photo should each trigger an appropriate error message 5. As a “Munir” user, if I save the changes, then the updated details should be stored in the database, and I should receive a confirmation message indicating the update was successful, with the new details immediately reflected in the Face Management list 6. As a “Munir” user, if I cancel the edit action, then no changes should occur, and the person’s details should remain as they were in the Face Management list

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
GP-11	As a 'Munir' user, I want to delete previously registered faces, so that the database remains accurate and up to date	2	Feature	Done	1. As a "Munir" user, if I navigate to "Face Recognition" from the Homepage, then I should have access to two options: "Face Management" and "Identify Person" 2. As a "Munir" user, if I select "Face Management," then a list of all previously added persons should be available 3. As a "Munir" user, if I click the delete button, then I should be prompted with a confirmation dialog asking me to confirm or cancel the deletion 4. As a "Munir" user, if I confirm the deletion, then the person's details should be permanently removed from the database, and I should receive a confirmation message indicating the deletion was successful, with the updated list immediately reflected in the Face Management list 5. As a "Munir" user, if I cancel the deletion, then no changes should occur, and the person's details should remain as they were in the Face Management list
GP-12	As a "Munir" user, I want to search for a stored person by name, so that I can easily find and manage them	3	Feature	Done	1. As a "Munir" user, if I navigate to "Face Recognition" from the Homepage, then I should have access to two options: "Face Management" and "Identify Person" 2. As a "Munir" user, if I select "Face Management," then a list of all previously added persons should be available 3. As a "Munir" user, if I type a name in the search bar, then the list should dynamically filter to show only the persons whose names match my input

ID	**PBI** (user story)	**Size** (Story points)	**Type** (Feature, defect, technical work, knowledge acquisition)	**Status** (to do, in progress, or done)	**Acceptance Criteria** (The conditions of satisfaction that must be met for that item to be accepted)
					4. As a "Munir" user, if there are no matches for my search, then I should receive a message indicating that no persons were found 5. As a "Munir" user, if I clear the search input, then the full list of stored persons should be displayed again
GP-13	As a "Munir" user, I want to create reminders with a title, date, time, and frequency (one-time, daily, weekly), so that I receive notifications at the right time on my phone	3	Feature	Done	1. As a "Munir" user, if I navigate to "Reminders" from the Homepage, then a list of all previously added reminders should be available 2. As a "Munir" user, I should have two options to add a reminder: manually by filling in the required fields, or via voice guidance 3. As a "Munir" user, if I click on the "Add Voice Reminder" button, then I should first hear a set of spoken preparation instructions. These instructions remain on screen for a few seconds before the voice guided process begins automatically 4. As a "Munir" user, once the preparation instructions end, I should be guided through spoken prompts to provide the reminder details (Title, Date, Time, and Frequency) one field at a time, with the system listening for my voice response after each prompt 5. As a "Munir" user, if I choose to add a reminder manually, I should be redirected to the "Add New Reminder" page where I can enter the reminder title, date, time, and frequency

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
					6. As a "Munir" user, if the reminder is added successfully, then it should be saved to the database, the notification scheduled, and I should receive a confirmation message 7. As a "Munir" user, if I miss or fail to respond to any required voice prompt while adding a reminder, then I should be given up to five attempts to respond. If no valid response is received after the fifth attempt, an error message should be played, and the process should end without creating the reminder 8. As a "Munir" user, if I am adding a reminder manually, then the "Add" button should not be enabled unless all required fields are properly filled in 9. As a "Munir" user, if I close or cancel the "Add Reminder" process, then no reminder should be created and all entered data should be discarded
GP-14	As a "Munir" user, I want to edit existing reminders, so that I can update them when my plans change	3	Feature	Done	1. As a "Munir" user, if I navigate to "Reminders" from the Homepage, then a list of all previously added reminders should be available 2. As a "Munir" user, if I click on "Edit" icon for existing reminder, then I should be presented with two options: to update the reminder manually or by voice 3. As a "Munir" user, if I update one or more fields, then the reminder should be updated in the database and a new notification scheduled according to the updated frequency, and I should receive a confirmation message

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
					4. As a "Munir" user, if I attempt to save a reminder without responding to one of the mandatory voice prompts (Title, Date, Time, or Frequency), then I should be given up to five attempts to provide the required input. If no valid response is received after the fifth attempt, an error message should be played, and the process should end without creating the reminder 5. As a "Munir" user, if I am editing a reminder manually, then the "Save Changes" button should not be enabled unless all required fields are properly filled in 6. As a "Munir" user, if I close or cancel the Edit Reminder process without saving, then no changes should be applied, and the original reminder should remain unchanged
GP-15	As a "Munir" user, I want to delete reminders, so that I can keep my reminder list organized	2	Feature	Done	1. As a "Munir" user, if I navigate to "Reminders" from the Homepage, then all previously added reminders should be available in the list 2. As a "Munir" user, if I select the "Delete" icon for existing reminder, then I should receive an accessible confirmation prompt asking me to confirm the deletion 3. As a "Munir" user, if I confirm the deletion, then the reminder should be removed from the database, any scheduled notifications should be cancelled, and a confirmation message indicating that the reminder was deleted

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
					4. As a "Munir" user, if I choose "Cancel" in the confirmation prompt or dismiss it, then no changes should be applied, and the reminder should remain unchanged in the list
GP-16	As a "Munir" user, I want to filter reminders by date, so that I can quickly find a specific reminder.	3	Feature	Done	1. As a "Munir" user, if I navigate to "Reminders" from the Homepage, then all previously added reminders should be available in the list 2. As a "Munir" user, when I tap the Filter option next to the search bar, I should see two filtering options: Quick filter options (Tomorrow, This week, This month) or search within a custom date range 3. As a "Munir" user, if I select "Custom Range," then I should be able to specify a start date and an end date, and the app should display all reminders that fall within the selected date range 4. As a "Munir" user, if I select a Quick Filter option (Tomorrow, This Week, or This Month), then the app should automatically apply the corresponding date range and display only the reminders that fall within it, without requiring any additional input from the user 5. As a "Munir" user, if I clear the applied filter, then the app should display all previously added reminders again

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
GP-17	As a “Munir” user, I want to add emergency contacts (name and phone), so that they are notified when I trigger the SOS alert	3	Feature	Done	1. As a “Munir” user, if I navigate to “Contacts” from the Footer, then a list of all previously added contacts should be available 2. As a “Munir” user, I should have two options to add a contact: manually by filling in the required fields, or via voice guidance 3. As a "Munir" user, if I click on the "Add Voice Contact" button, then I should first hear a set of spoken preparation instructions. These instructions remain on screen before the voice guided process begins automatically 4. As a "Munir" user, once the preparation instructions end, I should be guided through spoken prompts to provide the contact details (Name and Phone Number) one field at a time, with the system listening for my voice response after each prompt 5. As a “Munir” user, if I choose to add a contact manually, I should be redirected to the “Add New Contact” page where I can enter the contact’s name and phone number 6. As a "Munir" user, if I am adding a contact manually, then the "Add" button should not be enabled unless all required fields are properly filled in 7. As a "Munir" user, if I cancel the "Add Contact" process, then no contact should be saved, and I should be returned to the Contacts list

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
					8. As a “Munir” user, if I say a phone number that does not start with “05” or is not exactly 10 digits long, I get 5 attempts to provide a valid number. If all 5 attempts fail, an error message is played clearly explaining the error, and the contact is not added 9. As a “Munir” user, if the contact is added successfully, then their information should be saved in the database, and I should receive a confirmation message indicating the contact has been added
GP-18	As a “Munir” user, I want to edit existing emergency contacts, so that I can keep their information accurate	3	Feature	Done	1. As a “Munir” user, if I navigate to “Contacts” from the Footer, then a list of all previously added contacts should be available 2. As a “Munir” user, if I click on “Edit” icon for existing contact, then I should be presented with two options: to update the contact manually or by voice 3. As a “Munir” user, if I update one or more fields, then the contact should be updated in the database, and I should receive a confirmation message 4. As a “Munir” user, if I attempt to save the changes without responding to one of the required voice prompts, then I should be given up to five attempts to provide the missing input. If no valid response is received after the fifth attempt, an error message should be played, and the process should end without updating the contact

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
					5. As a “Munir” user, if I am editing a contact manually, then the “Save Changes” button should not be enabled unless all required fields are properly filled in. 6. As a “Munir” user, if I close or cancel the Edit Contact process without saving, then no changes should be applied, and the original contact should remain unchanged
GP-19	As a “Munir” user, I want to delete an emergency contact, so that my emergency list stays up to date and relevant	2	Feature	Done	1. As a “Munir” user, if I navigate to “Contacts” from the Footer, then all previously added contacts should be available in an accessible list 2. As a “Munir” user, if I select the “Delete” icon for existing contact, then I should receive an accessible confirmation prompt asking me to confirm the deletion 3. As a “Munir” user, if I confirm the deletion, then the contact’s details should be permanently removed from the database, and I should receive a confirmation message indicating the deletion was successful 4. As a “Munir” user, if I cancel the deletion, then no changes should be applied, and the contact’s details should remain unchanged

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
GP-20	As a “Munir” user, I want to use the mobile app camera to recognize faces and hear the person’s name spoken aloud, so that I can easily identify people around me when I use my phone	5	Feature	Done	1. As a “Munir” user, if I navigate to “Face Recognition” from the Homepage, then I should see two options: “Face Management” and “Identify Person” 2. As a "Munir" user, if I select "Identify Person," then I should first see and hear a set of spoken preparation instructions covering camera distance, lighting conditions, and face positioning then the camera opens automatically once the instructions finish 3. As a “Munir” user, if I capture or upload a face image, then the app should process the image and attempt to match it with the stored faces 4. As a “Munir” user, if the face is successfully recognized, then I should hear the person’s name spoken aloud 5. As a "Munir" user, if a face is not recognized, I should hear a spoken message such as "Unknown person" 6. As a “Munir” user, if a face is captured from a distance up to 3.5 meters, then the app should be able to recognize the person if their face is registered in the system 7. As a “Munir” user, if a face is captured from a distance greater than 3.5 meters (for example, 4 meters), then the app may not be able to recognize the person and should announce “No Face Detected” 8. As a “Munir” user, if multiple people appear in the same, then the app should identify each recognized person by speaking their names and announce “Unknown person” for any faces that are not recognized

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
					9. As a “Munir” user, if I capture or upload an image and no face is detected, I should hear a spoken message such as “No face detected”
GP-21	As a “Munir” user, I want to capture or upload an image with the mobile app camera and have the text read aloud, so that I can easily understand printed content on my phone	5	Feature	Done	1. As a “Munir” user, if I navigate to “Text Reading” from the Homepage, then I should be redirected to the camera screen 2. As a “Munir” user, if I am redirected to the camera screen, then I should have the option to either take a live photo or upload an existing image 3. As a “Munir” user, if I capture or upload an image, then the app should process the image, extract the text, and read it aloud 4. As a “Munir” user, if no readable text is found in the image, then I should hear a spoken message such as “No text recognized”
GP-22	As a “Munir” user, I want to capture or upload an image of a Saudi currency with the mobile app camera and hear its denomination announced aloud, so that I can easily know its value	5	Feature	Done	1. As a “Munir” user, if I navigate to “Currency identification” from the Homepage, then I should first see and hear a set of spoken preparation instructions covering camera positioning and how to place the currency note, then the camera should open automatically once the instructions finish. 2. As a “Munir” user, if I am redirected to the camera screen, then I should have the option to either take a live photo or upload an existing image

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
					3. As a "Munir" user, if I capture or upload an image of a currency, then the app should process the image, identify the currency denomination, and announce its value aloud 4. As a "Munir" user, if the denomination cannot be recognized, then I should hear a spoken message such as "Currency not identified" 5. As a "Munir" user, if an image contains up to two currency notes, then the app should identify each denomination and announce their values aloud 6. As a "Munir" user, if a currency note is partially cropped but enough features are visible, then the app should still be able to identify the denomination and announce its value aloud
GP-23	As a "Munir" user, I want to scan objects with the mobile app camera and hear the object's name and details such as its color spoken aloud, so that I can identify objects using my phone.	5	Feature	Done	1. As a "Munir" user, if I navigate to "Object detection" from the Homepage, then I should be redirected to the camera screen 2. As a "Munir" user, if I am redirected to the camera screen, then I should have the option to either take a live photo or upload an existing image 3. As a "Munir" user, if I capture or upload an image of an object, the app should process the image, identify the object, and announce its name and color aloud

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
GP-24	As a “Munir” user, I want to scan my surrounding environment using the mobile app camera and hear a spoken description of the scene, so that I can better understand my surroundings independently	5	Feature	Done	1. As a “Munir” user, if I navigate to “Scene Description” from the Homepage, then I should be redirected to the camera screen 2. As a “Munir” user, if I am redirected to the camera screen, then I should have the option to either take a live photo or upload an existing image 3. As a “Munir” user, if I capture or upload an image of my surroundings, the app should process the image, describe the scene and announce the details aloud
GP-25	As a “Munir” user, I want to trigger an SOS that shares my real-time location via SMS, so that my emergency contacts can reach me quickly	3	Feature	Done	1. As a "Munir" user, if I navigate to the Emergency icon from the footer for the first time, I should be asked to grant location permission. On subsequent visits, if permission has already been granted, the Emergency screen should open directly without prompting it again 2. As a “Munir” user, if I respond to the location permission prompt, then the Emergency screen with the large red “SOS” button should become available 3. As a “Munir” user, if I tap the “SOS” button and I have at least one saved emergency contact, I should be redirected to the phone’s messaging application with a pre-written alert message containing my current location, so that I only need to press the “Send” button to notify all saved contacts

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
					5. As a “Munir” user, if no emergency contacts are saved, the “SOS” button should be disabled, and a voice message should instruct me to add an emergency contact first. I should then be redirected automatically to the Contact page to add at least one contact before sending an alert 6. As a “Munir” user, if I add a new contact, I should be redirected back to the Emergency screen, and the “SOS” button should become enabled so that I can tap it to send an alert to my saved emergency contacts 7. As a "Munir" user, if I deny location permission permanently, the SOS button should still be accessible, but the alert message sent to emergency contacts will not include a location link
GP-26	As a “Munir” user, I want to connect or disconnect my smart glasses via Bluetooth, so that the application can manage the connection when needed	5	Feature	Done	1. As a “Munir” user, if I navigate to “Settings” from the Footer and open “Smart Glass”, there should be a button to connect the smart glasses via Bluetooth, along with the location of the last successful connection displayed below the button. 2. As a “Munir” user, if I tap the “Search for Glasses” button, the application should automatically search for nearby Munir smart glasses via Bluetooth and connect to them automatically once detected

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
					3. As a "Munir" user, If the smart glasses are detected, the application should connect automatically and display information such as the battery level and signal strength 4. As a "Munir" user, if no smart glasses are found, I should hear a message instructing me to make sure the smart glasses are turned on and Bluetooth is enabled 5. As a "Munir" user, if I want to disconnect my smart glasses, I should be able to navigate to "Settings," open "Smart Glass," and press the "Disconnect" button at the bottom of the page
GP-27	As a "Munir" user, I want to use the smart glasses camera to recognize faces and hear the person's name announced in real time, so that I can instantly know who me is around	5	Feature	Done	1. As a ''Munir'' user, if I want the smart glasses to listen to my voice command, I must first press the button on the glasses to activate listening mode 2. As a "Munir" user, if I say "Face" or "مين قدامي", then the smart glasses camera should automatically capture an image of the person 3. As a "Munir" user, if the captured face is within a distance of up to 3.5 meters and matches a previously registered person, I should receive a notification on my mobile phone, and the person's name should be announced through the mobile phone speaker 4. As a "Munir" user, if the captured face is more than 3.5 meters away or does not match any registered faces, I should receive a notification on my mobile phone, and the phone should announce a message such as "Unknown person"

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
					5. As a "Munir" user, if no face is detected in the captured image, I should receive a notification on my mobile phone, and the phone should announce a message such as "No face detected"
GP-28	As a "Munir" user, I want to use the smart glasses camera to capture printed text and hear it read aloud instantly, so that I can understand written content	5	Feature	Done	1. As a ''Munir'' user, if I want the smart glasses to listen to my voice command, I must first press the button on the glasses to activate listening mode 2. As a "Munir" user, if I say "Read" or "اقرا لي", the smart glasses camera should automatically capture an image of the printed text in front of me 3. As a "Munir" user, if text is successfully extracted from the captured image, I should receive a notification on my mobile phone, and the extracted text should be read aloud through the mobile phone speaker 4. As a "Munir" user, if no text is detected in the captured image, I should receive a notification on my mobile phone, and the phone should announce a message such as "No text detected"
GP-29	As a "Munir" user, I want to use the smart glasses camera to identify Saudi currency and hear their denomination spoken instantly, so that I can know its value	5	Feature	Done	1. As a ''Munir'' user, if I want the smart glasses to listen to my voice command, I must first press the button on the glasses to activate listening mode 2. As a "Munir" user, if I say "Money" or "فلوس", the smart glasses camera should automatically capture an image of the currency note in front of me

ID	**PBI** (user story)	**Size** (Story points)	**Type** (Feature, defect, technical work, knowledge acquisition)	**Status** (to do, in progress, or done)	**Acceptance Criteria** (The conditions of satisfaction that must be met for that item to be accepted)
					3. As a "Munir" user, if the currency denomination is successfully identified, I should receive a notification on my mobile phone, and the currency value should be announced through the mobile phone speaker 4. As a "Munir" user, if the system cannot determine the currency, I should receive a notification on my mobile phone, and the phone should announce a message such as "Currency not detected"
GP-30	As a "Munir" user, I want to use the smart glasses camera to detect objects and their colors and hear them announced instantly, so that I can recognize them in real time	5	Feature	Done	1. As a ''Munir'' user, if I want the smart glasses to listen to my voice command, I must first press the button on the glasses to activate listening mode 2. As a "Munir" user, if I say "Detect Item" or "تعرف على الشيء", the smart glasses camera should automatically capture an image of the object in front of me 3. As a "Munir" user, if the object and its color are successfully detected, I should receive a notification on my mobile phone, and the object name along with additional details should be announced through the mobile phone speaker
GP-31	As a "Munir" user, I want to use the smart glasses camera to scan my surroundings and hear a spoken description of the scene, so that I can better understand my environment in real time	5	Feature	Done	1. As a ''Munir'' user, if I want the smart glasses to listen to my voice command, I must first press the button on the glasses to activate listening mode 2. As a "Munir" user, if I say "Describe" or "اوصف", the smart glasses camera should automatically capture an image of the surrounding scene

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
					3. As a "Munir" user, if the scene is successfully analysed, I should receive a notification on my mobile phone, and a description of the scene should be announced through the mobile phone speaker
GP-32	As a "Munir" user, I want to receive a notification on my mobile phone when a scheduled reminder time arrives, so that I can be notified on time	5	Feature	Done	1. As a "Munir" user, if the current time matches a scheduled reminder, I should receive a notification on my mobile phone displaying the reminder title (e.g., "Take medicine") 2. As a "Munir" user, if multiple reminders are due at the same time, the system should display all reminder notifications sequentially on my mobile phone
GP-33	As a "Munir" user, I want to trigger SOS by saying a wake phrase, so that my emergency contacts receive my location via SMS	5	Feature	Done	1. As a ''Munir'' user, if I want the smart glasses to listen to my voice command, I must first press the button on the glasses to activate listening mode 2. As a "Munir" user, if I say "SOS" or "النجدة", I should receive a notification on my mobile phone that redirects me to the phone's messaging application with a pre-written SOS message prepared for all configured emergency contacts 3. As a "Munir" user, if no emergency contacts are configured, I should receive a notification on my mobile phone, and the phone should announce a message such as "No Contacts Found" If I tap the notification, I should be redirected to the Emergency Contact page to add a contact

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
GP-34	As a user, I want to access the User Manual from the footer, so that I can read information about the glasses, how to connect them, how to use them, and troubleshoot issues whenever needed.	3	Feature	Done	1. As a “Munir” user, if I need help understanding the glasses or pairing process, I should be able to access the User Manual from the “Guide’’ in the footer 2. As a user, when connecting the glasses for the first time, I should be automatically redirected to the User Manual 3. As a “Munir” user, I should have two options for reading the manual: Read the full list automatically or select specific cases to read 4. As a “Munir” user, if I choose the automatic reading option, the system should read the entire manual content sequentially 5. As a “Munir” user, if I choose specific cases, the system should only read the selected content
GP-35	As a user, I want to change the application language from the Settings page, so that I can use the application in either Arabic or English according to my preference	3	Feature	Done	1. As a “Munir” user, if I navigate to “Settings” from the footer, I should be able to access the language and choose between Arabic and English 2. As a “Munir” user, if I select Arabic, the application interface should be displayed in Arabic 3. As a “Munir” user, if I select English, the application interface should be displayed in English 4. As a “Munir” user, when I reopen the application, the system should remember and display the previously selected language

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
GP-36	As a Munir user, I want to change my account password so that I can protect my account from unauthorized access	3	Feature	Done	1. As a “Munir” user, if I navigate to “Settings” from the footer while signed in with email and password, then I should be able to see and access the “Change Password” page 2. As a “Munir” user, if I signed in using a Google account, then the “Change Password” option should not be displayed since no password is stored in the database 3. As a “Munir” user, if I open the “Change Password” page, then I should see fields for new password, current password, and confirm new password 4. As a “Munir” user, if I enter an incorrect current password, then I should receive an error message 5. As a “Munir” user, if the new password and confirm password do not match, then I should receive an error message 6. As a “Munir” user, if the new password does not meet security requirements (minimum 8 characters, including uppercase, lowercase, number, and special character), then I should receive an error message, and the password update should be prevented 7. As a “Munir” user, if all entered information is valid, then my password should be updated successfully, and I should receive a confirmation message

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
GP-37	As a "Munir" user, I want to access a "Privacy & Security" so that I can understand how my data is stored and protected	3	Feature	Done	1. As a "Munir" user, if I navigate to "Settings" from the footer, then I should be able to access the "Privacy & Security" page 2. As a "Munir" user, if I open the "Privacy & Security" page, then I should be able to view information about how my data is stored and protected, including encryption and security measures
GP-38	As a team member, I want to use the existing dataset for both face recognition and currency identification, so that I can develop and train AI-powered modules of the "Munir" application	2	knowledge acquisition	Done	1. The dataset, which includes both face images and currency notes, will be used for developing "Munir" AI powered recognition modules
GP-39	As a team member, I want to split the dataset, so that I can effectively train and test the AI-powered modules of the "Munir" application	3	knowledge acquisition	Done	1. The dataset will be split into training, testing and validating sets

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
GP-40	As a team member, I want to train face and Currency identification models using the acquired dataset, so that I can power the "Munir" system	3	knowledge acquisition	Done	1. The face model will be trained using the training dataset 2. The currency model will be trained using the training dataset 3. The trained models and label maps will be exported for the Munir system (e.g., TFLite)
GP-41	As a team member, I want to validate the face recognition and Currency identification models using a validation dataset, so that I can measure their performance	3	knowledge acquisition	Done	1. The face model and currency model will be evaluated using the validation dataset
GP-42	As a team member, I want to test notification delivery between the smart glasses and the mobile application, so that I can ensure notifications are received and function correctly	3	knowledge acquisition	Done	1. As a team member, if the mobile application is closed and a notification is sent from the smart glasses, then the notification should be delivered and displayed successfully 2. As a team member, if the mobile application is open and the user is currently viewing the application, then the notification should be received and displayed successfully 3. As a team member, if the mobile application is running in the background, then the notification should be delivered and displayed successfully

Table 3: Product Backlog

- Non-functional Requirements

To ensure a seamless and dependable experience, the following non-functional requirements define the quality standards that the **Munir** system must achieve across performance, usability, security, and reliability.

ID	**PBI** (user story)	**Size** (Story points)	**Type** (Feature, defect, technical work, knowledge acquisition)	**Status** (to do, in progress, or done)	**Acceptance Criteria** (The conditions of satisfaction that must be met for that item to be accepted)
GP-43	As a “Munir” user, I want captured images to be stored securely in the cloud with encryption, so that my personal data remains private and protected while still being accessible across devices	3	Feature	Done	1- Face recognition images are uploaded to the cloud in an encrypted format to ensure privacy and security
GP-44	As a “Munir” user, I want the application interface to be simple and intuitive, so that I can use it without errors after a short learning period of time	3	Feature	Done	1- The interface must remain simple and consistent across screens so users can navigate and complete core tasks with minimal errors 2- The application must provide clear success/error messages to reduce mistakes after a short learning period

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
GP-45	As a "Munir" user, I want the mobile phone to provide clear voice feedback for smart glasses responses, so that I can easily understand the system's responses without visual assistance	3	Feature	Done	1- The Munir system shall prevent overlapping voice prompts and provide clear and understandable audio feedback through the mobile phone speaker within less than 5 seconds of the triggering action.
GP-46	As a "Munir" user, I want reminders to be delivered at the exact scheduled time, so that I reliably receive alerts when I expect them	3	Feature	Done	1- The reminder fires within a tolerance of less than 1 minute of the user-set time
GP-47	As a "Munir" user, I want to receive email notifications whenever I log in or delete my account, so that I can monitor account access and detect any unauthorized login attempts	3	Feature	Done	1- An email notification shall be sent to the registered user email address upon every successful login 2- An email notification shall be sent to the registered user email address when the user deletes their account 3- Each email notification shall include the type of action performed (login or account deletion) and the corresponding date and time 4- Email notifications shall be sent within 2 minutes of the action being completed

ID	PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Status (to do, in progress, or done)	Acceptance Criteria (The conditions of satisfaction that must be met for that item to be accepted)
GP-48	As a user concerned about the security of my personal information, I want to ensure that the system enforces strong password requirements so that my account remains protected from unauthorized access	3	Feature	Done	1- As a user, if the password does not meet the required security rules (minimum 8 characters, uppercase letter, lowercase letter, number, and special character), I should receive an error message explaining the missing requirements 2- As a user, if the password satisfies all security requirements, the system should accept the password and proceed with account creation or password update.

Table 4: Non-functional Requirements

4.3 System Design

4.3.2 Architectural Diagram

In this section, we present the architectural design of the **Munir** system and how its components are organized and interact.

In designing **Munir**, selecting the right architecture was crucial to ensure the system's scalability, maintainability, and efficiency. After evaluating various options, we chose Client-Server Architecture, dividing the system into client-side and server-side components.

As shown in [**Figure 8**] The client-side consists of the **Munir** user interacting with the **Munir** System, which includes smart glasses and a mobile application. This component provides an intuitive interface where blind and visually impaired users can perform various tasks such as reading text, recognizing faces, detecting objects, describing scenes and identifying currencies, setting reminders, and sending emergency notifications. User input is provided through the mobile application or smart glasses, and results are presented primarily through audio feedback to ensure accessibility. All user requests are sent via the internet to the server for processing, and responses are received back through the same channel.

The server-side is responsible for processing all incoming requests from the **Munir** client application. It consists of a database that securely stores user data including profiles, reminders, and emergency contacts. The database also maintains registered person information with their associated face photos and facial embeddings. A central server processing unit handles business logic and coordinates data flow between components. The machine learning models are integrated to handle advanced recognition tasks such as face detection and recognition, as well as currency identification.

Additionally, the server connects to various external systems through APIs. Google Location Service provides the user's current location to be included in emergency alerts. The OCR API enables text extraction from captured images. The TTS API generates audio feedback for voice responses. An SMS service sends emergency alerts to registered contacts. Finally, Google Authentication ensures secure user sign-up and login processes. Vision API enables scene description and object detection by analyzing captured images. The STT API converts voice commands captured from the smart glasses into text, enabling the system to identify and execute the appropriate feature requested by the user.

This client-server architecture provides several advantages: it ensures easy maintenance by separating the user interface from business logic, allows scalability to support multiple users and devices concurrently, and enhances security and privacy by centralizing sensitive data processing and emergency notifications on the server while keeping the client-side experience straightforward and accessible. Additionally, smart glasses communicate with the mobile application via Bluetooth, transmitting captured data such as images and commands to the mobile application, which in turn sends requests to the server and receives the processed results. All responses, including the final output and audio feedback, are returned to the mobile application, which handles playback and delivers the experience to the user.

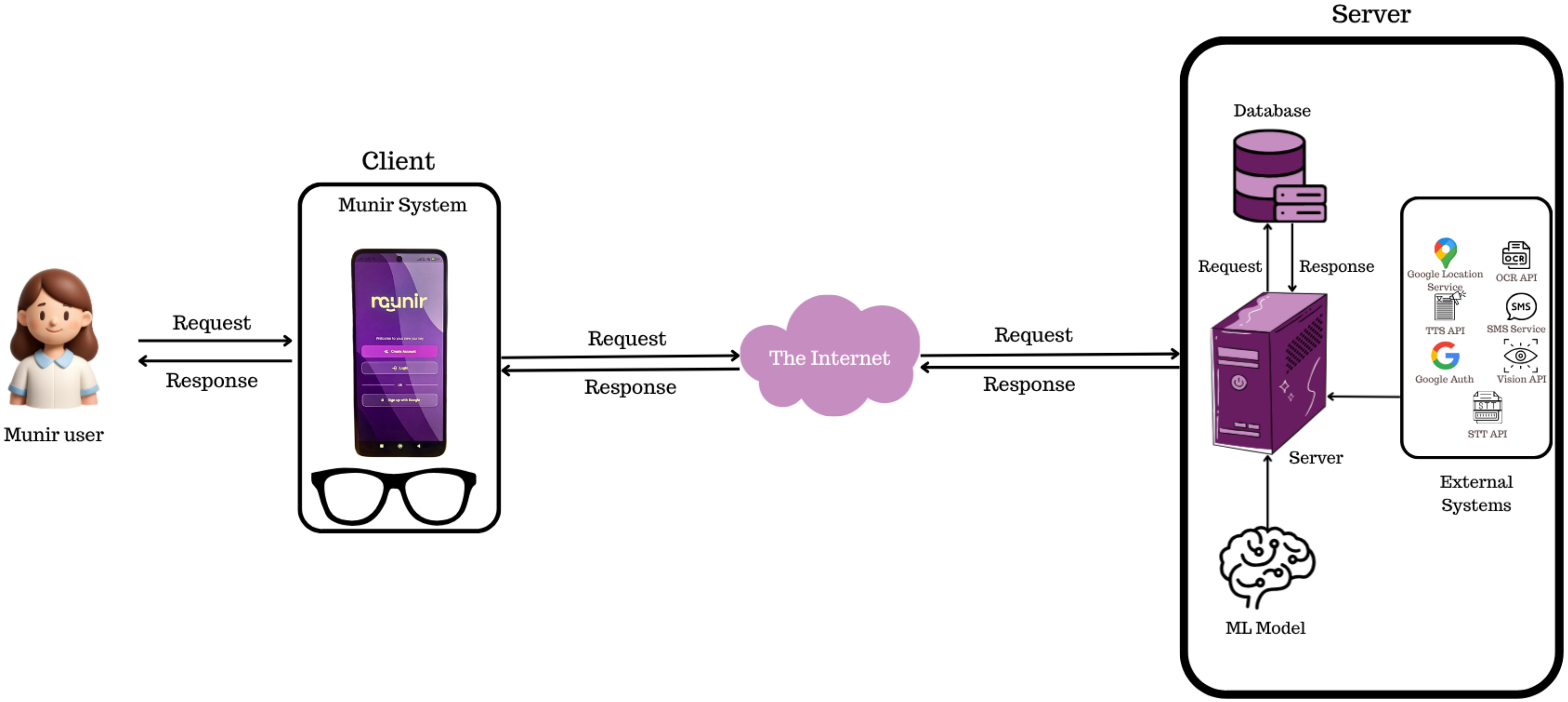


Figure 8: System Architecture

4.3.2 Class Diagram

In this section, we present the class diagram of the **Munir** system, which illustrates the structure, relationships, and responsibilities of the main classes that comprise the application.

In designing **Munir**, a comprehensive class diagram was crucial to represent the system's structure, illustrate the relationships between different components, and provide a clear blueprint of the application's data organization. The class diagram captures the static structure of the system by defining the main classes, their attributes, operations, and how they interact with one another. The class diagram consists of seven main classes that work together to provide the complete functionality of the **Munir** application:

As shown in [**Figure 9**] The **User** class serves as the central entity of the system, storing essential user information including **userID**, **email**, **password** and **user_fullName**. This class provides methods for account management such as signing up, logging in, editing profile information, sending SOS alerts, changing password, resetting password via email, logging out, and deleting the account. Each user can create and manage multiple reminders and emergency contacts, own one or more glasses devices, and register multiple persons for face recognition functionality.

The **Reminder** class enables users to set and manage personal reminders. It stores reminder details including **title**, **reminderDateTime** as a Timestamp, and **frequency** for recurring reminders. The **reminderID** is automatically generated by the system. Users can add new reminders by providing the title, date, time, and frequency, edit existing reminders, and delete them when no longer needed.

The **EmergencyContact** class manages emergency contacts for the user, storing **contact_name** and **contact_phone**. The **contactID** is automatically generated. Users can add new emergency contacts, edit their information, and delete them. These contacts receive SMS alerts during emergency situations.

The **RegisteredPerson** class represents individuals registered in the system for face recognition. Each person has a **registeredPerson_name** and stores **encryptedThumbnail** for privacy and security, and **embeddings_flat** as a list of doubles to support recognition accuracy. The **personID** is automatically generated. Users can add new persons with their photos, update their names, and remove persons entirely from the system.

The **AIService** class provides AI-powered features including text detection (OCR), object detection, currency identification, face recognition, and scene description. All methods process image files captured by the user and return results as text that is converted to audio feedback. The User class is directly associated with the AIService class through a uses relationship, where one user may invoke zero or more AI-powered operations, reflecting the on-demand nature of these features within the application.

The **Glasses** class represents the smart glasses device paired with the user's account. It stores the **macAddress** as a unique device identifier and **lastConnectedAt** timestamp to track the most recent connection. Users can pair a new device using pairGlasses() or unpair it via unpairGlasses(). Each user owns zero or more glasses devices, establishing a clear ownership relationship between the user and their hardware.

The **LastLocation** class is associated with the Glasses entity and records the device's last known geographic position when the glasses go offline, storing **latitude** and **longitude** coordinates along with a **disconnectedAt** timestamp. This enables the user to track where the glasses were last active. The class provides saveLastLocation() to persist coordinates and getLastLocation() to retrieve them.

This class diagram ensures clear organization of system components, proper data relationships, and well-defined responsibilities for each class to support all features of the **Munir** application efficiently.

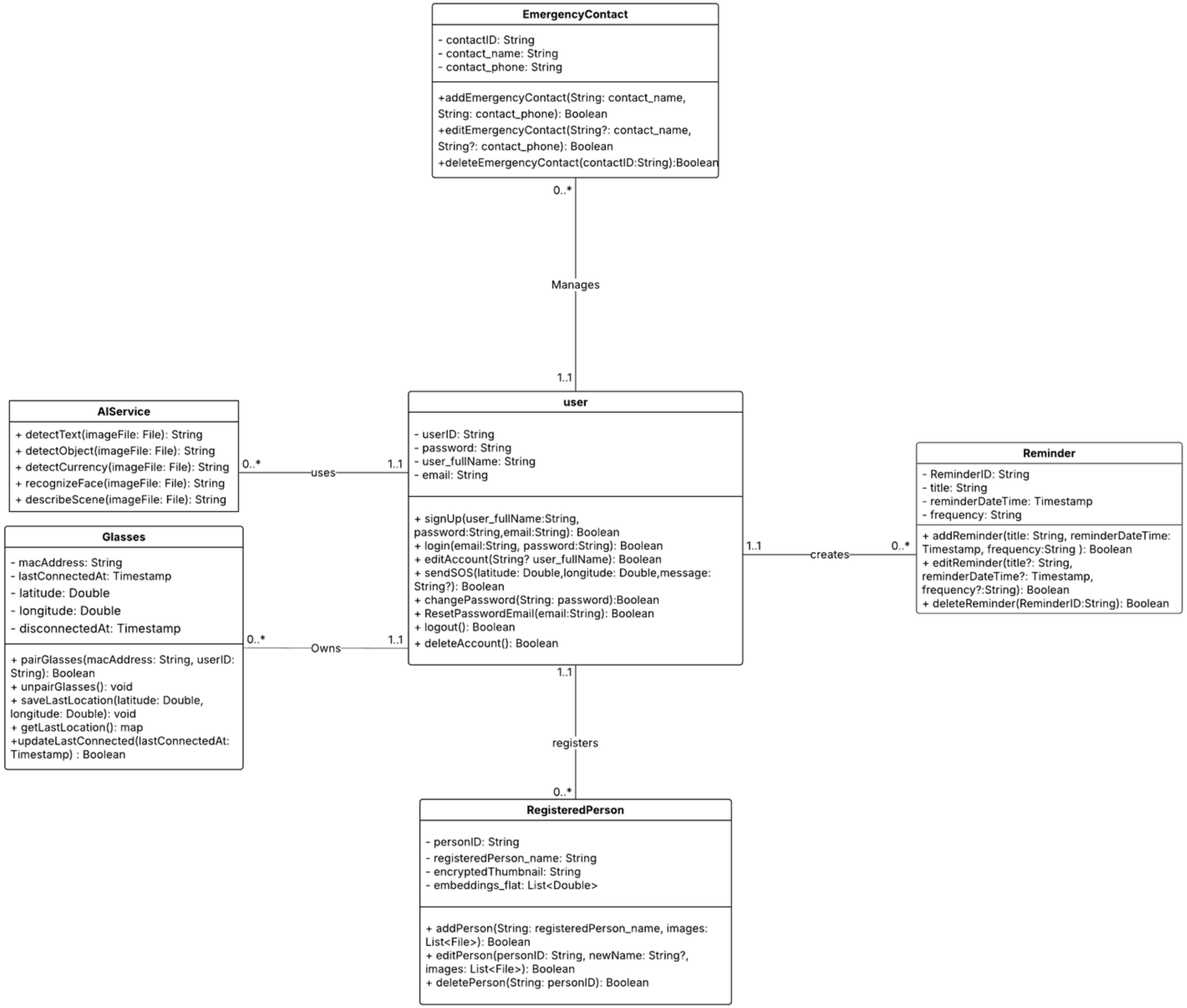


Figure 9: Class Diagram

4.3.1 Component Level Design

In this section illustrates the implementation of the key components of the **Munir** through detailed activity diagrams. These diagrams visualize how different modules interact to achieve the app's overall functionality, highlighting user interactions, system responses, and data flow across major processes. This representation provides a comprehensive understanding of the system's dynamic behavior and operational logic.

- **Activity Diagram of Face Recognition and Add Person Process in Face Management**

The activity diagram illustrated in [**Figure 10**] represents the flow of actions in the Face Recognition and Add Person processes within the Face Management module of the **Munir** app. Upon successfully logging in, the user accesses the homepage and navigates to the Face Recognition section.

From there, the process branches into two main paths:

- **Identify Person:** The user is redirected to the camera for facial identification. If the person is recognized, their name is displayed on the screen and spoken aloud for accessibility. If not recognized, an alert and voice prompt notify the user to try again.

- **Face Management:** The user is redirected to the Add New Person page, where a name and photo are required. Photos may be provided through two methods: **Uploading** three photos (front, left, and right) in any order, or using the **Take Photo** feature, which guides the user through capturing each angle sequentially. Any incorrect direction during capture triggers an error message. If any required field is missing, the submission button remains disabled. Once all fields are complete, the person is registered and the user is redirected to the Face Management page.

This diagram illustrates the interaction flow between the user and the system, ensuring a smooth and accessible experience for blind and visually impaired users during both recognition and registration processes.

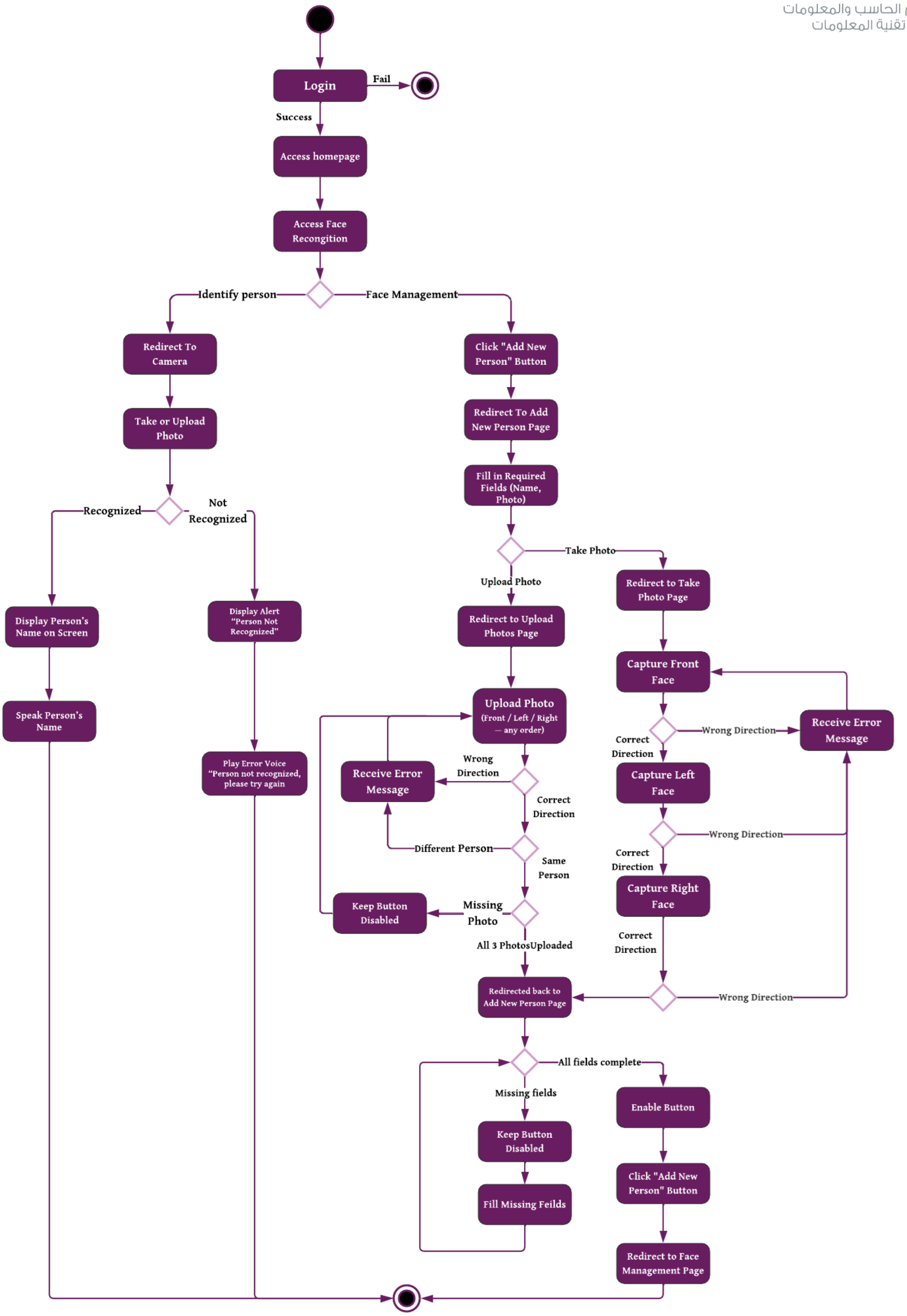


Figure 10: Activity Diagram of Face Recognition and Add Person

- **Activity Diagram of Emergency SOS Process**

The activity diagram illustrated in [**Figure 11**] outlines the flow of actions involved in triggering the Emergency SOS feature in the **Munir** app. After successfully logging in, the user accesses the homepage and navigates to the Emergency SOS section.

When the user clicks the "SOS" button, the system checks whether any emergency contacts have been added:

- **If contacts exist**, the system immediately sends an SOS message containing the user's location to all registered contacts.
- **If no contacts** exist, the user is redirected to the Emergency Contacts page, where a new contact must be added before proceeding. The system offers two distinct methods for adding contact: Add by Voice and Add Manually.

In the voice input pathway, the user activates voice input, and the system sequentially prompts for the contact's name followed by the phone number. If the voice input is not captured at any stage, the system re-prompts accordingly. Should the phone number fail to conform to the required format, the system permits up to three re-entry attempts. If the correct format is not provided within the permitted attempts, the process is terminated and the contact is not saved.

In the manual input pathway, the user is redirected to the Add Contact page, where the required fields (name and phone number) must be completed. If any fields remain incomplete, the submission button is kept disabled, and the user is prompted to fill in the missing information. Once all fields are completed, the button becomes enabled and the user may proceed to submit.

Upon successful contact addition through either pathway, the contact is saved, and the user is **redirected to the Emergency SOS** page. The system then redirects the user to the **SMS application**, where they are required to tap the Send button to dispatch the SOS message to all registered contacts.

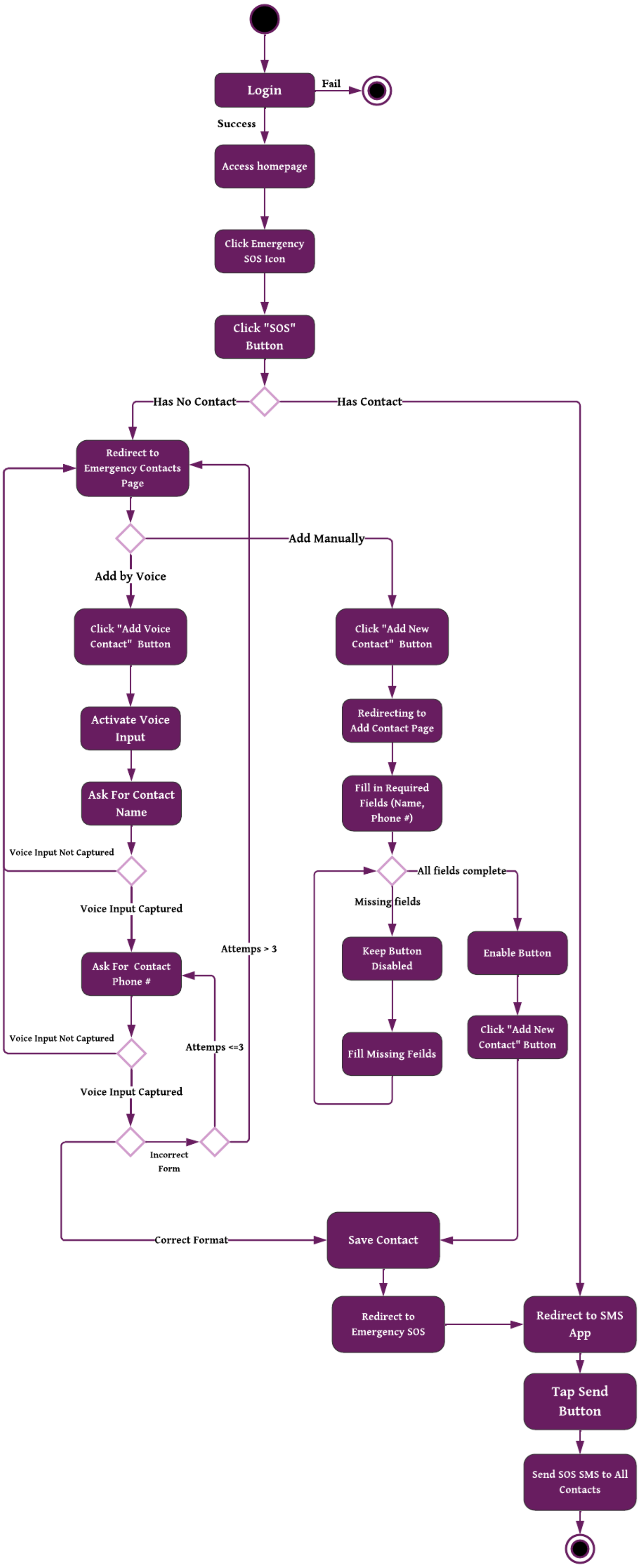


Figure 11: Activity Diagram of Emergency SOS Process

- **Activity Diagram of Edit Emergency Contact Process**

The activity diagram illustrated in [**Figure 12**] describes the flow of actions involved in editing an existing emergency contact within the Emergency SOS module of the **Munir** app. After successfully logging in, the user gains access to the homepage and navigates to the Emergency Contacts section.

Upon selecting the Edit icon associated with the desired contact, the system presents two distinct editing pathways: **Edit by Voice** and **Edit Manually**.

In the manual editing pathway, the user is redirected to the Edit Contact page, where the contact's name and phone number may be modified. The Save Changes button remains disabled unless at least one field has been altered. Once a modification is made, the button becomes enabled, and upon submission, the user is redirected to the Emergency Contacts page, where the contact is successfully updated.

In the voice input pathway, the system activates voice input and sequentially prompts the user to provide the contact's name followed by the phone number. If the voice input is not captured at any prompt, the user may either retry or cancel the operation. Should the phone number fail to meet the required format, the system permits up to three attempts before terminating the process. Upon successful capture of both fields within the permitted attempts, the contact information is updated accordingly.

If the user **cancels at any stage** throughout either pathway, the system returns to the Emergency Contacts page without applying any modifications. This diagram reflects a structured and accessible interaction flow designed to accommodate the needs of blind and visually impaired users, ensuring both data integrity and ease of use.

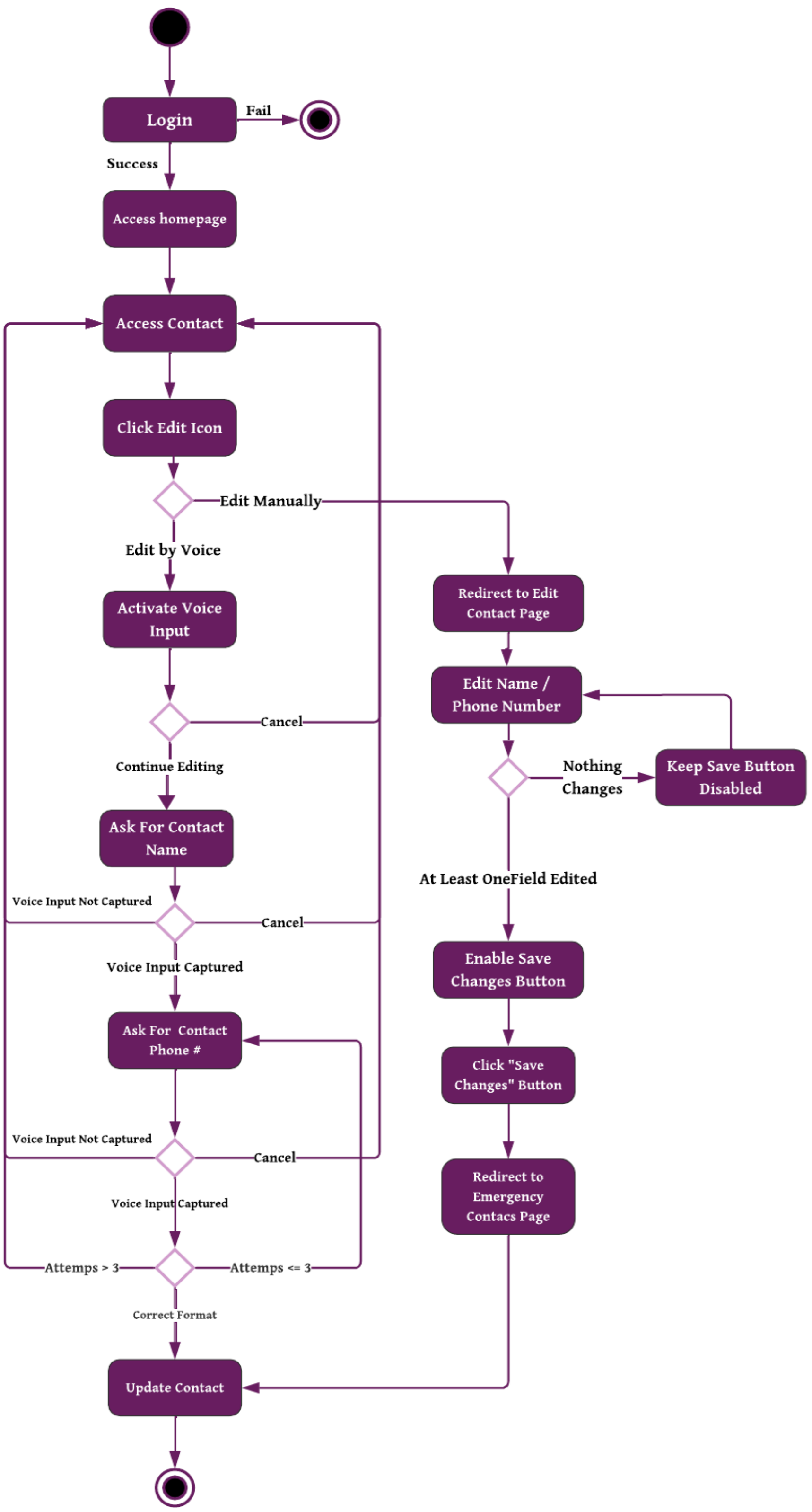


Figure 12: Activity Diagram of Edit Emergency Contact Process

- **Activity Diagram of Smart Glasses Settings Process**

The activity diagram illustrated in [**Figure 13**] represents the sequence of actions involved in the Smart Glasses settings within the **Munir** application. Upon successful authentication, the user navigates to the Settings section and selects the Smart Glasses option, where the process branches into two primary pathways.

View Last Known Location: The system displays the last connected location and time. The user may then click the "Open in Google Maps" button, upon which they are redirected to Google Maps to view the location.

Connect to Smart Glasses: The user clicks the "Search for Glasses" button, and the system begins searching automatically. If a glass is found, the user selects it from the list and the connection is established, after which the Glassese Info Page is displayed showing the battery level and signal strength. If no glass is found, the user receives a voice message stating "No glasses found nearby, make sure your glasses are turned on", and the system resumes searching automatically.

If the glass is **still not detected**, the system continues searching silently until a glass is found, at which point the user is prompted to select it and proceed with the connection.

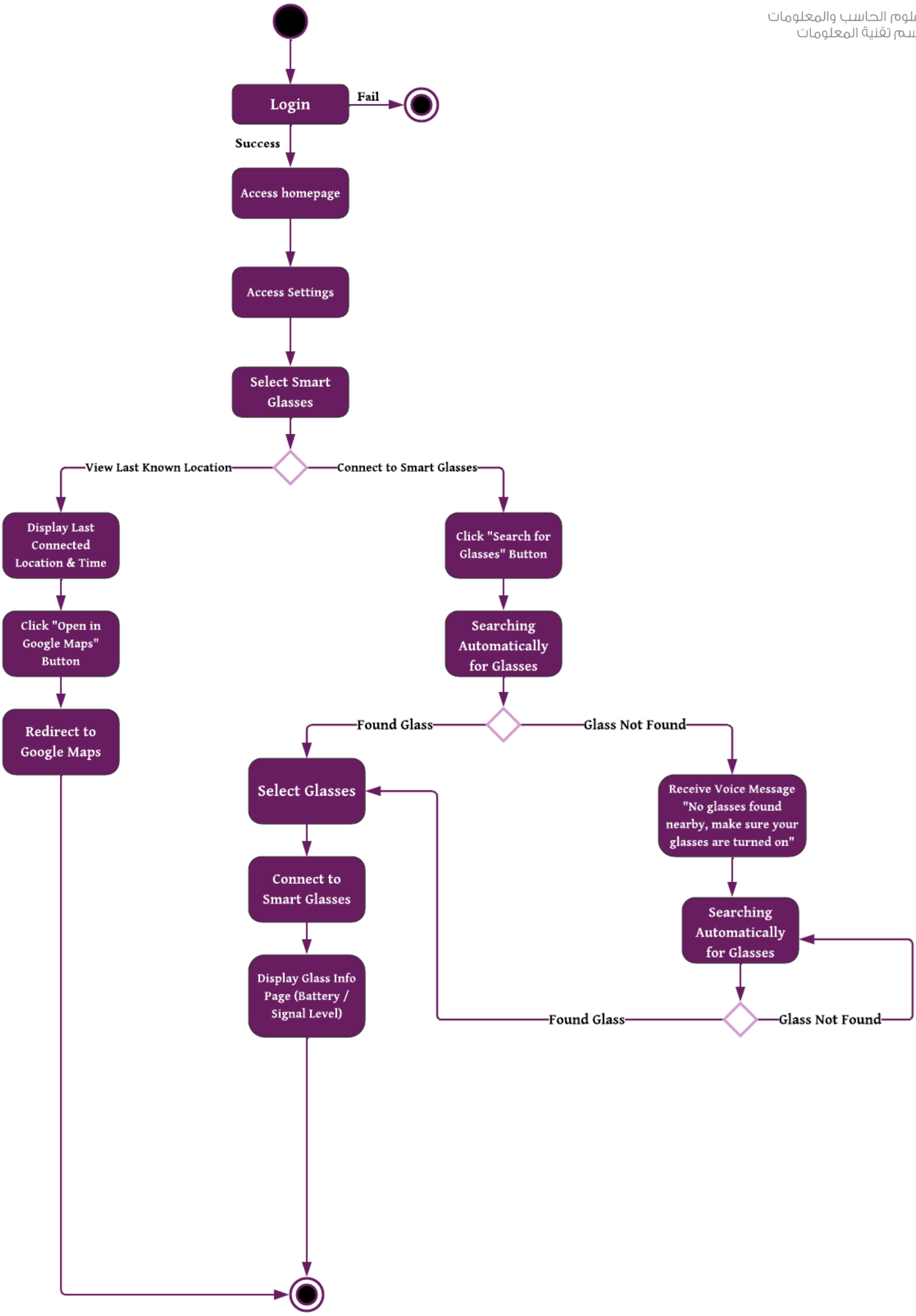


Figure 13: Activity Diagram of the Smart Glasses Connection process

4.4 Data Design

4.4.1 Data Models

In this section, we present the database design for **Munir**, including the Entity-Relationship (ER) diagram that defines the conceptual data model, the non-relational data model that illustrates the NoSQL structure.

A) ER Diagram

The ER diagram for **Munir**, as shown in [**Figure 14**], consists of six main entities: **User**, **EmergencyContact**, **Reminder**, **RegisteredPerson**, **Glasses,** and **LastLocation**. The **User** entity serves as the central component and stores essential information about each user, including their userID (primary key), full name, email, and password.

The **EmergencyContact** entity represents the user's emergency contact list, storing information about people who should be notified in case of emergencies or urgent situations. Each contact contains a unique contact ID (primary key), name, phone number. The relationship between User and Contact is a one-to-many association labelled "manages," where one user can maintain zero or more contacts (0..*), allowing users to build and manage their emergency contact list with multiple trusted individuals who can be reached when immediate assistance or notification is needed.

The **Reminder** entity stores scheduled alerts and notifications created by users to help them remember important tasks or events. Each reminder includes a reminder ID (primary key), title, reminder DateTime and frequency (one-time, daily, weekly, or monthly). The "creates" relationship between User and Reminder is one-to-many (0..*), allowing users to create and manage multiple reminders for different purposes.0

The **RegisteredPerson** entity represents individuals whose facial data has been enrolled in the system for facial recognition purposes. Each person has a personID (primary key), registeredPerson name and stores encryptedThumbnail for privacy and security, and embeddings flat as a list of doubles to support recognition accuracy. This entity has a one-to-many relationship (0..*) with the User entity through the "registers" association, indicating that each user can register multiple individuals whose faces the system should recognize.

The **Glasses** entity represents the smart glasses hardware paired with the user's account. Each glasses record contains a macAddress (primary key) and lastConnectedAt timestamp. The "owns" relationship between User and Glasses is one-to-many (0..*), meaning each user can own zero or more glasses devices, and each glasses device belongs to exactly one user. A user may use the **Munir** mobile application independently without pairing a glasses device. All of the core features remain fully accessible through the phone camera alone.

The **LastLocation** entity stores the last known GPS coordinates recorded when the glasses were disconnected. It contains Latitude, Longitude, and disconnectedAt timestamp. Since LastLocation cannot exist independently of a glasses device, it is a weak entity identified through its "has" relationship with Glasses. This relationship is one-to-zero-or-one (0..1), meaning a glasses device may have at most one last known location stored, recorded only after its first disconnection event.

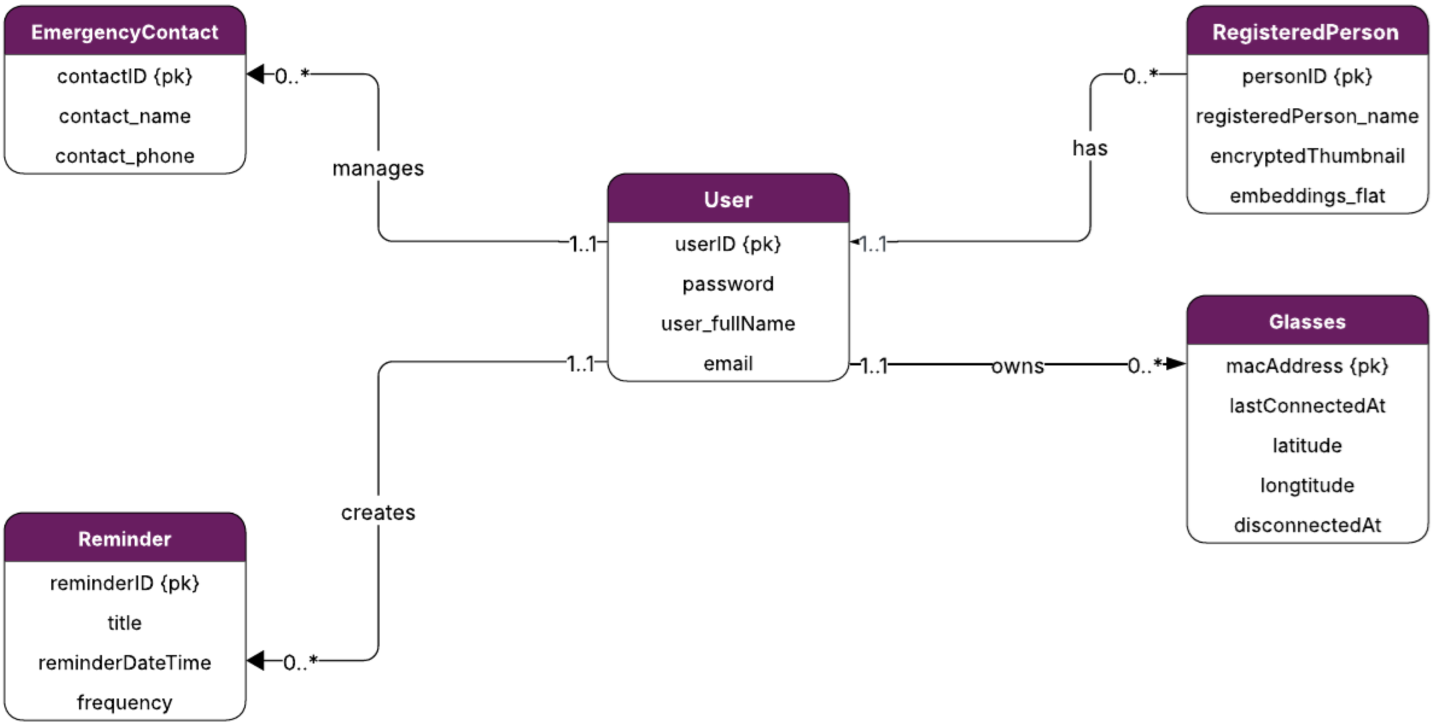


Figure 14: ER Diagram

B) Non-relational Data Model

The non-relational data model shown in [**Figure 15**] represents **Munir's** database structure and follows a reference-based design using sub-collections. Instead of embedding large objects into a single document, the model uses hierarchical parent-child relationships through linked sub-collections, allowing each part of the system to be queried, updated, or deleted independently without impacting the parent user document.

The **User** collection serves as the root, with four first-level sub-collections: **EmergencyContact**, **Reminder**, **RegisteredPerson, and Glasses**. EmergencyContact and Reminder are placed directly under the user, enabling emergency contacts and reminders to be retrieved either alone or alongside user information.

The **RegisteredPerson** sub-collection stores facial recognition data directly within each document, including the person's name, an encrypted thumbnail image, and a flattened face embedding vector. By embedding these fields directly the design allows face recognition data to be retrieved in a single read operation, which is essential for real-time identification performance.

The **Glasses** sub-collection stores the paired smart glasses device information, including the MAC address and last connected timestamp. It contains a nested **LastLocation** sub-collection that records the GPS coordinates and disconnection timestamp of the glasses device, forming a two-level hierarchy: User → Glasses → LastLocation. This structure allows location data to be stored and queried per device independently.

This design ensures that the database scales efficiently as users accumulate more contacts, reminders, registered faces, and paired devices over time, while maintaining consistent read and write performance across all collections.

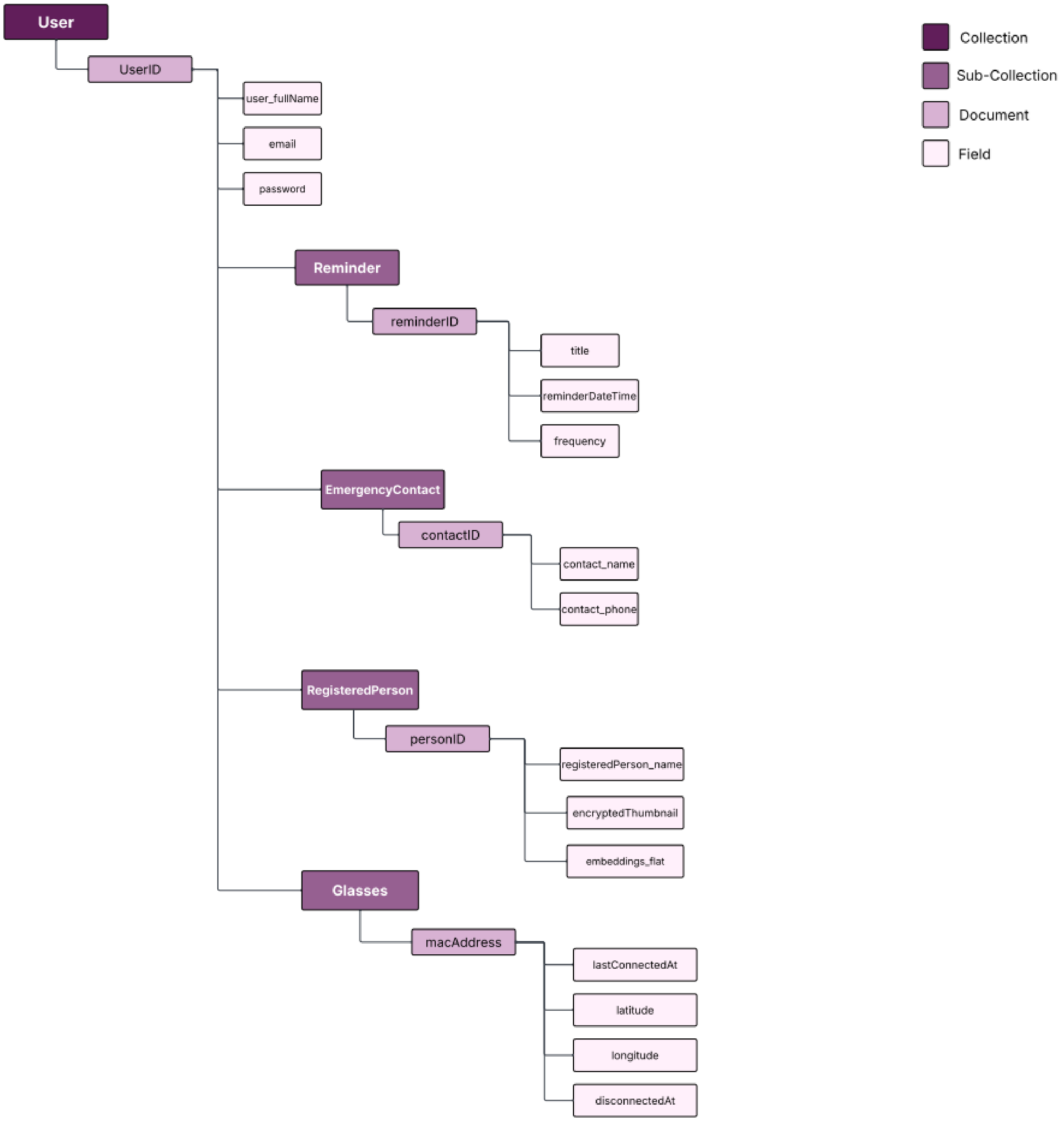


Figure 15: Non-relational Data Model

4.4.2 Data Collection and Preparation

In this section, we describe the datasets and preprocessing pipelines used to prepare training data for **Munir's** face recognition and currency identification models, including strategies to address class imbalance and ensure data quality.

- **Datasets:**

- **VGGFace2 Dataset:**

For face recognition model training, this research utilized the VGGFace2 dataset, a large-scale face recognition benchmark originally developed by the University of Oxford's Visual Geometry Group [38]. The full VGGFace2 collection contains over 3.3 million images across 9,000+ identities, capturing extensive variations in pose, age, illumination, ethnicity, and facial expressions. Due to its size, diversity, and quality, VGGFace2 has become one of the most widely adopted datasets in face recognition research, providing a rich and challenging resource for training and evaluating robust face recognition models capable of performing well in real-world scenarios.

For this research, a curated subset of VGGFace2 available through the Kaggle platform was utilized [7]. This Kaggle subset was selected over the full VGGFace2 dataset to achieve an optimal balance between research objectives and practical constraints. While the complete dataset offers greater scale, training on 3.3 million images across 9,000+ identities would require substantial computational resources and extended training periods beyond the scope of this academic project. The selected subset, containing 540 identities with approximately 198,000 images, provides sufficient diversity and scale to develop robust face recognition models while maintaining computational feasibility within the constraints of available hardware resources and project timeline. This configuration enables thorough experimentation, model comparison, and evaluation while retaining the essential characteristics of the full dataset, including diverse variations in pose, age, illumination, ethnicity, and facial expressions.

The dataset exhibits significant diversity in several key aspects. Images capture subjects across various age ranges, from young adults to elderly individuals, and represent multiple ethnic backgrounds, ensuring broad demographic coverage. Pose variation is extensive, including frontal views, profile shots, and various head orientations. Additionally, the dataset includes images taken under different lighting conditions, ranging from well-lit studio photographs to challenging low-light scenarios, as well as diverse backgrounds from controlled studio settings to cluttered real-world environments. Images are provided in JPEG format with varying resolutions, typically ranging from 224×224 to higher resolutions, reflecting real-world capture conditions. [**Figure 16**] illustrates sample images from the dataset, demonstrating the variety of individuals and visual conditions present in the data.

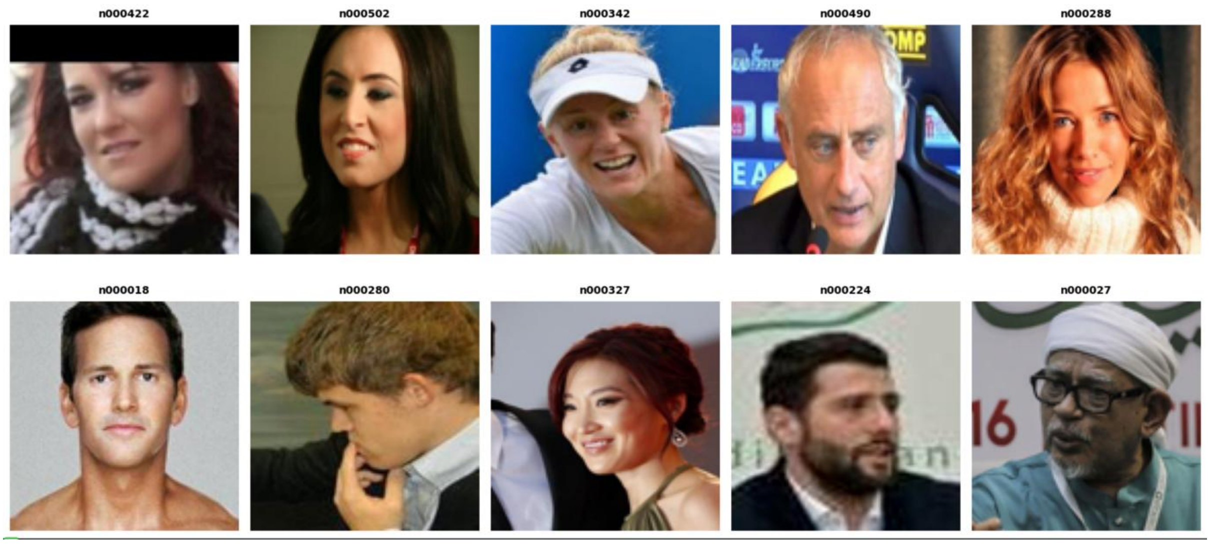


Figure 16: Sample images from VGGFace2 Dataset

The dataset is organized with a straightforward structure where each identity is represented by a folder containing all images belonging to that individual. Each folder is labeled with a unique identifier corresponding to a specific person, and the images within represent multiple captures of that identity under varying conditions. This organization facilitates efficient data loading and class-based training procedures.

The version obtained from Kaggle came in a pre-split configuration containing 540 unique identities. This distribution consisted of 480 classes in the training set with 176,706 images and 60 classes in the validation set with 21,404 images, representing an 89.2% to 10.8% train-validation split. [Table 5] summarizes the key statistics comparing the training and validation sets.

Table 5: VGGFace2 Dataset Statistics

Metric	Training Set	Validation Set
Number of Identities	480	60
Total Images	176,706	21,404
Avg. Images per Identity	368.1	356.7
Median Images per Identity	370	338
Min–Max Images per Identity	102–720	136–644
Standard Deviation	103.4	124.8

As shown in [Figure 17], the training set distribution exhibits considerable variation across identities. Each class in the training set contained an average of 368.1 images with a median of 370 and a standard deviation of 103.4, ranging from 102 to 720 images per class. Similarly, the validation set showed variation with an average of 356.7 images per identity, a median of 338, and a standard deviation of 124.8, ranging from 136 to 644 images. This substantial variation in the number of images per identity, with some classes having up to 7 times more images than others, indicates significant imbalance in the original dataset distribution.

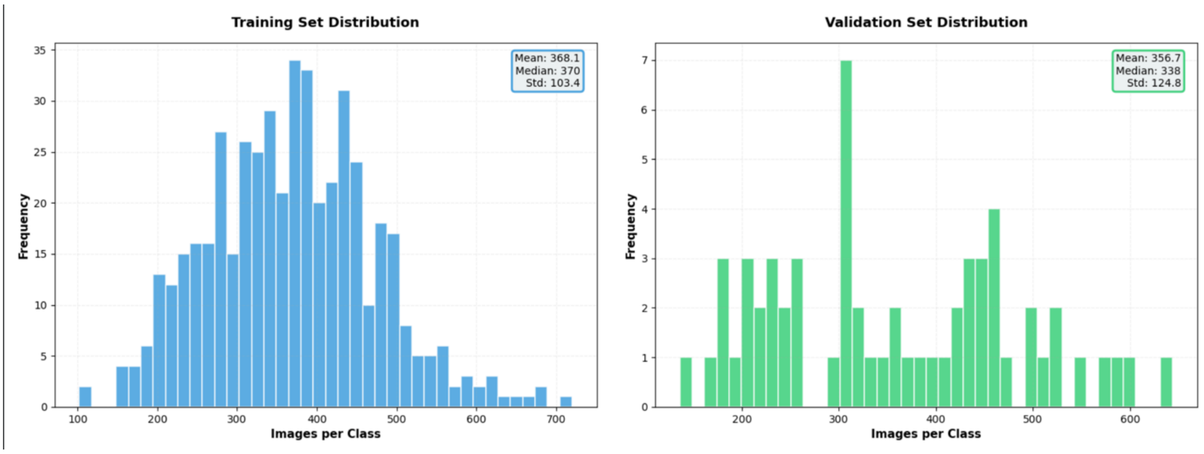


Figure 17: VGGFace2 Dataset Distribution Analysis

In summary, the VGGFace2 subset used in this work represents an appropriate choice for the research objectives, offering a diverse and challenging dataset suitable for developing robust face recognition models. The combination of 540 identities with substantial variation in pose, illumination, age, and ethnicity provides adequate complexity for training models that can generalize effectively to real-world face recognition scenarios, while the dataset's manageable scale ensures computational tractability within the constraints of this academic research.

- **Saudi Currency Dataset (AlFloos):**

The AlFloos dataset [39], created by Dr. Ghassan F. Bati from the Department of Computer and Network Engineering at Umm Al-Qura University and publicly available on the Kaggle repository [8], is a balanced and comprehensive dataset specifically designed for Saudi Arabian Currency identification research. It represents the first-ever dataset dedicated to the sixth issue of Saudi currency banknotes, which is the current series officially in circulation under the reign of King Salman.

The dataset was developed to address the significant gap in Arabic-language datasets for machine learning research and educational purposes, particularly given the prevalence of English-language resources and the relative scarcity of Arabic alternatives [39]. It contains images of Saudi banknotes across multiple denominations, including 5, 10, 50, 100, 200, and 500 Saudi Riyals. [**Figure 18**] presents representative samples from the AlFloos dataset, showcasing the visual characteristics of different Saudi Riyal denominations. The dataset was carefully curated to ensure balanced class distribution and high-quality labelling suitable for training robust Currency identification models.

Figure 18: Sample images from the AlFloos dataset showing different Saudi Riyal denominations

However, during our review of the dataset, we identified that the 20 Saudi Riyal denomination was not included in the original collection. To ensure comprehensive coverage of all currently circulating Saudi currency denominations, we manually collected and labeled additional images of the 20 Riyal note, incorporating them into the complete dataset for model development and evaluation. [**Figure 19**] presents sample images of the manually collected 20 SAR banknote, demonstrating the quality and variety of the images added to complete the dataset coverage.

3120x4160px

3120x4160px

Figure 19: Sample images of the manually collected 20 SAR banknote

Additionally, the AlFloos dataset originally contained two variants of the 5 Saudi Riyal note: cotton-based and polymer-based versions. To determine whether these variants should be treated as separate classes, we conducted a similarity analysis using deep feature extraction with MobileNetV2 and cosine similarity measurement. The analysis revealed a 98.67% similarity between the two variants, indicating that they are visually nearly identical from a machine learning perspective. Based on this finding, we selected the polymer variant for our dataset, as it represents the more commonly circulated version in current use. [**Figure 20**] illustrates the cosine similarity heatmap between the cotton-based and polymer-based 5 SR variants, demonstrating the high similarity score of 0.9867 between the two versions. This decision reduced potential classification confusion while maintaining comprehensive coverage of Saudi currency denominations.

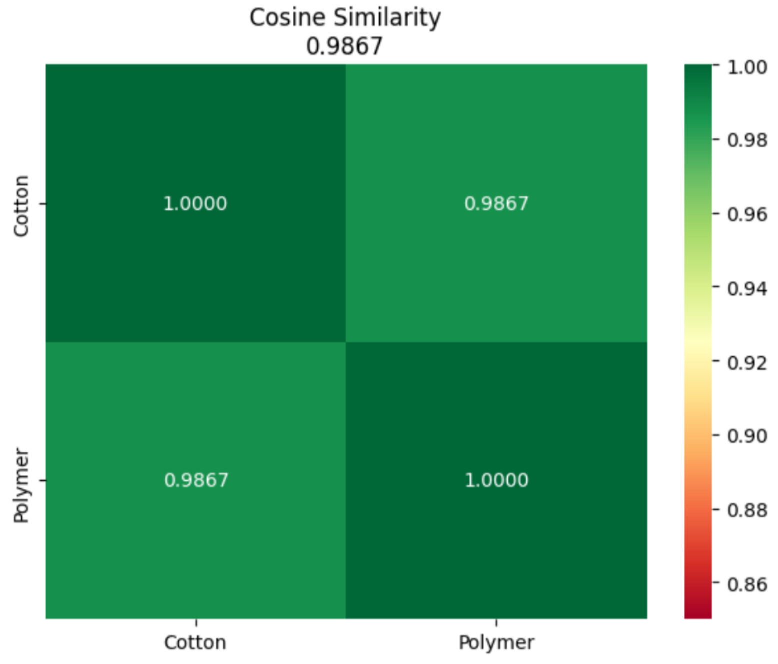


Figure 20 : Cosine similarity analysis between 5 SR cotton and polymer variants

AlFloos presents several inherent challenges due to its real-world diversity. Large intra-class variation exists as the same denomination may appear under different lighting conditions, camera angles, and levels of physical wear. Illumination and background variability further complicate recognition tasks, while the visual similarities between certain denominations require models to learn subtle distinguishing features. These characteristics make the dataset particularly suitable for developing and evaluating robust Currency identification systems in real-world scenarios encountered by blind and visually impaired users.

Due to its balanced structure, real-world diversity, and focus on an underrepresented currency system, AlFloos has proven valuable for developing assistive technology applications for Arabic-speaking communities, providing a practical and challenging resource for exploring the performance of different classification algorithms in Currency identification tasks. [**Figure 21**] presents the class distribution across all Saudi Riyal denominations in the final dataset, demonstrating the balanced representation of each currency class used for model training and evaluation.

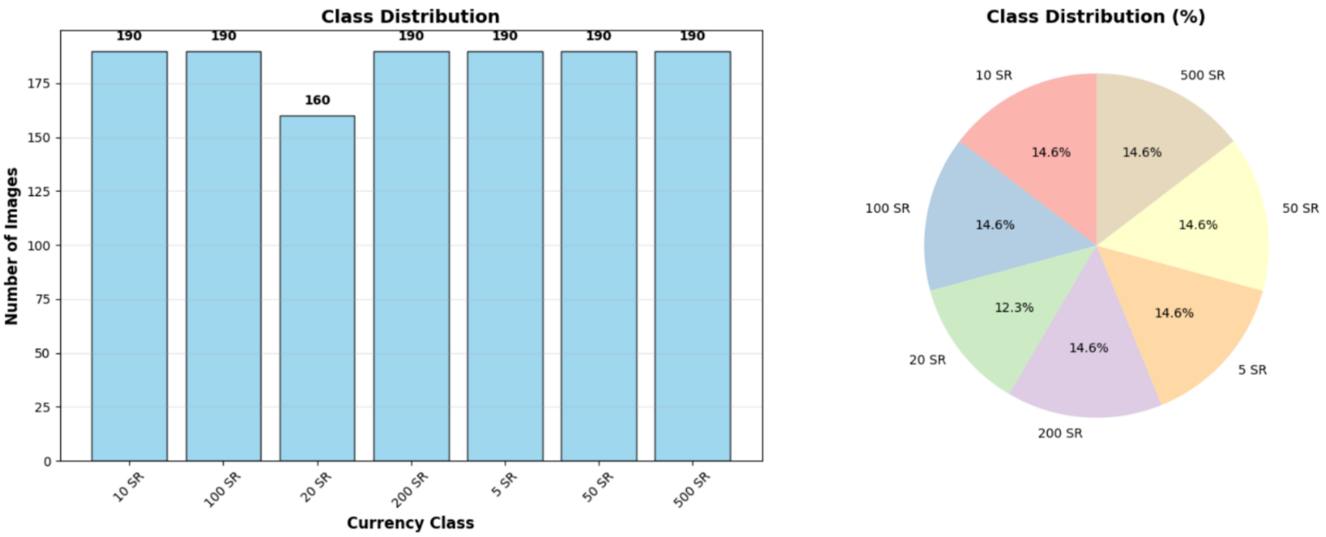


Figure 21: Class distribution for Saudi Riyals denominations

- ## Data preprocessing:

Data preprocessing is a fundamental stage in preparing visual datasets for machine learning models, ensuring that all images are clean, standardized, and properly formatted for model input. In computer vision applications, particularly face recognition systems, preprocessing plays a critical role in reducing noise, normalizing variations, and enhancing feature consistency across the dataset [40]. Effective preprocessing directly impacts model performance by ensuring that the learning algorithm receives high-quality, uniform data that facilitates robust feature extraction and generalization [41].

- **Face Recognition:**

Dataset Collection and Standardization

To ensure computational efficiency and consistent class representation across all registered individuals, a systematic collection and preprocessing pipeline standardizes the dataset by selecting a fixed number of images per person. Research in face recognition has established varying optimal sample sizes depending on dataset characteristics: benchmark datasets such as AT&T (ORL) utilize 10 images per person [42], while CMU PIE experiments have employed 25-60 images per person [43]. For this study, 60 images per person was selected based on three key considerations: sufficient sample diversity to capture variations in pose, lighting, and expression; balanced class distribution to prevent model bias toward over-represented individuals [44]; and computational tractability within available hardware constraints while maximizing representation quality for the face recognition model.

The preprocessing pipeline implements a six-step standardization process to ensure uniformity and eliminate redundancy across the entire facial image dataset:

Step 1: Duplicate Prevention - Duplicate file paths across source directories are eliminated using set-based deduplication, and file existence checks prevent re-processing of already-standardized images. This ensures each facial image appears exactly once in the final dataset, maintaining data integrity and preventing redundant samples that could artificially inflate class representation.

Step 2: Image Collection - Images are gathered from source directories containing varying numbers of samples per individual. The system collects exactly 60 images per person through random selection from the deduplicated image pool, ensuring diverse representation while maintaining fixed class sizes across all 540 registered individuals.

Step 3: Corruption Detection - Each image file is validated to verify it can be properly opened and decoded. Files failing validation are identified as corrupted and excluded from the dataset. This step prevents runtime errors during subsequent preprocessing and model training phases.

Step 4: Color Space Standardization - All facial images are converted to RGB (3-channel) format. Grayscale images are expanded to three channels through channel replication, while RGBA images have their alpha (transparency) channel removed. This ensures uniform color representation across the entire dataset.

Step 5: Unified Resizing - Images are resized to 224×224 pixels using Lanczos interpolation, which provides high-quality resampling while preserving facial features critical for recognition.

Step 6: Format Optimization - All facial images are saved as JPEG with 95% quality compression, reducing storage from approximately 22 GB (PNG format) to 2.8 GB—an 87% reduction—while maintaining perceptually lossless quality [45].

Result: The preprocessing pipeline successfully processed all 540 person classes, collecting and standardizing 32,400 facial images (540 classes × 60 images). Zero duplicate images were included in the final dataset due to the multi-level deduplication mechanism. Zero corrupt files were detected, and zero processing errors occurred. All images conform to uniform specifications: 224×224 pixels, RGB color space, JPEG format with 95% quality. The resulting dataset maintains perfect class balance with exactly 60 images per person, eliminating potential bias from unequal class representation.

- **Currency Identification:**

The data preprocessing pipeline was designed to ensure data integrity, standardization, and reproducibility. It consisted of four sequential stages: (1) duplicate detection, (2) data cleaning, (3) train–test splitting, and (4) data augmentation. Each stage contributed to improving dataset quality and preparing it for reliable model training and evaluation.

1) Duplicate Detection

To prevent data leakage and redundant learning, duplicate detection was performed to identify visually similar images that may exist across the dataset. Two images were considered duplicate if they exhibited more than 92% visual similarity, even after compression or minor modifications.

Method: Duplicate detection was conducted using perceptual hashing (pHash) with a similarity threshold of 92%.

Result: A total of 44 duplicate or near-duplicate images were identified and removed, reducing the dataset from 1,300 to 1,256 unique images. Class-level duplicate rates ranged from 0% (20 SR) to 5.26% (5 SR), with an overall rate of 3.38%. This cleaning step effectively prevented redundancy that could inflate accuracy metrics or cause overlap between the training and test sets.

2) Data Cleaning

A four-step cleaning pipeline was implemented to ensure all images were valid, standardized, and compatible with downstream preprocessing requirements.

Step 1: Corrupt File Removal - Each image was verified for proper readability. Corrupted or partially loaded files were removed to prevent training interruptions.

Step 2: HEIC/HEIF Conversion - Images in Apple's HEIC/HEIF formats were converted to JPEG for universal compatibility.

Step 3: Color Space Standardization - All images were converted to RGB (3-channel) format to align with the expected input for MobileNetV2.

Step 4: Filename Conflict Resolution - Files with duplicate names were automatically renamed with unique identifiers to prevent overwriting.

Result: All 1,256 images passed validation, with zero corrupt files detected after cleaning.

3) Train-Test Split

The cleaned dataset was divided using stratified random sampling [46] to maintain proportional representation across classes.

- **Training Set:** 70% (876 images)
- **Test Set:** 30% (380 images)

A fixed random seed (42) ensured reproducibility [47]. Stratification was critical to prevent class imbalance and bias in performance evaluation. [**Figure 22**] illustrates the stratified train-test split, demonstrating the proportional class distribution maintained across both subsets.

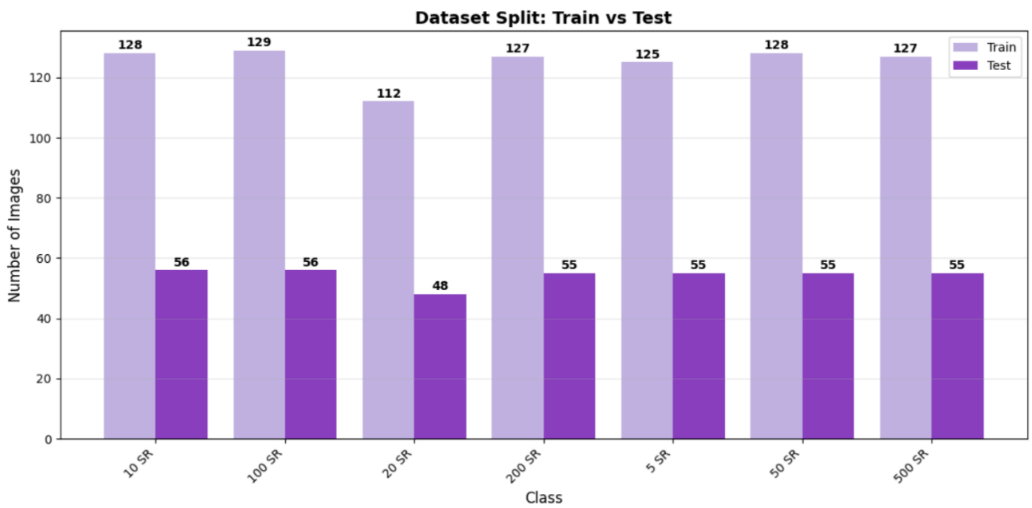


Figure 22:Train-Test split visualization showing stratified class distribution

4) Data Augmentation

Data augmentation was applied exclusively to the **training set** to (1) address class imbalance and (2) enhance model generalization for real-world currency identification under variable capture conditions.

Visually impaired users often capture currency images under unpredictable conditions such as:

- Lighting variability: indoor/outdoor light, shadows, low illumination
- Camera angles: non-frontal or tilted orientations
- Distance and framing inconsistencies
- Orientation diversity: rotated or flipped banknotes

To counter these challenges, augmentation simulated realistic variations to make the model invariant to these conditions. Prior works (Shorten & Khoshgoftaar [48]; Buda et al. [49]) demonstrated that augmentation improves performance on limited and imbalanced datasets by introducing label-preserving transformations.

(a) Applied Techniques

Six transformation types were applied probabilistically, each addressing a specific real-world variation

Table 6: Data Augmentation Techniques and Rationale

Technique	Parameter	Real-World Scenario Simulated	Reference
Random Rotation	±12° to ±25°	Tilted camera angles during non-perpendicular capture by users with limited vision	[48]
Horizontal/Vertical Flip	Probability = 0.5	Different banknote orientations and placement directions (face-up/down, left-to-right)	
Color Jitter	Brightness ±8-15%, Contrast ±8-12%	Indoor/outdoor lighting variations, different times of day, shadow conditions	
Random Affine	Translate ±10-12%, Scale 85-115%, Shear ±8-10°	Off-center framing, varying capture distances, slight hand movements	
Perspective Transform	Scale (0.02, 0.10)	Non-frontal viewing angles when camera planes are not parallel to banknote surface	[50]
Gaussian Noise	var_limit (5.0, 18.0)	Smartphone camera sensor noise, particularly in low-light conditions	[48]

These transformations are widely adopted in computer vision literature for realistic simulation of environmental and geometric variation. All images were resized to **224×224 pixels** (ImageNet standard) prior to feature extraction.

(b) Target Configuration

Given 876 training images across seven classes (112–129 per class), augmentation expanded each class to **350 samples**, yielding an average **2.8× increase**. This target ensured balanced representation, diversity, and computational efficiency without overfitting.

(c) Key Findings

1. **Configuration Notes:** The augmentation pipeline applied 15 transformation combinations, enhancing exposure to diverse conditions.
2. **Selected Setup (350 samples/class):** Achieved the **lowest generalization gap (1.84%)** and a **CV standard deviation ± 0.29%**, demonstrating high stability.
3. **Design Rationale:** A 2.8× ratio provided sufficient diversity while maintaining feasible training time (24.48 s per fold).

Results: The final augmented dataset achieved complete class balance. [Figure 23] presents representative examples of augmented images, demonstrating the variety of transformations applied to simulate real-world capture conditions. [Table 7] summarizes the augmentation results for each class.

Figure 23: Sample augmentation results

Table 7: Data Augmentation Results and Final Dataset Statistics

Currency Class	Original Images	Additional Images Generated	Augmented Images (Total)	Augmentation Ratio	Test Set (Unmodified)
5 SR	125	+225	350	2.80×	55
10 SR	128	+222	350	2.73×	56
20 SR	112	+238	350	3.13×	48
50 SR	128	+222	350	2.73×	55
100 SR	129	+221	350	2.71×	56
200 SR	127	+223	350	2.76×	55
500 SR	127	+223	350	2.76×	55
Total	**876**	**+1,571**	**2,447**	**2.79×**	**380**

Total: 876 → 2,447 training images (2.79× increase); test set = 380 (unaugmented).

5) Preprocessing Summary

The complete preprocessing pipeline successfully transformed the original dataset (1,256 images after duplicate removal, imbalance ratio 1.16) into a balanced, high-quality dataset suitable for training robust Currency identification models:

- **Total processed:** 2,827 images
- **Training set:** 2,447 images (350 per class)
- **Test set:** 380 images (original, unaugmented)
- **Duplicates removed:** 44 (3.38%)
- **Zero corrupt files:** after validation
- **All standardized to:** 224×224 RGB format

This finalized dataset was subsequently used for feature extraction via MobileNetV2, generating 1,280-dimensional feature vectors for classifier training.

4.5 Interface Design

In this section, we present the interface design for **Munir**, including the navigation flow diagram that outlines the main pages and subpages, and the UX guidelines that ensure accessibility, consistency, and usability for visually impaired users.

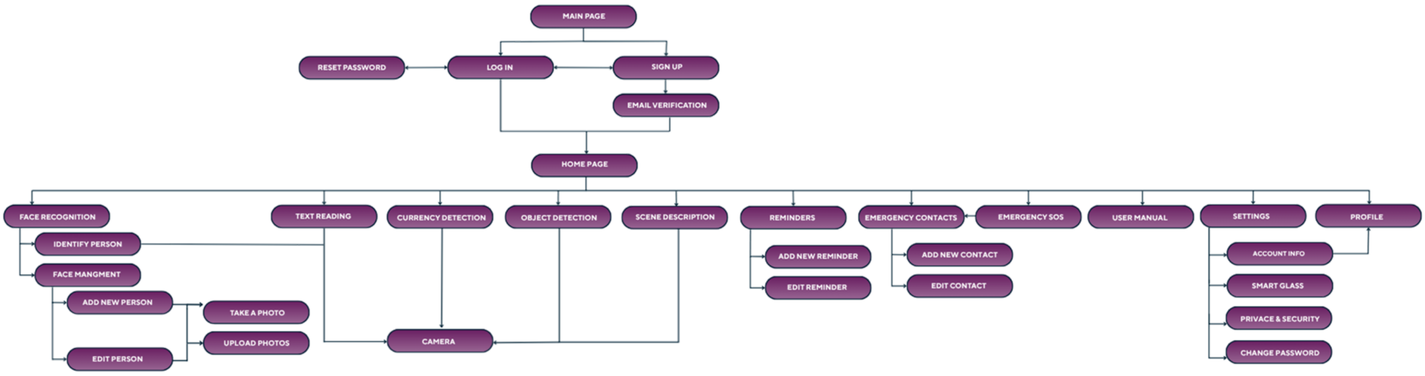


Figure 24: Navigation Flow Diagram

- ### UX guidelines

1- **Consistency:** is a fundamental UX principle that aims to provide uniform and predictable user experience across the entire application. It ensures the alignment of visual elements, interaction behaviors, terminology, and design patterns throughout the interface. Such coherence enhances user familiarity, reduces cognitive effort, and contributes to a more reliable and intuitive interaction experience [51].

- **Visual Consistency:** Consistent colors, typography, icons, and layout structures were applied across both the sign-up and login interfaces to maintain a unified visual identity and improve usability. This consistency enhances user familiarity and supports a seamless navigation experience, as illustrated in Figures [24 & 25] and Figure [26].

Figure 25: Visual Consistency 1

Figure 26: Continue Visual Consistency 1

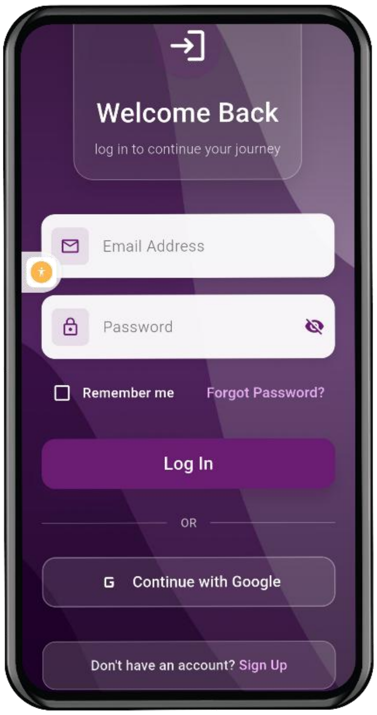


Figure 27: Visual Consistency 2

- **Functional Consistency:** The application maintains functional consistency by using the same voice-based interaction flow. For example, both the **Add Contact** and **Reminders** features follow an identical pattern in which the user provides information verbally, the system validates the input and requests correction when necessary and then completes the action. This consistent behavior supports intuitive learning and ensures a seamless experience for blind and visually impaired users, as illustrated in Figures [28 & 29] and Figures [30 & 31].

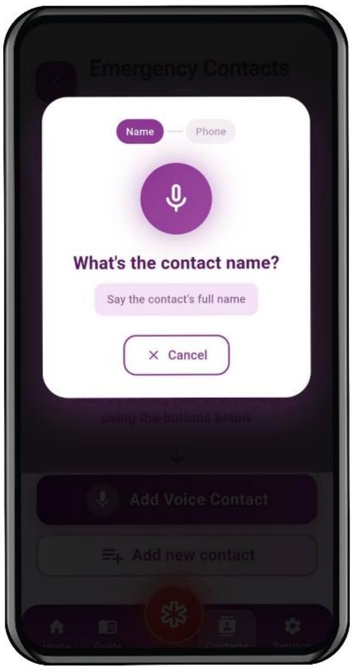


Figure 28: Functional Consistency1

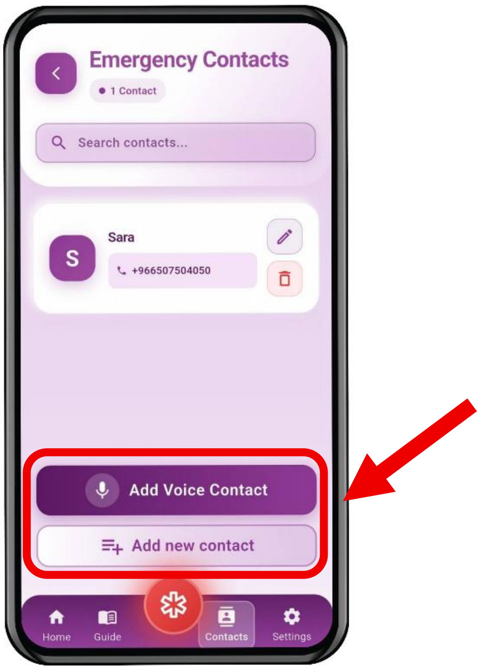


Figure 29: Continue Functional Consistency1

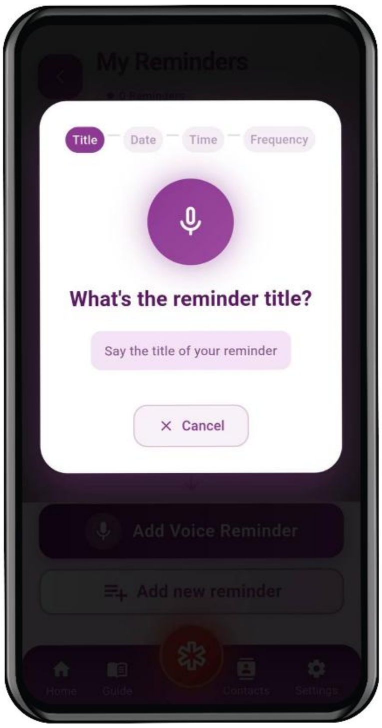


Figure 30: Functional Consistency2

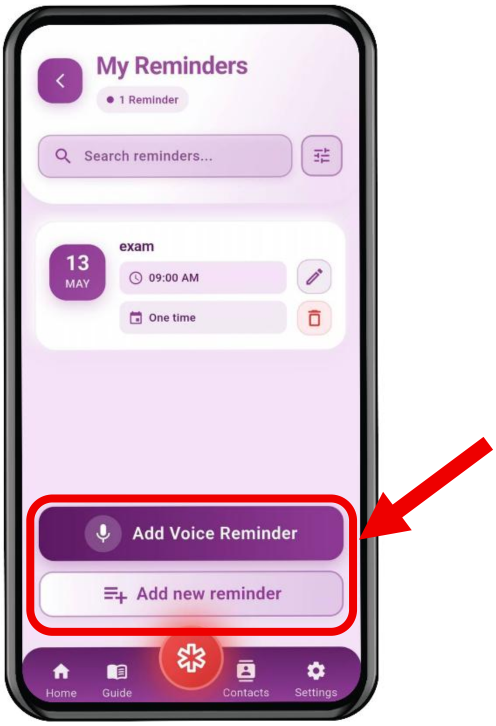


Figure 31: Continue Functional Consistency2

2- **Feedback:** Feedback is a core UX principle that ensures users receive clear, timely, and meaningful responses to their actions within the application. Its purpose is to keep users informed, reduce uncertainty, and reinforce confidence by providing system responses that align with user expectations and interaction context [52]

- **Immediate feedback:** The system provides real-time validation. As soon as the user begins typing in the input fields, error messages are displayed immediately and spoken aloud to inform the user of the incorrect input and guide them toward the correct format, as illustrated in [**Figure 32**].

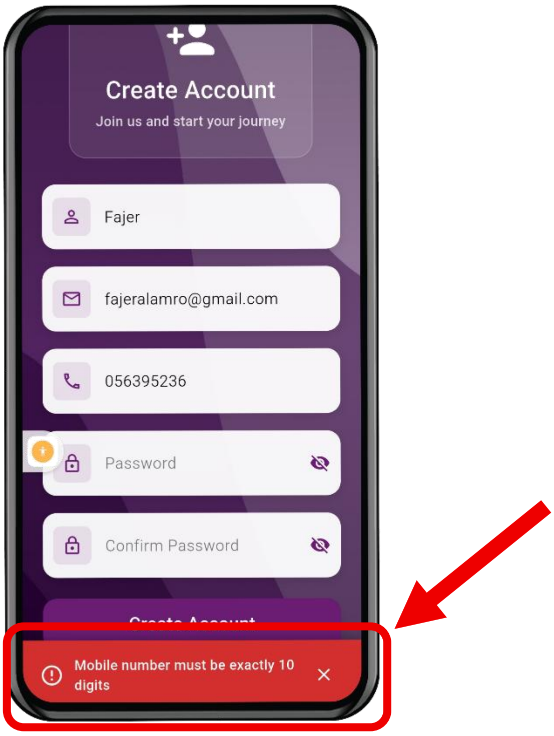


Figure 32: Immediate Feedback

- **Error feedback:** When an error occurs, the system provides a clear visual error message along with a spoken alert to help the user understand what went wrong and how to correct it, as illustrated in [Figure 33].

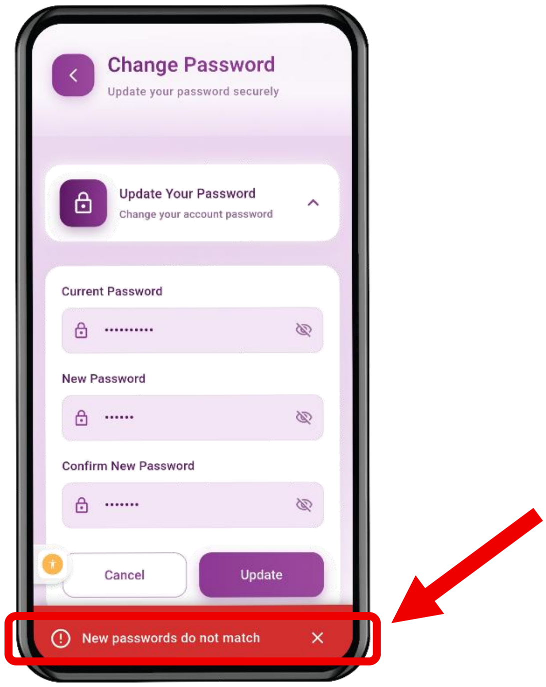


Figure 33: Error Feedback

3- **Robustness:** is a UX principle that focuses on designing an application capable of handling diverse user inputs and unexpected scenarios without malfunctioning. It ensures that the system remains stable, functional, and reliable even under unusual conditions [53].

- **Reachability:** The "**Munir**" bottom navigation bar ensures users can consistently access and switch to their desired pages at any time, allowing seamless navigation across essential features, as shown in [**Figure 34**].

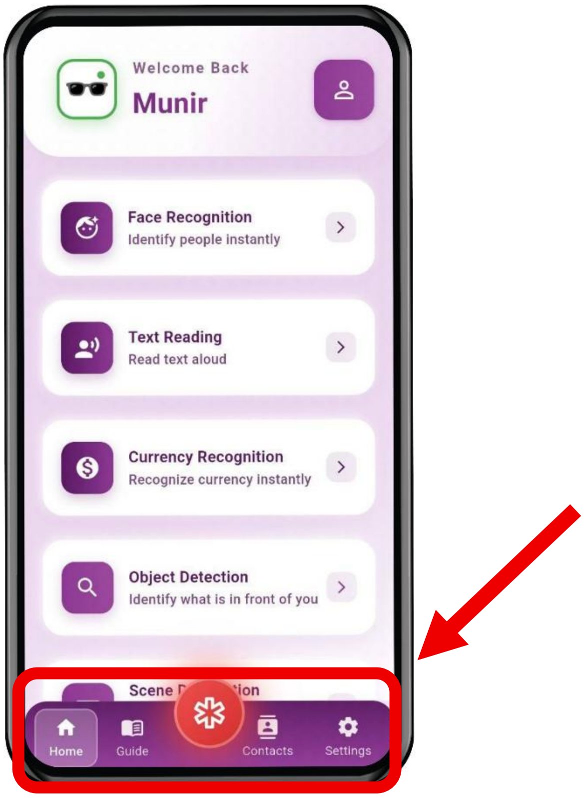


Figure 34: Reachability

- **Recoverability:** "**Munir**" ensures that users can easily correct mistakes without disrupting their progress. For example, on the sign-up page, the system provides real-time validation and spoken feedback, allowing users to identify and fix errors immediately rather than after submitting the form, as illustrated in [**Figure 35**].

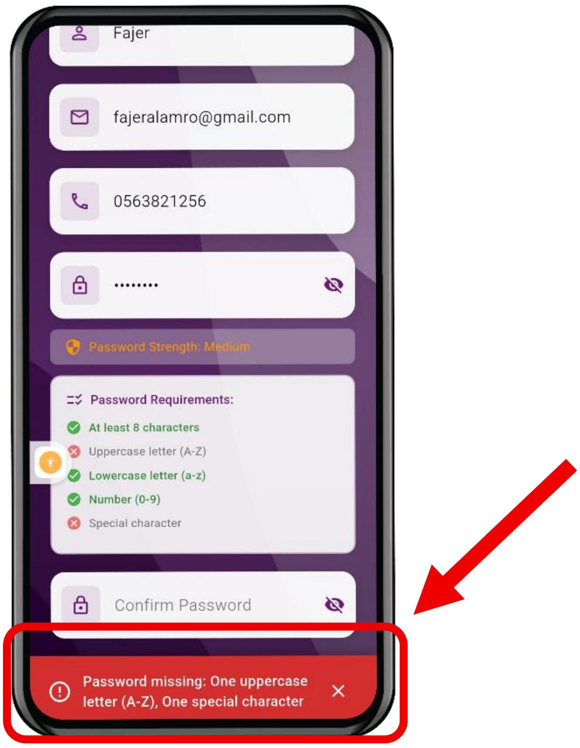


Figure 35: Recoverability

4- **Learnability:** is a UX principle that focuses on making an application easy for users to understand and operate, especially on their first use. It ensures that users can quickly grasp the core functions and features of the interface with minimal effort [54].

- **Predictability:** The system allows users to anticipate the outcome of their actions. When the user taps the Text Reading feature, the system activates the camera to capture the ticket and read it aloud, as illustrated in [**Figure 36**] and [**Figure 37**].

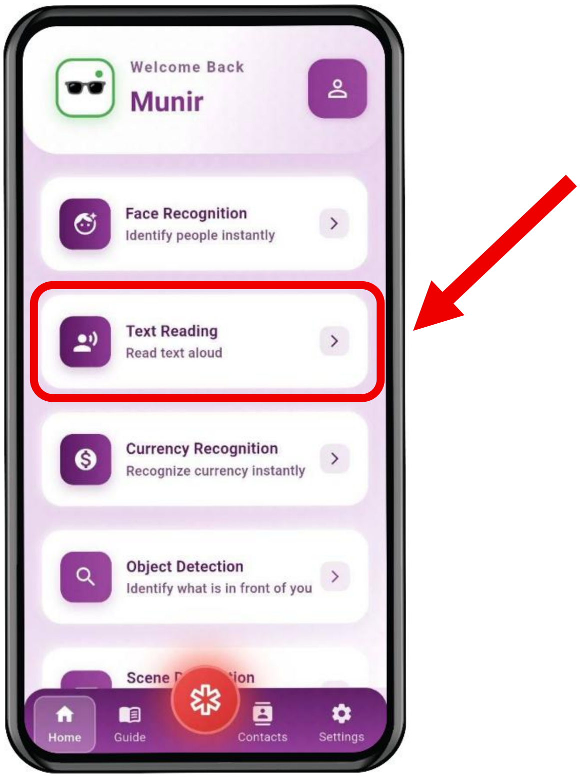


Figure 36: Predictability At Homepage

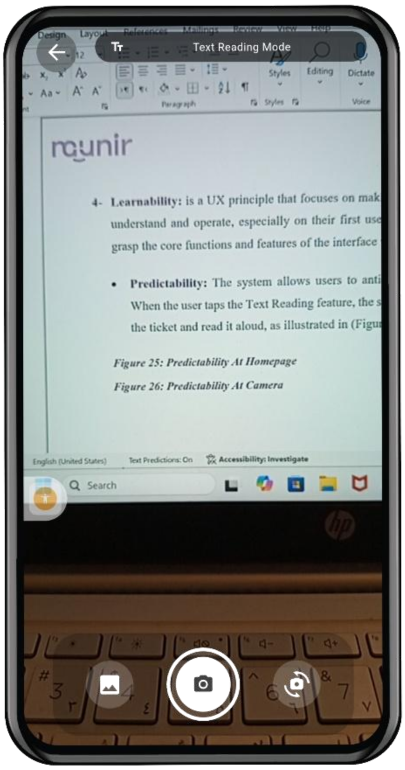


Figure 37: Predictability At Camera

5- **Discoverability:** is a UX principle that ensures users can easily identify and understand the features and actions available within the application, even without prior knowledge. In **Munir**, discoverability is particularly important to support blind and visually impaired users in exploring and recognizing interface functions through auditory cues rather than visual elements [55].

- **Auditory Awareness:** The system supports discoverability by announcing the name of each screen upon navigation (e.g., "Text Reading Page") and verbally identifying buttons when tapped (e.g., "Add Contact" or "Back"). This enables blind and visually impaired users to understand their current location in the app and recognize available actions through auditory cues.

6- Recognition Rather Than Recall: is a UX principle that emphasizes minimizing the user's memory load by making necessary information visible at the appropriate time. Instead of requiring users to remember how a feature works, the interface provides clear instructions and cues that guide them through the interaction [56].

- **Visible Guidance:** When the user activates the Voice Add feature, the system displays a concise set of instructions explaining how to use the voice input function effectively. The instructions appear in a modal overlay, ensuring that users can rely on presented information rather than recalling usage details from memory, as illustrated in [**Figure 38**].

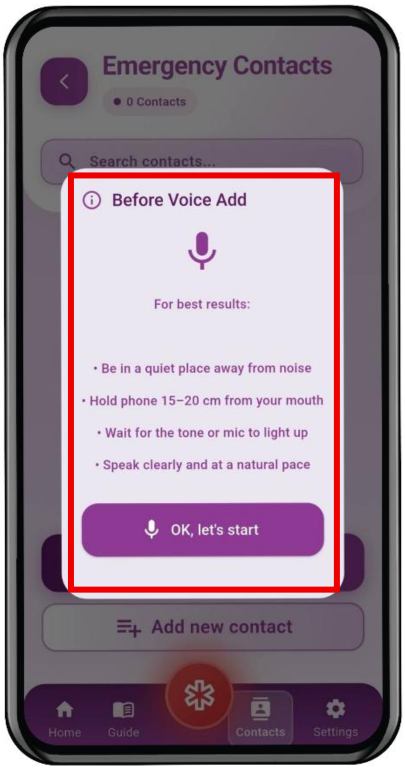


Figure 38: Visible Guidance

7- Visibility of System Status: is a UX principle that emphasizes keeping users informed about what is happening within the system through timely and clear feedback. The interface should continuously communicate the current status of processes and actions so that users always understand what the system is doing [57].

- **Real-Time Connection Feedback:** In the smart glass's connection screen, the application displays the current status (e.g., "Disconnected") and provides immediate feedback during device scanning. When the user taps "Search for Glasses," detected devices are listed with their names and signal strength. If no devices are found, the interface notifies the user and updates automatically when the glasses become available. This ensures that users are continuously informed about the connection progress and system state, as illustrated in [**Figure 39**] and [**Figure 40**].

Figure 39: Real-Time Connection Feedback

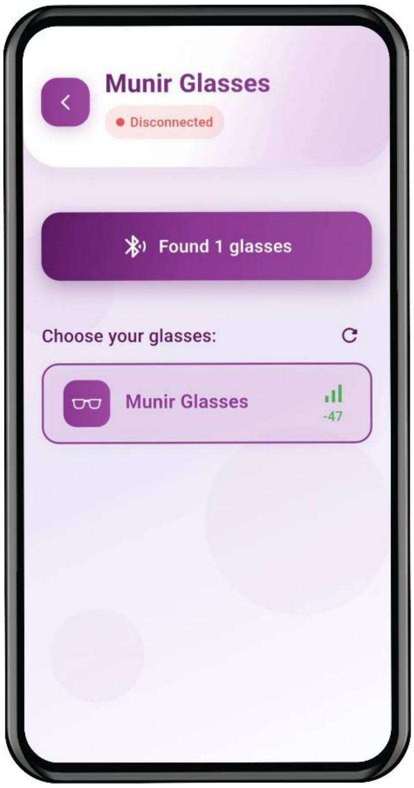


Figure 40: Continue Real-Time Connection Feedback

4.6 System Implementation

4.6.1 Third-Party Applications

A) Firebase Authentication Integration

This section explains how **Munir** integrates Firebase as a third-party Backend-as-a-Service. We use Firebase to manage authentication, user accounts, and data storage. Its ready-to-use APIs remove the need for building a custom backend, while still providing secure, scalable, and reliable user authentication for the application.

- **Firebase Configuration and Initialization**

Firebase requires platform-specific configuration files that contain the project credentials and service endpoints. For Android, the **(google-services. Json)** file is generated from the Firebase Console and placed in the appropriate project directory. After adding the configuration, the application initializes Firebase at startup [**Figure 41**]:

```
import 'package:flutter/material.dart';
import 'package:firebase_core/firebase_core.dart';
import 'package:firebase_app_check/firebase_app_check.dart';
import 'screens/splash_screen.dart';

Run | Debug | Profile
void main() async {
  WidgetsFlutterBinding.ensureInitialized();

  await Firebase.initializeApp();
```

Figure 41: Firebase Initialization in main. dart

This single initialization call establishes connections to all Firebase services (Authentication, Firestore, Cloud Functions) and configures the SDK with our project credentials.

- Core Dependencies

The signup implementation relies on four primary Firebase packages [**Figure 42**]:

```
dependencies:
  flutter:
    sdk: flutter
  firebase_auth: ^6.1.1
  cloud_firestore: ^6.0.3
  cloud_functions: ^6.0.3
  google_sign_in: ^6.2.1
  flutter_secure_storage: ^9.2.1
```

Figure 42: Firebase Packages Dependencies

- User Registration Flow

The signup process consists of three main steps: account creation, email verification, and profile storage.

- ### Step 1: Creating Firebase Authentication Account

When the user submits the registration form, we call Firebase Authentication API to create the account [**Figure 43**]:

```
void _registerUser() async {
  _speak("Starting registration process. Please wait...");
  final fieldErrors = _validateAllFields();
  if (fieldErrors.isNotEmpty) {
    await _announceAllErrors(fieldErrors);
    return;
  }

  setState(() => _isLoading = true);

  try {
    final cred = await FirebaseAuth.instance.createUserWithEmailAndPassword(
      email: emailController.text.trim(),
      password: passwordController.text.trim(),
    );

    final user = cred.user;
    if (user == null) throw Exception("Failed to create user");
```

Figure 43: Firebase account creation process

During account creation, Firebase Authentication securely hashes the password using bcrypt, generates a unique User ID (UID), and returns a UserCredential object. The user is then automatically signed in, and a session token is stored locally by the Firebase SDK. If the user object is null, an exception triggers the error-handling flow.

- **Step 2: Sending Verification Email**

After successful account creation, the system implements a dual-approach email verification strategy [**Figure 44**];

```
var emailSent = false;
try {
  final callable = FirebaseFunctions.instance.httpsCallable(
    'sendVerificationEmail',
  );
  await callable.call({
    'email': emailController.text.trim(),
    'displayName': nameController.text.trim(),
  });
  emailSent = true;
} catch (_) {
  try {
    await user.sendEmailVerification();
    emailSent = true;
  } catch (_) {
    emailSent = false;
  }
}

if (!emailSent) {
  await user.delete();
  throw FirebaseAuthException(
    code: 'email-send-failed',
    message: 'Failed to send verification email. Please try again later.',
  );
}
```

Figure 44: Email verification sending process

- **Email sending workflow:**

1. **Primary Method (Cloud Function):**

The primary method for sending emails uses a Cloud Function, where the app calls FirebaseFunctions.instance.httpsCallable('sendVerificationEmail') and passes the user's email and display name as parameters. The Cloud Function then runs server-side logic to send a customized HTML verification message, allowing the application to deliver branded emails with fully tailored templates.

2. Fallback Method (Firebase Default):

If the Cloud Function fails, the app falls back to user.sendEmailVerification(), which uses Firebase Authentication's built-in verification system. In this case, Firebase sends its default verification email template, providing a reliable backup even if the custom server-side email fails [**Figure 45**].

```dart
try {
  setState(() => _isLoading = true);
  _speak("Sending new verification email...");
  final user = FirebaseAuth.instance.currentUser;
  await user?.sendEmailVerification();

  _startResendCooldown();
  _showSuccessWithSpeech(
    "New verification email sent successfully! Please check your inbox and spam folder.",
  );
  HapticFeedback.mediumImpact();
} catch (e) {
  var msg = "Failed to resend email";
  if (e is FirebaseAuthException && e.code == 'too-many-requests') {
    msg = "Too many requests. Please wait and try again.";
  }
  _showErrorWithSoundAndBanner(msg);
} finally {
  if (mounted) setState(() => _isLoading = false);
}
```

Figure 45: Fallback verification email

3. Safety Rollback:

If both email-sending methods fail, the system calls user.delete() to remove the authentication account, preventing any orphaned accounts that cannot be verified. At this point, a FirebaseAuthException with the code email-send-failed is thrown to notify the user that the verification process could not be completed.

- **Step 3: Storing User Profile in Firestore**

Once authentication succeeds and the verification email is sent, the system stores the user's profile data in Cloud Firestore [**Figure 46**]:

```dart
await FirebaseFirestore.instance.collection('users').doc(user.uid).set({
  'full_name': nameController.text.trim(),
  'email': emailController.text.trim(),
  'email_verified': false,
  'lastVerificationEmailSentAt': FieldValue.serverTimestamp(),
});

setState(() {
  _showVerificationScreen = true;
  _isLoading = false;
});
_cancelAllSpeech(cancelTypingTimer: true);
_startCheckingVerification();
_showSuccessWithSpeech(
  lp.isArabic
      ? "اح! من فضلك تفقد بريدك الوارد ومجلد الرسائل المزعجة لرابط التحقق. أرسلنا بريداً إلى
      : "Account created successfully! Please check your inbox and spam folder for the verificati
);
```

Figure 46: Firestore user profile storage

- **Firestore database structure:**

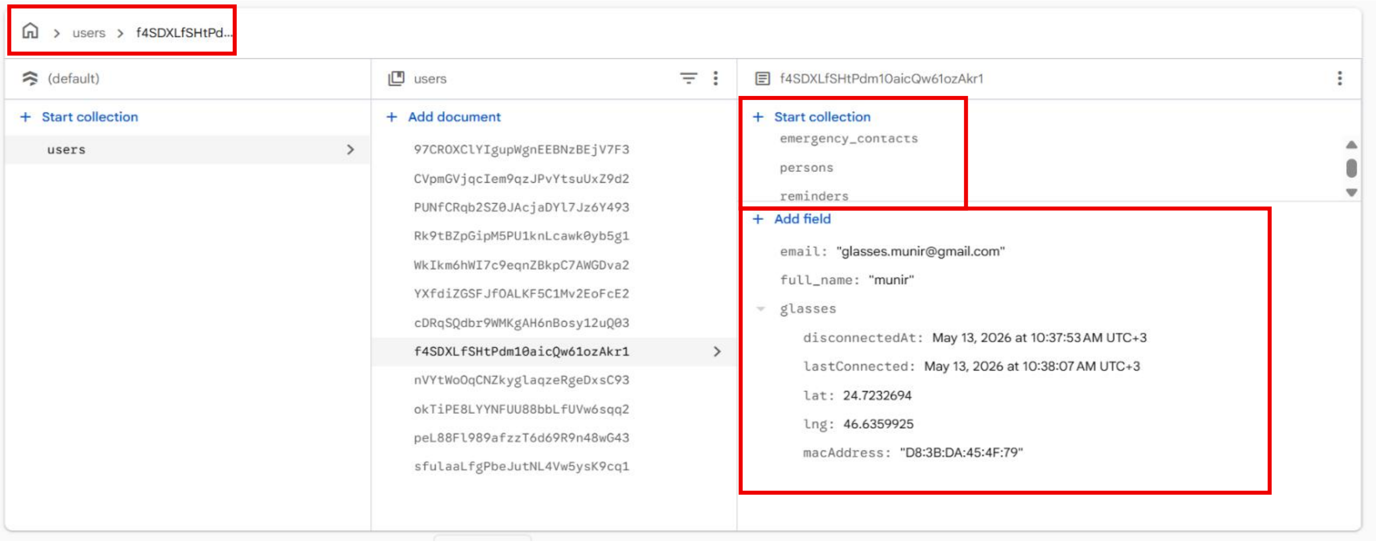


Figure 47: User Collection

- **Collection**: users (contains all registered user documents)
- **Document ID:** user.uid (the Firebase Authentication UID for each user)
- **Main Fields Stored:**
  - **email:** User's email address
  - **full_name:** User's full name entered during registration
- **Nested Map: glasses**
  - **lastConnected**: Timestamp of the most recent connection between the glasses and the system
  - **disconnectedAt:** Timestamp of the most recent disconnection
  - **lat:** Latitude of the glasses' last known location
  - **lng:** Longitude of the glasses' last known location
  - **macAddress:** MAC address of the smart glasses device
- **Subcollections:**
  - **persons:** Stores people associated with the user, including each person's name, facial embedding vector, and encrypted front-face thumbnail.
  - **reminders:** Stores reminder records created by the user, including details such as the reminder title, frequency, and scheduled date/time.
  - **emergency_contacts:** Stores emergency contact information, including the contact's name and phone number.

- Google Sign-In

```
Future<void> _signUpWithGoogle() async {
  try {
    _speak("Starting Google sign up process. Please wait.");
    setState(() => _isLoading = true);

    try {
      await GoogleSignInHandler.signInWithGoogle(context);
      await Future.delayed(const Duration(seconds: 2));
      final user = FirebaseAuth.instance.currentUser;
      if (user != null) {
        _showSuccessWithSpeech("Google sign up completed successfully!");
      } else {
        throw Exception("User not created");
      }
    } catch (_) {
      _showErrorWithSoundAndBanner(
        "Google sign-up canceled. Please try again.",
      );
      return;
    }
  } catch (_) {
    _showErrorWithSoundAndBanner("Google sign-up failed. Please try again.");
  } finally {
    if (mounted) setState(() => _isLoading = false);
  }
}
```

Figure 48: Google Sign-In

The GoogleSignInHandler manages the entire OAuth flow by opening the Google account picker, where the user selects their account and grants the required permissions. Google then returns an OAuth token, which is exchanged for Firebase credentials using signInWithCredential(). Firebase automatically creates a new user account or retrieves an existing one based on these credentials [**Figure 48**].

4.6.2 Delete Account

In this section, we describe the account deletion mechanism implemented in the **Munir** application, including the reauthentication process for different authentication providers, Firestore data removal, and the security measures that ensure safe and irreversible account deletion.

The account deletion mechanism in the application is designed to securely remove a user's account through Firebase Authentication and delete the associated data from Firestore. The process begins by displaying a danger confirmation dialog to ensure the user understands the irreversible nature of this action.

Upon confirmation, the method retrieves the currently authenticated user from FirebaseAuth and checks their authentication provider. The app supports two primary authentication methods: **Email/Password and Google Sign-In**, each requiring different reauthentication approaches before account deletion can proceed.

- **Email/Password Authentication**

For users who signed in with email and password, the **_showPasswordDialog** method is called to display an input dialog, allowing the user to confirm their password as shown in [**Figure 49**]. The dialog includes a TextField with password visibility toggle and validation to ensure the password field is not empty.

```
Future<String?> _showPasswordDialog() async {
  final TextEditingController passwordController = TextEditingController();
  bool isPasswordVisible = false;

  return await showDialog<String>(
    context: context,
    useRootNavigator: true,
    barrierDismissible: false,
    barrierColor: Colors.black.withOpacity(0.55),
    builder: (context) {
      bool announced = false;
      String? errorText;

      return _fixedDialog(
        StatefulBuilder(
          builder: (context, setState) {
            if (!announced) {
              announced = true;
              WidgetsBinding.instance.addPostFrameCallback((_) {
                _speakNow(
                  'Password confirmation. Please enter your password. Buttons: Confirm on the t
                );
              });
            }
```

Figure 49: Email/Password Authentication

Once the password is obtained, the app uses the **_verifyPassword** method shown in [**Figure 50**] to reauthenticate the user. This method creates an AuthCredential object with EmailAuthProvider.credential and passes it to the reauthenticateWithCredential method. The operation includes timeout handling to prevent indefinite waiting.

```
Future<bool> _verifyPassword(String email, String password) async {
  try {
    final cred = EmailAuthProvider.credential(
      email: email,
      password: password,
    );
    await _auth.currentUser!
        .reauthenticateWithCredential(cred)
        .timeout(const Duration(seconds: 12));
    return true;
  } on TimeoutException {
    _showErrorSnackBar('Network timeout while verifying password.');
    _speak('Network timeout while verifying password.');
    return false;
  } on FirebaseAuthException catch (e) {
    if (e.code == 'wrong-password' || e.code == 'invalid-credential') {
      return false;
    }
    _showErrorSnackBar('Auth error: ${e.code}');
    _speak('Authentication error.');
    return false;
  } catch (_) {
    _showErrorSnackBar('Unexpected error during verification.');
    _speak('Unexpected error during verification.');
    return false;
```

Figure 50: Password Verification Method

After successful reauthentication, the deletion process proceeds as shown in [**Figure 51**]. The user's Firestore document is first deleted from the **user's collection**, followed by the deletion of the Firebase Authentication account using the delete method. The entire operation is wrapped in the **_withBlocking** utility to display a loading overlay and prevent user interaction during the process.

```dart
await _withBlocking(() async {
  if (isGoogleOnly) {
    final ok = await _reauthenticateWithGoogle();
    if (!ok) {
      return;
    }
  } else if (hasPasswordProvider) {
    final ok = await _verifyPassword(email!, password!);
    if (!ok) {
      _showErrorSnackBar('Invalid password. Please try again');
      await _speakNow('Invalid password. Please try again.');
      return;
    }
  }

  try {
    final userDoc = _firestore.collection('users').doc(user.uid);
    final snap = await userDoc.get().timeout(const Duration(seconds: 8));
    if (snap.exists) {
      await userDoc.delete().timeout(const Duration(seconds: 8));
    }
  } on TimeoutException {
    _showErrorSnackBar('Network timeout while deleting your data.');
    await _speakNow('Network timeout while deleting your data.');      You,
    return;
```

Figure 51: Account Deletion for Email/Password Users

- ## Google Sign-In Authentication

For users who signed in with Google, the app uses a different approach. Instead of requesting a password, the **_reauthenticateWithGoogle** method is called as shown in [Figure 52]. This method first signs out from GoogleSignIn to force account selection, then prompts the user to sign in again. The fresh authentication tokens are used to create a GoogleAuthProvider.credential and reauthenticate with Firebase.

```dart
Future<bool> _reauthenticateWithGoogle() async {
  try {
    final googleSignIn = GoogleSignIn();
    await googleSignIn.signOut();

    final googleUser = await googleSignIn.signIn();
    if (googleUser == null) {
      _showErrorSnackBar('Reauthentication cancelled');
      await _speakNow('Reauthentication cancelled.');
      return false;
    }
    final googleAuth = await googleUser.authentication;
    final credential = GoogleAuthProvider.credential(
      accessToken: googleAuth.accessToken,
      idToken: googleAuth.idToken,
    );
    await _auth.currentUser!
        .reauthenticateWithCredential(credential)
        .timeout(const Duration(seconds: 12));      You, 1 second ago • Uncommitte
    return true;
  } on TimeoutException {
    _showErrorSnackBar('Network timeout while reauthenticating with Google.');
    await _speakNow('Network timeout while reauthenticating with Google.');
    return false;
  } on FirebaseAuthException catch (e) {
    _showErrorSnackBar('Auth error: ${e.code}');
```

Figure 52: Google Reauthentication Method

The main **_deleteAccount** method determines which authentication approach to use by examining the user's provider data, as demonstrated in [Figure 53]. For Google-only users, it calls the reauthentication method and provides audio feedback before proceeding with the deletion process. The same Firestore and Firebase account deletion steps are then executed.

```
final providers = user.providerData.map((p) => p.providerId).toList();
final bool isGoogleOnly =
    providers.length == 1 && providers.contains('google.com');
final bool hasPasswordProvider = providers.contains('password');
String? password;
String? email;
if (hasPasswordProvider && !isGoogleOnly) {
  password = await _showPasswordDialog();
  if (password == null || password.isEmpty) {
    _showErrorSnackBar('Deletion cancelled');
    await _speakNow('Deletion cancelled');
    return;
  }
  email = user.email;
  if (email == null || email.isEmpty) {
    for (final info in user.providerData) {
      if ((info.email ?? '').isNotEmpty) {
        email = info.email!;
        break;
      }
    }
  }
  if (email == null || email.isEmpty) {
    _showErrorSnackBar('Email not found. Please log in again.');
    await _speakNow('Email not found. Please log in again.');
    return;
```

Figure 53: Google Account Deletion

The method includes comprehensive error handling for network timeouts, invalid credentials, and the requires-recent-login error code. The **_withBlocking** utility displays a loading overlay during the entire operation to prevent user interaction. Finally, the app displays a success message using a SnackBar and navigates the user back to the login screen using Navigator.pushAndRemoveUntil to clear the navigation stack. Text-to-Speech provides audio feedback throughout the process for accessibility.

4.3.4 Text Reading Feature

In this section, we describe the Text Reading feature implemented in the **Munir**, which allows visually impaired users to capture or select an image, extract any present text, and have it read aloud using text-to-speech technology in both Arabic and English.

- **Image Capture and Gallery Selection**

Users can either capture a new image using the camera button or select an existing image from the device gallery using the **ImagePicker** package. Once an image is obtained from either source, it is passed to the processing pipeline for text extraction [**Figure 54**].

```
Future<void> _captureImage() async {
  if (_controller?.value.isInitialized != true) return;
  final file = await _controller!.takePicture();
  setState(() => _selectedImagePath = null);
  await _processImage(file.path, fromGallery: false);
}

Future<void> _pickImage() async {
  final file = await picker.pickImage(source: ImageSource.gallery);
  if (file == null) return;
  await _processImage(file.path, fromGallery: true);
}
```

Figure 54: Image capture and gallery selection methods

- **Text Extraction Using Cloud Vision API**

After an image is captured or selected, the application converts it into **Base64** format and sends it to a Firebase Cloud Function via an **HTTP POST** request. If no text is detected, the application notifies the user with an appropriate message [**Figure 55**].

```
Future<String> _extractTextFromImage(File imageFile) async {
      final languageProvider = Provider.of<LanguageProvider>( context, listen: false,);
  final Uint8List imageBytes = await imageFile.readAsBytes();
  final String base64Image = base64Encode(imageBytes);

  final response = await http.post(
    Uri.parse('https://us-central1-munir-21f4a.cloudfunctions.net/extractText'),
    headers: {'Content-Type': 'application/json'},
    body: jsonEncode({
      'imageBase64': base64Image,
      'language': languageProvider.isArabic ? 'ar' : 'en',
    }),
  );

  if (response.statusCode == 200) {
    final data = jsonDecode(response.body);
    if (data['success'] == true) {
      return data['text'] ?? "";
    }
  }
  return "";
}
```

Figure 55: Text extraction method

- **Firebase Cloud Function**

On the backend, the Cloud Function receives the image and passes it to the **Google Cloud Vision API**, which performs text detection and returns the recognized text as a **JSON** response to the mobile application [Figure 56].

```
const visionClient = new vision.ImageAnnotatorClient();

exports.extractText = onRequest(
  { cors: true },
  async (req, res) => {
    const { imageBase64 } = req.body;

    const [result] = await visionClient.textDetection({
      image: { content: imageBase64 }
    });

    const detections = result.textAnnotations;
    const extractedText = detections.length > 0
      ? detections[0].description
      : "";

    return res.status(200).json({
      success: true,
      text: extractedText.trim(),
    });
  }
);
```

Figure 56: Firebase Cloud Function

- **Text-to-Speech Conversion**

Once text is successfully extracted, the application detects the language based on Arabic Unicode character frequency. Arabic text is synthesized using **Amazon Polly**, while English text is handled by the device-native **FlutterTTS** engine. After playback completes, the interface resets automatically for the next capture [Figure 57].

```
Future<void> _speak(String text) async {
  if (_isArabicText(text)) {
    await _pollyTts.speak(text);
    final estimatedMs = (text.length * 80).clamp(1500, 20000);
    Future.delayed(Duration(milliseconds: estimatedMs), _resetForNextCapture);
  } else {
    _flutterTts.setCompletionHandler(_resetForNextCapture);
    await _flutterTts.speak(text);
  }
}
```

Figure 57: Text-to-speech method

4.4.4 Emergency SOS

This section describes the Emergency SOS feature in the **Munir**, which allows visually impaired users to send an emergency alert with their current location to their registered contacts, enabling them to respond quickly and efficiently.

- ### Contact Verification

When the user navigates to the SOS screen, the application checks whether emergency contacts have been registered in Firestore. If none are found, the user is notified via text-to-speech and redirected to the Contacts page. Once contacts are added, the SOS button is enabled upon returning to the screen [Figure 58].

```dart
final snapshot = await _firestore
    .collection('users')
    .doc(user.uid)
    .collection('emergency_contacts')
    .limit(1)
    .get();

if (snapshot.docs.isEmpty) {
  setState(() => _hasContacts = false);

  await _speak(
    lp.isArabic
        ? 'لا توجد جهات اتصال. سيتم تحويلك لصفحة جهات الإتصال'
        : 'No contacts found. You will be redirected to the Contacts page',
  );

  _goToContactsThenReturn();
} else {
  setState(() => _hasContacts = true);
  await _speakWelcome();
}
```

Figure 58: Checking emergency contacts availability

- ### Location Retrieval

When the SOS button is pressed, the application requests the device's current location using the **Geolocator package**. If permission is granted, the location is retrieved with high accuracy and converted into a Google Maps link. If location access is unavailable, the alert is still sent without the location link [Figure 59].

```dart
static Future<Position?> _getLocation() async {
  try {
    LocationPermission permission = await Geolocator.checkPermission();
    if (permission == LocationPermission.denied) {
      permission = await Geolocator.requestPermission();
    }
    if (permission == LocationPermission.deniedForever) return null;
    return await Geolocator.getCurrentPosition(
      desiredAccuracy: LocationAccuracy.high,
      timeLimit: const Duration(seconds: 10),
    );
  } catch (_) {
    return null;
  }
}
```

Figure 59: Retrieving the user's current location

- ### Emergency Contact Retrieval

The application retrieves the user’s registered emergency contacts from Firebase Firestore using the authenticated user ID. The retrieved contact data is converted into a list structure for further processing and communication features within the application. [Figure 60].

```dart
static Future<List<Map<String, dynamic>>> _getContacts() async {
  final user = FirebaseAuth.instance.currentUser;
  if (user == null) return [];
  final snap = await FirebaseFirestore.instance
      .collection('users')
      .doc(user.uid)
      .collection('emergency_contacts')
      .get();
  return snap.docs.map((doc) => doc.data()).toList();
}
```

Figure 60: Fetching emergency contacts from Firestore

- **Sending the SOS Alert**

Once the location and contacts are ready, the application constructs an emergency message containing a Google Maps link and passes all contact numbers to the **SMS: URI scheme**. The URI is then launched using the **url_launcher package**, which opens the native SMS app with all contacts and the message pre-filled, requiring the user only to press send [Figure 61].

```dart
static Future<void> sendEmergencyAlert() async {
  final position = await _getLocation();
  final mapsLink = position != null
      ? 'https://maps.google.com/?q=${position.latitude},${position.longitude}'
      : '';

  final contacts = await _getContacts();
  if (contacts.isEmpty) throw Exception('No contacts found');

  final phones = contacts
      .map((c) => _normalizePhone(c['phone'] as String? ?? ''))
      .where((p) => p.isNotEmpty)
      .join(';');

  if (phones.isEmpty) return;

  final message = mapsLink.isNotEmpty
      ? 'EMERGENCY ALERT\n\nThis is an automated SOS message.\nThe sender requires immediate assistance.\n\n📍 Location: $mapsLink\n\nPlea
      : 'EMERGENCY ALERT\n\nThis is an automated SOS message.\nThe sender requires immediate assistance.\n\nPlease try to contact them as

  final smsUrl = Uri.parse('sms:$phones?body=${Uri.encodeComponent(message)}');
  if (await canLaunchUrl(smsUrl)) {
    await launchUrl(smsUrl, mode: LaunchMode.externalApplication);
  }
}
```

Figure 61: Sending the emergency message via the SMS app

4.5.4 Face Recognition Feature

This section describes the Face Recognition feature in the **Munir**, which enables visually impaired users to identify people in their surroundings by capturing an image and receiving an immediate audio response with the recognized person's name.

- **Face Registration**

Before recognition can occur, the user must first register the people they wish to identify. Through the face registration interface, multiple images of each person are captured and sent to the **FastAPI** backend, where the **InsightFace buffalo_l** model extracts a 512-dimensional facial embedding for each image. These embeddings are stored in Firebase Firestore alongside the person's name, forming the reference database used during recognition. To ensure privacy, the frontal image of each registered person is stored as an **AES-256-GCM** encrypted thumbnail in Firebase Firestore. Additionally, images are transmitted to the **FastAPI** backend over an encrypted channel to protect biometric data during transfer [**Figure 62**].

```
munir-api / main.py
Code  Blame  836 lines (699 loc) · 29.3 KB
async def enroll_person(
        for idx, file in enumerate(files):
            try:
                img_bytes = await file.read()
                img = read_image(img_bytes)
                emb, error = extract_embedding(img)

                if emb is not None:
                    embeddings.append(emb.tolist())
                    success_count += 1
                    logger.info(f" ✅ Image {idx + 1}: Face detected")
                else:
                    failed_count += 1
                    logger.warning(f" ⚠ Image {idx + 1}: {error}")
            except Exception as e:
                failed_count += 1
                logger.warning(f" ❌ Image {idx + 1}: Error - {e}")
                continue

        if len(embeddings) < 3:
            raise HTTPException(400, f"Only {len(embeddings)} valid faces found, need at least 3")

        flattened_embeddings = []
        for emb in embeddings:
            flattened_embeddings.extend(emb)

        person_data = {
            'name': name,
            'embeddings_flat': flattened_embeddings,
            'embedding_dim': 512,
            'num_embeddings': len(embeddings),
            'num_angles': len(embeddings),
            'updated_at': firestore.SERVER_TIMESTAMP
        }
```

Figure 62:Face enrollment and embedding extraction

- **Face Recognition**

When the user captures an image, it is sent to the **FastAPI** backend, which extracts the facial embedding using **InsightFace**. The application then computes the cosine similarity between the captured embedding and all registered embeddings stored in Firestore. While an SVM classifier was trained and evaluated in [**Chapter 5**] to assess closed-set recognition accuracy, it was not adopted in the application as closed-set models require retraining whenever a new person is added.

Instead, cosine similarity enables open-set recognition, allowing new people to be registered at runtime without retraining. A match is confirmed when the similarity score exceeds a threshold of 0.40; otherwise, the face is classified as unknown [**Figure 63**].

```
munir-api / main.py
Code  Blame  836 lines (699 loc) · 29.3 KB

@app.post("/recognize")
async def recognize(
    user_id: str = Form(...),
    file: UploadFile = File(...)
):
    """Recognize a single face in an image"""
    try:
        if not face_app or not db:
            raise HTTPException(503, "Service not ready")
        start_time = time.time()
        img_bytes = await file.read()
        img = read_image(img_bytes)
        query_emb, error = extract_embedding(img)

        if error:
            raise HTTPException(400, error)

        match, score = find_match(query_emb, user_id)

        THRESHOLD = 0.40

        if match and score >= THRESHOLD:
            logger.info(f"✅ Recognized: {match['person_name']} (score: {score:.3f})")
            inference_time = time.time() - start_time
            return {
                "success": True,
                "recognized": True,
                "person_name": match["person_name"],
                "person_id": match["person_id"],
                "confidence": round(score, 4),
                "similarity_score": round(score * 100, 2),
                "message": f"Welcome {match['person_name']}!"
            }
        else:
            logger.info(f"❌ Unknown person (best score: {score:.3f})")
            return {
                "success": True,
```

Figure 63:Recognize person

- **Voice Response**

Once recognition is complete, the result is communicated to the user through text-to-speech. If a match is found, the recognized person's name is announced aloud. If no match is found, the user is notified that the person is unknown [**Figure 64**].

```
} else {
  if (known.isEmpty) {
    textToSpeak = total == 1
        ? "1 face detected, unknown person"
        : "$total faces detected, all unknown";
  } else if (unknownCount == 0) {
    textToSpeak = known.length == 1
        ? "Recognized ${known[0]}"
        : "Recognized ${known.length} people: ${known.join(', ')}";
  } else {
    textToSpeak =
        "$total faces detected. Known: ${known.join(', ')}. "
        "$unknownCount unknown ${unknownCount == 1 ? 'person' : 'people'}";
  }
}

_showMultiFaceResultDialog(multiResult, imagePath);
}
```

Figure 64: Voice response construction based on recognition result

4.6.4 Currency Identification

This section describes the Currency Identification feature in the **Munir**, which enables visually impaired users to identify Saudi Riyal banknotes by capturing an image and receiving an immediate audio response with the detected denomination.

- **Banknote Detection Using Roboflow**

When the user captures an image, it is sent to the **Roboflow** object detection model, which detects, localizes all banknotes present in the image. Each detected banknote is cropped with a 15% padding margin to ensure the full note is captured [Figure 65].

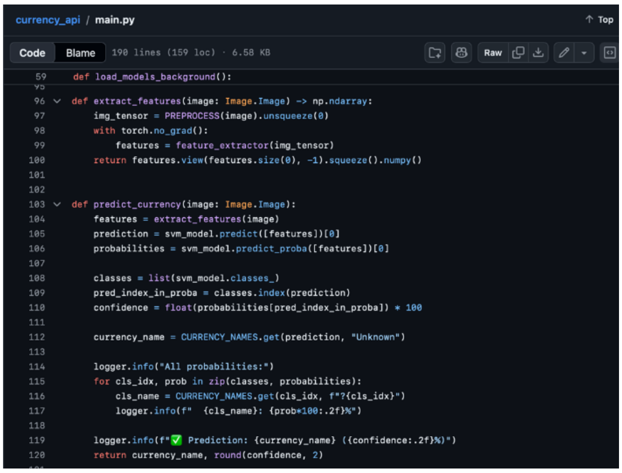


Figure 65: Banknote detection using Roboflow

- **Classification Verification Using MobileNetV2 and SVM**

Each cropped banknote image is sent to the **FastAPI** backend, where **MobileNetV2** is used solely for feature extraction, generating a 1280-dimensional feature vector. This vector is then passed to a trained **SVM classifier** with an RBF kernel to predict the denomination [Figure 66].

```dart
final validPredictions = predictions.where((p) {
  final conf = (p['confidence'] as num).toDouble();
  final w = (p['width'] as num).toDouble();
  final h = (p['height'] as num).toDouble();

  print('    Found: ${p['class']} | Confidence: ${(conf * 100).toStringAsFixed(1)}% | Dimensions: ${w}x${h}');
  return conf >= 0.40;
}).toList();
print('-------------------------------------------');

if (validPredictions.isEmpty) return [];

final imgWidth = (data['image']['width'] as num).toDouble();
final imgHeight = (data['image']['height'] as num).toDouble();
final image = img.decodeImage(bytes)!;
final tempDir = await getTemporaryDirectory();

final List<CroppedImage> croppedItems = [];

for (int i = 0; i < validPredictions.length; i++) {
  final pred = validPredictions[i];
  final x = (pred['x'] as num).toDouble();
  final y = (pred['y'] as num).toDouble();
  final w = (pred['width'] as num).toDouble();
  final h = (pred['height'] as num).toDouble();

  const padding = 0.15;

  final left = ((x - w / 2 - w * padding) / imgWidth * image.width).round().clamp(0, image.width);
  final top = ((y - h / 2 - h * padding) / imgHeight * image.height).round().clamp(0, image.height);
  final right = ((x + w / 2 + w * padding) / imgWidth * image.width).round().clamp(0, image.width);
  final bottom = ((y + h / 2 + h * padding) / imgHeight * image.height).round().clamp(0, image.height);

  if (right <= left || bottom <= top) continue;
```

Figure 66:Feature extraction and currency classification using MobileNetV2 and SVM

Once the recognition pipeline is complete, the detected denomination is announced to the user through text-to-speech. If no banknote is detected, the user is notified with an appropriate message prompting them to retake the image [Figure 67].

```dart
setState(() {
  _processingStatus = languageProvider.isArabic
      ? 'التعرف على العملة...'
      : 'Recognizing currency...';
});

try {
  final multiResult = await CurrencyApiService.recognizeAllCurrencies(
    imageFile: File(imagePath),
    fromCamera: !fromGallery,
  );

  if (!mounted) return;

  if (multiResult.success) {
    setState(() {
      _currencyResult = multiResult.toCombinedResult();
      _extractedText = "";
      _detectedColor = "";
      _faceResult = null;
      _smartColorResult = null;
    });

    textToSpeak = languageProvider.isArabic
        ? multiResult.textArabic
        : multiResult.textEnglish;

    print('✅ Currency detected: ${multiResult.count} banknote(s)');
```

Figure 67: Voice response for detected currency denomination

4.7.4 Smart Glass

This section describes the Smart Glasses feature in the **Munir** application, which enables visually impaired users to interact with the hands-free system through voice commands issued via the Omi Glass ESP32-S3 smart glasses, receiving audio responses based on the activated feature.

The Omi Glass ESP32-S3 was utilized as the wearable hardware component. As described in [Chapter 2], the device's firmware is provided by the Omi open-source project; the **Munir** team's contribution focused on adapting and integrating it with the **Munir** system to support the application's assistive features [58].

- **BLE Connection**

The **Munir** application connects to the **Omi Glass ESP32-S3** smart glasses via Bluetooth Low Energy (BLE). Upon pairing, the application discovers the device's services and characteristics, enabling unidirectional communication through dedicated channels for photo data, audio streaming, and voice commands [Figure 68].

```
class OmiGlassService {
  Future<void> _connectToDevice(
      await device.connect(
        timeout: const Duration(seconds: 30),
        autoConnect: false,
      );
      _connectedDevice = device;

      _isPoweredOff = false;

      _photoNotifySubscription?.cancel();
      _photoNotifySubscription = null;
      _audioNotifySubscription?.cancel();
      _audioNotifySubscription = null;
      _resetCharacteristics();

      try {
        final mtu = await device.requestMtu(517);
        print('📏 MTU: $mtu');
      } catch (e) {
        print('⚠ MTU failed: $e');
      }

      List<BluetoothService> services = await device.discoverServices();

      bool serviceFound = false;

      for (var service in services) {
        String svcUUID = service.uuid.toString().toLowerCase();

        if (svcUUID == serviceUUID) {
          serviceFound = true;
          print('✅ Found Omi Glass Service');

          for (var char in service.characteristics) {
            String charUUID = char.uuid.toString().toLowerCase();
```

Figure 68: BLE connection and service discovery

- ### Button-Activated Listening

The smart glasses are equipped with a physical button that the user presses to activate the voice command listening mode. Once the button is pressed, the application begins recording the user's voice command. [**Figure 69**] [**Figure 70**]

```
void _onButtonActivated() async {
  if (_isProcessing) {
    print('⚠ Still processing, resetting...');
    _isProcessing = false;
    _pendingCmd = 0;
  }
  print('✅ Wake button detected!');
  _recordingCommand = false;
  await _speak(_isArabic() ? 'تكلم' : 'Speak now');
  await Future.delayed(const Duration(milliseconds: 500));
  _recordingCommand = true;
  _commandBuffer.clear();
  _commandTimer?.cancel();
  _commandTimer = Timer(const Duration(seconds: 5), () {
    _finishCommandRecording();
  });
}
```

Figure 70: Button activation and voice command recording

```
void _listenToPhotos() {
  _photoSubscription?.cancel();
  _photoSubscription = _omiService.photoStream.listen((chunk) {
    print(
      '📷 photo chunk arrived: ${chunk.length}b | ${chunk.take(3).toList()}',
    );
    if (chunk.isEmpty) return;

    if (chunk.length == 2 && chunk[0] == 0xFF && chunk[1] == 0xFF) {
      _photoTimeout?.cancel();
   if (_photoBuffer.isNotEmpty) {
    final photo = List<int>.from(_photoBuffer);
    _photoBuffer = [];
    _receivingPhoto = false;

    if (_pendingCmd > 0) {
      final cmd = _pendingCmd;
      _pendingCmd = 0;
      _processPhotoWithCommand(photo, cmd);
    } else {
      print('🔘 Button press → starting voice command recording');
      _onButtonActivated();
    }
  }

      return;
    }

    if (chunk.length >= 3 && chunk[0] == 0x00 && chunk[1] == 0x00) {
      _photoBuffer = [];
      _receivingPhoto = true;
      _photoBuffer.addAll(chunk.sublist(3));
      return;
    }

    if (_receivingPhoto && chunk.length > 2) {
      _photoBuffer.addAll(chunk.sublist(2));
    }
  });
}
```

Figure 69 Button-activated listening and photo stream handling

- ### Voice Command Processing

Once the he button is pressed, the application records the user's voice command for five seconds and sends it to Firebase Cloud Functions, where **OpenAI Whisper**, a speech recognition model — transcribes the audio and maps it to one of the supported commands: face recognition, currency identification, text reading, scene description, object detection, or SOS. If the command is not recognized, the user is prompted to try again [**Figure 71**].

```
Future<void> _sendToWhisper(String audioPath) async {
  try {
    _speak(_isArabic() ? 'جاري التحليل' : 'Analyzing...');
    final audioBytes = await File(audioPath).readAsBytes();
    final audioBase64 = base64Encode(audioBytes);
    try {
      await File(audioPath).delete();
    } catch (_) {}

    final callable = FirebaseFunctions.instance.httpsCallable(
      'transcribeAudio',
    );
    final result = await callable.call({
      'audioBase64': audioBase64,
      'languageCode': _isArabic() ? 'ar' : 'en',
    });

    final data = result.data as Map<String, dynamic>;
    final command = data['command'] as String? ?? 'unknown';
    final text = data['text'] ?? data['transcript'] ?? '';
    print('XXXXX WHISPER OK: text="$text" command=$command');

    if (command == 'unknown') {
      await _speak(
        _isArabic()
            ? 'لم أفهم الأمر، قولي مرة ثانية'
            : 'Command not understood, try again',
      );
      _isProcessing = false;
      return;
```

Figure 71: Voice command transcription and processing

- **Command Execution**

Once the command is identified, the application captures a photo from the glasses and routes it to the appropriate processing pipeline based on the requested command. The result is then announced to the user through text-to-speech [**Figure 72**].

```
class GlassesCommandService {
  Future<void> _processPhotoWithCommand(List<int> photoBytes, int cmd) async {
          case CMD_CURRENCY:
            _speak(
              _isArabic() ? 'جاري التعرف على العملة' : 'Recognizing currency...',
            );
            print('XXXXX CURRENCY: calling CurrencyApiService...');
            try {
              // ✅ wakeUpServer كـ أيضاً هنا safety net لو ما اشتغل السابق
              await CurrencyApiService.wakeUpServer();
              final currencyResult = await CurrencyApiService.recognizeCurrency(
                imageFile: file,
                fromCamera: true,
              );
              print('XXXXX CURRENCY: success=${currencyResult.success}');
              result = currencyResult.success
                  ? (_isArabic()
                        ? currencyResult.textArabic
                        : currencyResult.textEnglish)
                  : (_isArabic()
                        ? 'لم أتعرف على العملة'
                        : 'Could not recognize currency');
            } catch (e, stack) {
              print('XXXXX CURRENCY CRASHED: $e');
              print('XXXXX CURRENCY STACK: $stack');
              result = _isArabic()
                  ? 'خطأ في التعرف على العملة'
                  : 'Currency recognition error';
            }
            print('XXXXX CURRENCY RESULT: $result');
            break;

          case CMD_READ:
            _speak(_isArabic() ? 'جاري قراءة النص' : 'Reading text...');
            final imageBytes = await file.readAsBytes();
            final base64Image = base64Encode(imageBytes);
```

Figure 72: Command execution and result delivery

4.7 Challenges

In this section, we describe the key technical challenges encountered during the development of the **Munir** application, including email delivery issues, dataset imbalance and computational constraints, and image format inconsistencies, along with the solutions implemented to address each challenge.

4.7.1 Email Delivery Challenge

During the development of the email verification and password-reset features, the team encountered a recurring issue where all system-generated emails were consistently marked as spam. This problem occurred when relying on Firebase's default email-sending mechanism, which uses generic templates and shared domains that are more likely to be filtered by spam-detection systems. As a result, users were unable to easily locate verification and password-reset messages, causing delays in account activation and password recovery during testing.

To overcome this challenge, a custom email-sending workflow was implemented using Firebase Cloud Functions in combination with a dedicated Gmail SMTP service through Nodemailer. Instead of depending on the built-in Firebase templates, the application now calls two callable functions (**sendVerificationEmail and sendCustomPasswordReset**) each responsible for generating secure verification or reset links via admin.auth() and sending branded HTML emails from the project's authenticated Gmail account. This approach provided a trusted sender identity, improved domain reputation, and enabled full control over the email structure and styling.

After deploying the solution, both the verification and reset-password emails consistently reached users' inboxes rather than their spam folder. This significantly increased reliability, reduced user confusion, and improved the overall onboarding and account-recovery experience.

Additionally, several **security considerations** were incorporated into the implementation, including structured error handling to avoid exposing sensitive information, and relying on Firebase's built-in verification link expiration to ensure that both verification and reset-password URLs remain valid only for a limited time. These measures collectively enhanced the security and integrity of the email-delivery workflow.

4.7.2 Face Recognition Dataset Imbalance and Computational Constraints

During the initial stages of face recognition dataset preparation, the team encountered significant variability in the number of available images per registered individual. The dataset exhibited extreme imbalance, with some individuals having as few as 102 images, while others had over 700 images in the source directories. This imbalance posed a dual challenge: first, unequal class representation could introduce bias during model training, causing the recognition system to favor individuals with more training samples; second, including all available images would result in a dataset far exceeding computational capacity, which would significantly strain hardware resources during feature extraction and model training phases.

To address this challenge, a systematic preprocessing pipeline was implemented that enforced strict class balance by selecting exactly 60 images per person through random sampling from the deduplicated image pool. This specific number was chosen based on three considerations: sufficient sample diversity to capture variations in facial expressions, lighting conditions, and head poses; balanced class distribution to prevent model bias toward over-represented individuals; and computational tractability within available hardware constraints. The random selection mechanism ensured that for individuals with more than 60 images, the system would choose a representative subset rather than simply taking the first 60 images sequentially, thereby maximizing variation within each class.

Since all individuals in the source dataset had at least 102 images (well above the 60-image threshold), the selection process could maintain perfect class balance without excluding any registered individuals. For those with significantly more images (up to 700), the random sampling approach prevented the systematic bias that would occur from simply selecting the first 60 images, ensuring diverse representation across different capture sessions, lighting conditions, and time periods.

The implementation included a multi-level deduplication mechanism using set-based path filtering to eliminate duplicate file paths across source directories before the random selection process. This prevented the same image from being counted multiple times if it appeared in different folders. Additionally, file existence validation was performed to ensure that selected image paths were accessible and readable before proceeding to the next preprocessing stages.

After applying this standardization approach, the final dataset achieved perfect class balance with 32,400 facial images (540 classes × 60 images per class), with zero duplicate images and zero processing errors. This balanced representation ensured that the face recognition model would not develop bias toward individuals with originally larger sample sizes, while the reduced dataset size (approximately 2.8 GB after JPEG compression) remained manageable within the available computational infrastructure.

4.7.3 Image Format and Color Space Inconsistency

Both face recognition and currency identification datasets exhibited significant inconsistencies in image formats and color spaces that threatened preprocessing pipeline stability and model training reliability. The face recognition dataset contained images in multiple formats including PNG, JPEG, and occasionally BMP files, with some images stored in RGBA format (4-channel with transparency) and others in grayscale (1-channel). Similarly, the currency dataset included images captured from iPhones in Apple's proprietary HEIC/HEIF format, which is not natively supported by standard Python image processing libraries such as PIL and OpenCV.

These inconsistencies posed several technical challenges: neural network architectures like MobileNetV2 expect uniform RGB (3-channel) input with consistent dimensions, making mixed color spaces incompatible with standard preprocessing workflows; HEIC format files would cause runtime errors when processed by PIL without additional conversion libraries; grayscale images represented facial features with only luminance information, potentially losing important color-based discriminative features; and RGBA images contained unnecessary alpha channel data that would increase memory consumption and cause dimension mismatches during model input.

To address these challenges, a systematic color space standardization pipeline was implemented for both datasets. For the face recognition dataset, all images were converted to RGB format through two conditional transformations: grayscale images were expanded to three channels through channel replication, while RGBA images had their alpha channel removed. For the currency dataset, the preprocessing pipeline incorporated a dedicated HEIC/HEIF conversion step that detected Apple-format images and converted them to standard JPEG format using the pillow-heif library, ensuring compatibility with standard PIL operations.

After implementing these standardization measures, both datasets achieved complete color space uniformity: all face recognition images conformed to 224×224 RGB specifications, and all currency images were standardized to 224×224 RGB format compatible with MobileNetV2 feature extraction. This consistency eliminated potential preprocessing errors, ensured stable model training, and guaranteed that neural networks received properly formatted input throughout the entire machine learning pipeline.

4.8.4 SMS Notification Delivery Constraints

During the development of the SOS feature, the team initially planned to deliver SMS alert messages using a registered sender domain associated with the project name "**Munir**." This approach was intended to provide professional and branded experience for end users. However, after consulting the Communication and Space Technology Commission (CST) service portal and reviewing the regulatory requirements for SMS sender ID registration in Saudi Arabia, it became evident that obtaining a custom sender domain required a formal registration process that could not be completed within the project timeline. [59]

As an alternative, the team evaluated using the Twilio API to deliver SMS messages through WhatsApp [60], which offered a more accessible integration path. However, this approach introduced a significant technical constraint: the Twilio WhatsApp sandbox environment requires that all recipient phone numbers be pre-registered and explicitly opted in before any messages can be received. This meant that the system could not send WhatsApp-based notifications to arbitrary users who had not previously enrolled in the sandbox, rendering the solution unsuitable for a dynamic user base in which individuals register independently without prior coordination with the development team.

To address these constraints while preserving the intended notification functionality, the team adopted a client-initiated messaging approach. Rather than having the server dispatch SMS messages programmatically, the system generates a pre-composed message containing the relevant notification content and prompts the user to send it directly from their own device by tapping a clearly labeled send button. This approach ensures that all users can receive the intended SMS, while maintaining the core communication objective of the system.

4.7.5 Thumbnail Encryption Key Security

During the development of the face recognition feature, the team initially implemented a client-side encryption mechanism to protect registered persons' thumbnails stored in Firebase Storage. The original approach derived the encryption key directly from the user's userId, applying it as the secret key for AES-256-CBC encryption. While functional, this approach introduced a significant security vulnerability: since the userId is a predictable and relatively low-entropy string, an adversary with access to the encrypted files could potentially reconstruct the key and compromise the confidentiality of the stored images.

To address this weakness, the team evaluated stronger encryption configurations and identified two primary limitations of the initial design. First, the absence of a key derivation function meant that the encryption key lacked sufficient cryptographic strength. Second, AES-256-CBC does not provide authenticated encryption, meaning that tampering with the ciphertext could go undetected.

As a resolution, the team migrated the encryption scheme to AES-256-GCM combined with PBKDF2 key derivation. Under this approach, a cryptographically random 256-bit salt is generated for each encryption operation and passed alongside the userId through 100,000 iterations of SHA-256 to derive a strong and unpredictable encryption key. The GCM mode additionally provides an authentication tag that ensures the integrity of the encrypted data, making any unauthorized modification detectable upon decryption. The final encrypted file follows the structure [salt(32 bytes)] + [IV(12 bytes)] + [ciphertext + authentication tag], ensuring that all parameters required for decryption are self-contained within the file while the key itself is never stored anywhere in the system.

4.7.6 Image Transmission Over BLE

During the development of the smart glasses integration, the team encountered a significant challenge related to transmitting captured images from the ESP32-S3 glasses to the mobile application over Bluetooth Low Energy (BLE). BLE is designed for low-power communication and imposes strict limitations on the maximum transmission unit (MTU), meaning that image data cannot be sent as a single payload. Attempting to transmit a full image in one packet resulted in data loss and communication failures.

To address this, a chunked transmission protocol was implemented on the ESP32-S3 firmware side, where the captured image is divided into fixed-size packets and transmitted sequentially over BLE. On the Flutter application side, a reassembly mechanism was developed to collect incoming chunks, detect the end-of-transmission signal, and reconstruct the complete image before passing it to the processing pipeline. This approach ensured reliable image delivery regardless of image size, while maintaining compatibility with BLE's power and bandwidth constraints.

After implementing this solution, image transmission became stable and consistent, enabling the mobile application to reliably receive and process images captured by the smart glasses in real time.

- GitHub link: https://github.com/aljawharah44/2025_GP_1/tree/main

CHAPTER 5:
SYSTEM EVALUATION

Chapter 5: System Testing

In this section, we present the final evaluation of the system, covering the performance of the implemented machine learning models and the functionality of the mobile application. The testing ensures that the system meets all functional and non-functional requirements, operates reliably, and offers smooth user experience. It includes experimental testing of face and Currency identification models to verify their accuracy and robustness, along with User Acceptance Testing (UAT) to evaluate usability, accessibility, and overall satisfaction.

5.1 Experimental Results

This section presents the experimental results obtained from the development and evaluation of the face recognition and currency identification models. It summarizes the datasets used, feature extraction methods, model configurations, and performance analyses conducted to select the most suitable models for integration into the **Munir** application.

5.1.1 Face Recognition Model

A. Dataset Description and Preprocessing

This section describes the dataset used to train and evaluate the face recognition model and outlines the preprocessing procedures applied to the images, such as data splitting, face detection, and face alignment, to prepare them for feature extraction and classification.

- **Dataset Description**

The face recognition system was developed using a subset of the VGGFace2 dataset, comprising 32,400 images distributed across 540 identity classes with 60 images per class. This dataset provides diverse facial variations including different poses, ages, lighting conditions, and ethnicities, making it suitable for training robust face recognition models. A detailed description of the dataset characteristics, acquisition process, and preprocessing procedures is provided in Chapter 5, (**Section 4.4.2**).

- **Data Splitting**

To ensure rigorous model evaluation and prevent data leakage, the dataset was partitioned using a stratified split approach before feature extraction. The split allocated 80% of the data for training (25,920 images, 48 images per class) and 20% for testing (6,480 images, 12 images per class). Stratified splitting ensures that each class maintains its proportional representation in both training and testing sets, which is critical for balanced model evaluation across all identity classes.

The separation between training and testing data was strictly maintained throughout the entire pipeline, with feature extraction performed independently on each subset to prevent any information leakage that could artificially inflate performance metrics.

B. Feature Extraction and Embedding Generation:

Prior to model training, face embeddings were extracted using **InsightFace's ArcFace model (buffalo_l)** [61]. This state-of-the-art face recognition framework employs a three-stage pipeline:

- **Face Detection**: The SCRFD (Sample and Computation Redistributed Face Detector) [62] detects faces in 224×224 preprocessed images, identifying facial regions and five key landmarks (eyes, nose, and mouth corners).
- **Face Alignment:** Detected faces are normalized to a standard 112×112 format using the identified landmarks, ensuring consistent facial orientation and scale.
- **Embedding Generation**: The ArcFace model, based on ResNet-100 architecture, processes the aligned 112×112 face images and generates 512-dimensional embedding vectors [56]. These embeddings are L2-normalized to unit length, which enables effective similarity comparison using cosine distance.

This feature extraction and embedding generation process transformed face images into compact **512-dimensional representations** for both training and testing sets, significantly reducing computational complexity while preserving discriminative facial features.

C. Model Training Configuration:

Three distinct machine learning classifiers were trained and evaluated: **Support Vector Machine with Linear kernel (SVM Linear), Support Vector Machine with RBF kernel (SVM RBF)**, and **Logistic Regression**. Each model was assessed using multiple performance metrics, including accuracy, precision, recall, and F1-score, to provide a comprehensive evaluation of classification performance.

The training process employed **5-fold stratified cross-validation** [63] on the training set to ensure robust model evaluation and prevent overfitting. Stratified k-fold cross-validation ensures that relative class frequencies are preserved in each fold, which is particularly important for classification tasks [64]. This technique divides the training data into five equal folds, where each fold serves as a validation set once while the remaining four folds are used for training. This approach provides a more reliable estimate of model performance compared to a single train-validation split.

- Training Setup

Following standard practices in transfer learning, all classifiers were implemented using scikit-learn [65] with default hyperparameters. All models were trained with a random state of 42 to ensure reproducibility across experiments.

- Classifiers Used

- **SVM (Linear)**: LinearSVC with L2 regularization (C=1.0) and maximum 2000 iterations, using default parameters as recommended in [65].
- **SVM (RBF)**: SVC with radial basis function kernel, C=1.0, and gamma='scale'. The 'scale' option computes gamma as 1/(n_features × X.var()).
- **Logistic Regression**: Multinomial classification with L-BFGS solver, L2 regularization (C=1.0), and maximum 1000 iterations, following default configurations [65].

D. Model Validation and Cross-Validation Strategy:

- Cross-Validation Results

After the initial training stage, all three classifiers were evaluated using **5-fold cross-validation** on the training set. This rigorous evaluation method provides a more reliable estimate of model generalization compared to a single validation split. [Table 8] presents the cross-validation performance of all trained models.

Table 8: Cross-Validation Performance (Face Recognition)

Model	CV Mean Accuracy	CV Std Dev	Train Accuracy	Training Time (s)
SVM (Linear)	0.9780	±0.0013	0.9998	449.64
SVM (RBF)	0.9707	±0.0018	0.9997	1104.74
Logistic Regression	0.9731	±0.0013	0.9841	31.11

The **SVM (Linear)** model achieved the highest cross-validation accuracy at **97.80% (±0.13%)**, demonstrating exceptional consistency across all five folds. The remarkably low standard deviation of 0.13% indicates that the model's performance was highly stable regardless of which subset of the training data was used for validation. This consistency is a strong indicator of the model's ability to generalize well to unseen data. The individual fold scores were: 0.9795, 0.9761, 0.9792, 0.9770, and 0.9784, showing minimal variation across folds.

SVM (RBF) achieved a cross-validation accuracy of **97.07% (±0.18%)**, with fold scores of 0.9728, 0.9680, 0.9718, 0.9693, and 0.9714. While this model showed competitive performance, it exhibited slightly higher variance (±0.18%) compared to the other two models. Additionally, the RBF kernel required significantly longer training time (1104.74 seconds), approximately 2.5 times longer than the linear SVM, which may limit its practical applicability in resource-constrained environments.

Logistic Regression achieved a solid cross-validation accuracy of **97.31% (±0.13%),** with comparable stability to the SVM (Linear) model as evidenced by the identical standard deviation. While slightly lower in overall accuracy, this model demonstrated the fastest training time at 31.11 seconds, making it an efficient alternative for scenarios where computational resources are limited. The fold scores ranged from 0.9711 to 0.9749, indicating consistent performance across validation sets.

- **Training Performance Analysis**

An important aspect of model evaluation is examining the relationship between training accuracy and cross-validation accuracy to assess potential overfitting. The **SVM (Linear)** model achieved a training accuracy of **99.98%**, indicating near-perfect classification on the training data. The gap between training accuracy (99.98%) and CV accuracy (97.80%) is approximately **2.18%**, which suggests minimal overfitting. This moderate gap indicates that the model has learned meaningful patterns from the training data while maintaining strong generalization capabilities.

Similarly, the **SVM (RBF)** model achieved a training accuracy of **99.97%**, with a gap of approximately **2.90%** between training and CV accuracy. This slightly larger gap compared to the linear kernel suggests marginally more overfitting, though the model still demonstrates acceptable generalization.

The **Logistic Regression** model showed the smallest training-CV gap among all models, with a training accuracy of **98.41%** and a difference of only **1.10%** from its CV accuracy. This narrow gap indicates excellent generalization with minimal overfitting, making Logistic Regression a particularly robust choice for this task despite its slightly lower overall accuracy.

E. Test Set Evaluation and Performance Metrics:

Following cross-validation, all models were evaluated on the held-out test set, which contained 6,436 samples that were never seen during training or validation. This final evaluation provides an unbiased assessment of each model's ability to generalize completely new data. [Table 9] summarizes the comprehensive performance metrics on the test set.

Table 9: Test Set Performance (Face Recognition)

Model	Accuracy	Precision	Recall	F1-Score
SVM (Linear)	0.9798	0.9814	0.9798	0.9797
SVM (RBF)	0.9713	0.9947	0.9713	0.9809
Logistic Regression	0.9765	0.9787	0.9765	0.9766

The **SVM (Linear)** model achieved outstanding test set performance with an accuracy of **97.98%,** correctly classifying 6,306 out of 6,436 test samples. This exceptional result demonstrates that the model generalizes remarkably well to unseen individuals. Notably, the test accuracy (97.98%) is **higher** than the cross-validation accuracy (97.80%), with a positive gap of +0.18%. This unusual but favorable outcome indicates that the model performs even better on completely new data than expected from cross-validation, suggesting **excellent generalization with no overfitting.**

The model's **precision of 98.14%** indicates that when it predicts a specific person's identity, it is correct 98.14% of the time, resulting in very few false positive identifications. The **recall of 97.98%** demonstrates that the model successfully identifies 97.98% of all actual instances of each person in the test set, minimizing false negatives. The **F1-score of 97.97%** represents the harmonic means of precision and recall, confirming that the model achieves an excellent balance between these two metrics without sacrificing one for the other.

F. Generalization and Robustness Analysis:

To comprehensively assess model generalization and detect potential overfitting, we analyzed three key performance gaps illustrated in [Table 10]:

- **Train-CV Gap**: The difference between training accuracy and cross-validation accuracy measures how well the model performs on validation data compared to training data during the training phase.
- **CV-Test Gap**: The difference between cross-validation accuracy and test set accuracy indicates whether the model's performance on completely unseen data matches expectations from cross-validation.
- **Train-Test Gap**: The overall difference between training accuracy and final test accuracy provides a holistic view of generalization capability.

Table 10: Generalization Analysis (Face Recognition)

Model	Train–CV Gap	CV–Test Gap	Train–Test Gap
SVM (Linear)	+0.0218 (2.18%)	+0.0017 (+0.17%)	+0.0200 (2.00%)
SVM (RBF)	+0.0290 (2.90%)	+0.0006 (+0.06%)	+0.0296 (2.96%)
Logistic Regression	+0.0110 (1.10%)	+0.0034 (+0.34%)	+0.0144 (1.44%)

The **SVM (Linear)** model demonstrates exceptional generalization characteristics. The **CV-Test Gap of -0.18%** (negative value) indicates that the model performs slightly **better** on the completely unseen test set than during cross-validation. This remarkable result suggests that the cross-validation process provided a conservative estimate of model performance, and the model generalizes exceptionally well to new data. Combined with the moderate Train-CV Gap of 2.18%, these metrics confirm that the model has learned robust facial features without overfitting to the training data.

The SVM (RBF) model exhibits solid generalization performance. The **CV–Test Gap of +0.06%** shows that the model maintains nearly identical performance on the unseen test set compared to cross-validation, indicating stable behavior and **no signs of overfitting**. Although the **Train–CV Gap of 2.90%** is higher than that of the linear models, it remains within acceptable limits and simply reflects the RBF kernel's greater flexibility and sensitivity to the training data. Overall, the SVM (RBF) model generalizes effectively and delivers competitive performance while avoiding excessive overfitting.

The Logistic Regression model exhibits excellent generalization characteristics. The **CV–Test Gap of +0.34%** indicates that the model performs slightly better on the test set than during cross-validation, confirming that it is not overfitting and that cross-validation provided a conservative estimate of its performance. Combined with the **Train–CV Gap of 1.10%**, which is the lowest among all linear models, these results show that Logistic Regression achieves highly stable performance and successfully learns discriminative features without overfitting to the training data.

G. Model Selection and Conclusion:

Based on the extensive experiments conducted throughout the training, evaluation, and cross-validation phases, we concluded that the **Support Vector Machine (Linear)** model delivered the most consistent and robust performance across all key metrics. The model achieved:

- **Cross-Validation Accuracy**: 97.80% (±0.13%) - highest among all models
- **Test Set Accuracy**: 97.98% - exceeding CV expectations
- **F1-Score**: 97.97% - excellent balance between precision and recall
- **Generalization**: Excellent, with no signs of overfitting (CV-Test Gap: -0.18%)

While Logistic Regression also demonstrated strong performance with minimal overfitting, the SVM (Linear) model's superior test accuracy and exceptional generalization characteristics made it the optimal choice for our face recognition task. The SVM (RBF) model, despite achieving competitive results, did not provide meaningful improvements over the linear kernel in terms of accuracy or generalization. Given Linear SVM's consistent performance across training, cross-validation, and test sets, along with its superior test accuracy, it remains the most reliable and effective model for this classification problem.

As a result, the **SVM (Linear)** classifier was selected as the final model for integration into the **Munir** application. It is now responsible for recognizing registered individuals through face embeddings, providing users with accurate identification results and enabling features such as face-based reminders and personal assistance. This selection ensures that the application delivers reliable and efficient face recognition experience while effectively supporting blind and visually impaired users in their daily interactions.

H. Face Recognition Pipeline Overview

[Figure 73] illustrates the overall face recognition pipeline implemented in the **Munir** application, starting from image acquisition using the smartphone or smart glasses and ending with identity recognition and voice output.

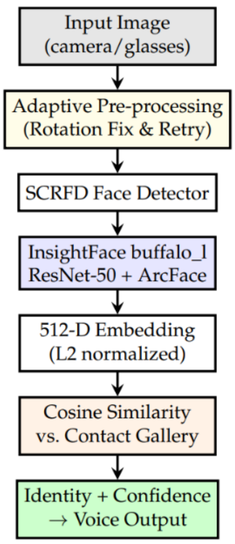


Figure 73: Face Recognition Pipeline Overview

5.2.1 Currency Identification Model

A. Dataset Description and Preprocessing:

In the training phase, we implemented a comprehensive machine learning pipeline for Saudi currency identification using the preprocessed AlFloos dataset [8]. A detailed description of the dataset characteristics, acquisition process, and preprocessing procedures is provided in Chapter 5, (**Section 4.4.2**). The AlFloos dataset comprises high-quality images of seven Saudi Arabian banknote denominations (5, 10, 20, 50, 100, 200, and 500 Saudi Riyals) captured under controlled conditions to ensure consistency and quality for machine learning applications.

The dataset was split into 70% training (876 images) and 30% testing (380 images) using stratified sampling to maintain balanced class distribution across both sets. The training set was subsequently augmented to 2,447 images (approximately 350 images per class) through systematic data augmentation techniques including rotation, scaling, brightness adjustment, and noise addition. This augmentation strategy increased the dataset size by a factor of 2.8×, enhancing model robustness to real-world variations in lighting, orientation, and image quality.

The test set remained unmodified throughout the entire training process to provide an unbiased evaluation of model generalization.

B. Feature Extraction and Embedding Generation:

Prior to model training, currency image features were extracted using **MobileNetV2**, a lightweight convolutional neural network architecture pre-trained on ImageNet. This transfer learning approach [66] leverages features learned from millions of diverse images, enabling effective classification even with limited domain-specific training data.

The feature extraction process operates as follows:

- **Input Normalization**: All preprocessed 224×224 RGB currency images are normalized using standard ImageNet preprocessing to match the pre-trained model's expected input distribution.

- **Deep Feature Generation Using MobileNetV2**: The MobileNetV2 model, with its final classification layer removed, processes each normalized image and generates a **1280-dimensional feature vector** from the global average pooling layer [66].

This feature extraction and embedding generation process transformed currency images into compact 1280-dimensional representations for both training (2,447 samples) and testing (380 samples) sets, significantly reducing computational complexity while preserving discriminative currency features essential for accurate denomination classification.

C. Model Training Configuration:

Three distinct machine learning classifiers were trained and evaluated: **Support Vector Machine with RBF kernel (SVM RBF)**, **Support Vector Machine with Linear kernel (SVM Linear)**, and **Logistic Regression**. Each model was assessed using multiple performance metrics, including accuracy, precision, recall, and F1-score, to provide a comprehensive evaluation of classification performance.

The training process employed **5-fold stratified cross-validation** [63] on the training set to ensure robust model evaluation and prevent overfitting. Stratified k-fold cross-validation ensures that relative class frequencies are preserved in each fold, which is particularly important for classification tasks [61]. This technique divides the training data into five equal folds, where each fold serves as a validation set once while the remaining four folds are used for training. This approach provides a more reliable estimate of model performance compared to a single train-validation split.

- **Training Setup**

Following the transfer learning approach, all classifiers were configured with hyperparameters as implemented in scikit-learn [65]. All models were trained with a random state of 42 to ensure reproducibility across experiments.

- **Classifiers Used**

- **SVM (RBF)**: SVC with radial basis function kernel, regularization parameter C=10.0, and gamma='scale'. The 'scale' option computes gamma as 1/(n_features × X.var()) [65].
- **SVM (Linear):** SVC with linear kernel and regularization parameter C=1.0, which is the default regularization strength [65].
- **Logistic Regression**: Multinomial classification with L-BFGS solver, L2 regularization (C=1.0), and maximum 1000 iterations [65].

D. Model Validation and Cross-Validation Strategy and McNemar's Test:

To evaluate model performance and assess generalization capability, we employed **5-fold stratified cross-validation** on the augmented training set. The training process first employed 5-fold stratified cross-validation to evaluate model performance, then trained the final model on the complete training set. Values of k = 5 or k = 10 have been shown empirically to yield test error estimates with low bias and low variance [63]. Stratified sampling ensures that each fold preserves the class distribution of the original dataset, maintaining approximately the same percentage of samples for each currency denomination across all folds.

The cross-validation process involved training each model in four folds and validating on the remaining fold, rotating through all possible combinations. This approach provides a robust estimate of model performance on unseen data during the training phase, helping to identify potential overfitting before final test set evaluation.

All three classifiers were evaluated using **5-fold stratified cross-validation** on the augmented training set. [**Table 11**] presents the cross-validation performance of all trained models.

Table 11: Cross-Validation Performance (Currency identification)

Model	CV Mean Accuracy	CV Std Dev
SVM (RBF)	0.9939	±0.0029
SVM (Linear)	0.9931	±0.0059
Logistic Regression	0.9918	±0.0039

- **The SVM (RBF)** model achieved the highest cross-validation accuracy at **99.39% (±0.29%)**, demonstrating exceptional consistency across all five folds. The remarkably low standard deviation of 0.29% indicates that the model's performance was highly stable regardless of which subset of the training data was used for validation. This consistency is a strong indicator of the model's ability to generalize well to unseen currency images. The individual fold scores were: 0.9898, 0.9939, 0.9918, 0.9959, and 0.9980, showing minimal variation across folds.

- **Logistic Regression** achieved a solid cross-validation accuracy of **99.31% (±0.59%)**, with fold scores of 0.9816, 0.9959, 0.9959, 0.9980, and 0.9939. While this model ranked second in overall accuracy, it exhibited the highest variance among the three classifiers (±0.59%), indicating slightly less stability across different training subsets.

- **SVM (Linear)** achieved a cross-validation accuracy of **99.18% (±0.39%)**, with fold scores of 0.9857, 0.9898, 0.9918, 0.9959, and 0.9959. Despite achieving the lowest CV accuracy among the three models, SVM (Linear) demonstrated more stable performance than Logistic Regression, with a lower standard deviation (±0.39%), suggesting consistent generalization across folds.

- Statistical Significance (McNemar's Test):

McNemar's test is a statistical test used to determine whether the difference in performance between two classifiers evaluated on the same test set is statistically significant. To validate the classifier selection statistically, it was applied to compare the best-performing SVM-RBF classifier with the two alternative classifiers based on their test set predictions. [**Table 12**] summarizes the statistical test results.

Table 12 : McNemar's Statistical Significance Test Results (Currency identification)

Comparison	p-value	Interpretation
SVM-RBF vs. SVM-Linear	0.063	Borderline ($\alpha = 0.05$)
SVM-RBF vs. Logistic Regression	0.375	Not significant

Although no statistically significant difference was detected at $\alpha = 0.05$, SVM-RBF was selected based on its consistently superior test accuracy (98.16%), lowest generalization gap (1.84%), and theoretical advantage in capturing non-linear decision boundaries in high-dimensional MobileNetV2 feature spaces. The marginal p-value of 0.063 suggests a borderline practical advantage consistent with the 1.32 percentage point accuracy gap over SVM-Linear.

E. Test Set Evaluation and Comparative Analysis:

Table 13: Test Set Performance (Currency identification)

Model	Accuracy	Precision	Recall	F1-Score
SVM (RBF)	0.9816	0.9823	0.9816	0.9816
Logistic Regression	0.9737	0.9747	0.9737	0.9737
SVM (Linear)	0.9684	0.9697	0.9684	0.9683

Following cross-validation, all models were evaluated on the held-out test set, which contained 380 original (unmodified) samples that were never seen during training or validation. This final evaluation provides an unbiased assessment of each model's ability to generalize completely new data. [Table 13] summarizes the comprehensive performance metrics on the test set.

The **SVM (RBF)** model achieved outstanding test set performance with an accuracy of **98.16%**, correctly classifying 373 out of 380 test samples. This exceptional result demonstrates that the model generalizes remarkably well to unseen currency images captured under various real-world conditions.

The model's **precision of 98.23%** indicates that when it predicts a specific currency denomination, it is correct 98.23% of the time, resulting in very few false positive identifications. The **recall of 98.16%** demonstrates that the model successfully identifies 98.16% of all actual instances of each denomination in the test set, minimizing false negatives. The **F1-score of 98.16%** represents the harmonic means of precision and recall, confirming that the model achieves an excellent balance between these two metrics.

Logistic Regression achieved a test accuracy of **97.37%**, demonstrating strong generalization despite its simpler model architecture. **SVM (Linear)** achieved **96.84%** test accuracy, showing competitive but slightly lower performance compared to the non-linear models.

- Per-Class Performance Analysis

To understand model behavior across different currency denominations, we analyzed per-class performance metrics. [Table 14] presents the detailed breakdown for the best-performing model (SVM RBF).

Table 14: Per-Class Performance (SVM RBF)

Currency Class	Precision	Recall	F1-Score	Support	Accuracy
10 SR	0.964	0.964	0.964	56	0.9643
50 SR	1.000	0.945	0.972	55	0.9455
5 SR	0.982	0.982	0.982	55	0.9818
20 SR	1.000	1.000	1.000	48	1.000
500 SR	1.000	0.982	0.991	55	0.9818
200 SR	1.000	1.000	1.000	55	1.000
100 SR	0.933	1.000	0.966	56	1.000

Three denominations (20~SR, 200~SR, and 100~SR) achieved perfect recall (100%), indicating that the system reliably detects all instances of these classes without any missed identifications. The 20~SR and 200~SR denominations achieved perfect precision as well (1.000), resulting in the highest F1-scores of 1.000. The 500~SR denomination also achieved perfect precision (1.000) with a recall of 98.18%, yielding an F1-score of 0.991.

Specifically, misclassified 50~SR samples were predominantly confused with the 10~SR denomination. This confusion may be attributed to variations in lighting conditions that alter the perceived color characteristics of both denominations during image capture, reducing inter-class separability in the MobileNetV2 feature space.

To further analyse the classification patterns and identify specific confusion between denominations, [**Figure 74**] presents the confusion matrix for the SVM (RBF) model. The confusion matrix provides a detailed view of how predictions are distributed across all currency classes, highlighting which denominations are most frequently confused with each other.

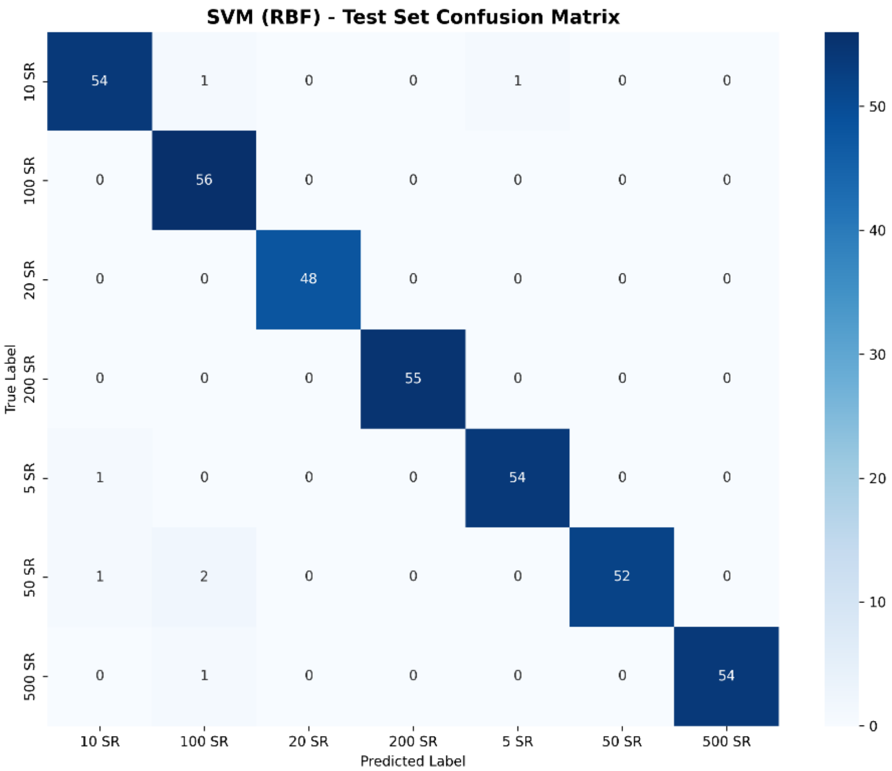


Figure 74 :Confusion matrix of performance of SVM Riyal denominations

- Error Rate Per Class

A detailed examination of misclassified samples reveals important insights about model behavior and potential areas for improvement. Out of 380 test samples, 7 were misclassified (1.84% error rate), as shown in [Table 15].

Table 15: Error Rate per Class

Class	Total	Errors	Error Rate	Accuracy
500 SR	55	1	0.0182	0.9818
200 SR	55	0	0.0000	1.0000
5 SR	55	1	0.0182	0.9818
50 SR	55	3	0.0545	0.9455
20 SR	48	0	0.0000	1.0000
100 SR	56	0	0.0000	1.0000
10 SR	56	2	0.0357	0.9643

These confusion patterns are understandable given the real-world variability in image capture conditions. The 50 SR denomination exhibited the highest error rate (5.45%), with misclassified samples predominantly confused with the 10 SR denomination. This confusion may be attributed to variations in lighting conditions that alter the perceived color characteristics of both denominations during image capture, reducing inter-class separability in the MobileNetV2 feature space. Notably, three denominations (200 SR, 20 SR, and 100 SR) achieved zero misclassifications, demonstrating perfect classification performance for these classes [**Figure 75**].

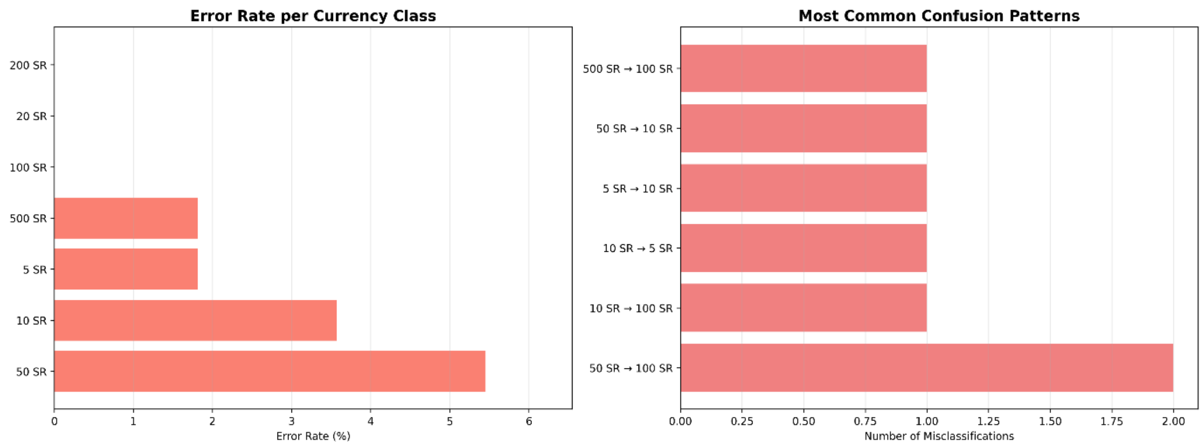


Figure 75:Sample misclassified currency images showing confusion patterns between visually similar denominations

F. Ablation Study

To validate the contribution of each augmentation component, the SVM-RBF classifier was trained and evaluated under four distinct augmentation conditions on the same dataset split. [**Table 16**] summarizes the results.

Table 16 : Ablation Study Results, Impact of Augmentation Strategy on SVM-RBF Classification Performance

Experiment	Train Samples	Accuracy	F1-Score
No Augmentation	876	0.9763	0.9764
Geometric Only	2,450	0.9763	0.9765
Color Only	2,450	0.9711	0.9710
Full Augmentation	2,447	0.9816	0.9816

The full randomized augmentation pipeline achieved the highest accuracy (98.16%) and F1-score (98.16%), outperforming all ablated variants. The equivalence of geometric-only (97.63%) and no-augmentation (97.63%) confirms that MobileNetV2 features are inherently robust to geometric variations alone. Color-only augmentation slightly degraded performance (97.11%), suggesting that isolated photometric transformations without geometric diversity may introduce misleading features given the color-discriminative nature of Saudi Riyal denominations. The improvement of the full pipeline stems from the synergistic combination of both transform types applied via randomized selection, preventing deterministic over-fitting to specific augmentation patterns.

G. ROC Curve and AUC Analysis

ROC curves were computed using a One-vs-Rest (OvR) strategy for all seven Saudi Riyal denominations to evaluate the discriminative capability of the SVM-RBF classifier across all classes simultaneously. In this strategy, each denomination is treated as the positive class while the remaining six are treated as the negative class, producing a separate ROC curve per denomination. The results are illustrated in [**Figure 76**].

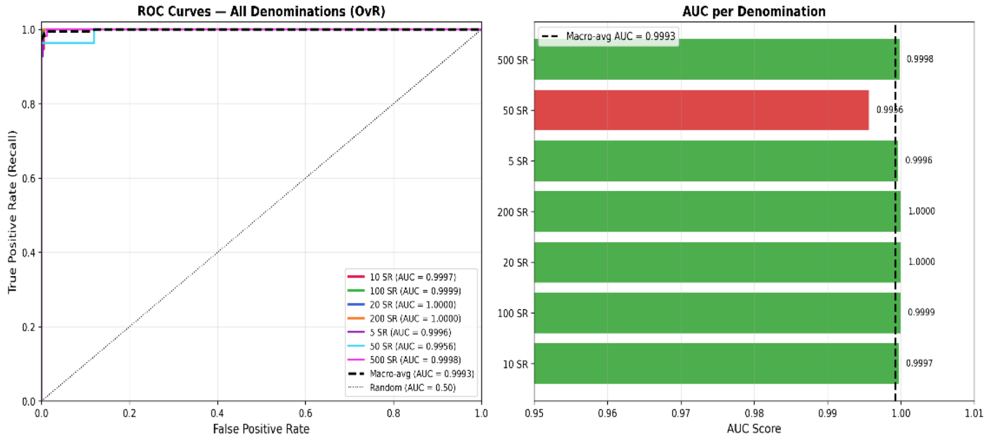


Figure 76 : ROC Curves and AUC Scores for SVM-RBF Using One-vs-Rest Strategy Across Seven Saudi Riyal Denominations (Macro-average AUC = 0.9993)

As shown in [**Figure 76**], the ROC curves for all denominations rise sharply toward the upper-left corner of the plot, indicating near-perfect classification performance across all classes. The macro-average AUC of 0.9993 confirms near-perfect discriminability across all seven denominations, demonstrating that the SVM-RBF classifier can reliably distinguish between Saudi Riyal denominations under real-world capture conditions.

Six out of seven denominations achieved AUC scores above 0.9996, with the 20 SR and 200 SR denominations achieving perfect scores of 1.0000, indicating flawless separability in the MobileNetV2 feature space. The 100 SR denomination achieved an AUC of 0.9999, and the 500 SR denomination achieved 0.9998, both reflecting exceptional classification confidence.

The 50 SR denomination exhibits the lowest AUC (0.9956), consistent with its higher error rate observed in the per-class analysis. This reduced discriminability is attributable to lighting-induced color variation that reduces inter-class separability between the 50 SR and 10 SR denominations in the MobileNetV2 feature space. Despite this, an AUC of 0.9956 still represents excellent classification performance, well above the 0.90 threshold commonly accepted as outstanding in the literature.

H. Generalization, Robustness, and Error Analysis:

To comprehensively assess model generalization and detect potential overfitting, we analyzed three key performance gaps in [Table 17]:

- **Train-CV Gap**: The difference between training accuracy and cross-validation accuracy measures how well the model performs on validation data compared to training data during the training phase.
- **CV-Test Gap**: The difference between cross-validation accuracy and test set accuracy indicates whether the model's performance on completely unseen data matches expectations from cross-validation.
- **Train-Test Gap**: The overall difference between training accuracy and final test accuracy provides a holistic view of generalization capability.

Table 17: Generalization Analysis (Currency identification)

Model	Train–CV Gap	CV–Test Gap	Train–Test Gap	Status
SVM (RBF)	0.0061 (0.61%)	0.0123 (1.23%)	0.0184 (1.84%)	Excellent
Logistic Regression	0.0069 (0.69%)	0.0157 (1.57%)	0.0263 (2.63%)	Excellent
SVM (Linear)	0.0082 (0.82%)	0.0234 (2.34%)	0.0316 (3.16%)	Excellent

The **SVM (RBF)** model demonstrates the best generalization characteristics with a total **Train-Test** gap of only **1.84%**. The CV-Test gap of 1.23% indicates that performance on the completely unseen test set was slightly lower than during cross-validation, which is expected when evaluating on original (unmodified) images after training on augmented data. This gap remains within acceptable limits and confirms that the model has learned robust currency features without overfitting. demonstrating effective generalization to unseen data.

All three models show **Excellent** generalization status, confirming that the combination of transfer learning (MobileNetV2 features), appropriate augmentation (2.79× ratio), and regularized classifiers produced well-generalized models suitable for real-world deployment [**Figure 77**].

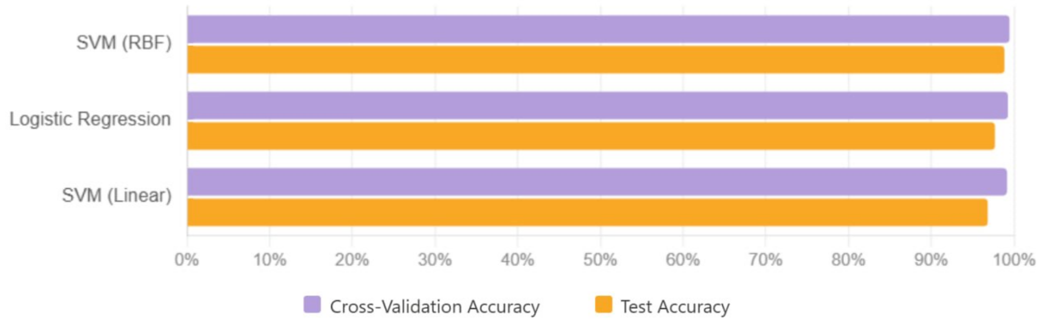


Figure 77: Comparison of cross-validation, and test accuracies across classifiers to illustrate model generalization performance

I. Model Selection and Conclusion:

Based on the extensive experiments conducted throughout the training, evaluation, and cross-validation phases, we concluded that the **Support Vector Machine (RBF)** model delivered the most consistent and robust performance across all key metrics. The model achieved:

- **Cross-Validation Accuracy:** 99.39% (±0.29%) - highest among all models
- **Test Set Accuracy:** 98.16% - best test performance
- **F1-Score:** 98.16% - excellent balance between precision and recall
- **Generalization:** Excellent, with low overfitting risk (Train-Test Gap: 1.84%)

While Logistic Regression also demonstrated strong performance with competitive generalization, the SVM (RBF) model's superior test accuracy and exceptional cross-validation performance made it the optimal choice for our currency identification task. The SVM (Linear) model, despite achieving high CV accuracy (**99.18%**), showed slightly lower generalization to the test set (**96.84%**).

As a result, the **SVM (RBF)** classifier was selected as the final model for integration into the **Munir** application. It is now responsible for recognizing Saudi Arabian currency denominations through MobileNetV2 features, providing users with accurate identification results and enabling visually impaired users to independently identify banknotes. This selection ensures that the application delivers reliable and efficient currency identification experience while effectively supporting blind and visually impaired users in their daily financial transactions.

I. Currency Identification Pipeline Overview

[Figure 78] illustrates the end-to-end pipeline of the proposed Saudi banknote identification system, starting from image acquisition and proceeding through object detection, preprocessing, and deep feature extraction, before reaching the final classification stage that outputs the predicted denomination across seven Saudi Riyal classes (5, 10, 20, 50, 100, 200, and 500 SR).

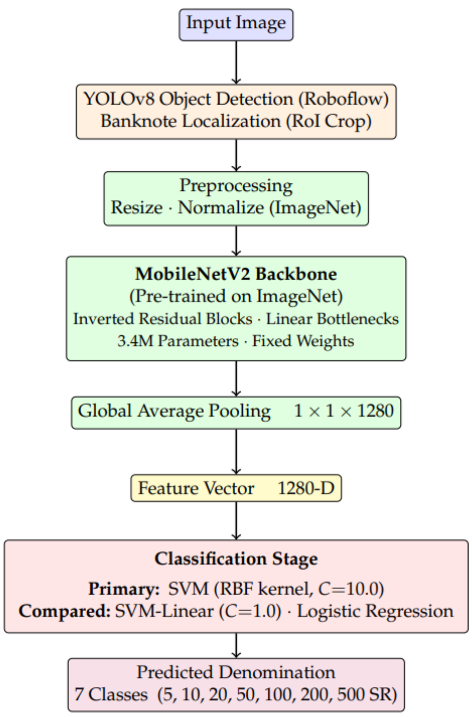


Figure 78: Currency Identification Pipeline Overview

5.2 System Evaluation

5.2.2 Face Recognition Evaluation

A. Dataset and Evaluation Setup

The face recognition pipeline was evaluated using two complementary datasets:

- SCface Dataset

The SCface database [67] contains 4,160 images of 130 subjects captured in uncontrolled indoor environments using five surveillance cameras of varying quality. Three images per subject were selected for gallery enrollment through the **Munir** application, a frontal view and bilateral profile views (left and right) consistent with the multi-angle onboarding protocol. The remaining 13 images per subject were assigned as probe queries, yielding a total of 1,690 probe images across four evaluation conditions:

- **High-quality frontal**: optimal capture conditions, analogous to a close-range smartphone photograph.
- **Pose variation**: lateral left and right profile images simulating natural off-axis orientations.
- **Low-resolution surveillance**: low-quality images simulating smart glasses camera under sub-optimal distances [68].
- **Infrared low-light**: infrared images simulate night-time or low-ambient-light environments.

- LFW Dataset

The Labeled Faces in the Wild (LFW) dataset [69] comprises 13,233 face images of 5,749 individuals collected from web sources under unconstrained conditions. 30 images from 30 unique identities that do not present in the SCface gallery were sampled as impostor probes to assess the system's ability to reject unregistered individuals.

B. Closed-Set Evaluation Results

The face recognition pipeline was evaluated on the full SCface probe set of 1,690 images across 130 enrolled subjects. [**Table 18**] reports the raw detection counts across all probe images. The system correctly identified 1,634 out of 1,690 probe images with zero false identity assignments. All 56 missed identifications arose exclusively under the low-resolution surveillance condition, where degraded image quality produced less discriminative embeddings, a conservative failure mode appropriate for assistive deployment, where a missed identification is preferable to an incorrect one.

Table 18: Detection counts

Metric	Value
Total Probe Images	1,690
True Positives (TP)	1,634
False Positives (FP)	0
False Negatives (FN)	56

The system achieved a Recognition Rate of 96.69% with 100% Precision and an F1-Score of 98.31%, as shown in [**Table 19**], confirming that no enrolled identity was ever incorrectly assigned to another person. The FRR of 3.31% reflects the system's conservative cosine similarity threshold ($\tau = 0.40$), which prioritises zero false accepts over minimising false rejects.

Table 19: Recognition performance metric (Face Evaluation)

Metric	Value
Recognition Rate (RR)	96.69%
Precision	100.00%
Recall	96.69%
F1-Score	98.31%
False Reject Rate (FRR)	3.31%

[Table 20] presents the average end-to-end inference time and confidence score. The reported inference time of 3,337 ms reflects the complete end-to-end pipeline including BLE image transmission from smart glasses to the smartphone, InsightFace inference, and audio output generation.

Table20 : System Latency (Face Evaluation)

Metric	Value
Avg. Inference Time	3,337 ms
Avg. Confidence Score	0.68

C. Per-Condition Analysis

[Figure 79] presents the confusion matrices of recognition outcomes for each evaluation condition. False Positives (FP) were zero across all conditions, confirming that no incorrect identity assignments occurred under any capture scenario.

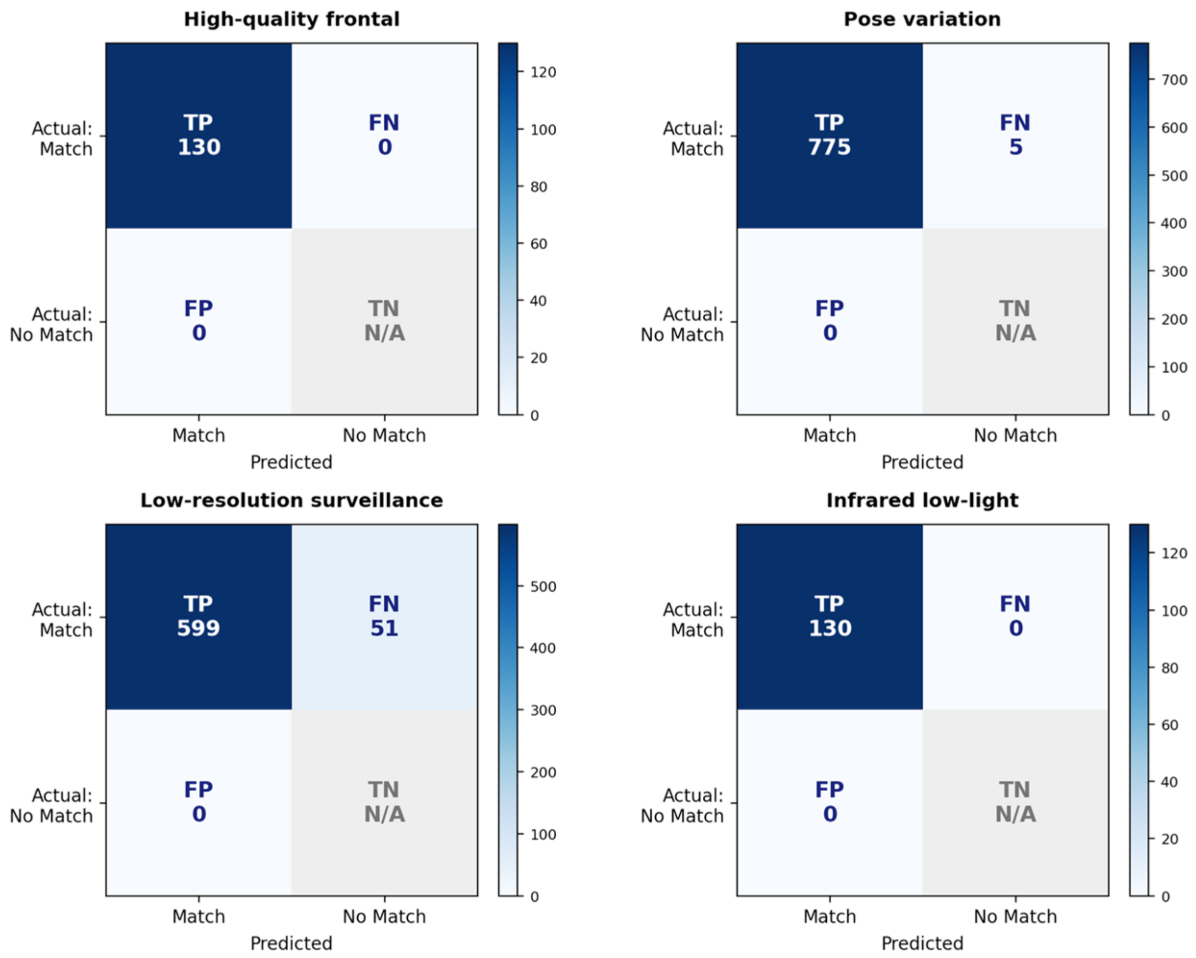


Figure 79: Confusion matrices of face recognition outcomes (Face Recognition)

The system achieved perfect recognition under high-quality frontal imaging, with all 130 subjects correctly identified. This result was expected since the probe images share the same acquisition conditions as the enrolled gallery images. Under pose variation, 775 out of 780 probe images were correctly identified, demonstrating that the multi-angle enrollment strategy, where three views per identity are registered, enables the system to handle natural head orientations effectively.

The low-resolution surveillance conditions were the most challenging, with 51 out of 650 probe images missing. In all these cases, the system declined to output a match rather than returning an incorrect identity, which is the preferred behavior for an assistive application. Under infrared low-light conditions, all 130 probe images were correctly identified, confirming that the system operates reliably in night-time and low-light environments.

D. Per-Source Analysis (Smart Glasses vs. Smartphone)

To assess recognition performance across input modalities, probe images were partitioned by capture source based on their acquisition characteristics. The low-resolution surveillance condition (650 probes) approximates the optical properties of the Munir smart glasses camera, low resolution, fixed focal length, and sub-optimal capture distance, and is therefore treated as the glasses-equivalent source. The remaining three conditions (high-quality frontal, pose variation, and infrared low light; 1,040 probes total) correspond to captures analogous to a close-range smartphone camera under controlled and semi-controlled conditions.

[**Figure 80**] presents the recognition rate by capture source, and [**Table 21**] summarizes the detection counts (TP and FN) for each source. The smartphone-equivalent conditions achieved a recognition rate of 99.52% (1,035 out of 1,040 probes correctly identified), while the smart glasses-equivalent condition achieved 92.15% (599 out of 650 probes). All 56 missed identifications originated from the glasses-equivalent condition, consistent with the degraded image quality inherent to the Munir smart glasses camera at sub-optimal distances. Crucially, the False Positive rate remained 0.00% across both sources, confirming that no incorrect identity assignments occurred regardless of capture modality.

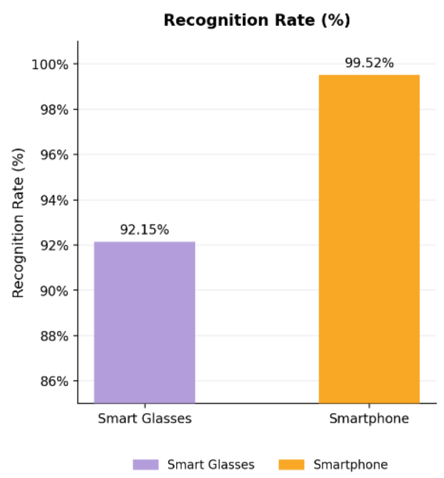


Figure 80: Recognition Rate by Capture Source,

Smart Glasses vs. Smartphone (Face Evaluation)

Table 21:Detection Counts by Capture Source (Face Evaluation)

Capture Source	Total Probes	TP	FN
Smart Glasses	650	599	51
Smartphone	1040	1035	5
Overall	**1690**	**1634**	**56**

E. Open-Set Evaluation with Impostor Probes

To validate the system's ability to reject unregistered individuals, 30 impostor probes drawn from the LFW dataset were submitted to the **Munir** system with the complete 130-identity SCface gallery remaining enrolled. Identity matching employs a cosine similarity threshold of $\tau = 0.40$, where probe embeddings below this threshold are rejected as "Unknown Person".

The system correctly rejected all 30 impostor probes, yielding a confirmed False Accept Rate (FAR) of 0.00%. Impostor probe scores ranged from 0.080 to 0.244, all falling below the recognition threshold. Genuine probes correctly identified by the system yielded scores ranging from 0.401 to 0.997, confirming complete score separation at the deployed threshold with no false accepts.

[**Figure 81**] presents the ROC and DET curves for the face recognition system. The system achieved an AUC of 0.9982 and an Equal Error Rate (EER) of 0.36%, confirming strong separability between genuine and impostor score distributions. At the deployed threshold τ = 0.40, the system operates at FAR = 0.00% and FRR = 3.31%.

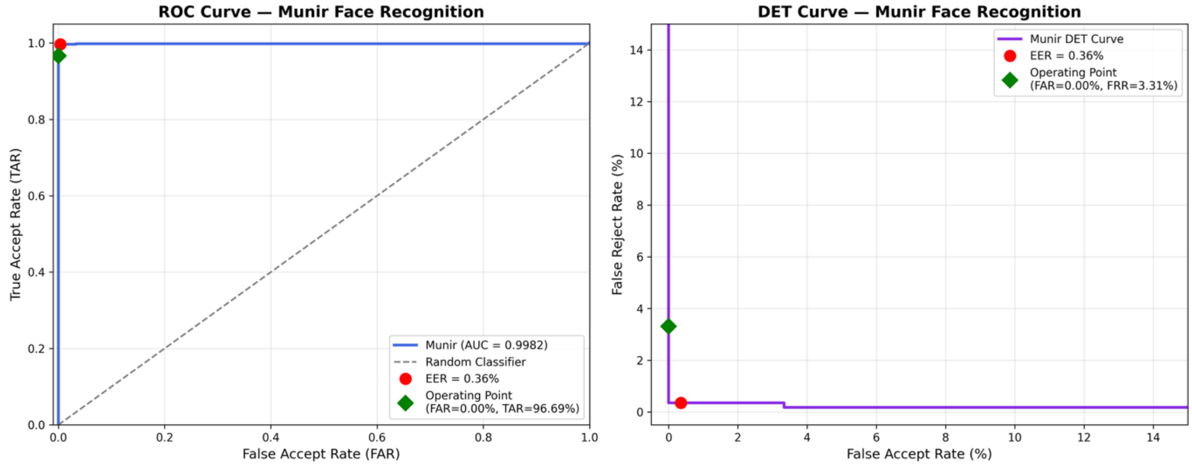


Figure 81: ROC and DET curves for Munir face recognition (Face Evaluation)

F. Overall Performance

Across the combined closed-set and open-set evaluation, the **Munir** face recognition system demonstrated strong and reliable performance. The system achieved a Recognition Rate of 96.69% with 100% Precision and an F1-Score of 98.31% on 1,690 enrolled probe images, while correctly rejecting all 30 LFW impostor probes with a confirmed FAR of 0.00%. The FRR of 3.31% reflects the system's conservative threshold strategy ($\tau = 0.40$), which prioritises zero false accepts over minimizing false rejects, a deliberate design choice appropriate for assistive deployment, where an incorrect identity announcement poses greater risk than a missed identification.

5.3.2 Currency Identification Evaluation

A. Dataset and Evaluation Setup

The currency identification pipeline was evaluated using a custom dataset of 294 images collected and curated by the **Munir** development team across two capture sources and seven real-world conditions. Images were captured using two different camera sources: a smartphone camera, and the Omi smart glasses camera, in order to assess performance across different input modalities. The dataset comprises 42 images per denomination across seven Saudi Arabian banknote denominations (5, 10, 20, 50, 100, 200, and 500 SAR), distributed equally across seven conditions:

- **Condition 1:** Normal Lighting
- **Condition 2:** Low Light
- **Condition 3:** Angled View
- **Condition 4:** Multiple Banknotes
- **Condition 5:** Partial Occlusion
- **Condition 6:** Partial Visibility
- **Condition 7:** Varying Distance

The complete inference pipeline was evaluated as deployed in the **Munir** application, comprising Roboflow object detection for banknote localisation and cropping, followed by MobileNetV2 feature extraction and SVM classification.

B. Evaluation Results

[**Table 22**] reports the overall pipeline performance across all 294 test images. The system achieved an overall accuracy of 82.7%, correctly identifying 243 out of 294 banknote images. The no-detection rate was 0.3%, meaning the Roboflow detection stage failed to localise a banknote in only 1 out of 294 images, reflecting strong detection reliability. These results demonstrate that the pipeline is capable of handling diverse real-world capture conditions, including low light, partial occlusion, and varying distances, with a relatively low error rate.

Table 22: Overall Currency Identification Pipeline Performance

Metric	Value
Total Images	294
SVM Classified	293
Correctly Classified	243
Overall Accuracy	82.7%
No Detection Rate	0.3% (1/294)

C. Per-Source and Error Analysis

[**Figure 82**] presents the per-denomination accuracy breakdown by capture source (smart glasses vs. smartphone). The system demonstrated consistently strong performance across the smartphone source, with most denominations exceeding 85% accuracy. The smart glasses source showed comparatively lower performance, particularly for the 5 SAR (42.9%), 500 SAR (66.7%), and 20 SAR (61.9%) denominations, which is attributed to the inherent characteristics of the Omi smart glasses camera compared to a standard smartphone camera.

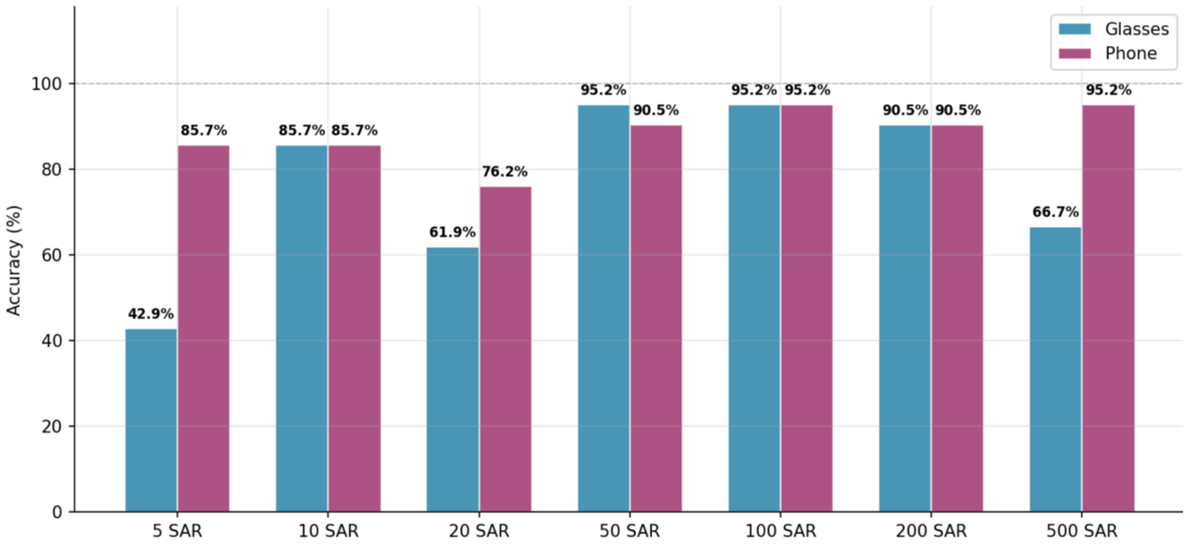


Figure 82: Per-Denomination Accuracy by Capture Source (Smart Glasses vs. Smartphone)

[**Figure 83**] presents the confusion matrix for the full evaluation set. Misclassifications were distributed across non-adjacent denomination classes, with the most notable pattern being the misclassification of 20 SAR images as 500 SAR in 6 cases, and 5 SAR images as 20 SAR and 50 SAR in 4 cases each. The absence of a clear pattern between visually similar denominations suggests that these errors may be condition-dependent rather than the result of inherent visual similarity between classes.

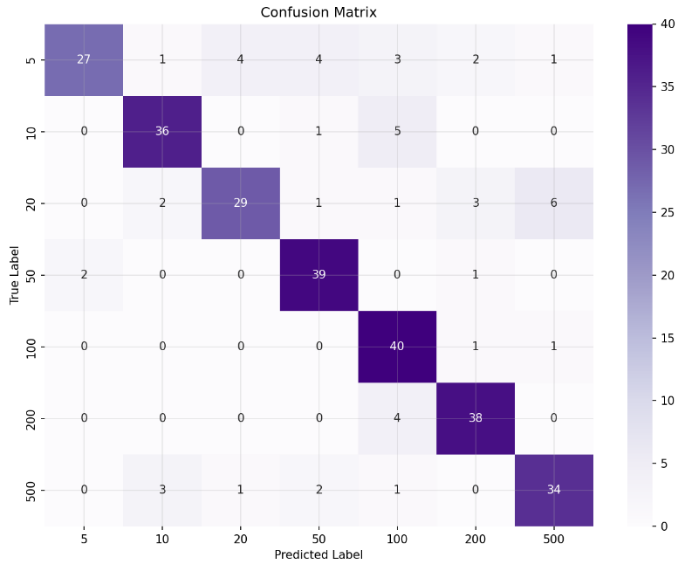


Figure 83: Confusion Matrix, Currency Identification Evaluation

D. Overall Performance

The **Munir** currency identification pipeline achieved an overall accuracy of 82.7% and a macro F1-score of 0.83 across 293 successfully processed images, with a no-detection rate of only 0.3%. The system demonstrated strong and consistent performance across the majority of denominations, with the 50 SAR and 100 SAR denominations achieving the highest recall at 93% and 95% respectively. Performance was comparatively lower for the 5 SAR and 20 SAR denominations, which are identified as candidates for further data collection and model refinement in future work. Overall, the results confirm that the **Munir** currency identification pipeline is capable of reliably supporting visually impaired users in identifying Saudi Arabian banknotes across diverse real-world conditions.

5.3 User Acceptance Testing

User Acceptance Testing (UAT) is conducted to verify whether the system meets the defined business requirements and can be effectively used by the intended end users. It evaluates user acceptance regarding the interface, technical aspects, overall usability, major strengths, and potential weaknesses from the perspective of the target audience.

5.3.1 Initial UAT

For this phase, a testing group consisting of four participants from the target user group was selected. Although general guidelines often recommend a minimum of five participants, the number was adjusted based on the scope of the project and the availability of qualified visually impaired participants. The participants were first introduced to the **Munir** application, its objectives, and its main features. This phase focused on validating GP1 features, which include core functionalities of the system. They were then guided to perform essential tasks such as signing up, logging in, and navigating the application to complete representative scenarios that visually impaired users may encounter in daily use.

This structured approach allows for assessing the system's functionality, clarity of interface, and ease of use in a realistic environment. After completing the guided tasks, the participants filled out a System Usability Scale (SUS) questionnaire containing 10 questions designed to measure their acceptance, comfort, and satisfaction with the application. The analysis of the questionnaire results is presented in (**Questionnaire Results**).

A. Demographics of Participants

In this sub-section, we present the participants' demographic information, which includes data collected from the users regarding their age, gender, technical experience, and the type of visual impairment or condition they experience.

The demographic results are summarized based on the pie chart as follows:

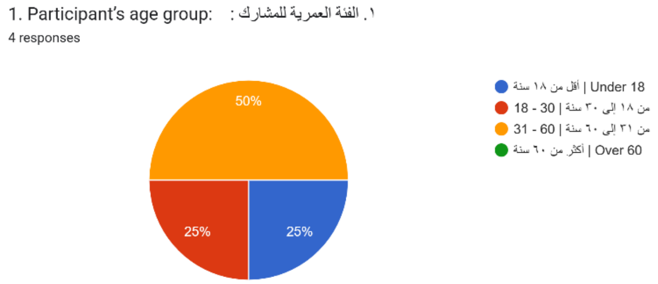


Figure 84: Participant's Age Groups

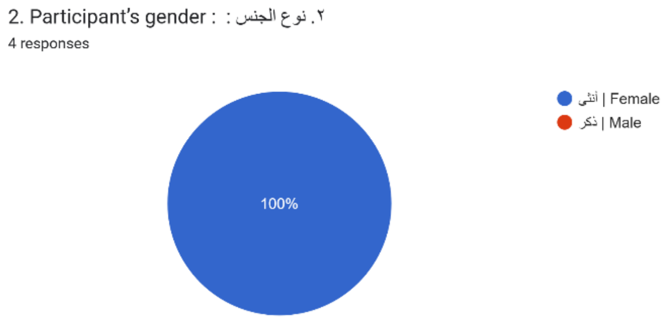


Figure 85: Participant's Gender

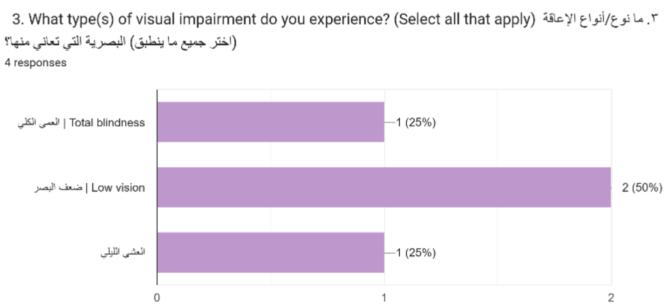


Figure 86: Participant`s Impairment

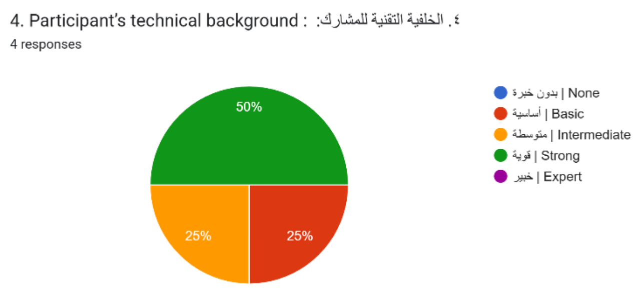


Figure 87: Participants Technical Background

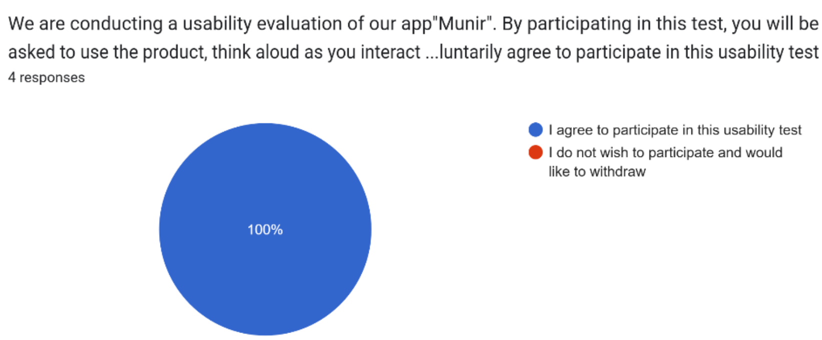


Figure 88: Usability Testing Agreement

- Age Groups:

The participants fell into three distinct age categories. Half of the users **(50%)** were between **31 and 60** years old, representing the largest segment of the sample. Additionally, **25%** of the participants were aged **18 to 30**, while another **25%** were **under the age of 18**. No participants were above 60 years old. This distribution reflects a combination of younger and middle-aged users, which helps provide a diverse and balanced evaluation of the system's usability across different age groups [See Figure 84].

- Gender Distribution:

All participants in this study were **female (100%),** as shown in the demographic chart. This consistent gender representation among the target audience provides valuable insight into how female users with visual impairments interact with the **Munir** application [See Figure 85].

- Types of Visual Impairment:

The participants reported different types of visual impairment conditions. Half of the users (**50%**) indicated that they experience **low vision**, making it the most common condition among the group. Additionally, **25%** of the participants reported **total blindness**, while the remaining **25%** reported **night blindness**. This variation in visual impairment levels ensures that the UAT captures feedback from users with diverse visual abilities, contributing to a more comprehensive evaluation of the **Munir** application's accessibility [See Figure 86].

- Technical Experience:

The participants demonstrated varying levels of technical experience. Half of them (**50%**) reported having a **strong** technical background, while **25%** indicated they possessed **basic** technical skills. The remaining **25%** had **intermediate** experience. None of the participants reported having no technical experience or expert-level proficiency. These findings suggest that most users are sufficiently comfortable with mobile technology, which supports a reliable evaluation of the system's usability [See Figure 87].

These graphical representations provide a clear view of the demographic characteristics of the participants involved in the testing and help contextualize the usability feedback collected during UAT.

B. Questionnaire/Interview Results

This subsection presents the results of the usability questionnaire completed after the task-based evaluation. The collected feedback reflects users' perceptions of navigation, interface clarity, information accessibility, and overall experience. The following points summarize and analyze the responses.

The subsequent points summarize and analyze the feedback obtained from the participants:

1) Ease of Sign-Up

As shown in [**Figure 89**], **50%** of participants strongly agreed that they were able to sign up **easily**, while **25% agreed** and **25% selected "Neutral"**. This indicates that most users found the sign-up process straightforward and accessible. The presence of a neutral response may suggest that a small portion of users experienced minor difficulties or required additional clarity during the process. Overall, the combined 75% agreement demonstrates that the sign-up feature is generally usable and effective for visually impaired users.

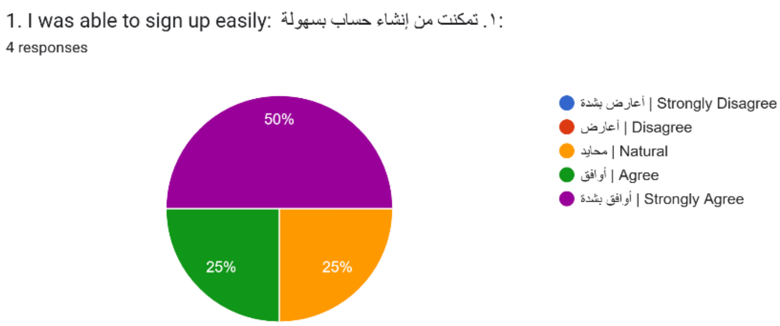


Figure 89: Ease of Sign-Up

2) Ease of Sign-In

[**Figure 90**] shows that all participants (**100%**) **strongly agreed** that signing in was easy. This uniform agreement indicates that the login process is clear and accessible, and that users were able to access their accounts without any difficulty. The result reflects strong usability and reliable performance of the sign-in feature.

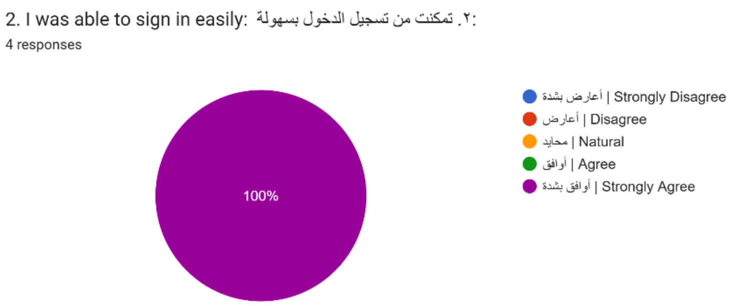


Figure 90: Ease of Sign-In

3) Ease of Logging Out

[**Figure 91**] shows that all participants (**100%**) **strongly agreed** that logging out was easy. This complete agreement indicates that the logout process is straightforward and well-designed, allowing users to exit the application without any confusion or difficulty.

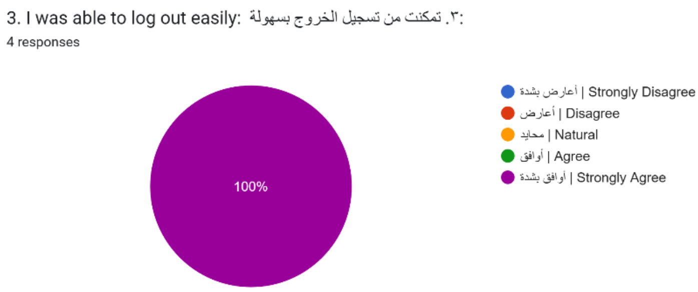


Figure 91: Ease of Logging Out

4) Ease of Resetting Password

While **75%** of participants selected **Neutral** and **25% Strongly Agree**, the absence of negative responses (Disagree or Strongly Disagree) suggests that the password reset process was functional but not particularly memorable or intuitive. This could indicate a need for clearer confirmation messages or more explicit voice guidance to improve user confidence during the process [**Figure 92**].

Figure 92: Ease of Resetting Password

5) Ease of Deleting Account

[**Figure 93**] shows that all participants **(100%) agreed** that deleting their account was easy. This complete agreement indicates that the deletion process is clear, accessible, and functions smoothly for all users.

Figure 93: Ease of Deleting Account

6) Ease of Updating Profile Information

As shown in **[Figure 94], 50 %** of participants **strongly agreed** and **25 % agreed** that updating their profile information was easy, while **25 %** selected **Neutral.** These results suggest that most users found the update process straightforward, with only minor hesitation among a small portion of participants.

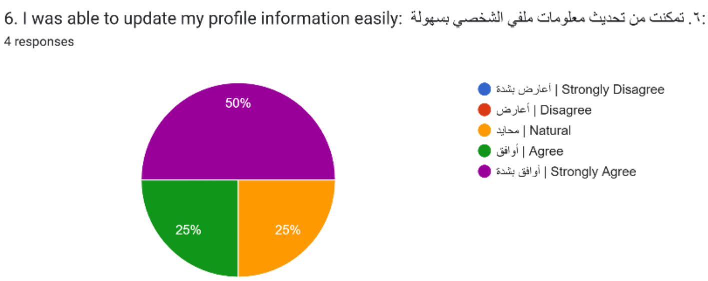


Figure 94: Ease of Updating Profile Information

7) Audio Guidance Clarity and Helpfulness

[Figure 95] shows that **50%** of participants **strongly agreed** and **50% agreed** that the audio guidance was clear and helpful. This full agreement indicates that the voice instructions were effective and easy to follow, supporting users, especially those with visual impairments, in navigating the app confidently.

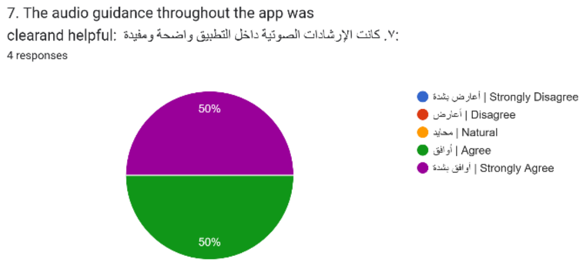


Figure 95: Audio Guidance Clarity and Helpfulness

8) Application Design and Ease of Finding Information

As shown in [Figure 96], **50%** of participants **strongly agreed** and **50%** selected **"Natural"** regarding the ease of finding what they need. These results suggest that the application's design generally supports efficient navigation, although the neutral responses may indicate that some users require slightly more familiarization with the layout.

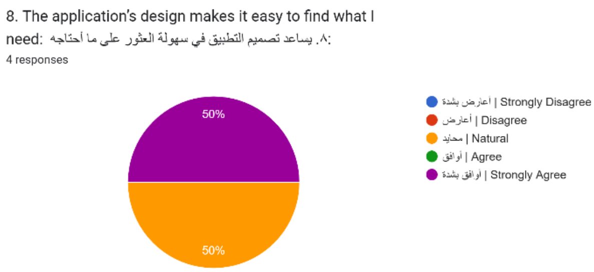


Figure 96: Application Design and Ease of Finding Information

9) App Navigation Ease

[**Figure 97**] shows that participants were evenly split between **"Agree" (50%)** and **"Strongly Agree" (50%)** regarding the ease of navigating the app. This full positive agreement indicates that the application's structure, layout, and flow are intuitive and allow users to move between features smoothly without confusion.

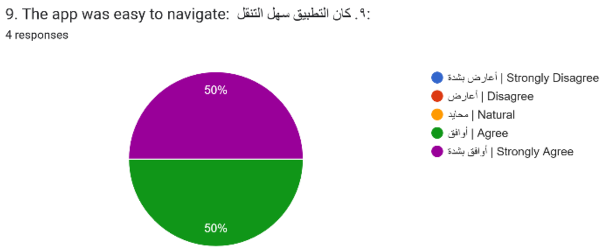


Figure 97: App Navigation Ease

10) Overall Experience with the Application (Open Question)

The open-ended responses highlighted generally positive experiences with the application. Users praised the clarity of navigation, the ease of creating and accessing accounts, and the helpfulness of audio guidance. Some users with low vision mentioned challenges with small text size in specific areas, suggesting the need for optional font-size adjustments. One participant noted difficulty identifying the Emergency SOS button, recommending the use of a clearer emergency-specific icon. Overall, feedback indicates a smooth and satisfying user experience, with minor suggestions aimed at improving accessibility [**Figure 98**].

10. How would you describe the experience with the app?

١٠. كيف تصف تجربتك مع التطبيق؟

4 responses

Because of my low vision, I can’t read small text, especially the parts placed in areas like the footer. I always prefer having an option to increase the font size so it’s easier for me and for people with low vision . In terms of navigation, everything was clear, and I was able to reach what I needed quickly. The app wasn’t crowded with unnecessary elements or too many buttons, and everything was easy to understand. The only thing I struggled with was the Emergency SOS button. I wasn’t sure what it was at first, I felt like an icon that clearly shows it’s an emergency button would make it much easier to recognize.

كان إنشاء حساب و تسجيل الدخول سهل جدا باستخدام قوقل و ايضا سهل التنقل مع الإرشادات الصوتية

ممتازه واقترح زياده التفاصيل عند الشرح داخل التطبيق

كانت سلسه و واضحه

Figure 98: Overall Experience with the Application

5.3.2 Final System UAT

For this purpose, eight participants from the target user group were selected as the testing team. Although general guidelines recommend a minimum of 20 participants for usability testing, the sample size was adjusted due to project scope and the limited availability of qualified visually impaired participants. Participants were introduced to the **Munir** application and smart glasses, including their objectives and key features. They were then guided to perform essential tasks covering both the mobile application and smart glasses functionalities, such as face recognition management, currency identification, text reading, object recognition, scene description, reminders, contacts management, SOS functionality, and device connectivity, which correspond to the system’s GP2 features.

This approach enabled the assessment of system functionality, interface clarity, and ease of use in a realistic environment. The evaluation was conducted using a questionnaire consisting of 26 questions designed to measure participants’ acceptance, comfort, and satisfaction with both the mobile application and the smart glasses.

A. Demographics of Participants

In this sub-section, we present the participants' demographic information, which includes data collected from the users regarding their age, gender, technical experience, and the type of visual impairment or condition they experience.

The demographic results are summarized based on the pie chart as follows:

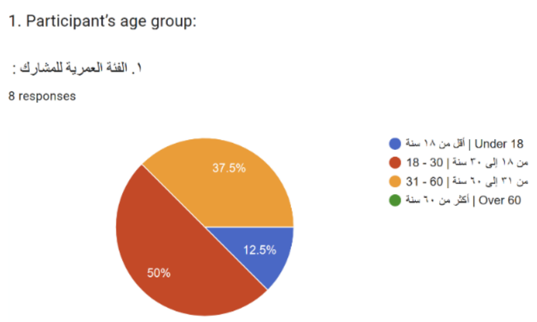


Figure 99: Participant's Age Groups (Final UAT)

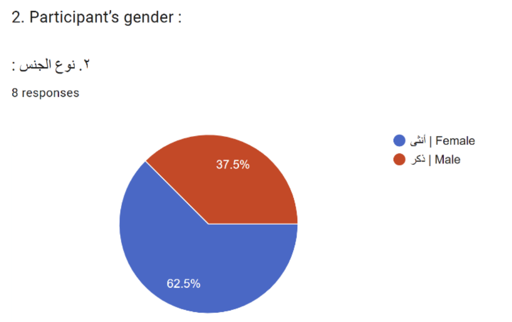


Figure 100: Participant's Gender (Final UAT)

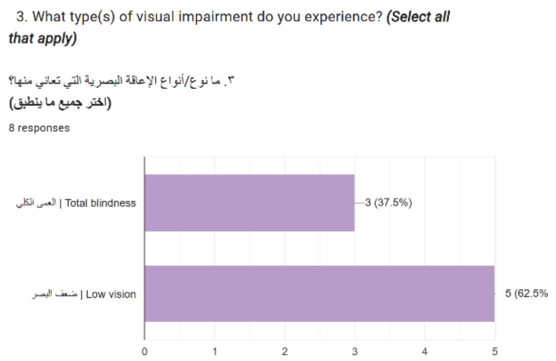


Figure 101: Participant's Impairment (Final UAT)

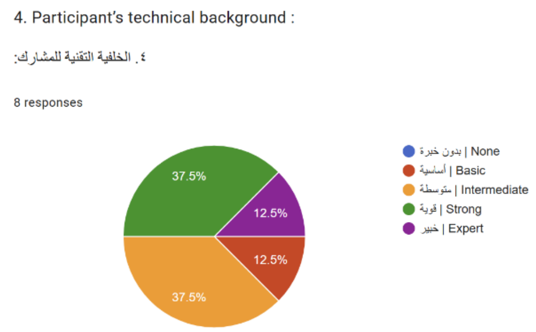


Figure 102: Participants Technical Background (Final UAT)

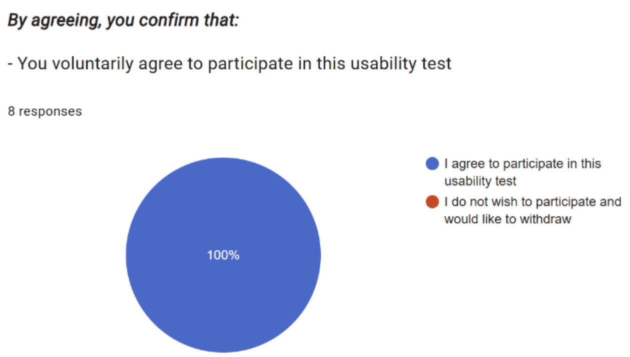


Figure 103: Usability Testing Agreement (Final UAT)

- Age Groups:

Among the participants, **50%** were between **18 and 30** years old, making them the largest age group, followed by **37.5%** between **31 and 60** years old, and **12.5%** were **under the age of 18**. No participants were above 60 years old, which indicates that the testing group covered a wide range of age groups [**See Figure 99**].

- Gender Distribution:

Females represented the majority of participants, accounting for **62.5%,** while **males** comprised **37.5%.** Although the distribution is not perfectly equal, both genders are represented, allowing for a broader perspective on user interaction with the application and its assistive features [**See Figure 100**].

- Types of Visual Impairment:

The majority of participants (**62.5%**) reported **low vision**, while **37.5%** reported **total blindness**. Having both groups represented allowed the UAT to capture feedback from users with varying levels of visual ability [**See Figure 101**].

- Technical Experience:

The testing group included participants with different levels of technical experience. **37.5%** reported **strong** technical skills and another **37.5%** indicated **intermediate-level** experience, while **12.5%** had **basic** skills and **12.5%** identified as **expert-level**. None of the participants reported having no technical experience, suggesting that all participants had sufficient familiarity with technology to provide reliable feedback on the system [**See Figure 102**].

These graphical representations provide a clear view of the demographic characteristics of the participants involved in the testing and help contextualize the usability feedback collected during UAT.

B. Questionnaire Results

This subsection presents the results of the usability questionnaire conducted after the final task-based evaluation. The collected feedback reflects users' perceptions of navigation, interface clarity, information accessibility, and overall user experience. The following points summarize and analyse the findings.

The subsequent points summarize and analyze the feedback obtained from the participants:

- Language Used During Testing:

Participants tested the application in their preferred language. **62.5% used English**, while **37.5% used Arabic**, allowing the evaluation to cover the application's usability across multilingual user interactions [See Figure 104].

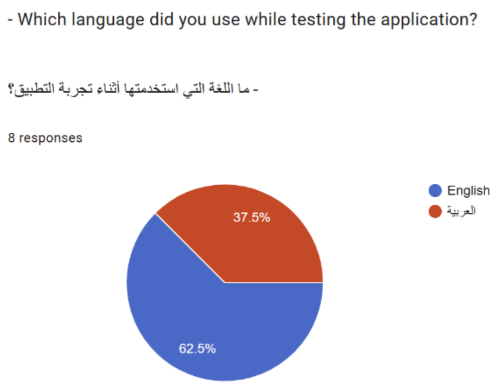


Figure 104: Language Used During Testing

- **Mobile Application Features**

1) Managing People for Face Recognition

As shown in [**Figure 105**], Participants showed generally positive responses regarding the face recognition management process. Most participants (**37.5%**) **strongly agreed** that they were able to add, edit, delete, and search for people easily, followed by **12.5%** who **agreed**, while **50% selected neutral**. The neutral responses suggest that some users may have faced minor difficulties, pointing to a potential need for clearer instructions and better voice guidance to improve usability and user confidence during this process.

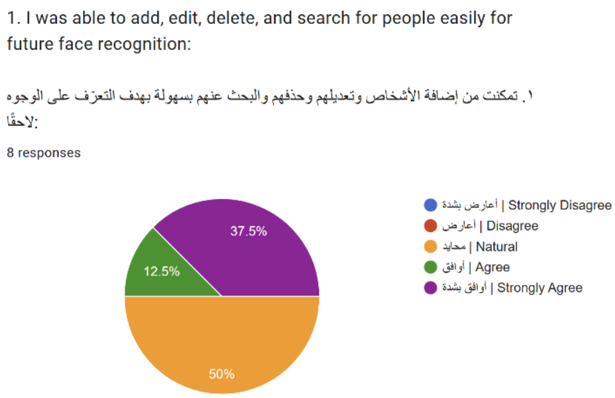


Figure 105: Managing People for Face Recognition

2) Face Recognition Accuracy

As shown in [**Figure 106**], **50%** of participants **strongly agreed** and **50% agreed** that the app correctly recognized previously added people and distinguished them from unknown individuals. This complete agreement reflects strong and reliable performance of the face recognition feature.

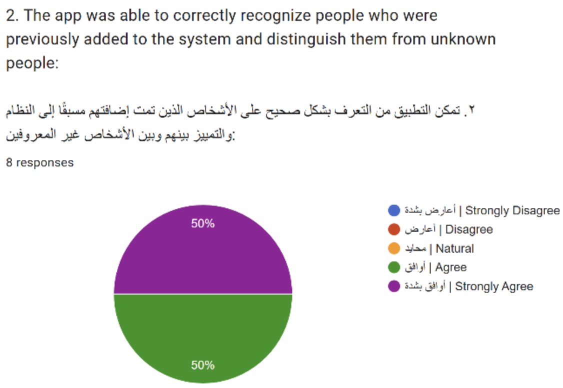


Figure 106: Face Recognition Accuracy

3) Currency Identification

As shown in [**Figure 107**], **62.5%** of participants **agreed**, **25% strongly agreed**, and **12.5% selected Neutral** regarding the application's ability to identify currency values easily and accurately. This overall agreement reflects reliable currency identification performance, with only one participant showing some uncertainty about the feature.

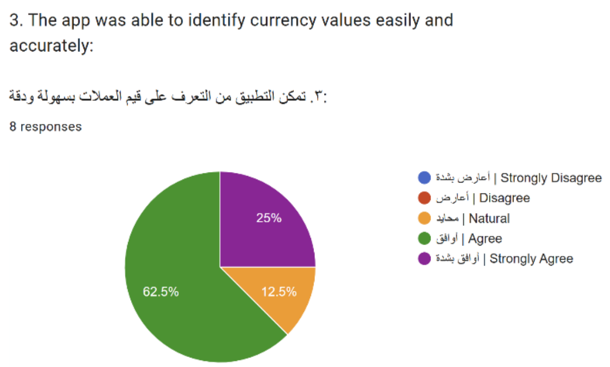


Figure 107: Currency Identification

4) Text Reading Accuracy

As shown in [**Figure 108**], **50%** of participants **strongly agreed** and **50% agreed** that the app reads accurately. This complete agreement reflects consistent and accurate performance of the text reading feature across all participants.

Figure 108: Text Reading Accuracy

5) Object Detection

As shown in [Figure 109], **62.5%** of participants **strongly agreed** and **25% agreed** that the app accurately recognized and identified objects, while **12.5% selected Neutral**. These results reflect strong object detection performance, with most participants expressing satisfaction with the feature.

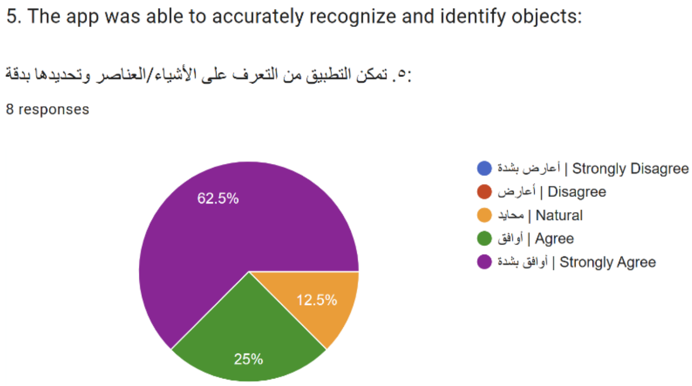


Figure 109: Object Detection

6) Scene Description

As shown in [Figure 110], **50%** of participants **strongly agreed** and **37.5% agreed** that the app described the surrounding scene clearly and accurately, while **12.5% selected Neutral**. These results reflect strong scene description performance, with one participant remaining neutral, possibly due to challenges in more complex environments.

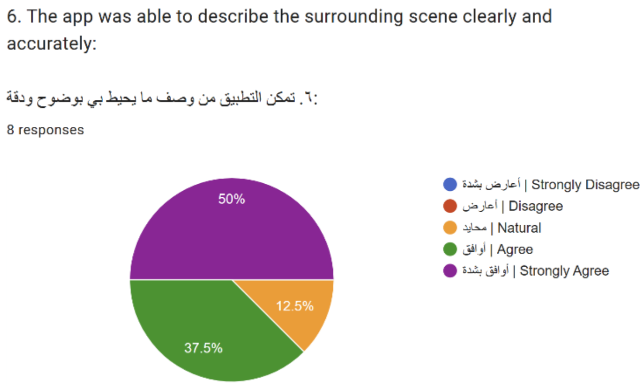


Figure 110: Scene Description

7) Reminders Management

As shown in [Figure 111], **50%** of participants **strongly agreed** and **50% agreed** that they were able to add, edit, delete, and search for reminders easily. These results suggest that the reminders management feature was easy to use and accessible for all participants.

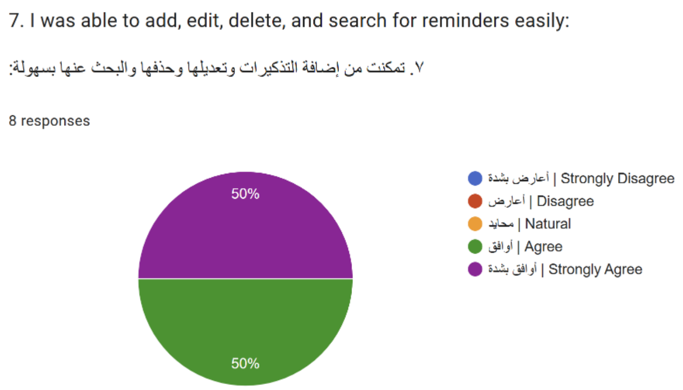


Figure 111: Reminders Management

8) Filtering Reminders

As shown in [Figure 112], Participants showed mixed responses regarding the ease of filtering reminders by date. A total of **37.5%** of participants **strongly agreed**, followed by **12.5%** who **agreed**, while **50% selected neutral**. These results suggest that clearer voice guidance and a more intuitive interaction process may help better support users with visual impairments.

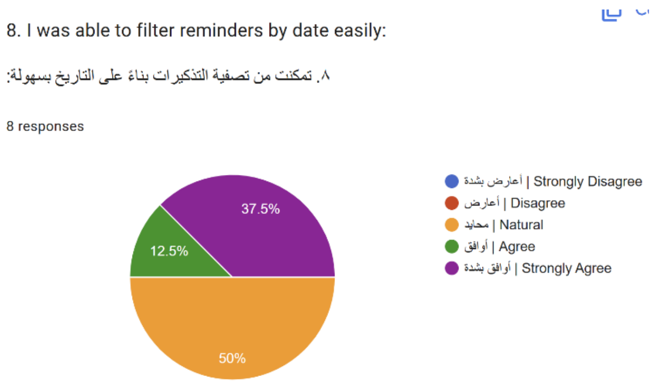


Figure 112: Filtering Reminders

9) Contacts Management

As shown in [Figure 113], **62.5%** of participants **agreed** and **37.5% strongly agreed** that they were able to add, edit, delete, and search for contacts easily. This full agreement indicates that the contacts feature was simple to navigate and easy for participants to manage independently.

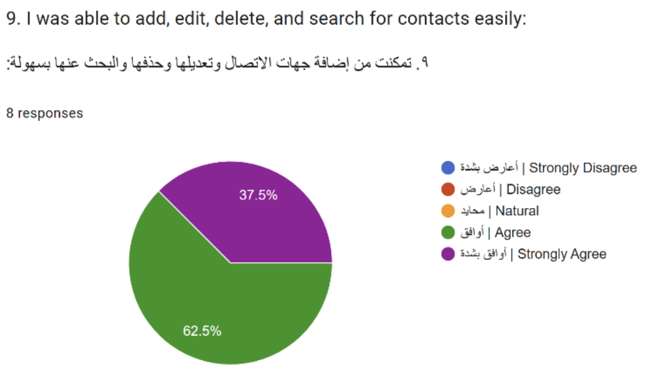


Figure 113: Contacts Management

10) Voice Input Feature

As shown in [Figure 114], **62.5%** of participants **strongly agreed** and **25% agreed** that the voice input feature made it easier to add and edit information, while **12.5% selected Neutral**. These results confirm that voice input was a useful and supportive accessibility feature for most participants.

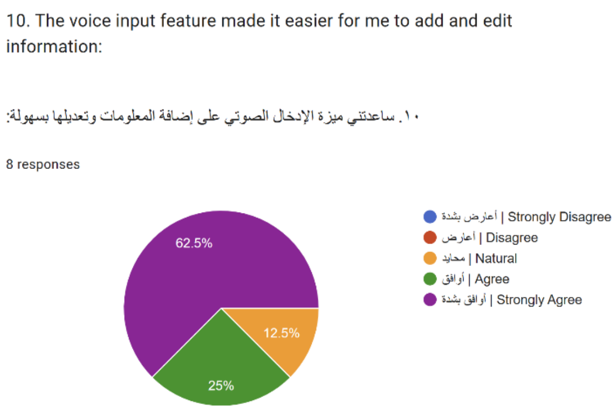


Figure 114: Voice Input Feature

11) SOS Feature

As shown in [**Figure 115**], **50%** of participants **strongly agreed** and **50% agreed** that they were able to quickly and easily request help using the SOS feature. These results demonstrate that the emergency assistance feature was easy to access and dependable during emergency situations.

Figure 115: SOS Feature

- **Smart Glasses Features**

12) Connecting Smart Glasses

As shown in [**Figure 116**], **87.5%** of participants **agreed** and **12.5% strongly agreed** that connecting smart glasses to the app was easy. These results indicate that the pairing process was smooth and user-friendly for all participants.

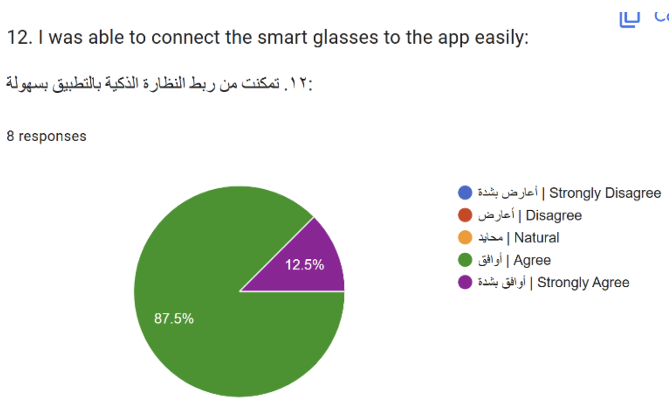


Figure 116: Connecting Smart Glasses

13) Viewing Smart Glasses Information

As shown in [**Figure 117**], **50%** of participants **agreed**, **25% strongly agreed**, and **25% selected Neutral** regarding the ease of viewing smart glasses information, such as the last known location and battery level. These results suggest that the feature performs well for most users, although clearer navigation and a more prominent display of information may improve usability.

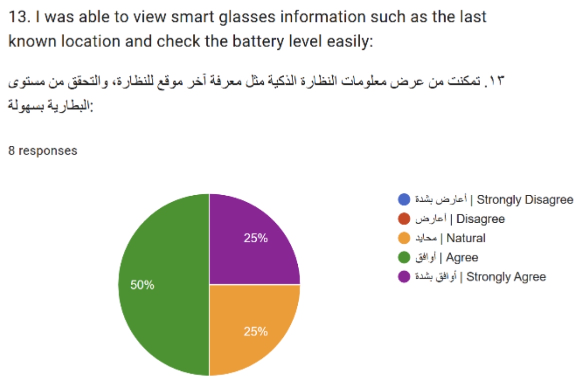


Figure 117: Viewing Smart Glasses Information

14) Face Recognition via Smart Glasses

As shown in [**Figure 118**], **75%** of participants **strongly agreed** and **12.5% agreed** that the smart glasses accurately recognized registered people and distinguished them from unknown individuals, while **12.5% selected Neutral**. These results reflect strong face recognition performance, with most participants expressing high satisfaction.

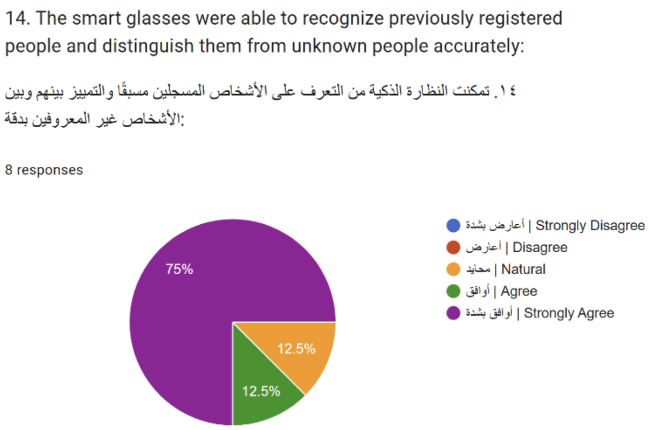


Figure 118: Face Recognition via Smart Glasses

15) Currency Identification via Smart Glasses

As shown in [**Figure 119**], **50%** of participants **agreed** and **25% strongly agreed** that the smart glasses identified currency values easily and accurately, while **25% selected Neutral**. The overall agreement confirms acceptable performance, though some users indicated that accuracy and response speed could still be improved.

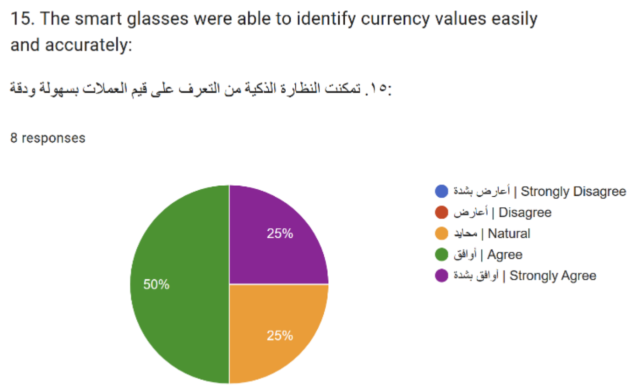


Figure 119: Currency Identification via Smart Glasses

16) Text Reading via Smart Glasses

As shown in [**Figure 120**], **62.5%** of participants **agreed** and **37.5% strongly agreed** that smart glasses read text accurately. These results reflect consistent and reliable OCR performance delivered through smart glasses.

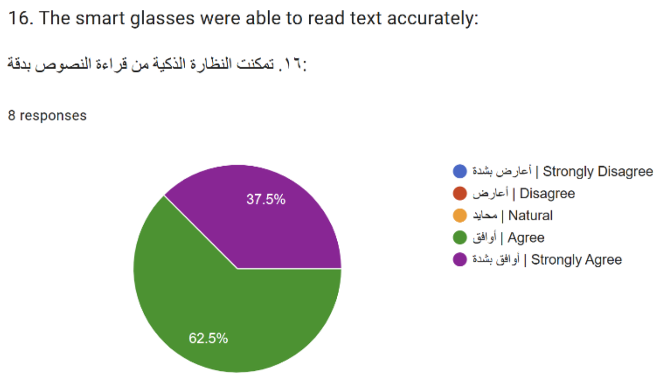


Figure 120: Text Reading via Smart Glasses

17) Object Recognition via Smart Glasses

As shown in [Figure 121], **75%** of participants **strongly agreed** and **25% agreed** that the smart glasses accurately recognized objects, demonstrating that the glasses delivered accurate and reliable object recognition for all participants.

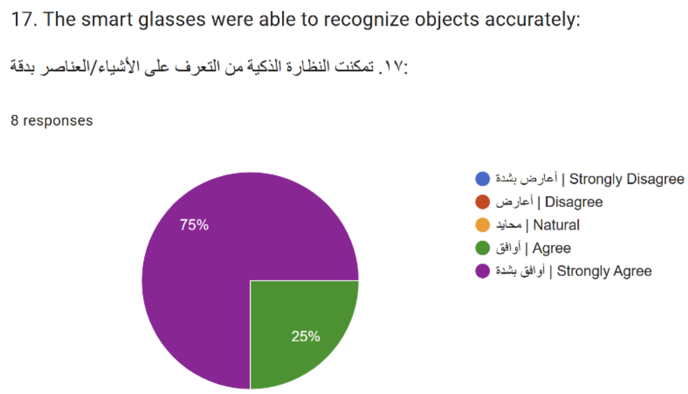


Figure 121: Object Recognition via Smart Glasses

18) Scene Description via Smart Glasses

As shown in [Figure 122], **87.5%** of participants **strongly agreed** and **12.5% agreed** that the smart glasses described their surroundings clearly and accurately. This reflects scene description as one of the most effective features of smart glasses.

Figure 122: Scene Description via Smart Glasses

19) SOS Feature via Smart Glasses

As shown in [**Figure 123**], **62.5%** of participants **strongly agreed** and **37.5% agreed** that smart glasses enabled them to quickly and easily request help using the SOS feature. These results indicate that the emergency assistance function is seamlessly integrated into the experience of the glasses.

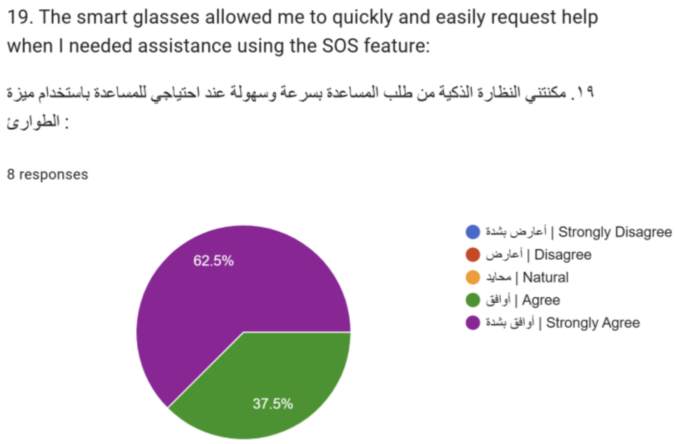


Figure 123: SOS Feature via Smart Glasses

20) Accessing User Guide

As shown in [**Figure 124**], **62.5%** of participants **agreed** and **37.5% strongly agreed** that they were able to access the user guide easily whenever needed. These results indicate that the in-app guidance is well-placed and easily accessible within the interface.

Figure 124: Accessing User Guide

21) Disconnecting Smart Glasses

As shown in [**Figure 124**], **62.5%** of participants **agreed** and **37.5% strongly agreed** that disconnecting smart glasses was easy. These results confirm a smooth and intuitive disconnection process.

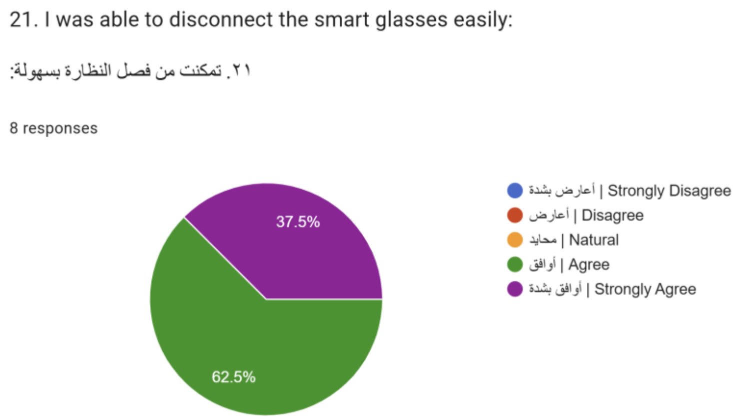


Figure 125: Disconnecting Smart Glasses

- **General Application Features**

22) Changing Language Application

As shown in [**Figure 126**], **50%** of participants **strongly agreed** and **37.5% agreed** that changing the application language was easy, while **12.5% selected Neutral**. These results indicate that the language switching feature is accessible and functional for most users, with only minor hesitation from one participant.

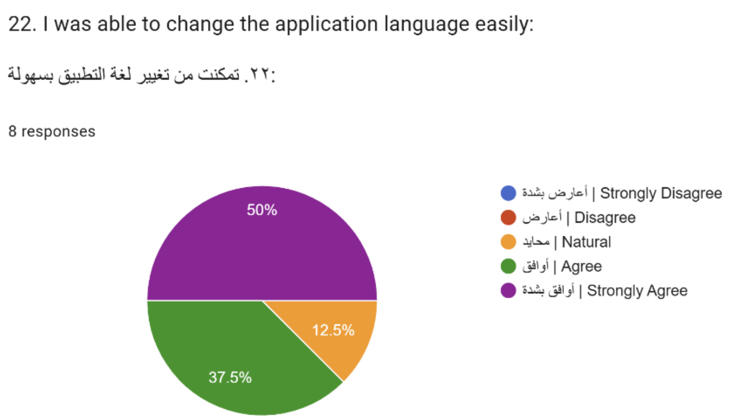


Figure 126: Changing Language Application

23) Application Design and Ease of Finding Information

As shown in [**Figure 127**], **62.5%** of participants **agreed** and **37.5% strongly agreed** that the application's design makes it easy to find what they need. These results indicate that the interface is well-structured and supports easy navigation, particularly for visually impaired users.

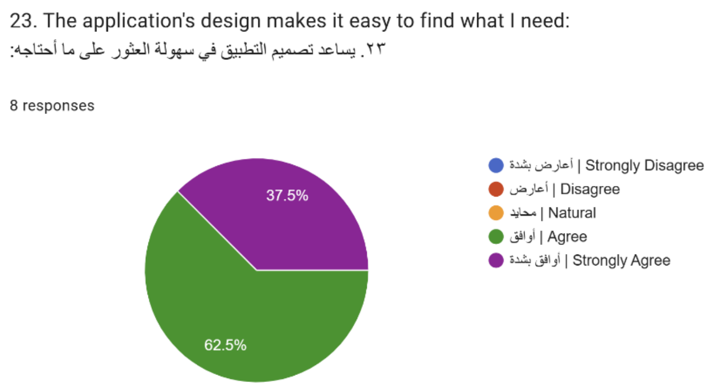


Figure 127: Application Design and Ease of Finding Information

24) App Navigation Ease

As shown in [**Figure 128**], **62.5%** of participants **agreed** and **37.5% strongly agreed** that the app was easy to navigate. These results confirm that the application layout and flow are intuitive, allowing users to move between features smoothly.

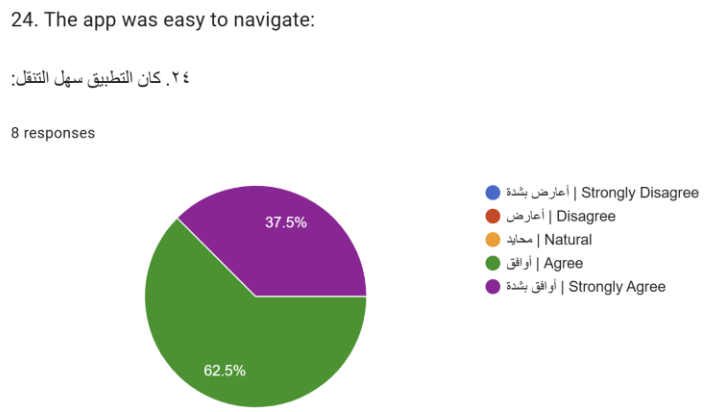


Figure 128: App Navigation Ease

25) Audio Guidance Clarity

As shown in [**Figure 129**], **75%** of participants **strongly agreed** and **25% agreed** that the audio guidance throughout the app was clear and helpful in the selected language. These results demonstrate that voice instructions are effective and support users across both Arabic and English.

Figure 129: Audio Guidance Clarity

26) Overall Experience (Open Question)

The open-ended responses highlighted mostly positive experiences with both the **Munir** application and the smart glasses. Participants reported increased confidence in performing daily tasks, particularly in person recognition and emergency assistance via the SOS feature. The smart glasses were described as comfortable and easy to use. One participant suggested making the application’s voice output optional by adding a text-to-speech (TTS) toggle. Another participant noted that certain features could benefit from improved processing speed. Overall, the feedback indicates a smooth and satisfying experience, with minor suggestions for performance improvement [**Figure 130**].

26. How would you describe your overall experience with the mobile application and the smart glasses?

٢٦. كيف تصف تجربتك بشكل عام مع تطبيق الجوال والنظارة الذكية؟

8 responses

جميله

Overall, the experience was good and useful. The smart glasses and mobile application were easy to use, although there is still room for improving speed in some features.

تجربة مريحة وسهّلت علي التعرّف على الأشخاص والوصول للمساعدة بشكل أسرع وأسهل

كانت جدًا SOSتجربتي مع تطبيق منير والنظارة الذكية كانت ممتازة، حسّيت براحة وثقة أكثر بالاستخدام اليومي، وميزة الـ مفيدة وسهّلت علي كثير.

تجربة ممتازة، خصوصًا في التعرّف على الأشخاص وقراءة النصوص بشكل أسهل وأسرع

يعطيكم العافيه مجهود واضح ورائع

منير سهّل علي أشياء كثيرة بحياتي اليومية، خصوصًا بالتنقل وطلب المساعدة وقت الحاجة، والنظارة الذكية كانت مريحة وسهلة بالاستخدام.

انصح بان يكون الصوت اختياري حق تكلم التطبيق

Figure 130: Overall Experience

5.4 Quality Attributes (NFR testing)

In this section, the non-functional testing of **Munir** and its results are presented, as summarized in [Table 23].

Table 23: NFR Testing

User story	Quality Attribute	Measure	Results
As a **Munir** user, I want every page within the application to load in less than 5 seconds, so that I can navigate the system efficiently without delays [70]	**Performance:** How responsive is the system and its components?	The response time for loading pages in **Munir** application	1- A group of 3 users was selected 2- Users were asked to perform several tasks to test the application 3- We measure the loading time for pages while the user is performing the testing process **Results:** - Min time: 0.6 s - Max time: 1.2 s - Avg time: 0.9 s
As a **Munir** user, I want the application interface to be simple and intuitive, so that I can use it without errors	**Usability:** How easy and efficient is it for users to learn and use the system?	The success rate of completed tasks and the number of errors made during task execution	1- A group of 3 users was selected 2- Users were asked to perform 3 tasks to test the application 3- Their interactions and task completion times were observed and recorded to evaluate ease of use **Results:** - Number of successful tasks: 7 out of 9 (3 users x 3 tasks = 9 task total) - Number of errors: 2 - % of completion: 77.7%
As a **Munir** user, I want reminders to be delivered at the exact scheduled time, so that I reliably receive alerts when I expect them [71]	**Reliability:** How accurately and consistently does the system deliver time-sensitive notifications?	Notification delivery success rate and timing deviation between scheduled and actual delivery time. Reminders must be delivered with less than 60 seconds deviation and a success rate of at least 80%	1- 30 controlled trials conducted across 3 app states: foreground, background, and closed (10 trials each) 2- A reminder was scheduled at a predefined time, and the delivery timestamp was recorded manually upon notification arrival **Results:** Success Rate: **100%** across all states - Min time: 0s - Max time: 12th - Avg: 4.4s

As a **Munir** user, I want to receive email notifications whenever I log in or delete my account, so that I can monitor account access and detect any unauthorized login attempts [72]	**Security:** How well does the system notify users of account activity to detect unauthorized access?	Email delivery success rate and delivery time between the action (login/account deletion) and email arrival. Emails must be delivered within 1 minute of the triggering action	1- 3 users were selected 2- Each user performed 2 actions: login and account deletion 3- Email delivery was confirmed and delivery time was recorded manually 4- The content of each email was verified to confirm it included the action type and corresponding date and time **Results:** Success Rate: **100%** (6/6 emails delivered) - Min: 40s - Max: 60s - Avg 50s Email Content: All 6 emails included the action type (login/account deletion) and corresponding date and time

5.5 Discussion

The system evaluation results presented in (Chapter 5) indicate that **Munir** successfully achieved its goal of offering an accessible, voice-driven assistant for visually impaired users, while also revealing areas that can benefit from further refinement across the process, design, and implementation.

- Process Evaluation: The combination of task-based UAT and usability questionnaires effectively assessed the system's functionality. Participants completed most tasks, such as signing up, logging in and navigating features, without difficulty, showing that the scenarios were appropriate and clear.

- Design and Implementation Reflections: Users found the interface clear and well-suited to their needs, supported by large buttons, consistent colors, and strong voice guidance. The TTS-based interaction was viewed as a major strength. Nonetheless, some participants experienced minor difficulty when switching between the email application and **Munir** during steps such as password reset or verification, which is expected in multi-app flows.

- Results Interpretation: Usability questionnaire responses were largely positive, with most items rated as "Agree" or "Strongly Agree." Task completion rates were similarly encouraging. These results confirm strong usability and effective support for the target users. However, some neutral responses, mainly relating to speed and clarity, show that improvements are still needed before considering the system fully mature.

- Areas for Improvement:

- **Process Improvements:** Involve a larger and more diverse group of participants and apply iterative evaluation cycles
- **Design Enhancements:** Strengthen voice prompts and confirmations and add personalization options such as speech rate
- **Implementation Improvements:** Maintain architectural readiness for future smart-glasses integration

The evaluation results show that **Munir** is reliable and highly usable, with a few areas that can be improved to enhance the process, design, and implementation.

CHAPTER 6:
CONCLUSIONS AND FUTURE WORK

Chapter 6: Conclusions and Future Work

This chapter concludes the development of **Munir**, an AI-powered system that combines both a mobile application and smart glasses to support visually impaired users through accessible, voice-driven features. This release focuses on the integrated solution, highlighting the core functionalities that enhance users' independence and usability in everyday life.

This document details our journey, starting with the Introduction, where we presented our mission to empower visually impaired users. We introduced key concepts related to visual impairment, the main problem, our solution, and product information such as the roadmap, scope, and details about the Scrum team.

Next, in the Background chapter, we examined visual impairment and the role of smart glasses in assistive technologies, along with key concepts such as OCR, face recognition, object detection, scene description, TTS, and Android accessibility. We also explored GPS and emergency alert systems. The Literature Review then analysed existing assistive applications and highlighted gaps that Munir aims to address.

The System Requirements chapter focused on user needs and requirement gathering. It included user interactions visualized as use case diagrams, along with a product backlog of functional and non-functional requirements and knowledge acquisition. In the System Design chapter, we used various diagrams to illustrate the system:

- **Class Diagram:** shows the system's structure through its main classes and their relationships.
- **Component-Level Design:** Explained key components via activity diagrams.
- **Data Design:** Focused on stored data using ER and non-relational models.
- **System Interfaces:** Highlighted UX guidelines followed during implementation.

The System Implementation chapter summarized the development of the **Munir** application, including third-party integrations and key implementation challenges. The Testing chapter outlined user acceptance testing, non-functional testing, questionnaire analysis, and participant demographics, followed by insights to improve the user experience.

6.1 Global and local Impact

6.1.1 Global Impact

Globally, **Munir** addresses the universal challenge of accessibility for blind and visually impaired individuals through an integrated assistive system that combines mobile and wearable technologies. The system is developed in English to ensure accessibility across different regions without language barriers, while also supporting Arabic for Arabic-speaking users. This multilingual and inclusive design enhances digital accessibility and promotes inclusion by enabling users to independently perform essential daily activities, including interacting with their environment through voice feedback. Furthermore, the initial domain analysis and requirements engineering phases established a foundation that considers universal accessibility beyond local requirements.

The face recognition model was developed and tested as part of the system's AI components to evaluate its performance and potential use cases. Although it demonstrated promising accuracy, it was not integrated into the final version of the **Munir** application. On the other hand, the currency identification model, currently trained to recognize Saudi Riyal banknotes, enables visually impaired tourists and visitors to independently identify local currency, supporting smoother financial transactions and improving the travel experience.

In addition, the smart glasses component extends the system into a fully wearable solution, enabling a hands-free assistive experience. It allows users to receive environmental feedback without relying on constant interaction with a mobile device, improving usability in dynamic and outdoor settings and enhancing user independence.

The authentication system with Google Sign-In provides a universally familiar and accessible login method. Combined with voice-based interaction, **Munir** delivers a comprehensive assistive solution that supports visually impaired users internationally and promotes accessibility-first design principles.

6.1.2 Local Impact

Saudi Arabia is advancing toward an inclusive society as part of its Vision 2030, where empowering people with disabilities represents a key pillar of social development. **Munir** contributes to this vision through a systematic development approach beginning with comprehensive requirements engineering and domain analysis to understand the specific needs of visually impaired users in the Saudi context.

The currency identification model has been developed and trained to recognize Saudi Riyal banknotes, supporting visually impaired users in performing daily financial activities such as shopping and transactions, thereby enhancing their independence and reducing reliance on assistance.

Although the face recognition model was developed and tested within the system's AI components, it was not integrated into the final application. However, it highlights potential future use cases in supporting social interaction and identity recognition for visually impaired users in the Kingdom.

The mobile application provides voice guidance and audio instructions, enabling users to interact naturally with the system and receive real-time feedback without visual input. In addition, the authentication system supports Google Sign-In, allowing quick and effortless access without manual credential entry, which simplifies the login process and improves accessibility.

Overall, **Munir** adopts an accessibility-focused design that enables visually impaired users in Saudi Arabia to navigate the application and utilize its features effectively.

6.2 Problems and challenges

During the development of **Munir**, the team faced several technical and implementation challenges across different system components. One of the main challenges was related to the SOS communication feature, where the intended approach of using a registered SMS sender ID was not feasible due to regulatory constraints within the project timeframe. This led to adopting a user-initiated messaging approach, where the system prepares the alert message and allows users to send it manually.

Another challenge was encountered in the development of the currency identification model, where certain Saudi banknote variations, particularly the 20 Saudi Riyal note, were not available in existing datasets. This required manually collecting and photographing real-world samples to ensure complete coverage of local currency types, which increased the effort required during the data preparation phase.

In addition, the integration of smart glasses introduced communication limitations due to the constraints of Bluetooth Low Energy, where large data such as images could not be transmitted in a single transfer. This was resolved by splitting the data into smaller segments and reconstructing it on the mobile side after reception, ensuring stable and reliable transmission between devices.

Furthermore, some emulators experienced unexpected crashes during testing, and the resource-intensive nature of the application placed strain on development machines. An additional challenge was related to email delivery, where system-generated messages for verification and password reset were frequently marked as spam when using Firebase's default mechanism; this was resolved by implementing a custom email-sending workflow. Designing the user interface for visually impaired users also proved more complex than typical applications, requiring careful consideration to balance simplicity, usability, and non-visual interaction.

6.3 Limitations of the system

While **Munir** provides comprehensive AI-powered assistance for blind and visually impaired users, it has several limitations.

First, the smart glasses do not include a built-in speaker; therefore, all audio feedback is delivered through the mobile device, reducing convenience and requiring users to always carry their phones to receive system responses.

Second, the SOS functionality requires manual activation through the mobile application, meaning users must open the app and trigger the feature themselves rather than it being automatically initiated in emergency situations.

Third, the system does not include GPS-based navigation assistance, turn-by-turn walking directions, or real-time obstacle detection for mobility support, though emergency location sharing is available.

Fourth, the currency recognition module requires that banknotes be presented individually with clear separation between them, as overlapping notes significantly reduce recognition accuracy.

Finally, cloud-based architecture requires stable internet connectivity for AI processing features such as face recognition and currency identification, meaning these functions cannot operate offline.

6.4 The main contribution of the project

We currently live in a digital age where assistive technologies have reshaped the way blind and visually impaired individuals interact with their environment and achieve greater independence. Recent advancements in artificial intelligence and computer vision have significantly improved how individuals with visual impairments perform daily tasks, engage in social interactions, and manage financial activities.

Munir enhances users' lives by addressing the critical gap in affordable, regionally adapted assistive technology for blind and visually impaired individuals. By leveraging AI-powered currency identification through a mobile application integrated with smart glasses, the system provides a seamless assistive experience that enables users to handle financial transactions independently.

Smart glasses serve as the primary hardware component, equipped with a built-in camera that captures the user's surroundings and streams visual data to the mobile application for instant AI processing. This hands-free, wearable design allows users to interact with the system naturally without requiring manual input, making the experience intuitive and non-intrusive in daily life.

With a projected cost of approximately 1,200 SAR, up to 19 times lower than existing solutions, **Munir** makes essential assistive technology accessible to a broader population. This combination of affordability and AI-powered assistance fosters greater independence and social participation for blind and visually impaired users in an increasingly connected world.

6.5 Future work

With the successful completion of the **Munir** system, the core functionalities of the mobile application, along with its integration with smart glasses, have been fully developed and implemented. This establishes a solid foundation for an assistive solution that enables blind and visually impaired users to interact with their surroundings in a seamless and hands-free manner.

Future work will focus on expanding and enhancing **Munir's** capabilities to provide a more comprehensive assistive experience. This includes the potential integration of advanced AI-powered features such as extending the currency identification module beyond the Saudi Riyal to support multiple currencies.

- On-Device Audio Feedback

Future development will focus on equipping the smart glasses with an integrated speaker to provide real-time audio feedback directly through the device. This enhancement will allow users to receive system responses without relying on a smartphone for sound output. By delivering audio directly through the smart glasses, the system will reduce dependence on external devices, offering users a more self-contained and independent assistive experience.

- On-Device Functionality Expansion

Future development will focus on enabling smart glasses to operate as a fully standalone system, where all functionalities are performed directly through the device. Currently, features such as the SOS function require manual activation via the mobile application.

Planned enhancements will allow users to trigger SOS alerts, manage contacts, add recognized individuals, and set reminders directly through the smart glasses, reducing reliance on external devices and improving overall independence.

- Wi-Fi Connectivity Integration

Future development will focus on integrating built-in Wi-Fi connectivity into the smart glasses, enabling the device to access the internet independently without relying on a paired smartphone. This enhancement will allow **Munir** to operate as a fully standalone assistive system, providing users with a more seamless and self-sufficient experience.

- Multi-Currency Support

Currently, the currency identification module supports the Saudi Riyal only. Future development will focus on expanding this capability to recognize and identify multiple international currencies, enabling the system to support users across different countries and regions. This enhancement will broaden **Munir's** accessibility and make it a more globally inclusive assistive solution.

- Improvement of Accessibility

Based on the testing findings, some participants indicated neutral responses regarding certain interaction processes, particularly in managing people and filtering reminders. This suggests that the current interface could be further improved by increasing button sizes, enhancing spacing between elements, and optimizing the overall layout, as well as providing clearer and more guided voice instructions to better support visually impaired users.

- Improvement of Filtering and Search Functionality

Based on the testing findings, filtering reminders by date received some neutral responses, suggesting usability challenges. Enhancing filtering mechanisms and making them more intuitive through expanded use of voice commands could improve accessibility and efficiency.

CHAPTER 7:
ACKNOWLEDGMENT

Chapter 7: Acknowledgment

We would like to begin by expressing our heartfelt gratitude to our team members, Rama Alomair, Fajer Alamro, Aljawharah Alsubaie, and Mashael Alammar, for their dedication, hard work, creativity, and collaboration throughout every stage of this journey. Their commitment and perseverance were the foundation of everything this project achieved. We also extend our deepest appreciation to their families, whose unwavering support, patience, and encouragement made all the difference. This achievement would not have been possible without them.

We are also sincerely grateful to our supervisor, Dr. Nora Alhammad, for her continued guidance, valuable feedback, and support throughout the development of this project. We extend our sincere gratitude to the thesis committee for their time and effort in reviewing this document and evaluating the project. Their insights and feedback are greatly valued.

We also had the opportunity to participate in the IBM AI Developer Program, a six-month initiative that provided us with hands-on experience and professional exposure in the field of artificial intelligence and technology. This experience enriched our technical knowledge and supported our professional growth throughout the project journey.

We also had the privilege of being invited to participate in several significant events and conferences, where we had the opportunity to present the Munir project to professionals and experts, receive valuable feedback, and engage in meaningful discussions that contributed to refining our ideas and strengthening the project's impact:

- Tamkeen X Training Forum:

We were invited to participate in the Tamkeen X Training Forum, held at King Saud University from January 19–21, 2025, under the theme "Your Skills Build Your Future." The forum focused on developing human capabilities through specialized training sessions, interactive workshops, and professional guidance. During the event, we presented the Munir project to attendees and experts, which allowed us to exchange insights, gain constructive feedback, and better understand real-world expectations and challenges [**Figure 131**].

- ICAN 2026 – International Conference on Data & AI Capacity Building:

We were also invited to attend and present at ICAN 2026 on January 28–29, 2026, an international conference that explored the future of education, work, and capacity building in the era of data and artificial intelligence. The conference brought together local and international experts, academics, and industry leaders. As part of our participation, we presented the Munir project to professionals and visitors, engaging in discussions that enriched our perspective and supported the project's continuous improvement [**Figure 132**].

- Publications:

In addition to our participation in events and conferences, a research paper related to the Munir project is currently under review in a peer-reviewed academic journal. This work presents the core concepts, system design, and evaluation findings of the project. The manuscript has been submitted to the journal Sensors (MDPI), under the section Intelligent Sensors, as part of the Special Issue Human–Computer Interaction in Sensor Systems. The study falls within the fields of intelligent sensing, human–computer interaction, and assistive technologies, with a particular focus on accessibility solutions for visually impaired users through AI-driven systems

In addition, a second research paper focusing on the currency identification component of the system has also been submitted and is currently awaiting review. This work has been submitted to the journal Applied Sciences (MDPI), under the section Computing and Artificial Intelligence, as part of the Special Issue AI-Based Supervised Prediction Models.

Nora Alhammad, Aljawharah Alsubaie, Rama Alomair, Fajer Alamro and Mashael Alammar, "Munir: A Multimodal Smart-Glass System for Enhancing Human-Computer Interaction and Accessibility for Visually Impaired Individuals," Sensors, 2026 | ISI-Q2 (under review)

Nora Alhammad, Aljawharah Alsubaie, Rama Alomair, Fajer Alamro and Mashael Alammar, "A Hybrid Framework for Real-Time Saudi Riyal Banknote Recognition in Assistive Applications," Applied Sciences, 2026 | ISI-Q2 (under review)

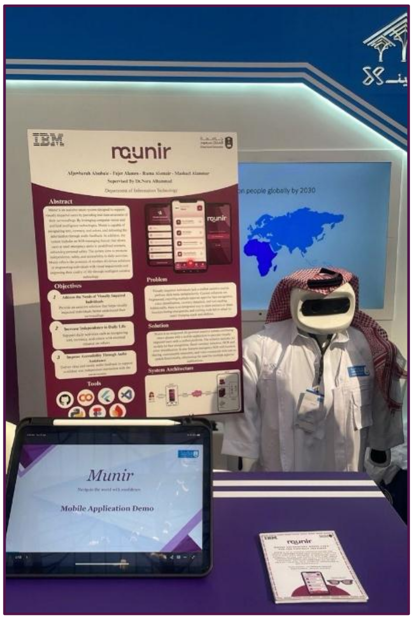


Figure 131: Munir at Tamkeen X

Figure 132: Munir at ICAN 2026

CHAPTER 8:
REFERENCES

8 References

[1] M. A. Hersh and M. A. Johnson, "Assistive Technology for Visually Impaired and Blind People," London, UK: Springer-Verlag, 2008. [Online]. Available: https://link.springer.com/book/10.1007/978-1-84628-867-8

[2] R. Elmannai and K. Elleithy, "Sensor-based assistive devices for visually-impaired people: Current status, challenges, and future directions," Sensors, vol. 17, no. 3, p. 565, Mar. 2017. [Online]. Available: https://doi.org/10.3390/s17030565

[3] General Authority for Statistics. (2022). *Population and housing census*. https://www.apd.gov.sa/en/statistics

[4] World Health Organization, "World Report on Vision," Geneva, Switzerland, Oct. 2019. [Online]. Available: https://www.who.int/publications/i/item/9789241516570

[5] OrCam MyEye – Wearable AI device for the blind, *OrCam Technologies*, 2024. [Online]. Available: https://www.orcam.com/en/myeye

[6] Envision Glasses – AI-powered smart glasses for the blind and visually impaired, *Let's Envision*, 2024. [Online]. Available: https://www.letsenvision.com/glasses

[7] VGGFace2 Dataset, 2023. [Online]. Available: https://www.kaggle.com/datasets/hearfool/ vggface2

[8] G. F. Bati, "AlFloos: Saudi Arabian Riyal Banknote Dataset," Kaggle, 2024. [Online]. Available: https://www.kaggle.com/datasets/gfbati/alfloos

[9] World Health Organization. (2023). *Blindness and visual impairment*. https://www.who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment

[10] Aldakhil, S. (2025). Prevalence, Causes, and Risk Factors of Visual Impairment: Evidence from Duhknah, a Rural Community in Saudi Arabia. *Healthcare*, *13*(15), 1927. https://www.mdpi.com/3438860

[11] World Health Organization. (2024, January 2). Assistive technology. https://www.who.int/news-room/fact-sheets/detail/assistive-technology

[12] Okolo, G. I., Althobaiti, T., & Ramzan, N. (2024). Assistive systems for visually impaired persons: Challenges and opportunities for navigation assistance. Sensors, 24(11), Article 3572. https://doi.org/10.3390/s24113572

[13] Kim, D., & Choi, Y. (2021). Applications of smart glasses in applied sciences: A systematic review. Applied Sciences, 11(11), 4956. https://42doi.org/10.3390/app11114956

[14] Based Hardware Inc. (2025). *omiGlass DevKit Hardware*. Omi Docs. Retrieved September 18, 2025, from https://docs.omi.me/doc/hardware/omiGlass

[15] Krishnamoorthy, S., & Kumar, A. (2024). IoT-driven accessibility: A refreshable OCR-Braille solution for visually impaired and deaf-blind users through WSN. *Journal of Engineering Research and Reports*, 26(4), 246-258. https://www.sciencedirect.com/science/article/pii/S2949948824000246

[16] Tesseract OCR Team. (2023). *Tesseract documentation*. https://tesseract-ocr.github.io/

[17] Google Cloud. (2023). *Vision API: Derive insights from your images in the cloud or at the edge*. Google LLC. https://cloud.google.com/vision

[18] Microsoft. (2023). *Azure Computer Vision: OCR and image analysis*. Microsoft Corporation. https://azure.microsoft.com/en-us/services/cognitive-services/computer-vision/

[19] Wang, M., & Deng, W. (2019). A survey of deep learning based face recognition. *Computer Vision and Image Understanding*, 189, 102805. https://doi.org/10.1016/j.cviu.2019.102805

[20] Zhao, W., Chellappa, R., Phillips, P. J., & Rosenthal, A. (2003). Face recognition: A literature survey. *ACM Computing Surveys*, 35(4), 399-458. https://doi.org/10.1145/954339.954342

[21] Bonakdarpour, P., Moghadam, P., & Faust, O. (2024). Assistive systems for visually impaired people: A survey on current requirements and advancements. *Neurocomputing*, 582, 127501. https://doi.org/10.1016/j.neucom.2024.127501

[22] Tan, X., Chen, J., Liu, H., Cong, J., Zhang, C., Liu, Y., ... & Qin, T. (2021). A survey on neural speech synthesis. *arXiv preprint arXiv:2106.15561*. https://arxiv.org/abs/2106.15561

[23] Gonzalez, R., Collins, J., Bennett, C., & Azenkot, S. (2024). Investigating use cases of AI-powered scene description applications for blind and low vision people. In Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24). ACM. https://doi.org/10.1145/3613904.3642211

[24] Valipoor, M. M., & de Antonio, A. (2023). Recent trends in computer vision-driven scene understanding for VI/blind users: a systematic mapping. Universal Access in the Information Society, 22, 983–1005. https://link.springer.com/article/10.1007/s10209-022-00868-w

[25] Oviedo, F., Ferres, J. L., Budd, S., Zerman, E., Llach, J., & De Nigris, S. (2022). BankNote-Net: Open dataset for assistive universal currency recognition. arXiv preprint arXiv:2204.03738. https://arxiv.org/abs/2204.03738

[26] Hamid, N. N. A., Edwards, S., Lim, C. G., & Sosa, N. (2021). Haptic feedback to assist blind people in indoor environment using vibration patterns. *Sensors*, 21(24), 8223. https://pmc.ncbi.nlm.nih.gov/articles/PMC8749676/

[27] Okolo, G.I., Althobaiti, T., & Ramzan, N. (2024). Assistive Systems for Visually Impaired Persons: Challenges and Opportunities for Navigation Assistance. *Sensors*, 24(11), 3572. https://doi.org/10.3390/s24113572

[28] Rathgeb, C., Uhl, A., Wild, P., & Hofbauer, H. (2024). An overview of privacy-enhancing technologies in biometric recognition. *ACM Computing Surveys*, *57*(3), Article 58. https://doi.org/10.1145/3664596

[29] National Institute of Standards and Technology. (2018). *Framework for improving critical infrastructure cybersecurity* (Version 1.1). U.S. Department of Commerce. https://doi.org/10.6028/NIST.CSWP.04162018

[30] Al-Hayani, B., & Ilhan, H. (2020). Image processing based intelligent traffic monitoring system using IoT cloud environment. *Sustainable Cities and Society, 61,* 102325. https://doi.org/10.1016/j.scs.2020.102325

[31] Voigt, P., & Von dem Bussche, A. (2017). *The EU General Data Protection Regulation (GDPR): A practical guide.* Springer International Publishing. https://doi.org/10.1007/978-3-319-57959-7

[32] "Seeing AI – Talking camera app for the blind community," *Microsoft*, 2024. [Online]. Available: https://www.microsoft.com/en-us/ai/seeing-ai

[33] OXSIGHT Products – Powerful Smart Glasses for the Visually Impaired, OXSIGHT, 2024. [Online]. Available: https://www.oxsight.co.uk/products/

[34] NuEyes – Empowering Your Vision, NuEyes Technologies, 2024. [Online]. Available: https://www.nueyes.com/

[35] Nuqtat Tahawol – Official website of the Nuqtat Tahawol Association,2024. [Online]. Available: https://www.turn-point.org/

[36] Ru'ya – Official website of Ru'ya Blind Association, 2024. [Online]. Available: https://blindroya.sa/

[37] K. S. Rubin, *Essential Scrum: A Practical Guide to the Most Popular Agile Process*. Upper Saddle River, NJ, USA: Addison-Wesley, 2012.

[38] VGGFace2 Paper, 2018. [Online]. Available: https://doi.org/10.1109/FG.2018.00020

[39] G. F. Bati, "AlFloos: a balanced dataset for the sixth issue of the Saudi Arabian currency banknote," Journal of Umm Al-Qura University for Engineering and Architecture, May 2024. [Online]. Available: https://doi.org/10.1007/s43995-024-00067-z

[40] E.Learned-Miller, G. B. Huang, A. RoyChowdhury, H. Li, and G. Hua, "Labeled faces in the wild: A survey," in *Advances in Face Detection and Facial Image Analysis*, Cham: Springer, 2016, pp. 189-248. [Online]. Available: https://doi.org/10.1007/978-3-319-25958-1_8

[41] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA: MIT Press, 2016. [Online]. Available: https://www.deeplearningbook.org/

[42] F. S. Samaria and A. C. Harter, "Parameterisation of a stochastic model for human face identification," in Proc. 2nd IEEE Workshop Appl. Comput. Vis., 1994, pp. 138-142. [Online]. Available: https://doi.org/10.1109/ACV.1994.341300

[43] T. Sim, S. Baker, and M. Bsat, "The CMU pose, illumination, and expression database," IEEE Trans. Pattern Anal. Mach. Intell., vol. 25, no. 12, pp. 1615-1618, Dec. 2003. [Online]. Available: https://doi.org/10.1109/TPAMI.2003.1251154

[44] N. Japkowicz and S. Stephen, "The class imbalance problem: A systematic study," Intell. Data Anal., vol. 6, no. 5, pp. 429-449, 2002. [Online]. Available: https://doi.org/10.3233/IDA-2002-6504

[45] G. K. Wallace, "The JPEG still picture compression standard," IEEE Trans. Consum. Electron., vol. 38, no. 1, pp. xviii-xxxiv, Feb. 1992. [Online]. Available: https://doi.org/10.1109/30.125072

[46] R. Kohavi, "A study of cross-validation and bootstrap for accuracy estimation and model selection," in Proc. 14th Int. Joint Conf. Artif. Intell. (IJCAI), vol. 2, Montreal, Canada, 1995, pp. 1137-1143. [Online]. Available: http://ai.stanford.edu/~ronnyk/accEst.pdf

[47] X. Bouthillier, C. Laurent, and P. Vincent, "Unreproducible research is reproducible," in Proc. 36th Int. Conf. Mach. Learn. (ICML), vol. 97, Long Beach, CA, USA, 2019, pp. 725-734. [Online]. Available: http://proceedings.mlr.press/v97/bouthillier19a.html

[48] C. Shorten and T. M. Khoshgoftaar, "A survey on image data augmentation for deep learning," J. Big Data, vol. 6, no. 1, Art. no. 60, 2019. [Online]. Available: https://doi.org/10.1186/s40537-019-0197-0

[49] M. Buda, A. Maki, and M. A. Mazurowski, "A systematic study of the class imbalance problem in convolutional neural networks," Neural Netw., vol. 106, pp. 249-259, Oct. 2018. [Online]. Available: https://doi.org/10.1016/j.neunet.2018.07.011

[50] A. Buslaev et al., "Albumentations: Fast and flexible image augmentations," Information, vol. 11, no. 2, Art. no. 125, Feb. 2020. [Online]. Available: https://doi.org/10.3390/info11020125

[51] “Guide to Design Consistency: Best Practices for UI/UX Designers,” *UXPin*, 2024. [Online]. Available: https://www.uxpin.com/studio/blog/guide-design-consistency-best-practices-ui-ux-designers/

[52] Z. C., “How the Principle of Feedback Affects your UX Design,” *UX Planet*, Aug. 19, 2022. [Online]. Available: https://uxplanet.org/how-the-principle-of-feedback-affects-your-ux-design-770325d299a

[53] “Robust Design definition & examples,” *RD8.tech*, Aug. 19, 2025. [Online]. Available: https://uxplanet.org/how-the-principle-of-feedback-affects-your-ux-design-770325d299a

[54] J. mind, "Learnability in UI design: making things easy," *UX Planet*, Jun. 29, 2020. [Online]. Available: https://uxplanet.org/learnability-in-ui-design-making-things-easy-5e404c3d66d6

[55] "Discoverability in UX and UI Design — 9 Techniques to Try," UXPin Studio Blog, [Online]. Available: https://www.uxpin.com/studio/blog/discoverability-in-ux/

[56] Interaction Design Foundation, "What is Recognition vs Recall?," *IxDF*. [Online]. Available: https://ixdf.org/literature/topics/recognition-vs-recall

[57] "All About Usability Heuristic #1: Visibility of System Status," *UX Planet*. [Online]. Available: https://uxplanet.org/all-about-usability-heuristic-1-visibility-of-system-status-50e252522e41

[58] BasedHardware, "omi," *GitHub Repository*. [Online]. Available: https://github.com/BasedHardware/omi/tree/main

[59] Communications, Space & Technology Commission (CST), "SMS Services Provision License – Details," *Mutasil Platform*. [Online]. Available: https://mutasilbus.cst.gov.sa/Services/Details/GFYJntOpuN0JH9MVIA5jCg--

[60] Twilio, "WhatsApp Business API Overview," *Twilio*. [Online]. Available: https://www.twilio.com/en-us/messaging/channels/whatsappAvailable: https://taqnyat.sa/en/offers/2025/Whatsapp-api. [Accessed: 21-May-2026].

[61] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive angular margin loss for deep face recognition," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), Long Beach, CA, USA, Jun. 2019, pp. 4690–4699. [Online]. Available: https://openaccess.thecvf.com/content_CVPR_2019/html/Deng_ArcFace_Additive_Angular_Margin_Loss_for_Deep_Face_Recognition_CVPR_2019_paper.html

[62] J. Yosinski, J. Clune, Y. Bengio, and H. Lipson, "How transferable are features in deep neural networks?," in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), Montreal, QC, Canada, Dec. 2014, vol. 27, pp. 3320–3328. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2014/hash/532a2f85b6977104bc93f8580abbb330-Abstract.html

[63] G. James, D. Witten, T. Hastie, and R. Tibshirani, An Introduction to Statistical Learning: With Applications in R. New York, NY, USA: Springer, 2013. [Online]. Available: https://www.statlearning.com

[64] J. Guo, J. Deng, A. Lattas, and S. Zafeiriou, "Sample and computation redistribution for efficient face detection," in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2022. [Online]. Available: https://arxiv.org/abs/2105.04714

[65] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay, "Scikit-learn: Machine learning in Python," J. Mach. Learn. Res., vol. 12, pp. 2825–2830, 2011. [Online]. Available: https://jmlr.org/papers/v12/pedregosa11a.html

[66] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), Salt Lake City, UT, USA, Jun. 2018, pp. 4510–4520. [Online]. Available: https://arxiv.org/abs/1801.04381

[67] M. Grgic, K. Delac, and S. Grgic, "SCface—Surveillance cameras face database," *Multimedia Tools and Applications*, vol. 51, no. 3, pp. 863–879, 2011, doi: 10.1007/s11042-009-0417-2. Available: https://link.springer.com/article/10.1007/s11042-009-0417-2.

[68] G. B. Huang, M. Ramesh, T. Berg, and E. Learned-Miller, "Labeled Faces in the Wild: A database for studying face recognition in unconstrained environments," University of Massachusetts Amherst, Tech. Rep. 07-49, 2007. Available: http://vis-www.cs.umass.edu/lfw/.

[69] N. K. Mishra, M. Dutta, and S. K. Singh, "Multiscale parallel deep CNN (mpdCNN) architecture for the real low-resolution face recognition for surveillance," *Image and Vision Computing*, vol. 115, p. 104290, 2021, doi: 10.1016/j.imavis.2021.104290. Available: https://www.sciencedirect.com/science/article/pii/S0262885621001487.

[70] Philip Walton, "Time to Interactive (TTI)," *web.dev*. [Online]. Available: https://web.dev/articles/tti

[71] A. Author(s) not specified, "Article," *ScienceDirect*. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1574119222000074?via%3Dihub

[72] University of Washington Information School, "Office 365 SLA (Service Level Agreement)," *iSchool Knowledge Base*. [Online]. Available: https://www.kb.ischool.uw.edu/2017/08/04/office-365-sla-service-level-agreement/

CHAPTER 9:
APPENDIX

9 Appendix

9.1 Appendix A: interview

This Appendix will represent all the interview-related supplements, including the interview questions and transcriptions.

9.1.1 Interview Questions

Q1: What are the main challenges you face in your daily life due to visual impairment or loss of sight, and wish there was a technology that could help you with them?

Q2: Have you ever used any assistive technologies, such as screen readers, smart glasses, voice-based applications, or others? How was your experience? Do you think they are sufficient? What did you like or dislike about them?

Q3: If you encounter difficulty in a situation, whom do you usually turn to? Or how do you usually handle such cases?

Q4: When you need to read a street sign, a book, or a bill, what method do you usually rely on? Do you find it effective in all situations?

Q5: If a device could tell you the identity of the person in front of you, how could that help you in your daily life?

Q6: Would you prefer to use a device like "Munir" all the time, or only in specific situations? And when exactly do you feel you would need it?

Q7: What is the most suitable way for you to interact with the device (e.g., voice, vibration, touch, or others)? And why do you prefer this method?

Q8: Would you prefer that the device interacts only with you, or that it can also alert people around you in certain cases (such as warning them if there is danger)?

Q9: Do you think having a companion mobile app for the device (like the "Munir" app) would be useful for you? What features would you like to support?

Q10: If you could design or develop a wearable device to be used daily, what are the three most important features you would like it to have? Do you have any suggestions to make it more useful or easier to use?

9.1.1 Interview's Transcriptions

Interview transcriptions provide a detailed record of conversations between interviewees and interviewers, including participant names, location, time, objectives, agenda, observations, and uncovered topics. They also include ten thought-provoking questions for in-depth analysis and understanding of perspectives and strategies.

Table 24: Interview #1 Overview

Interview #1	
Interviewee: Huda	**Interviewer:** Rama Alomair
Location / Medium: Zoom meeting - online	**Appointment Date:** 4 Aug 2025 **Start Time:** 1:00 pm **End Time:** 1:43 pm
Objectives: **1-** Identify main challenges visually impaired users face and how Munir could help **2-** Understand preferred features and interaction methods (voice, color recognition, navigation) **3-** Gauge interest in participation and feedback for improving the device	**Reminders**: The interviewee struggles with colors and finds color recognition very important.
Agenda:	**Approximate Time:**
- Introduction	4 min
- Background in project	3 min
- Overview of interview	1 min
- Topic to be covered	2 min
- Permission to record	1 min
- Question1	3 min
- Question2	3 min
- Question3	1 min
- Question4	2 min
- Question5	3 min
- Question6	2 min
- Question7	3 min
- Question8	3 min
- Question9	2 min
- Question10	3min
- Summary of major points	2 min
- Questions from interviewee	3 min
- Closing	2 min

Table 25: Interview #1 Transcript

<table>
<tr><th colspan="2">Interview #1</th></tr>
<tr><td colspan="2">General Observations:
She strongly values independence but still relies on family, assistants, or AI tools when necessary, highlighting a balance between self-reliance and external support.
She emphasized the importance of practical assistive features, including color recognition, text reading, and navigation assistance, to enhance confidence and manage daily tasks effectively.
She prefers voice feedback for alerts and notifications, as it ensures clarity, reliability, and helps her remember planned activities such as medications or tasks.</td></tr>
<tr><td colspan="2">Topic not covered:
Battery life expectations were not discussed.
Aesthetic preferences or style considerations beyond general comfort and lightweight design were not discussed.
Device setup, calibration, or ease of learning advanced features were not covered.</td></tr>
<tr><td>Interviewee: Huda</td><td>Date: 4 Aug 2025</td></tr>
<tr><td>Questions:</td><td>Answers and notes:</td></tr>
<tr><td>Q1: What are the main challenges you face in your daily life due to visual impairment or loss of sight, and wish there was a technology that could help you with them?</td><td>Answer: I believe that every blind person needs support when it comes to recognizing colors. Although there are applications that can identify colors, after certain updates they may become inaccurate. As women, we also like to know the colors of our clothing. Furthermore, when walking on roads or through pathways, it would be very beneficial to have an assistant that can guide me regarding the objects and obstacles in front of me.

Observations: The interviewee emphasized that color recognition is not just functional but also tied to personal expression and confidence, particularly in clothing choices. She also highlighted the need for reliable navigation assistance in outdoor environments, showing that both appearance-related and safety-related features are equally important in her daily life.</td></tr>
</table>

Q2: Have you ever used any assistive technologies, such as screen readers, smart glasses, voice-based applications, or others? How was your experience? Do you think they are sufficient? What did you like or dislike about them?	**Answer:** I have never had the opportunity to use smart glasses before. Now, I mainly rely on screen readers in almost all aspects of my daily life, and I also make use of some assistive applications. While these tools are helpful, I find that they still require significant improvement. For instance, with Be My Eyes, the service is useful but limited, as I am unable to upload multiple images at once and instead must upload them one by one, which makes the process time-consuming and less practical in many situations. **Observations:** The interviewee has no prior experience with smart glasses and currently depends on screen readers and assistive apps, but she finds existing tools limited and inefficient, especially when tasks require handling multiple images.
Q3: If you encounter difficulty in a situation, whom do you usually turn to? Or how do you usually handle such cases?	**Answer:** At home, when it comes to colors or images, I usually rely on family or friends and ask a sighted person for assistance. This helps me understand and engage with visual content accurately. However, when I am outside, I have not yet encountered a situation where I needed such support, so I manage independently in public settings. **Observations:** The interviewee relies on family or friends at home for visual support but feels independent in public settings, suggesting that her main challenges occur in private rather than outdoor environments.
	Answer: In public places, I usually have an assistant with me who helps by reading bills and handling money, which allows me to manage daily tasks more

Q4: When you need to read a street sign, a book, or a bill, what method do you usually rely on? Do you find it effective in all situations?	smoothly. At work, if I need to deal with a bill that is in the form of an image on my phone, I rely on AI tools to read and process it. However, if the bill is on paper, I ask a sighted person for help to ensure I understand the information accurately. **Observations:** The interviewee uses a combination of human assistance and AI tools to manage reading tasks, showing that no single solution fully meets her needs, especially when dealing with paper versus digital content.
Q5: If a device could tell you the identity of the person in front of you, how could that help you in your daily life?	**Answer:** If a device could tell me the identity of the person in front of me, it could describe the person's appearance, including details such as height, build, and whether they are male or female. If possible, it could also recognize the individual and tell me their name. This would help me identify people quickly and accurately in different situations, making social interactions easier. **Observations:** The interviewee views appearance and identity recognition as valuable for improving social interactions, emphasizing the importance of quickly and confidently identifying people in different situations.
Q6: Would you prefer to use a device like "Munir" all the time, or only in specific situations? And when exactly do you feel you would need it?	**Answer:** I will not use the glasses all the time. If the color recognition feature is available, I can use it whenever I need, for example, to choose or match my clothes. The glasses can also assist me at work by alerting me if someone is standing in front of me, and they help me read printed papers or digital images whenever necessary. **Observations:** The interviewee prefers selective use of the glasses, relying on them mainly for color recognition, obstacle alerts, and reading tasks,

	indicating that the device should be flexible and adaptable to different contexts.
Q7: What is the most suitable way for you to interact with the device (e.g., voice, vibration, touch, or others)? And why do you prefer this method?	**Answer:** I prefer using sound because if the information is delivered through vibration, I might not understand it correctly or could easily forget it. This could lead to misunderstandings or mistakes, especially in situations that require accuracy. With sound, I can receive alerts and notifications more clearly and reliably, which helps me stay aware, respond appropriately, and manage tasks more effectively. It also reminds me of things I planned to do, like taking medications or completing daily tasks. **Observations:** The interviewee prefers voice feedback over vibration for receiving information, noting that sound provides clearer and more reliable alerts and notifications. This ensures better task management and helps them remember planned activities, such as taking medications or completing daily tasks.
Q8: Would you prefer that the device interacts only with you, or that it can also alert people around you in certain cases (such as warning them if there is danger)?	**Answer:** Usually, I do not go out alone in public places, so I have not encountered this problem before. Currently, I go out alone for work, and I have almost memorized the area. However, in case an unexpected situation arises that requires alerting people I know, having this feature would be very helpful. It would improve my safety and allow me to navigate public spaces more confidently and independently. Therefore, I see no reason not to use it when needed. **Observations:** The interviewee rarely goes out alone but recognizes that optional alert

	features could enhance safety and confidence in unfamiliar or unexpected situations.
Q9: Do you think having a companion mobile app for the device (like the "Munir" app) would be useful for you? What features would you like it to support?	**Answer:** Certainly, if there is an application that includes a highly accurate text reader, it would be very useful for me. Although many text reader apps exist, most of them are not precise enough to rely on fully. Additionally, it would be extremely helpful if the application could read product labels and identify them for me, for example, informing me about the type and ingredients of items such as juices or chocolates. **Observations:** The interviewee values a highly accurate text-reading app and emphasizes the usefulness of reading product labels, showing the importance of precise and practical information for daily independence.
Q10: If you could design or develop a wearable device to be used daily, what are the three most important features you would like it to have? Do you have any suggestions to make it more useful or easier to use?	**Answer:** The three most important features for me are color recognition, text reading, and navigation assistance in public places. Color recognition helps me make choices in clothing and other daily tasks, text reading allows me to access printed or digital information independently, and navigation assistance ensures I can move around public spaces safely and confidently. Together, these features greatly enhance my independence and make everyday activities more manageable. **Observations:** The interviewee prioritizes color recognition, text reading, and navigation assistance, indicating that these features are essential for independence and confidence in both personal and public settings.

Table 26: Interview #2 Overview

<table>
<tr><th colspan="2">Interview #2</th></tr>
<tr><td>Interviewee: Mashari</td><td>Interviewer: Rama Alomair</td></tr>
<tr><td>Location / Medium: Zoom meeting - online</td><td>Appointment Date: 5 Aug 2025
Start Time: 1:00 pm
End Time: 1:48 pm</td></tr>
<tr><td>Objectives:

1- Identify main challenges visually impaired users face and how Munir could help.
2- Understand preferred features and interaction methods (voice, color recognition, navigation).
3- Gauge interest in participation and feedback for improving the device.</td><td>Reminders: The interviewee struggles with colors and uses voice-based tools and her phone to support daily tasks both at home and outside.</td></tr>
<tr><td>Agenda:
- Introduction
- Background in project
- Overview of interview
- Topic to be covered
- Permission to record
- Question1
- Question2
- Question3
- Question4
- Question5
- Question6
- Question7
- Question8
- Question9
- Question10
- Summary of major points
- Questions from interviewee
- Closing</td><td>Approximate Time:
4 min
3 min
1 min
2 min
1 min
4 min
2 min
3 min
2 min
4 min
3 min
3 min
2 min
3 min
4 min
2 min
3 min
2 min</td></tr>
</table>

Table 27: Interview #2 Transcript

<table>
<tr><th colspan="2">Interview #2</th></tr>
<tr><td colspan="2">General Observations:
He emphasized struggles with color distinction and reading texts, especially in public spaces or at work, indicating a need for features that enhance both safety and daily functioning.
He prefers flexible, user-controlled interaction features, such as adaptive use of sound and vibration, and values reminders and the ability to share his location with trusted contacts when needed.
He values independence but recognizes situations where assistance may be necessary, highlighting a balance between self-reliance and optional support.</td></tr>
<tr><td colspan="2">Topic not covered:
Pricing and affordability were not explored in this interview.
Expectations for software updates or feature improvements over time were not discussed.
Privacy, data security, and sharing personal information with the device were not mentioned.</td></tr>
<tr><td>Interviewee: Mashari</td><td>Date: 4 Aug 2025</td></tr>
<tr><td>Questions:</td><td>Answers and notes:</td></tr>
<tr><td>Q1: What are the main challenges you face in your daily life due to visual impairment or loss of sight, and wish there was a technology that could help you with them?</td><td>Answer: As a visually impaired person, one of the embarrassing situations I face is when stores have floors and stairs that are almost the same color, which makes it difficult for me to distinguish between them and sometimes creates uncomfortable situations. In addition, in the workplace, I struggle with reading texts, and in many cases, I need someone to be with me to provide support. These challenges highlight the areas where I could benefit from additional tools or technologies to increase my independence and confidence.

Observations: The interviewee has trouble distinguishing similar colors in public spaces and relies on assistance at work for reading texts, indicating a need for tools that enhance independence and confidence.</td></tr>
<tr><td>Q2: Have you ever used any assistive technologies, such as screen readers, smart glasses, voice-based applications, or others? How was your experience? Do</td><td>Answer: Yes, I have used voice-based applications. They are helpful in some cases, but they are not always sufficient. Sometimes the information is limited, so I still need to rely on others. What I like</td></tr>
</table>

you think they are sufficient? What did you like or dislike about them?	is the independence they can provide, but what I dislike is the lack of accuracy and detail. **Observations**: The interviewee uses voice-based applications for support, appreciating the independence they offer, but notes that limited information and lack of detail still require reliance on others.
Q3: If you encounter difficulty in a situation, whom do you usually turn to? Or how do you usually handle such cases?	**Answer:** If I need urgent help, my first option is usually to contact someone close to me, such as a family member or friend. When I am in a public place, I may also ask someone nearby for assistance. In many situations, I rely on my phone to check for directions, exits, or other important information. However, I do not consider this the best solution, because it depends heavily on internet connectivity and phone access, which are not always fast or reliable in urgent situations. This is why I believe having more direct and accessible solutions would be very helpful to increase both safety and independence. **Observations:** The interviewee relies on family, nearby people, or his phone for urgent help, but recognizes that these solutions can be unreliable, highlighting the need for more direct and accessible support to ensure safety and independence.
Q4: When you need to read a street sign, a book, or a bill, what method do you usually rely on? Do you find it effective in all situations?	**Answer:** I usually rely on my phone to assist me in many situations. Since I can read to some extent, one of the strategies I use is to take a picture of the text, zoom in on it, and then try to read it carefully. This method is not always perfect, but it often allows me

	to manage on my own without immediately needing external help. **Observations:** The interviewee uses his phone to capture and zoom in on text as a strategy to read independently, showing a preference for self-reliant solutions even if they are not always perfect.
Q5: If a device could tell you the identity of the person in front of you, how could that help you in your daily life?	**Answer:** Since I am visually impaired and not completely blind, I cannot answer this question based on my own personal experience. However, based on my knowledge of people who are blind, I believe this feature would be extremely useful for them in social events and gatherings. It could help them recognize the people around them, reducing the chances of awkward or embarrassing situations. In addition, it would allow them to interact more confidently and independently, making social participation easier and more comfortable. This feature could significantly enhance their ability to engage with others without constantly needing assistance. **Observations:** The interviewee recognizes that identity recognition features would greatly benefit completely blind individuals in social situations, helping them interact confidently and independently while reducing awkwardness.
Q6: Would you prefer to use a device like "Munir" all the time, or only in specific situations? And when exactly do you feel you would need it?	**Answer:** Depending on the features available, I may be able to use the glasses in different situations. Some features might be more useful outside the home, while others can generally be used both at home and outdoors. For example, outside the home, the glasses could alert me to obstacles in front of me, such as closed doors, curbs, or other hazards, helping

	me navigate safely. At home, they could assist me in locating light switches, using appliances like the microwave, or finding other objects I need. **Observations:** The interviewee prefers context-dependent use of the glasses, valuing navigation alerts outdoors and object/location assistance at home, highlighting the importance of versatile features for different environments.
Q7: What is the most suitable way for you to interact with the device (e.g., voice, vibration, touch, or others)? And why do you prefer this method?	**Answer:** Usually, having both sound and vibration together is preferable, as each method can be useful in different situations. For example, in public places, it might be difficult for me to hear the sound clearly due to background noise, so vibration works better in that context. On the other hand, in quieter environments, sound can be more effective and provide clearer guidance on what is required from me. Using both methods depending on the situation allows me to receive information more reliably and respond appropriately, enhancing my overall independence and confidence. **Observations:** The interviewee prefers both sound and vibration, using each method adaptively depending on the environment to ensure reliable guidance and maintain independence.
Q8: Would you prefer that the device interacts only with you, or that it can also alert people around you in certain cases (such as warning them if there is danger)?	**Answer:** It is preferable for this feature to be available, but it should be activated based on the user's choice, as people's abilities and needs vary. For example, I may require assistance and want to request help, but this should only happen with my explicit consent. Sometimes I get lost and face difficulty describing my exact location to others, so having the option to send a message with my location to people I know would make it much easier to receive help. At the same time, other

	individuals may not need this feature and can navigate situations independently. **Observations:** The interviewee values optional, user-controlled alert features, noting that assistance should only be activated with the user's explicit consent. They emphasized that having the ability to share their location with trusted contacts when lost would be especially useful, while still preserving independence for those who may not need such support.
Q9: Do you think having a companion mobile app for the device (like the "Munir" app) would be useful for you? What features would you like it to support?	**Answer:** It would be an excellent option because I can control the basic functions of the glasses through my phone. For us as visually impaired individuals, using a phone with a voice assistant is generally easier than manually operating the glasses. I would also like the application to include additional features beyond just adjusting the glasses' settings. For example, it could allow me to use my phone to read text, identify the name of a store, or provide other helpful information. **Observations:** The interviewee prefers controlling the glasses via a mobile app with a voice assistant, and values additional app features like reading text or identifying stores to enhance independence and convenience.
Q10: If you could design or develop a wearable device to be used daily, what are the three most important features you would like it to have? Do you have any suggestions to make it more useful or easier to use?	**Answer:** First, the device should be fully voice-enabled, so that any action I perform or any feature I use can be communicated to me through sound. This is essential for making the device accessible and intuitive for someone with visual impairment. Additionally, it should have a fast response time, which makes its use easier and more efficient, allowing me to complete tasks quickly without needing to look for a faster alternative. Together, these aspects would greatly enhance usability, independence, and confidence in using the device in various situations. **Observations:** The interviewee emphasizes the need for a fully voice-enabled device with fast response, highlighting that these features would improve usability, independence, and confidence for visually impaired users.

Table 28: Interview #3 Overview

<table>
<tr><th colspan="2">Interview #3</th></tr>
<tr><td>Interviewee: Mohammed</td><td>Interviewer: Aljawharah Alsubaie</td></tr>
<tr><td>Location / Medium: Zoom meeting - online</td><td>Appointment Date: 5 Aug 2025
Start Time: 2:00 pm
End Time: 2:37 pm</td></tr>
<tr><td>Objectives:

1. Identify main challenges visually impaired users face and how Munir could help.
2. Understand preferred features and interaction methods (voice, color recognition, navigation).
3. Gauge interest in participation and feedback for improving the device.</td><td>Reminders: The interviewee emphasizes the need for discreet, lightweight wearables with strong reading and brightness features, supported by vibration feedback and a customizable companion app to enhance independence in public and nighttime settings.</td></tr>
<tr><td>Agenda:
- Introduction
- Background in project
- Overview of interview
- Topic to be covered
- Permission to record
- Question1
- Question2
- Question3
- Question4
- Question5
- Question6
- Question7
- Question8
- Question9
- Question10
- Summary of major points
- Questions from interviewee
- Closing</td><td>Approximate Time:
2 min
1 min
2 min
1 min
1 min
4 min
3 min
3 min
3 min
3 min
2 min
2 min
2 min
2 min
2 min
2 min
1 min
1 min</td></tr>
</table>

Table 29: Interview #3 Transcript

Interview #3	
General Observations: • He showed genuine interest in the idea of smart glasses, especially for reading assistance and night vision. • He emphasized the importance of features like magnification, brightness adjustment, and color contrast. • He also highlighted that lightweight design and appearance are critical factors for everyday use.	
Topic not covered: • Battery life expectations were not discussed. • Pricing and affordability were not explored in this interview. • No detailed discussion on integration with other accessibility devices or apps.	
Interviewee: Mohammed	**Date:** 5 Aug 2025
Questions:	**Answers and notes:**
Q1: What are the main challenges you face in your daily life due to visual impairment or loss of sight, and wish there was a technology that could help you with them?	**Answer:** The biggest challenge for me is reading—whether it's documents, books, or signs. I also struggle with night vision because of my condition, which makes it difficult to walk safely without someone accompanying me in the dark. If there were a device that could brighten my surroundings and make text clearer, it would really help me feel more independent. **Observations:** The interviewee struggles with reading text and night vision, showing the need for brightness enhancement and clearer text support.
Q2: Have you ever used any assistive technologies, such as screen readers, smart glasses, voice-based applications, or others? How was your experience? Do you think they are sufficient? What did you like or dislike about them?	**Answer:** Yes, I have tried digital magnifiers and similar tools. They are helpful, especially when they allow me to enlarge the text or invert the colors to white on black, which makes reading much easier for me. However, they are sometimes bulky or not convenient to carry around, and they don't always work in every situation. They are useful, but I feel they are not enough on their own for everyday life.

	Observations: The interviewee benefits from digital magnifiers but finds them bulky and insufficient, highlighting a need for more portable solutions.
Q3: If you encounter difficulty in a situation, whom do you usually turn to? Or how do you usually handle such cases?	**Answer:** Most of the time, I try to handle it on my own, even if it takes extra effort. I prefer being independent, but in very difficult situations, I might ask someone nearby for help Still, I like having tools that give me confidence to manage things myself. **Observations:** The interviewee values independence, seeking help only when necessary, indicating a preference for self-reliance tools.
Q4: When you need to read a street sign, a book, or a bill, what method do you usually rely on? Do you find it effective in all situations?	**Answer:** I usually take a picture with my phone and then zoom in on it until it's clear enough to read. This method works most of the time, but it can be slow and inconvenient, especially if I need information quickly or if I'm outdoors at night. **Observations:** The interviewee relies on phone zoom for reading but finds it slow and inconvenient, especially at night, showing a need for faster reading support.
Q5: If a device could tell you the identity of the person in front of you, how could that help you in your daily life?	**Answer:** That would be very helpful because sometimes I don't recognize people until they start speaking. Having the device tell me who is in front of me would save me from awkward situations and make social interactions easier. **Observations:** The interviewee has difficulty recognizing people until they speak, showing the value of face identification for smoother social interactions.
Q6: Would you prefer to use a device like "Munir" all the time, or only in specific situations? And when exactly do you feel you would need it?	**Answer:** I would probably use it mostly in specific situations, like when I'm outside, walking at night, or trying to read something important. Indoors, I may not need it as much. So, it would be situational rather than all the time.

	Observations: The interviewee would use the device mainly outdoors or at night, reflecting situational but essential needs.
Q7: What is the most suitable way for you to interact with the device (e.g., voice, vibration, touch, or others)? And why do you prefer this method?	**Answer:** I think vibration or touch would be best, because they are simple, quick, and don't disturb others around me. Voice commands can be useful too, but sometimes it's not practical to speak out loud, especially in public. **Observations:** The interviewee prefers vibration or touch for privacy and simplicity, while considering voice as a secondary option.
Q8: Would you prefer that the device interacts only with you, or that it can also alert people around you in certain cases (such as warning them if there is danger)?	**Answer:** For me, it's enough if the device interacts only with me. I don't feel it's necessary to alert others, as long as I am aware of what's happening and can respond to it myself. **Observations:** The interviewee prefers the device to interact only with him, emphasizing personal and private use.
Q9: Do you think having a companion mobile app for the device (like the "Munir" app) would be useful for you? What features would you like it to support?	**Answer:** Yes, I think having an app would be very useful. It could provide extra settings like night mode, text magnification, and color adjustments. The app could also make it easier to customize the device to fit different situations. **Observations:** The interviewee sees a companion app as useful for customization, night mode, magnification, and color adjustments.
Q10: If you could design or develop a wearable device to be used daily, what are the three most important features you would like it to have? Do you have any suggestions to make it more useful or easier to use?	**Answer:** First, it should be lightweight and comfortable to wear all day. Second, it needs to look normal and not attract too much attention, so it doesn't feel embarrassing in public. Third, it should have strong features like magnification, brightness adjustment, and color contrast. These three would make it very practical for daily use. **Observations:** The interviewee prioritizes comfort, discreet design, and strong features like magnification and contrast, reflecting a need for practical and socially acceptable wearables.

Table 30:Interview #4 Overview

Interview #4	
Interviewee: Alzahra	**Interviewer:** Fajer Alamro
Location / Medium: Zoom meeting - online	**Appointment Date:** 17 Aug 2025 **Start Time:** 5:00 pm **End Time:** 5:41 pm
Objectives: 1. Identify main challenges visually impaired users face and how Munir could help. 2. Understand preferred features and interaction methods (voice, color recognition, navigation). 3. Gauge interest in participation and feedback for improving the device.	**Reminders**: The interviewee emphasizes independence and prioritizes features like color recognition, obstacle alerts, and face identification, supported by discreet vibration feedback and a companion app for daily usability.
Agenda:	**Approximate Time:**
- Introduction	3 min
- Background in project	2 min
- Overview of interview	1 min
- Topic to be covered	1 min
- Permission to record	1 min
- Question1	4 min
- Question2	4 min
- Question3	3 min
- Question4	3 min
- Question5	3 min
- Question6	3 min
- Question7	3 min
- Question8	3 min
- Question9	3 min
- Question10	3 min
- Summary of major points	2 min
- Questions from interviewee	1 min
- Closing	1 min

Table 31:Interview #4 Transcript

<table>
<tr><th colspan="2">Interview #4</th></tr>
<tr><td colspan="2">General Observations:
She expressed strong interest in technologies that improve independence in daily activities such as reading, mobility, and clothing selection.
She emphasized reading books fully, as current audio apps only provide summaries.
She placed high importance on vibration alerts for navigation and safety.</td></tr>
<tr><td colspan="2">Topic not covered:
No discussion on device cost or affordability.
Battery performance and durability were not explored.
Privacy and data security concerns were not addressed in this session.</td></tr>
<tr><td>Interviewee: Alzahra</td><td>Date: 17 Aug 2025</td></tr>
<tr><td>Questions:</td><td>Answers and notes:</td></tr>
<tr><td>Q1: What are the main challenges you face in your daily life due to visual impairment or loss of sight, and wish there was a technology that could help you with them?</td><td>Answer: One of my biggest challenges is entering new or unfamiliar places, since I don’t know what obstacles, I might face. Choosing clothes, especially identifying colors, is also difficult for me. If there were a technology that could alert me to barriers, help me avoid accidents in public spaces, and assist me in selecting the right clothing colors, it would make my life much easier.
Observations: The interviewee struggles with navigating unfamiliar places and choosing clothing colors, showing the need for obstacle alerts and color recognition.</td></tr>
<tr><td></td><td>Answer: Yes, I’ve used screen readers and audio book apps. They are very helpful because they allow me to access information without depending on others. However, I face problems with pronunciation, sometimes the reader mispronounces words even when I know they are written correctly. This makes me feel I cannot rely on them completely. Also, audio book apps</td></tr>
</table>

Q2: Have you ever used any assistive technologies, such as screen readers, smart glasses, voice-based applications, or others? How was your experience? Do you think they are sufficient? What did you like or dislike about them?	often give only summaries, not the full text, which limits my learning experience. **Observations**: The interviewee benefits from screen readers and audiobooks but is frustrated by mispronunciations and limited content, indicating a need for more reliable tools.
Q3: If you encounter difficulty in a situation, whom do you usually turn to? Or how do you usually handle such cases?	**Answer:** Honestly, I prefer not to depend on others. I like to manage situations on my own as much as possible. If it's something urgent and I can't handle it, then I might call a relative for help, but in general I try to be very independent. **Observations:** The interviewee prefers handling situations independently, seeking help only in emergencies, reflecting a desire for greater self-reliance.
Q4: When you need to read a street sign, a book, or a bill, what method do you usually rely on? Do you find it effective in all situations?	**Answer:** Usually, I take a picture with my phone and use an app that reads the text from the image. This works most of the time. If the sign is big and the text is clear, I sometimes try to read it myself. However, if the sign is far away, it becomes difficult. That's why I think smart glasses with a built-in camera and voice feedback could be very useful. **Observations:** The interviewee relies on phone apps to read text but struggles with distant signs, highlighting the need for wearable reading assistance.
	Answer: That would be useful, especially in public or social settings outside of family. With my relatives, I can usually recognize them by voice or even by their perfume. But in public spaces like universities, markets, or events, I

Q5: If a device could tell you the identity of the person in front of you, how could that help you in your daily life?	often don't know who is in front of me unless they speak. If the device could tell me who the person is, it would help me interact more confidently. **Observations:** The interviewee has difficulty recognizing people in public spaces, showing the value of face recognition for confident interactions.
Q6: Would you prefer to use a device like "Munir" all the time, or only in specific situations? And when exactly do you feel you would need it?	**Answer:** I would use it mainly in specific situations. At home, I don't need it much because I already know the environment. But in places like markets, the university, or when walking in public, it would be very helpful. I would also use it for reading full books, especially during exams, so I wouldn't need to wait for someone else to read for me. **Observations:** The interviewee would mainly use the device in public spaces and during study, suggesting situational but critical needs.
Q7: What is the most suitable way for you to interact with the device (e.g., voice, vibration, touch, or others)? And why do you prefer this method?	**Answer:** I prefer vibration alerts, because they are more private and I notice them immediately. Voice can be helpful too, but sometimes it's not convenient in public. Ideally, the device could combine both, but vibration is my top choice. **Observations:** The interviewee prefers vibration alerts for privacy, indicating the importance of discreet feedback methods.
	Answer: I would prefer both. For example, recently I was almost hit by people riding scooters in a crowded place, and I wished there had been a warning system. In such dangerous

Q8: Would you prefer that the device interacts only with you, or that it can also alert people around you in certain cases (such as warning them if there is danger)?	situations, it would be better if the device could alert me and also alert people around me, especially if they face the same risks. **Observations:** The interviewee wants alerts for herself and others in risky situations, highlighting the need for shared safety features.
Q9: Do you think having a companion mobile app for the device (like the "Munir" app) would be useful for you? What features would you like it to support?	**Answer:** Yes, absolutely. The app could help with checking my money balance when shopping, reading books, and selecting clothing colors. Sometimes it's easier to use the phone directly instead of wearing the glasses all the time, so having both options would be very convenient. **Observations:** The interviewee finds a companion app useful for money, books, and colors, suggesting the importance of flexible device integration.
Q10: If you could design or develop a wearable device to be used daily, what are the three most important features you would like it to have? Do you have any suggestions to make it more useful or easier to use?	**Answer:** First, the ability to read full books, not just summaries. Second, color recognition to help me choose clothes and shoes correctly. Third, navigation alerts to warn me of obstacles or dangerous objects ahead, especially in public spaces. These features would make it very practical and supportive for daily life. **Observations:** The interviewee prioritizes full book reading, color recognition, and obstacle alerts, reflecting her top functional needs.

Table 32:Interview #5 Overview

<table>
<tr><th colspan="2">Interview #5</th></tr>
<tr><td>Interviewee: Renad</td><td>Interviewer: Mashael Alammar</td></tr>
<tr><td>Location / Medium: Zoom meeting - online</td><td>Appointment Date: 18 Aug 2025
Start Time: 6:48 pm
End Time: 7:24 pm</td></tr>
<tr><td>Objectives:
1. Identify main challenges visually impaired users face and how Munir could help.
2. Understand preferred features and interaction methods (voice, color recognition, navigation).
3. Gauge interest in participation and feedback for improving the device.</td><td>Reminders: Prioritizes independence from companions, requires socially acceptable lightweight design for daily public use, strongly prefers voice feedback, wants customizable app features, values person recognition for social situations, eager to participate in development. Currently relies on family/video calls and screen readers (limited outdoor utility).</td></tr>
<tr><td>Agenda:</td><td>Approximate Time:</td></tr>
<tr><td>- Introduction</td><td>2 min</td></tr>
<tr><td>- Background in project</td><td>2 min</td></tr>
<tr><td>- Overview of interview</td><td>1 min</td></tr>
<tr><td>- Topic to be covered</td><td>1 min</td></tr>
<tr><td>- Permission to record</td><td>1 min</td></tr>
<tr><td>- Question1</td><td>3 min</td></tr>
<tr><td>- Question2</td><td>2 min</td></tr>
<tr><td>- Question3</td><td>3 min</td></tr>
<tr><td>- Question4</td><td>4 min</td></tr>
<tr><td>- Question5</td><td>2 min</td></tr>
<tr><td>- Question6</td><td>1 min</td></tr>
<tr><td>- Question7</td><td>3 min</td></tr>
<tr><td>- Question8</td><td>1 min</td></tr>
<tr><td>- Question9</td><td>2 min</td></tr>
<tr><td>- Question10</td><td>3 min</td></tr>
<tr><td>- Summary of major points</td><td>2 min</td></tr>
<tr><td>- Questions from interviewee</td><td>1 min</td></tr>
<tr><td>- Closing</td><td>1 min</td></tr>
</table>

Table 33: Interview #5 Transcript

<table>
<tr><th colspan="2">Interview #5</th></tr>
<tr><td colspan="2">General Observations:
She stressed the importance of independence and avoiding constant reliance on family or friends.
She showed strong interest in technologies that could replace the need for a human guide in public spaces.
She placed high importance on appearance and comfort of the glasses, saying users would prefer a normal, lightweight design over bulky or “strange looking” ones.</td></tr>
<tr><td colspan="2">Topic not covered:
No discussion on device cost or affordability.
Battery performance and durability were not explored.
Technical specifications (connectivity, storage, compatibility) were not explored.</td></tr>
<tr><td>Interviewee: Renad</td><td>Date: 18 Aug 2025</td></tr>
<tr><td>Questions:</td><td>Answers and notes:</td></tr>
<tr><td>Q1: What are the main challenges you face in your daily life due to visual impairment or loss of sight, and wish there was a technology that could help you with them?</td><td>Answer: I would expect them to replace the need for a human companion. For example, when I go to public places, I usually rely on video calls with family or use apps like Be My Eyes. But these require strangers to help, which can feel uncomfortable. Glasses that guide me directly would give me more independence and reduce the need to always have someone with me.
Observations: The interviewee faces emotional challenges (discomfort, embarrassment) when relying on strangers for assistance in public spaces, demonstrating the impact of dependency and the critical need for autonomous assistive technology.</td></tr>
<tr><td>Q2: Have you ever used any assistive technologies, such as screen readers, smart glasses, voice-based applications, or others? How was your experience? Do you think they are sufficient? What did you like or dislike about them?</td><td>Answer: Yes, I use screen readers like VoiceOver on my iPhone and Apple Watch, and I enable it on all my devices. They are very helpful, but only in certain situations. For example, they don’t help me much when I’m outside, because I still need someone physically with me. So they are useful, but limited.</td></tr>
</table>

	Observations: The interviewee experiences a clear limitation of current screen reader technology while useful for device interaction, they provide no assistance for outdoor navigation or physical mobility, highlighting the need for environmental awareness features
Q3: If you encounter difficulty in a situation, whom do you usually turn to? Or how do you usually handle such cases?	**Answer:** Usually, I call someone right away. For example, in the gym, I might need to know which exercise machine is free or where it's located. I send a picture or video to family groups or friends on Snapchat. But this can be embarrassing, since strangers around me might notice and misunderstand what I'm doing. So, it's not always comfortable. **Observations:** The interviewee's reliance on real-time photo/video sharing with family for simple tasks causes social awkwardness and embarrassment, indicating need for private, immediate, independent assistance technology.
Q4: When you need to read a street sign, a book, or a bill, what method do you usually rely on? Do you find it effective in all situations?	**Answer:** I usually ask someone for help, especially if I'm outside. At work, if I receive a document, I use apps like Envision AI to take a picture and it reads the text to me. More recently, I've started using ChatGPT as well by attaching an image, it reads the text right away. These tools are very useful, but they still require me to take out my phone and capture an image. **Observations:** The interviewee relies on apps and human help for reading text but finds taking photos with phone cumbersome, indicating need for seamless wearable solution.

Q5: If a device could tell you the identity of the person in front of you, how could that help you in your daily life?	**Answer:** That would be useful in some cases. For example, I could identify family or friends in a room without having to ask "Who is this?" out loud, which can be awkward. It would also help me avoid interacting with someone I don't want to talk to at the moment. So yes, it would save me from embarrassment and help me choose how to engage with people. **Observations:** The interviewee seeks to prevent awkward moments and manage social interactions independently, indicating need for private person identification technology.
Q6: Would you prefer to use a device like "Munir" all the time, or only in specific situations? And when exactly do you feel you would need it?	**Answer:** I would definitely use them daily, as long as the design looks normal and not strange. If they look like regular sunglasses or medical glasses, I wouldn't mind wearing them all the time. I would especially use them in public places like malls, restaurants, or when going out with friends. At home, I wouldn't need them much, except maybe for tasks like cooking or identifying colors. **Observations:** The interviewee requires normal-looking design for daily public use showing appearance as key factor for consistent adoption.
Q7: What is the most suitable way for you to interact with the device (e.g., voice, vibration, touch, or others)? And why do you prefer this method?	**Answer:** Voice feedback is the best option for me. It's clear, direct, and easy to understand. Vibrations could be useful in advanced versions, but they are more limited compared to voice. If possible, the device should allow users to choose between voice and vibration. But honestly, voice is the most effective method. **Observations:** The interviewee strongly prefers voice feedback as most effective but values having

	customizable options between voice and vibration, indicating need for user-controlled interaction preferences.
Q8: Would you prefer that the device interacts only with you, or that it can also alert people around you in certain cases (such as warning them if there is danger)?	**Answer:** I would prefer it to alert only me in normal situations. But in emergencies, it could notify specific trusted contacts through the mobile app, like family members. I wouldn't want the device to make loud noises or draw public attention, but sending notifications privately would be very helpful and safer. **Observations:** The interviewee wants personal alerts only, with private emergency notifications to family through app, indicating need for discrete communication avoiding public attention.
Q9: Do you think having a companion mobile app for the device (like the "Munir" app) would be useful for you? What features would you like it to support?	**Answer:** Yes, a mobile app would be very useful. It should allow me to customize the features, for example, turning off alerts I don't want, or enabling extra options for people who need them. The app would make it easier for users to control and personalize the glasses. The more options you add, the more valuable the app becomes. **Observations:** The interviewee values mobile app for customization and control of features, showing need for personalization options to enable/disable alerts based on user preferences.
Q10: If you could design or develop a wearable device to be used daily, what are the three most important features you would like it to have? Do you have any suggestions to make it more useful or easier to use?	**Answer:** First, it must be lightweight—if the glasses are heavy, they will be uncomfortable. Second, they should look normal and socially acceptable, like regular medical glasses rather than bulky VR headsets. Third, the design should be practical for everyday use. Aesthetic appearance is very important, because I don't want people to think I look strange when wearing them. **Observations:** The interviewee emphasizes lightweight comfort, socially-acceptable appearance, and practical design, indicating social judgment concerns outweigh feature priorities

9.2 Appendix B: Questionnaires

This Appendix will represent all the questionnaire-related supplements, including the questionnaire questions, and a representation of the responses we obtained through charts.

9.2.1 Requirements Gathering Questionnaire

A) Questionnaire Questions

Q1: Participant’s age group:

•Under 18 •18-30 • 31-60 •+60

Q2: Participant’s gender:

•Femail •Male

Q 3: What type(s) of visual impairment do you experience? ***(Select all that apply)***

•Total blindness • Low vision • other

Q4: Have you previously used any assistive technologies or tools designed to support visual impairments?

•Yes •No

Q5: Which of the following activities do you find challenging due to your visual impairment? ***(Select all that apply)***

• Distinguishing between colors • Reading text (e.g., books, labels, signs)

• Recognizing faces or individuals •Requesting assistance during emergencies

• Remembering tasks or responding to reminders •Identifying currency or reading prices

• Other

Q6: How often do you experience difficulties completing daily tasks due to your vision? ***(1 = Never, 5 = Always)***

•1 •2 •3 •4 •5

Q7: To what extent do visual challenges impact your independence or quality of life? ***(1 = Not at all, 5 = Extremely)***

•1 •2 •3 •4 •5

Q8: When encountering visual difficulties, how do you usually respond?

• I rely on assistive technology • I seek help from others

•I avoid the task or situation I • Other

Q9: Would you be open to using smart glasses integrated with a mobile application and accompanied by headphones, designed to assist individuals with visual impairments?

• Yes, definitely •Possibly • I'm not sure •No

Q10: How helpful do you think the following features would be if offered through Munir Glasses? ***Rate each feature from 1 (Not helpful at all) to 5 (Extremely helpful)***

• Reading printed and digital text aloud • Receiving reminders and task notifications

•Distinguishing between colors or currencies • Identifying people

• Receiving reminders and task notifications

Q11: If Munir Glasses included the features you need, how likely are you to use them in your daily life? ***(1 = Very Unlikely, 5 = Very Likely)***

•1 •2 •3 •4 •5

Q12: Do you believe a mobile app connected to the glasses would improve your ability to manage daily tasks? ***(1 = Strongly Agree, 5 = Strongly Disagree)***

•1 •2 •3 •4 •5

Q13: What feature or functionality do you wish to have on Munir glasses? (Open Question)

Q14: Do you have any additional suggestions? (Open Question)

B) Questionnaire answers

Q1:

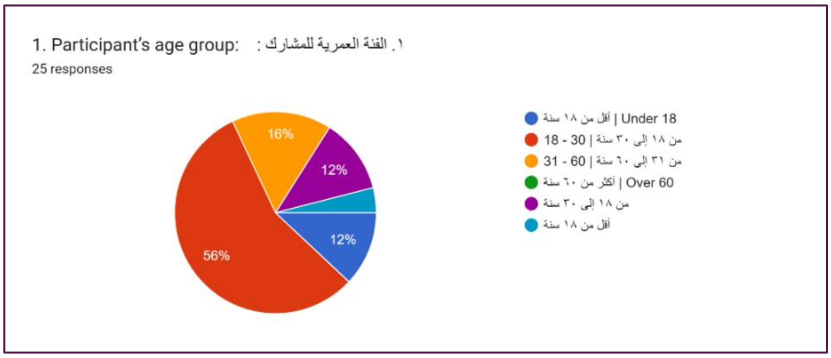


Figure 133: First question in questionnaire

Q2:

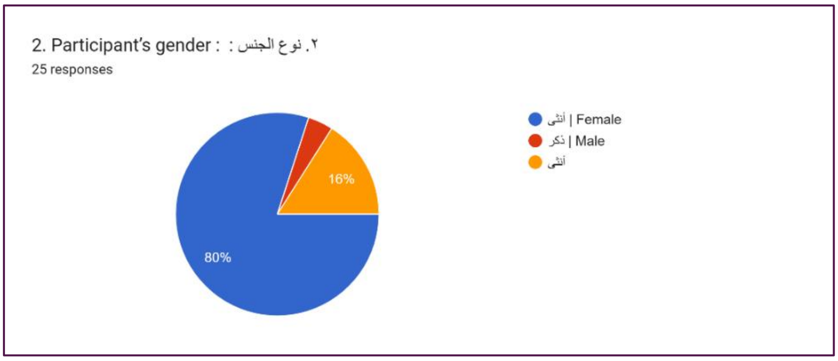


Figure 134: Second question in questionnaire

Q3:

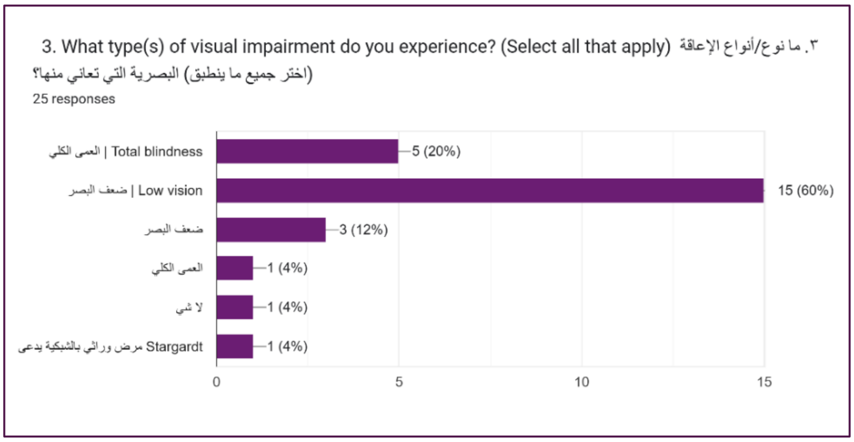


Figure 135: Third question in questionnaire

Q4:

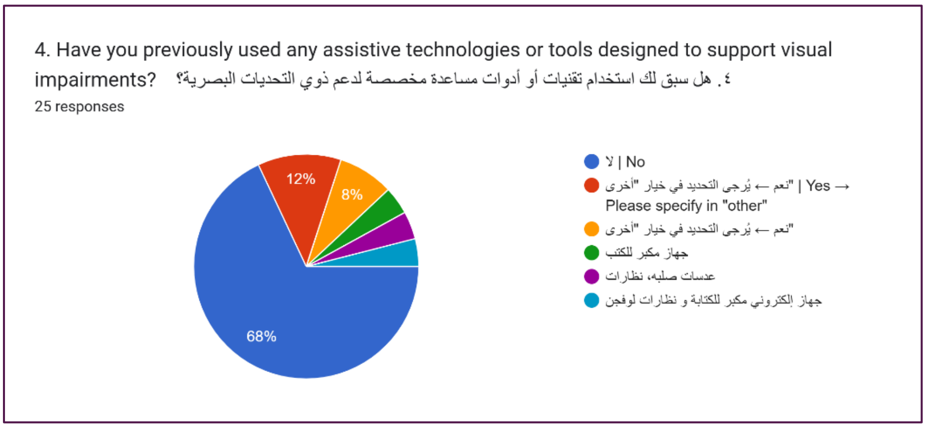


Figure 136: Fourth question in questionnaire

Q5:

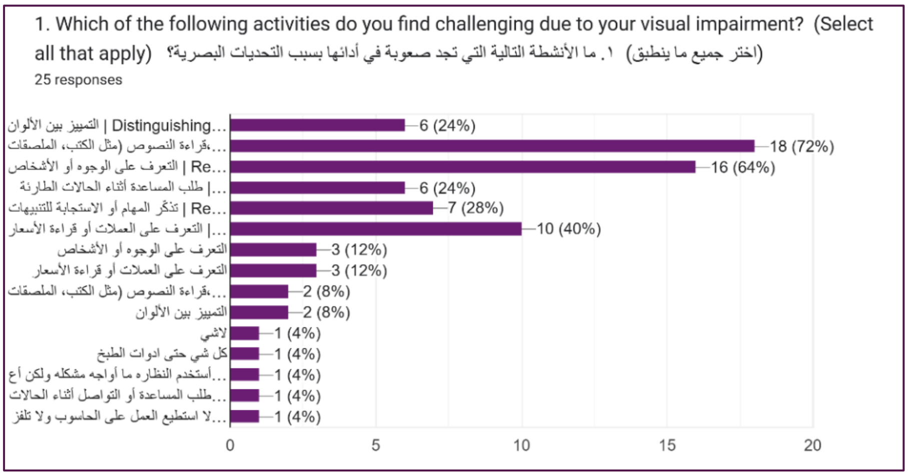


Figure 137: Fifth question in questionnaire

Q6:

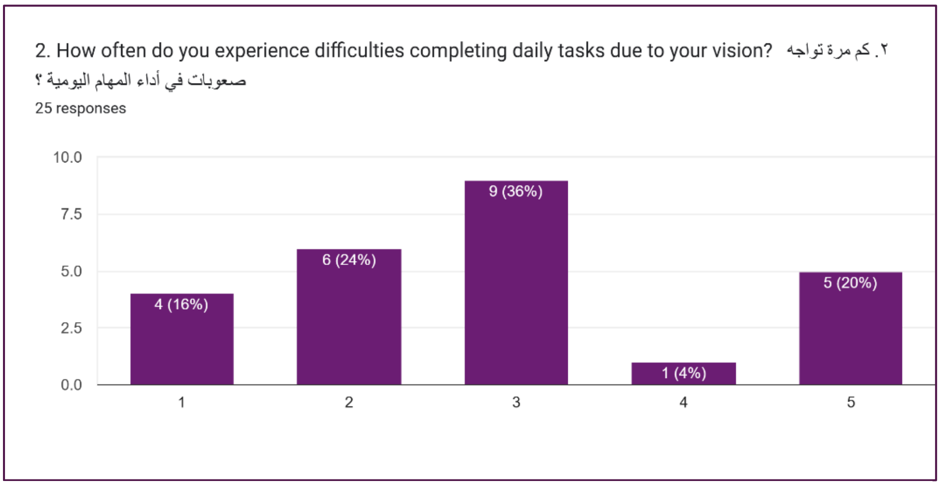


Figure 138: Sixth question in questionnaire

Q7:

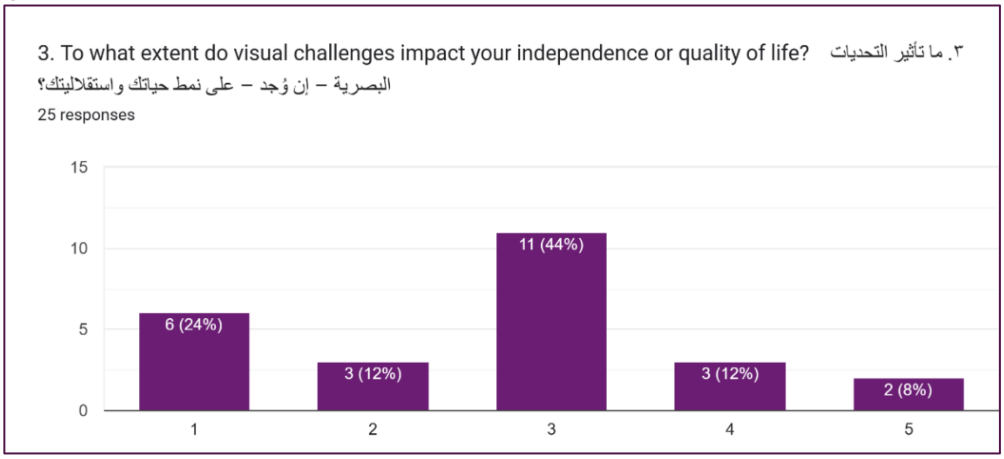


Figure 139: Seventh question in questionnaire

Q8:

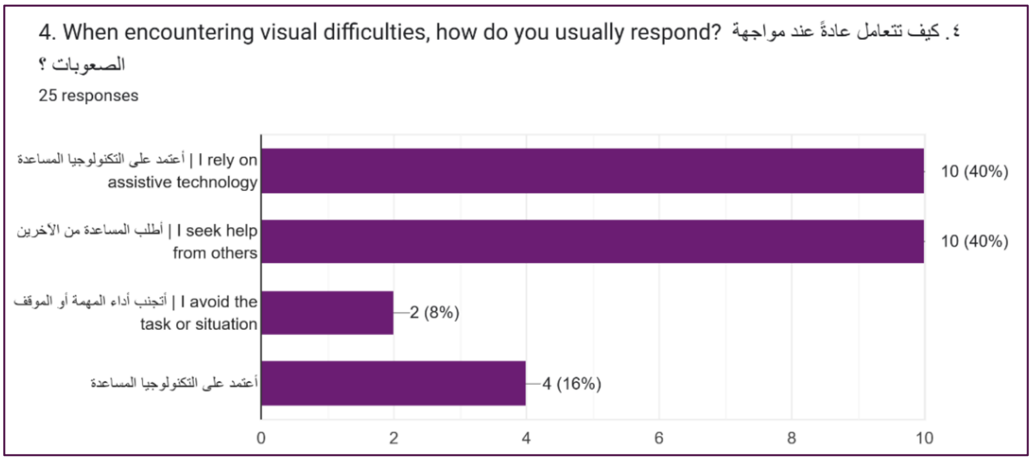


Figure 140: Eight question in questionnaire

Q9:

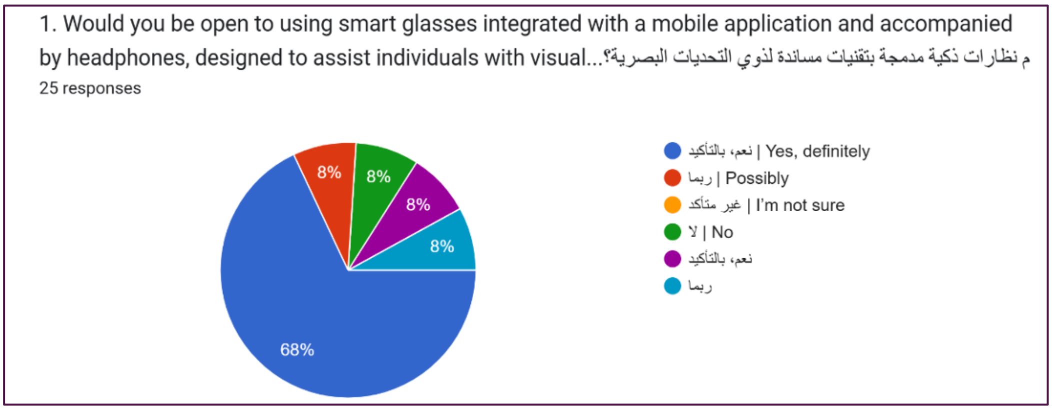


Figure 141: Ninth question in questionnaire

Q10:

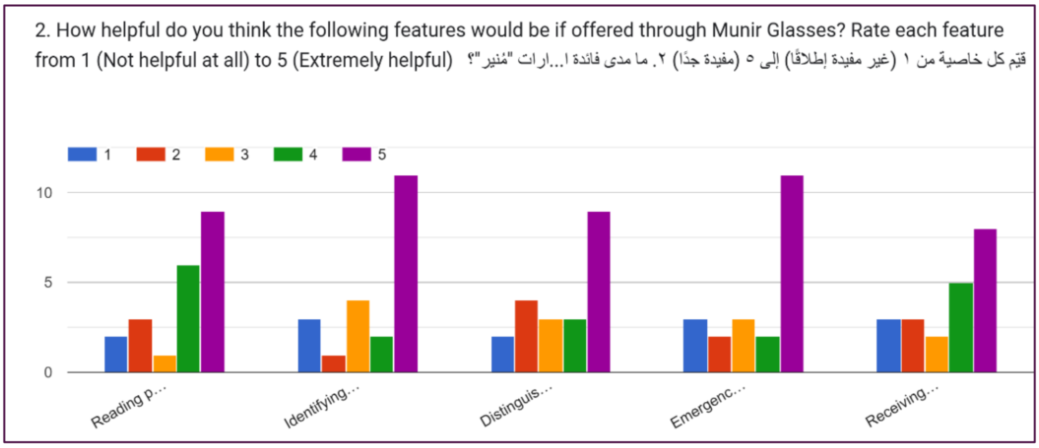


Figure 142: Tenth question in questionnaire

Q11:

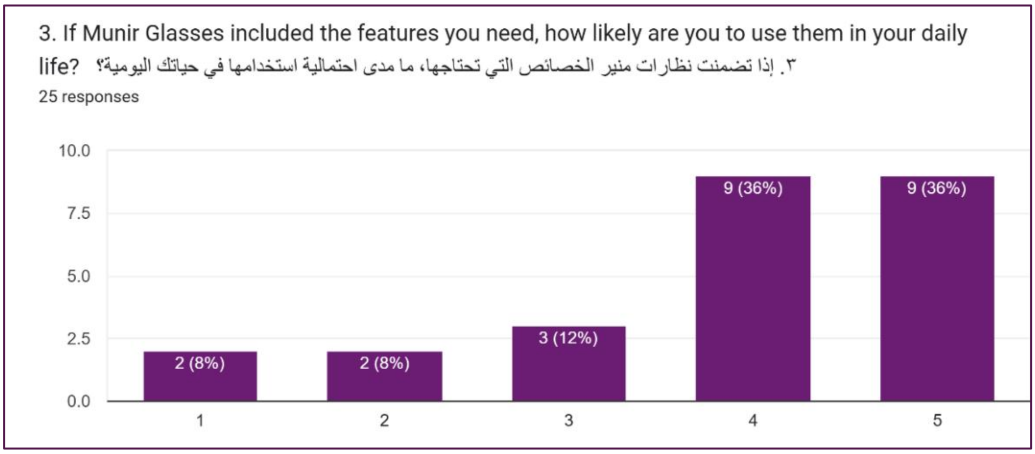


Figure 143: Eleventh question in questionnaire

Q12:

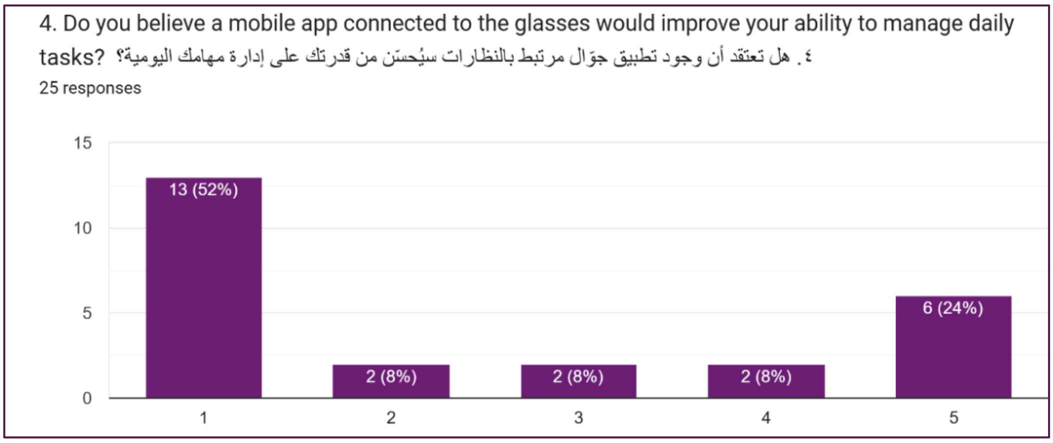


Figure 144: Twelfth question in questionnaire

Q13:

1. What feature or functionality do you wish to have on Munir glasses?

١. ما الخاصية أو الوظيفة التي تود وجودها في نظارات "منير"؟

15 responses

تحسين الجوده عند الاتجاهات

اريد جهاز مكبر للكتابه ما اريد نظارات

لا يوجد

لاشيء

توصح قراءة النصوص

Health Sensors (مستشعرات الصحة): قياس معدل ضربات القلب، متابعة النشاط مثل الخطوات والسعرات، مع تنبيهات عند قلة الحركة لدعم الصحة واللياقة اليومية.

لا اعلم

كامل برنامج مُنير بالتوفيق لكم

لا شيء

تعلمني كيف اتمكن من الحصول على اي شي ارغب في رؤيته

سهولة ووضوح الاستخدام لتخدم الجميع حتى وإن كانوا غير ملمين بالتقنية واستعمالاتها لتحقيق أقصى فائدة

تنبيه في حاله الاصطدام بشي وخاصية التعرف عاالالوان

التفاعل معي وتذكيري ببعض المهام وايضاً وجود منبه للصلاه

العثور بسهولة على النظارة أثناء فقدانها : من الممكن ان تكون مثل فكره Find my iphone

اتمنى ان يكون سعر اجهزة المكبرة للقراءة مناسب لشرائه

Figure 145: Thirteenth question in questionnaire

Q14:

2. Do you have any additional suggestions?

٢. هل لديك أي اقتراحات إضافية؟

14 responses

لا

توفير تسهيلات واستثناءات بالسماح للطلاب باستعمال هذه الاجهزه المساعده للبصر والتثقف بالتعامل مع الحالات الانسانيه

فكرة رائعة جدًا تخدم فئة كبيرة وتسهل حياتهم، تمنياتي لكم بالتوفيق والسداد وأسال الله أن يبارك بوقتكم وجهدكم ونرى الفكرة على أرض الواقع.

ان تكون مضيئه ويكون فيها جهاز تتبع من قبل الأهل وان يكون فيها كاميرا

ربط التطبيق بملف صحي فيما يتعلق بضعف البصر: واعطاء مثلا تنبيهات بان يجب زيارة الطبيب

Instant Translation (الترجمة الفورية):
ترجمة المحادثات واللافتات مباشرة إلى نص أو صوت، مما يسهل التواصل وفهم اللغات المختلفة

لا شكرًا

تكون ناطقه وواضحه

Figure 146: Fourteenth question in questionnaire

- **Link to questionnaire responses:**

https://docs.google.com/spreadsheets/d/1aPZMBNPy1TQEnmiJH2Y0A-er-ZZsXtGv3bp3AHyS_VTI/edit?usp=sharing

9.2.2 Initial Testing Questionnaire

A) Questionnaire Questions

1. I was able to sign up easily
   • Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree
2. I was able to sign in easily
   • Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree
3. I was able to log out easily
   • Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree
4. I was able to reset my password easily
   • Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree
5. I was able to delete my account easily
   • Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree
6. I was able to update my profile information easily
   • Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree
7. The audio guidance throughout the app was clear and helpful
   • Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree
8. The application's design makes it easy to find what I need
   • Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree
9. The app was easy to navigate
   • Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree
10. How would you describe the experience with the app? (the answer here should be text)

B) Link to questionnaire responses:

https://docs.google.com/spreadsheets/d/125E9Dz-9YL1nNLjvBTKWiF-Q9dDtAisaKFz2jcX3gDY/edit?usp=sharing

9.2.3 Final Testing Questionnaire

A) Questionnaire Questions

1. I was able to add, edit, delete, and search for people easily for future face recognition:

• Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree

2. The app was able to correctly recognize people who were previously added to the system and distinguish them from unknown people:

• Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree

3. The app was able to identify currency values easily and accurately:

• Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree

4. The app was able to read text accurately:

• Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree

5. The app was able to accurately recognize and identify objects:

• Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree

6. The app was able to describe the surrounding scene clearly and accurately:

• Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree

7. I was able to add, edit, delete, and search for reminders easily:

• Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree

8. I was able to filter reminders by date easily:

• Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree

9. I was able to add, edit, delete, and search for contacts easily:

• Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree

10. The voice input feature made it easier for me to add and edit information:

• Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree

11. I was able to quickly and easily request help when I needed assistance using the SOS feature:

• Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree

12. I was able to connect the smart glasses to the app easily:

• Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree

13. I was able to view smart glasses information such as the last known location and check the battery level easily:

• Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree

14. The smart glasses were able to recognize previously registered people and distinguish them from unknown people accurately:

• Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree

15. The smart glasses were able to identify currency values easily and accurately:

• Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree

16. The smart glasses were able to read text accurately:

• Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree

17. The smart glasses were able to recognize objects accurately:

• Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree

18. The smart glasses were able to describe and explain my surroundings clearly and accurately:

• Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree

19. The smart glasses allowed me to quickly and easily request help when I needed assistance using the SOS feature:

• Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree

20. I was able to access the user guide easily whenever I needed information about the smart glasses:

• Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree

21. I was able to disconnect the smart glasses easily:

• Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree

22. I was able to change the application language easily:

• Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree

23. The application's design makes it easy to find what I need:

• Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree

24. The app was easy to navigate:

• Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree

25. The audio guidance throughout the app was clear and helpful in the language you selected:

• Strongly Disagree •Disagree • Neutral •Agree •Strongly Agree

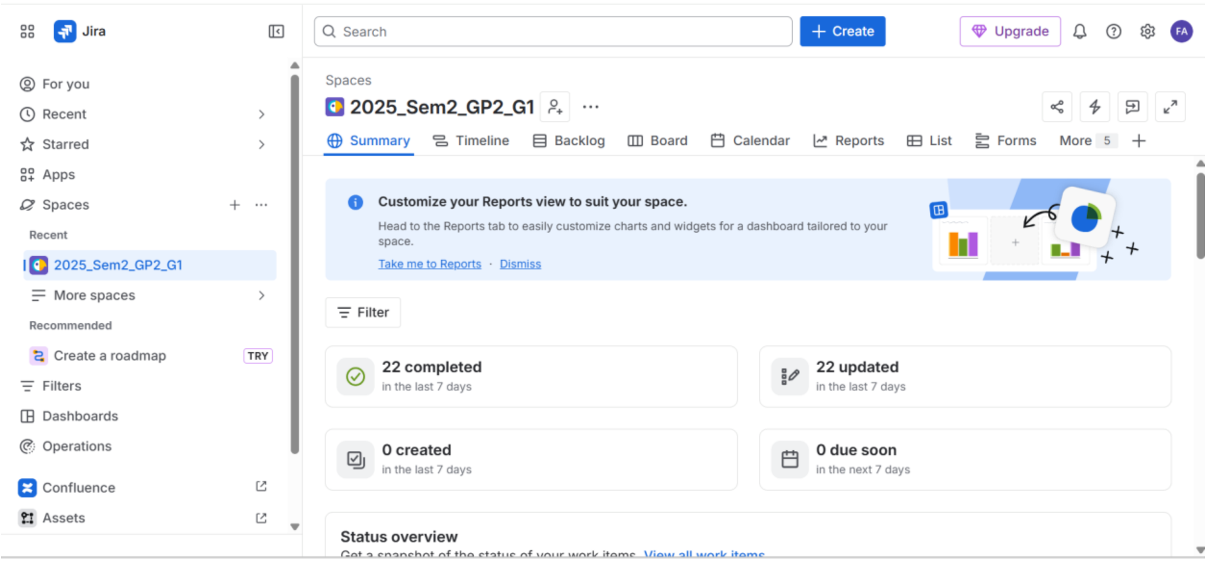
26. How would you describe your overall experience with the mobile application and the smart glasses? (the answer here should be text)

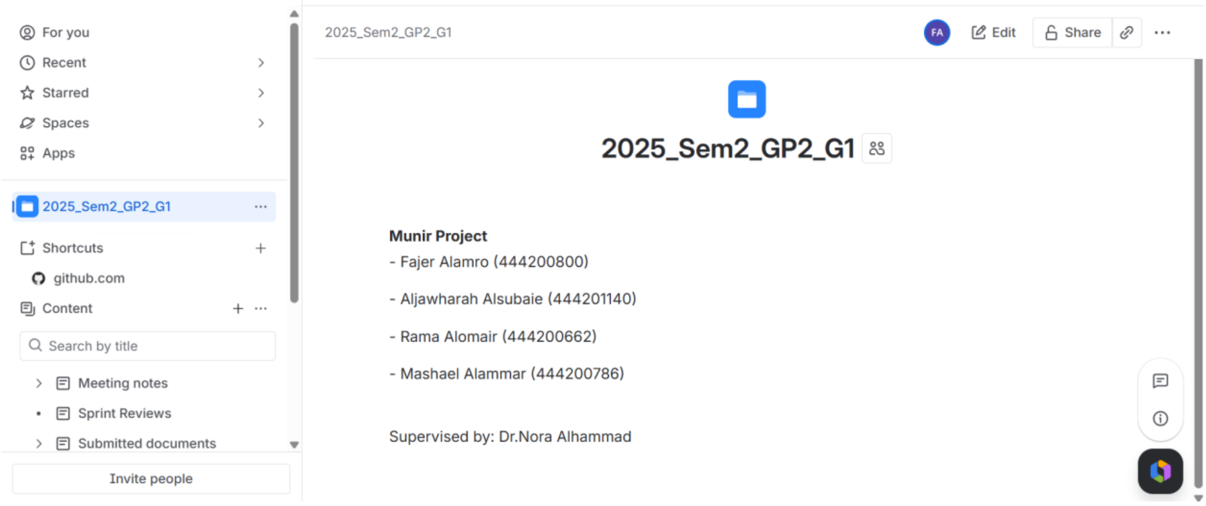
B) Link to questionnaire responses:

https://docs.google.com/spreadsheets/d/1TuRk7C6dRoIuGnm5QQTlpbl2EPyQRrMNPgpmMeYCdtk/edit?usp=sharing

9.3 Appendix C: JIRA

https://student-team-sxuclzqg.atlassian.net/jira/software/projects/GP/summary





9.4 Appendix D: GitHub

https://github.com/aljawharah-alsubaie/2025_GP2_G1